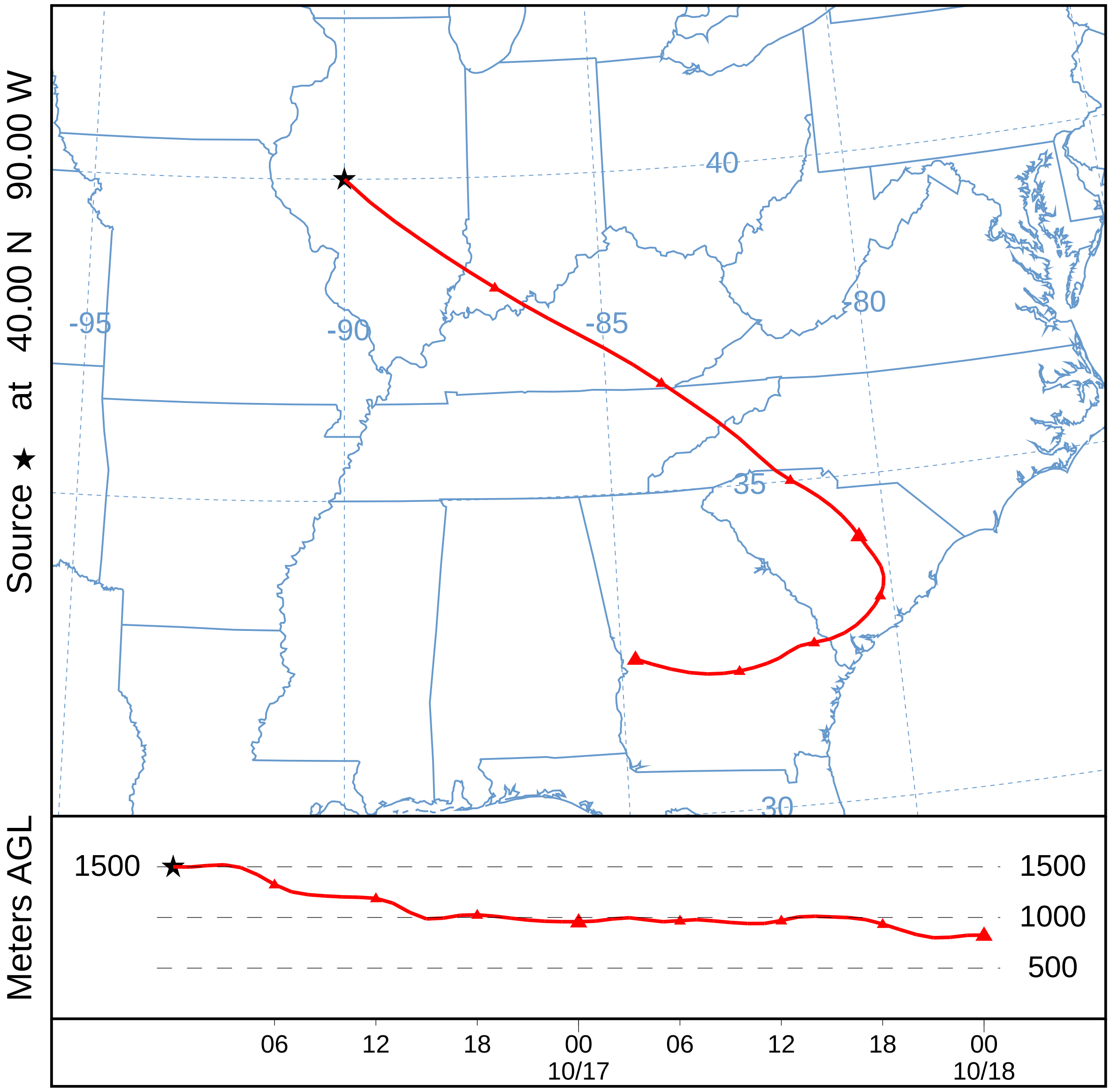
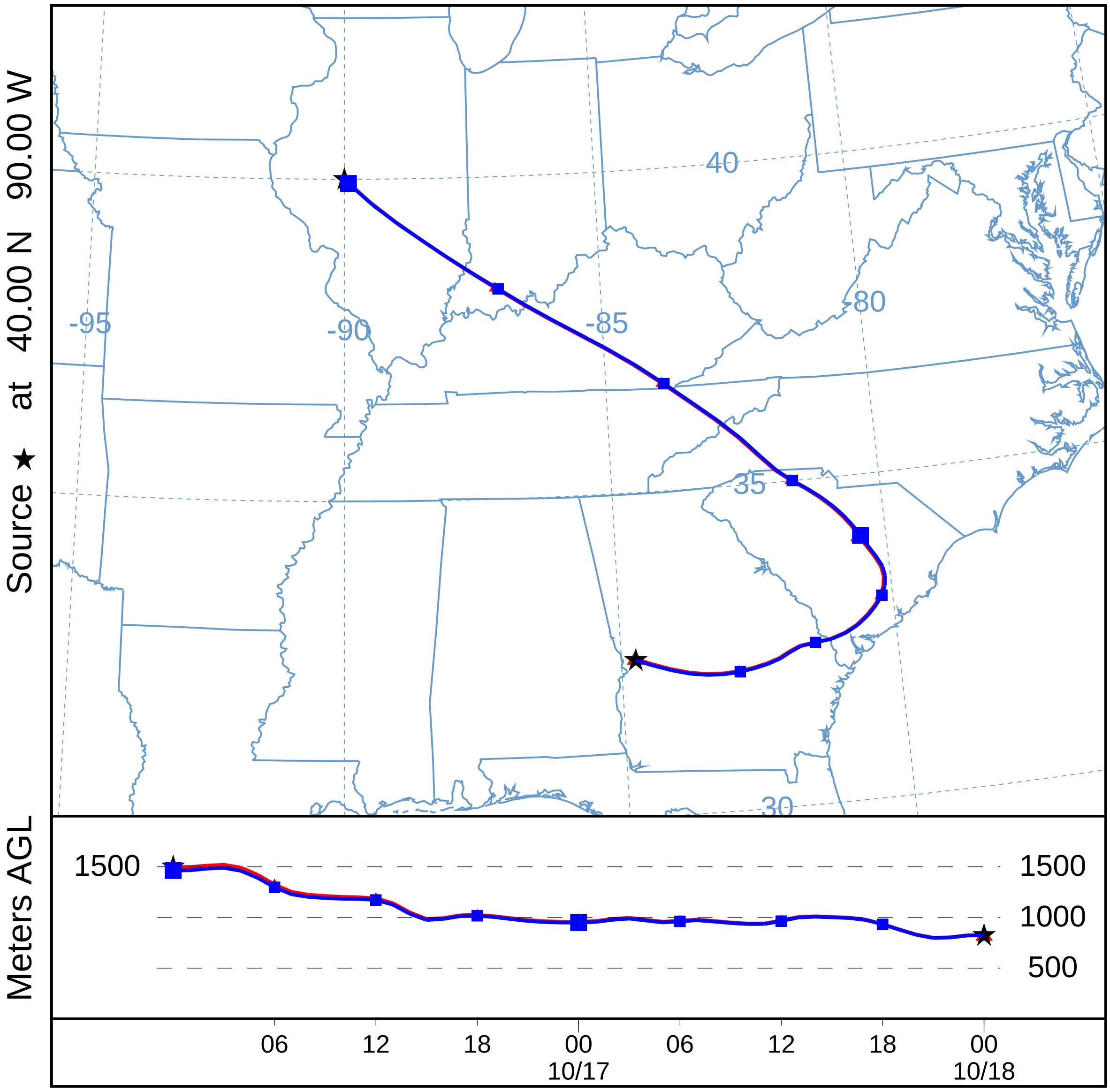
Forward Trajectory (001)
Forward trajectory starting at 0000 UTC 16 Oct 95
NGM Meteorological Data



Add Backward Trajectory (002)

Forward trajectory starting at 0000 UTC 16 Oct 95

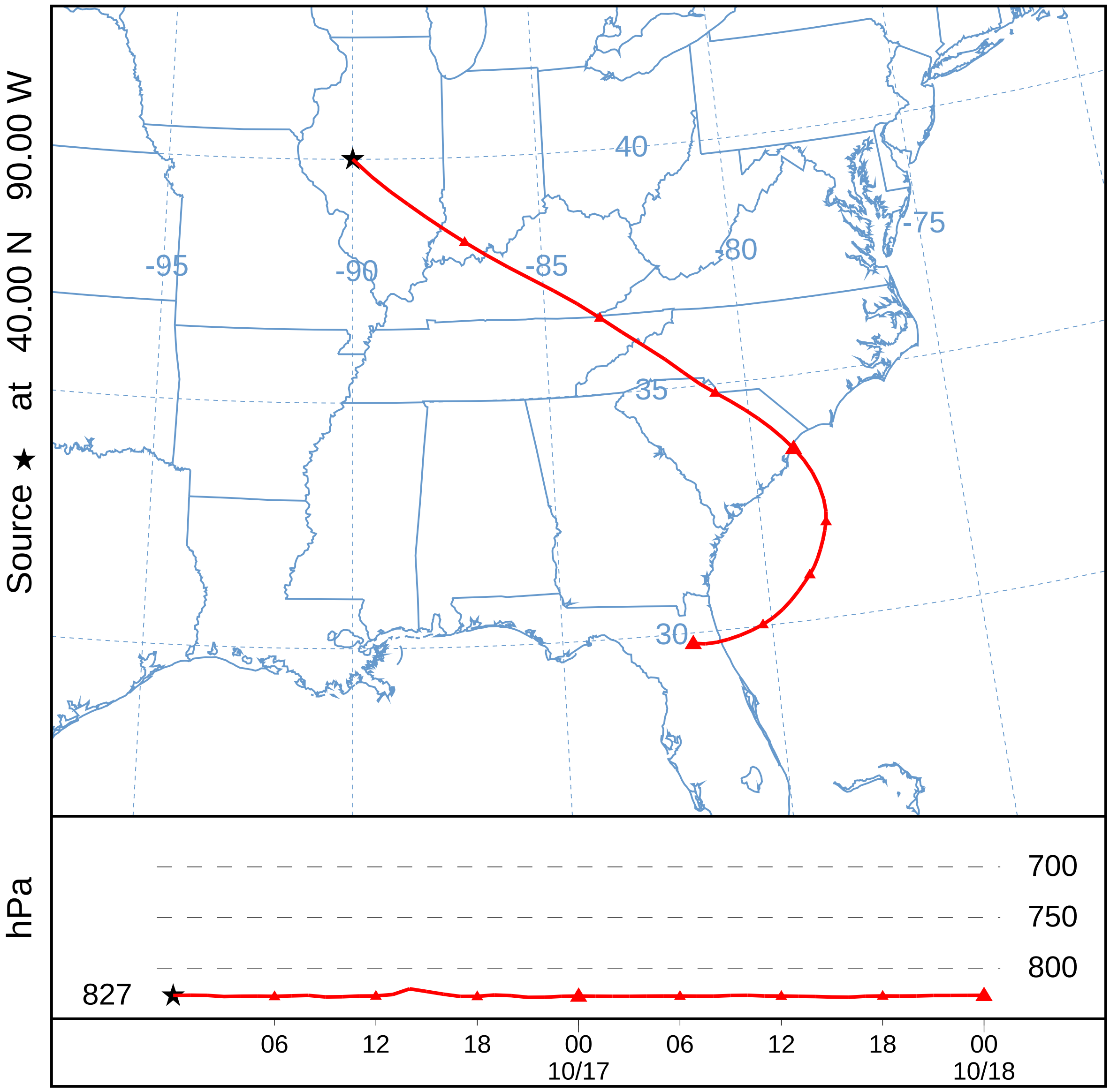
NGM Meteorological Data



Forward Isobaric Trajectory (003)

Forward trajectory starting at 0000 UTC 16 Oct 95

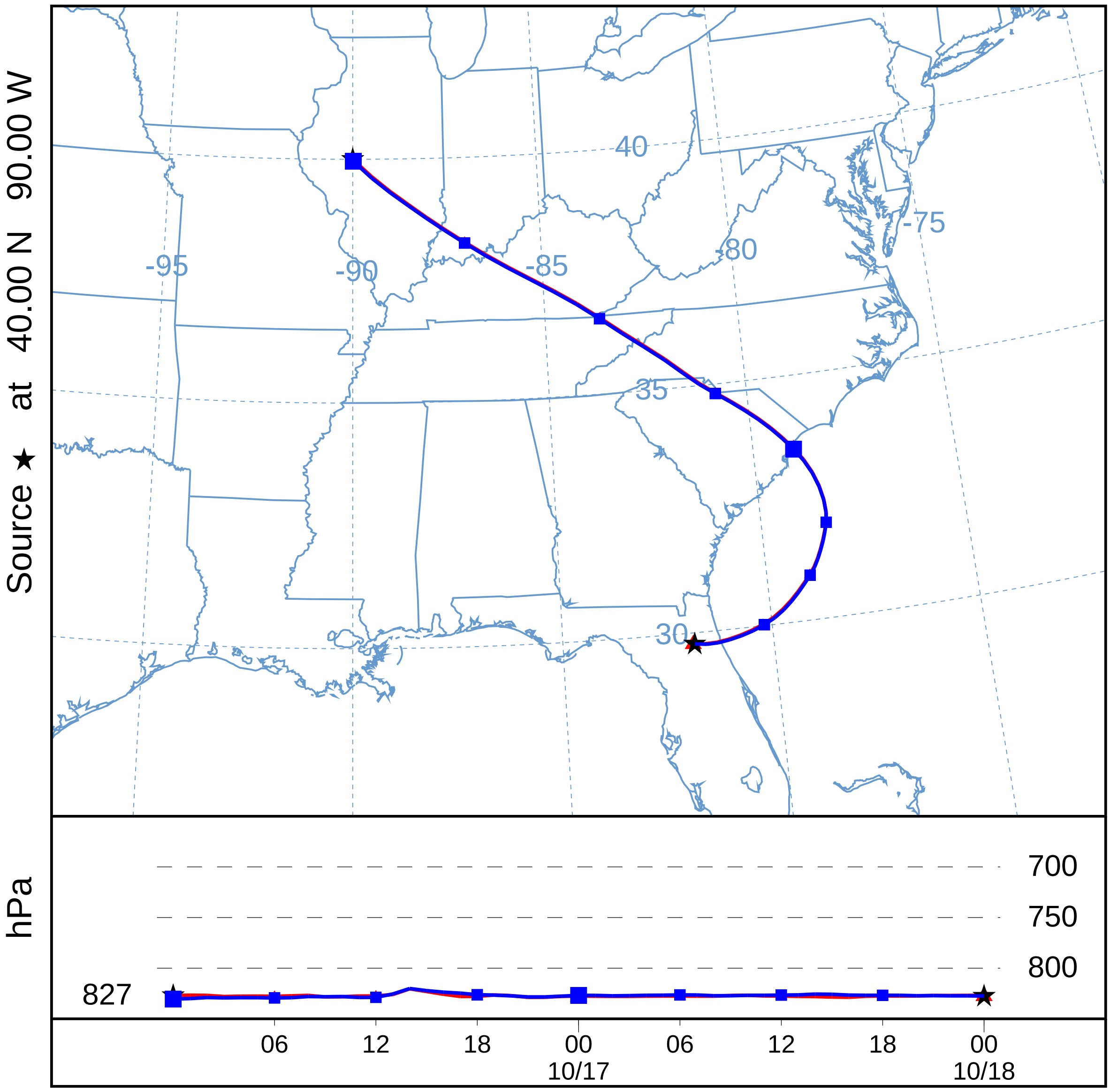
NGM Meteorological Data



Add Backward Isobaric Trajectory (004)

Forward trajectory starting at 0000 UTC 16 Oct 95

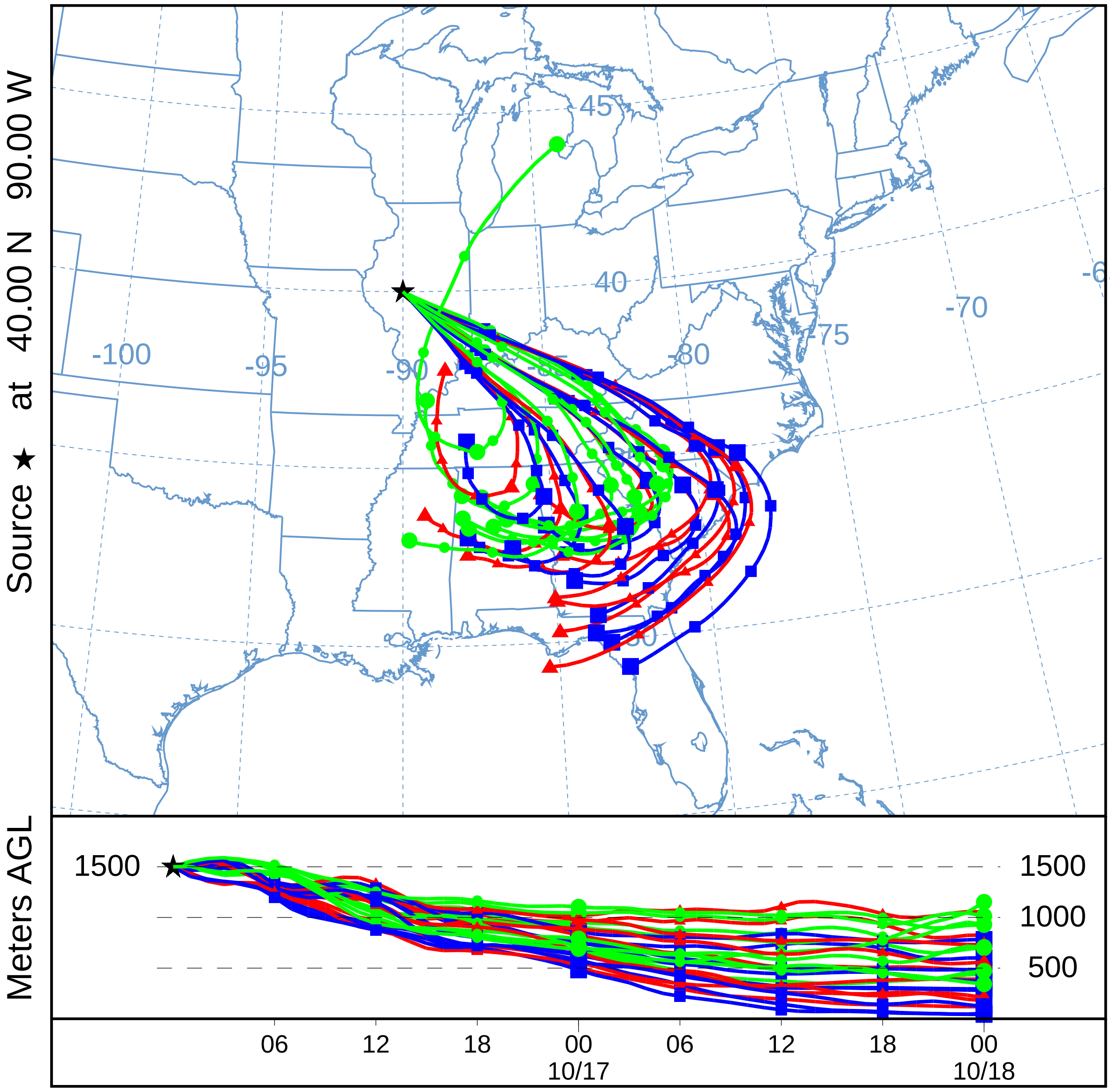
NGM Meteorological Data



Trajectory Ensemble (005)

Forward trajectories starting at 0000 UTC 16 Oct 95

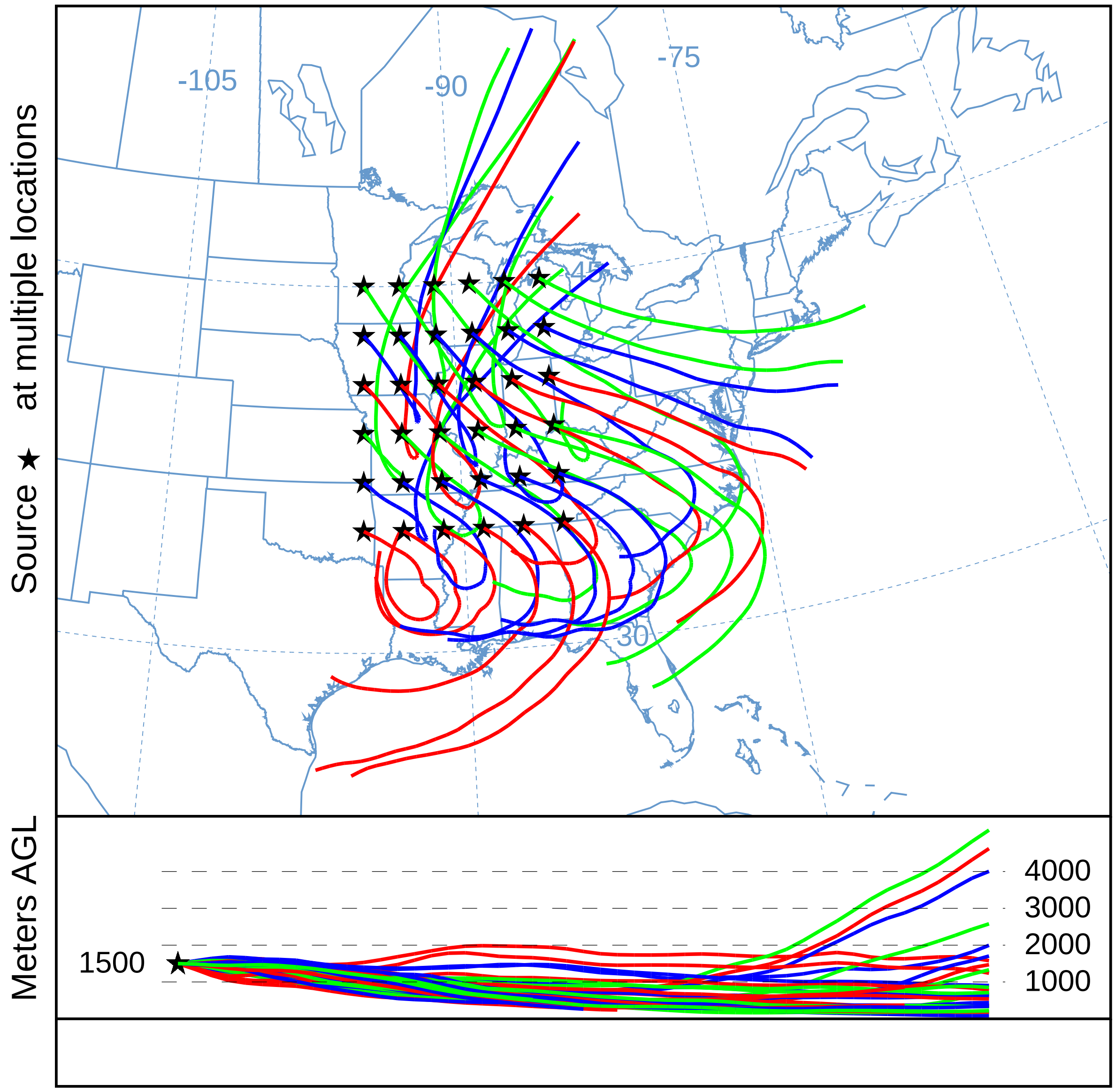
NGM Meteorological Data



Trajectory Matrix (006)

Forward trajectories starting at 0000 UTC 16 Oct 95

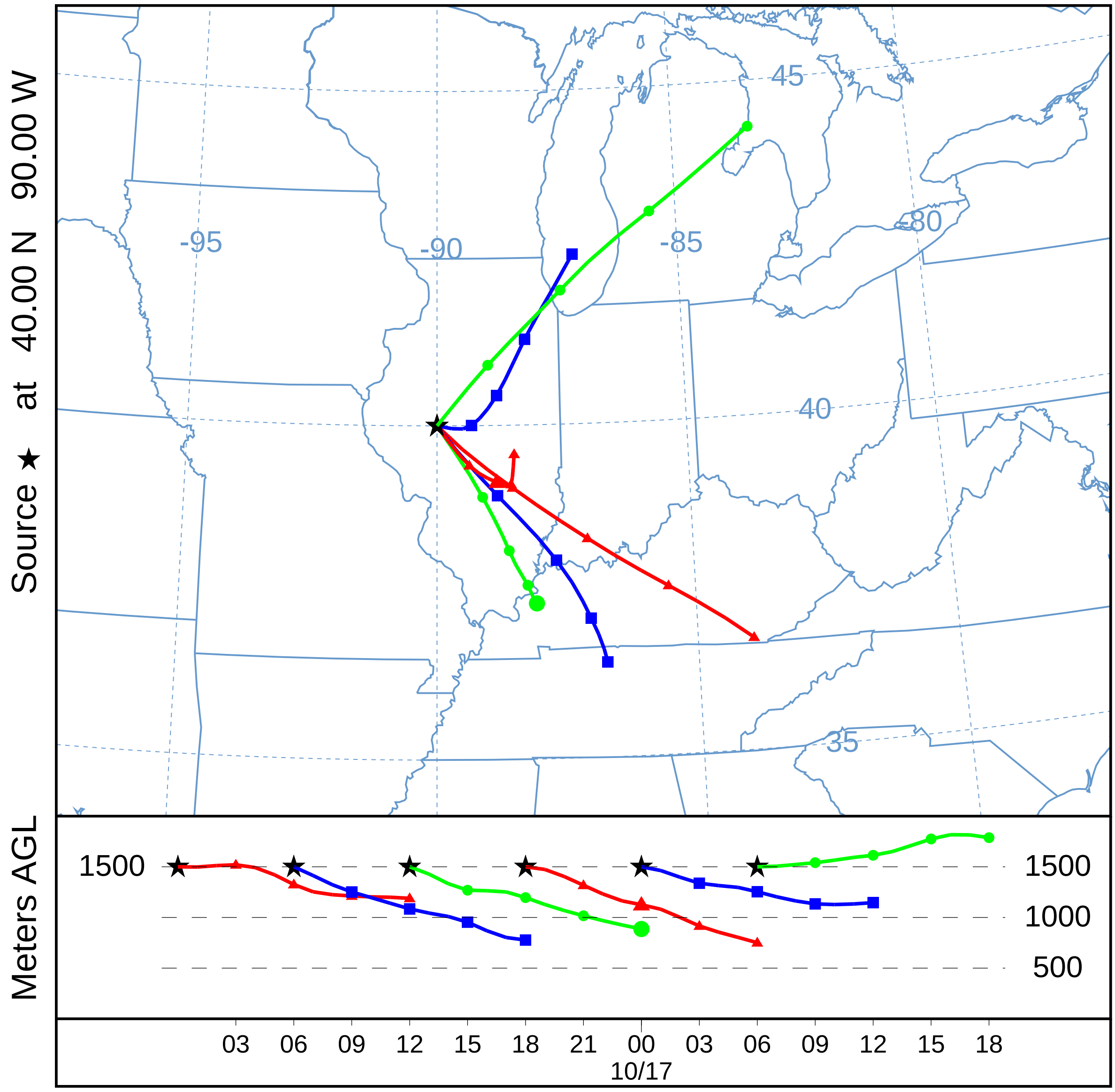
NGM Meteorological Data



Multiple-in-Time Trajectory (007)

Forward trajectories starting at 0000 UTC 16 Oct 95

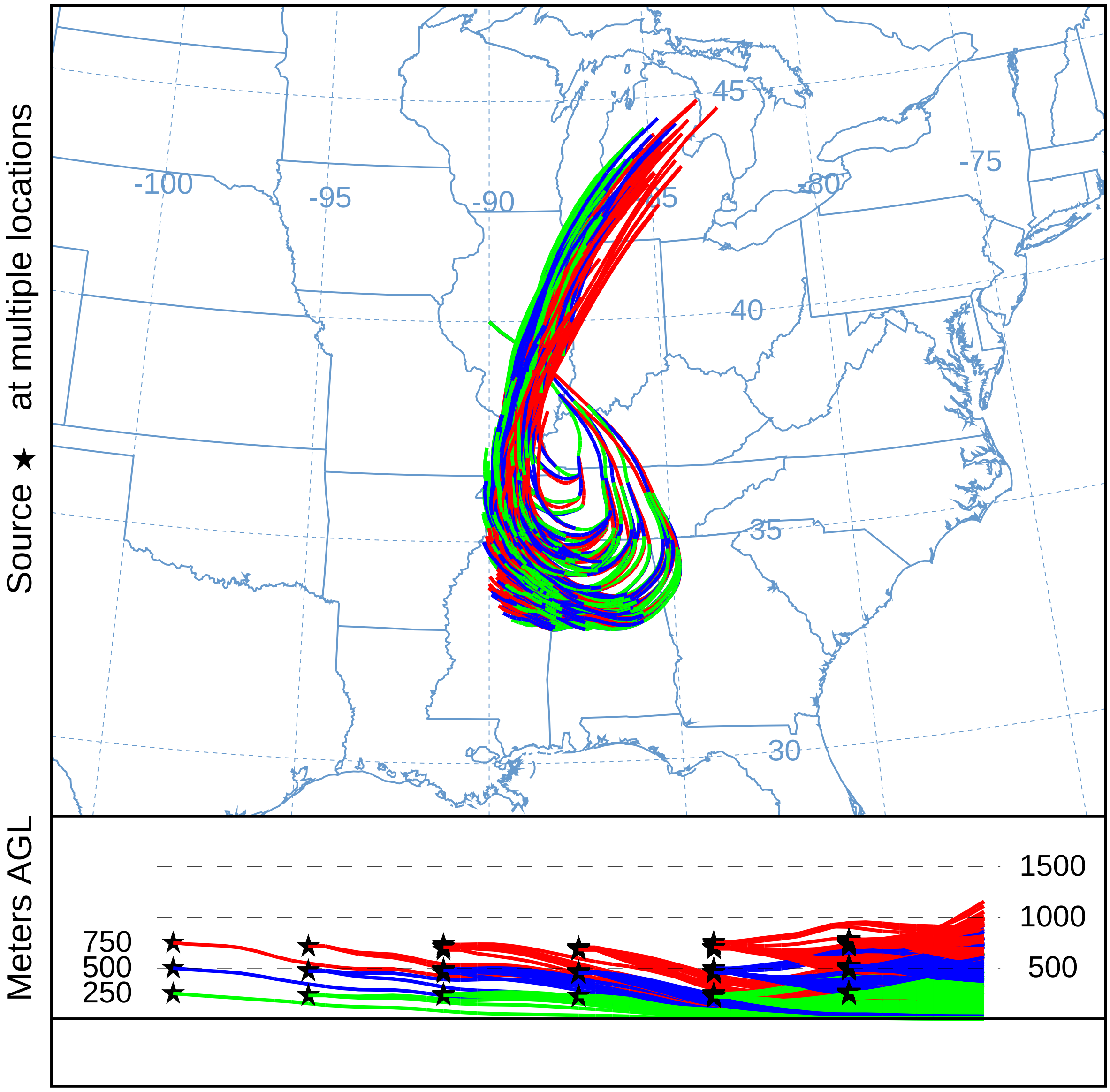
NGM Meteorological Data



Splitting-in-Time Trajectory (008)

Forward trajectories starting at 0000 UTC 16 Oct 95

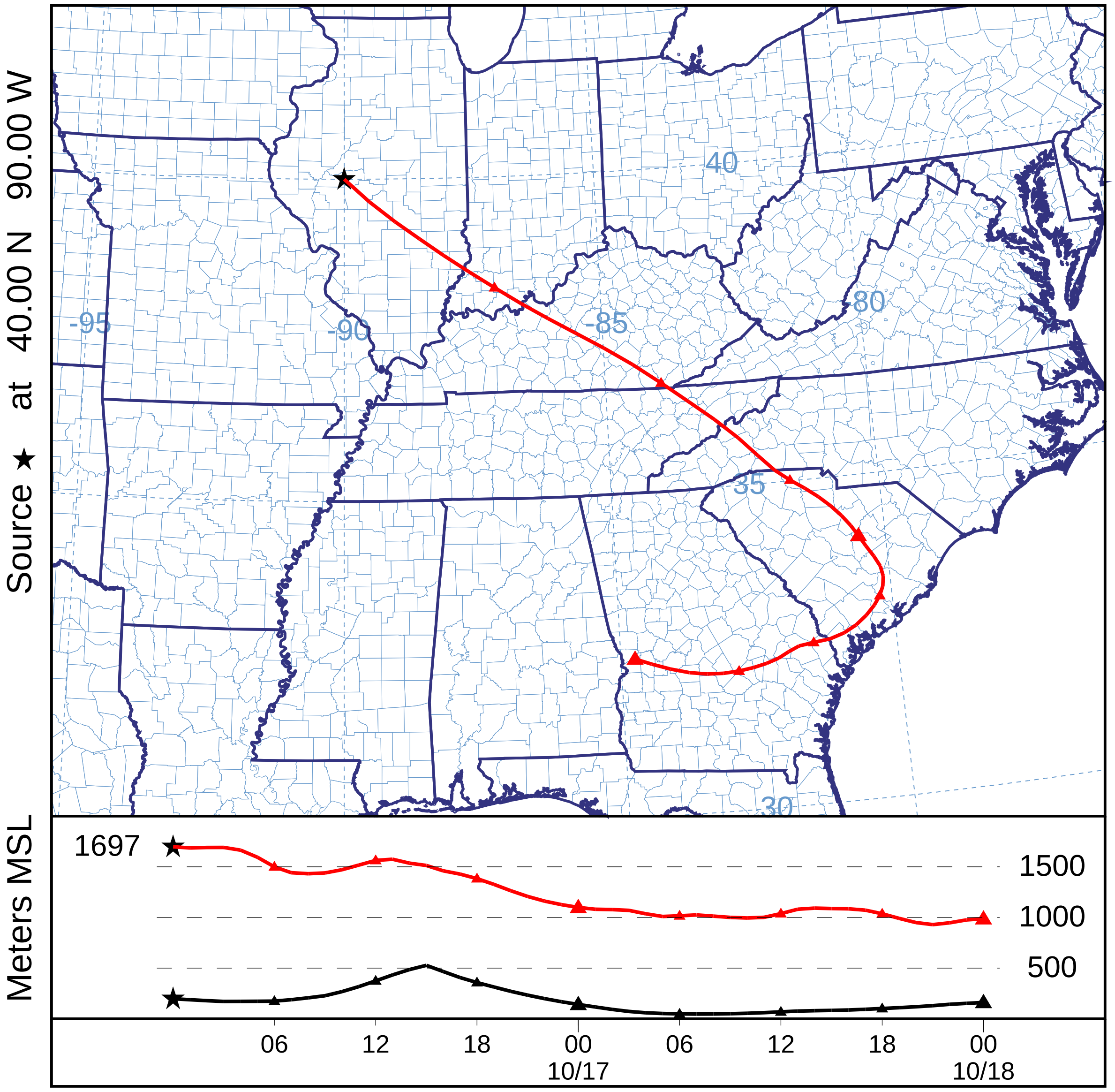
NGM Meteorological Data



Terrain below Trajectory (009)

Forward trajectory starting at 0000 UTC 16 Oct 95

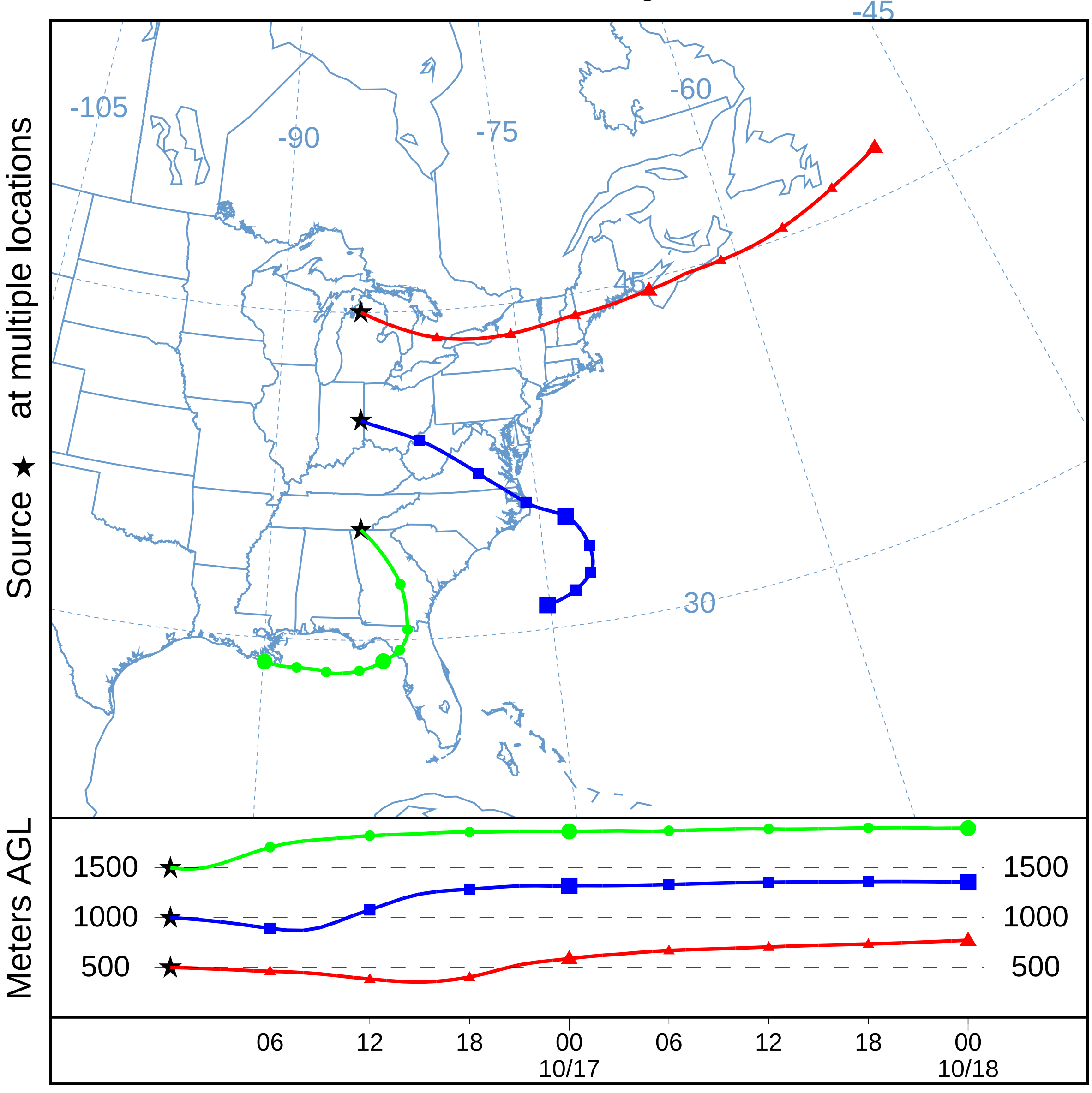
NGM Meteorological Data



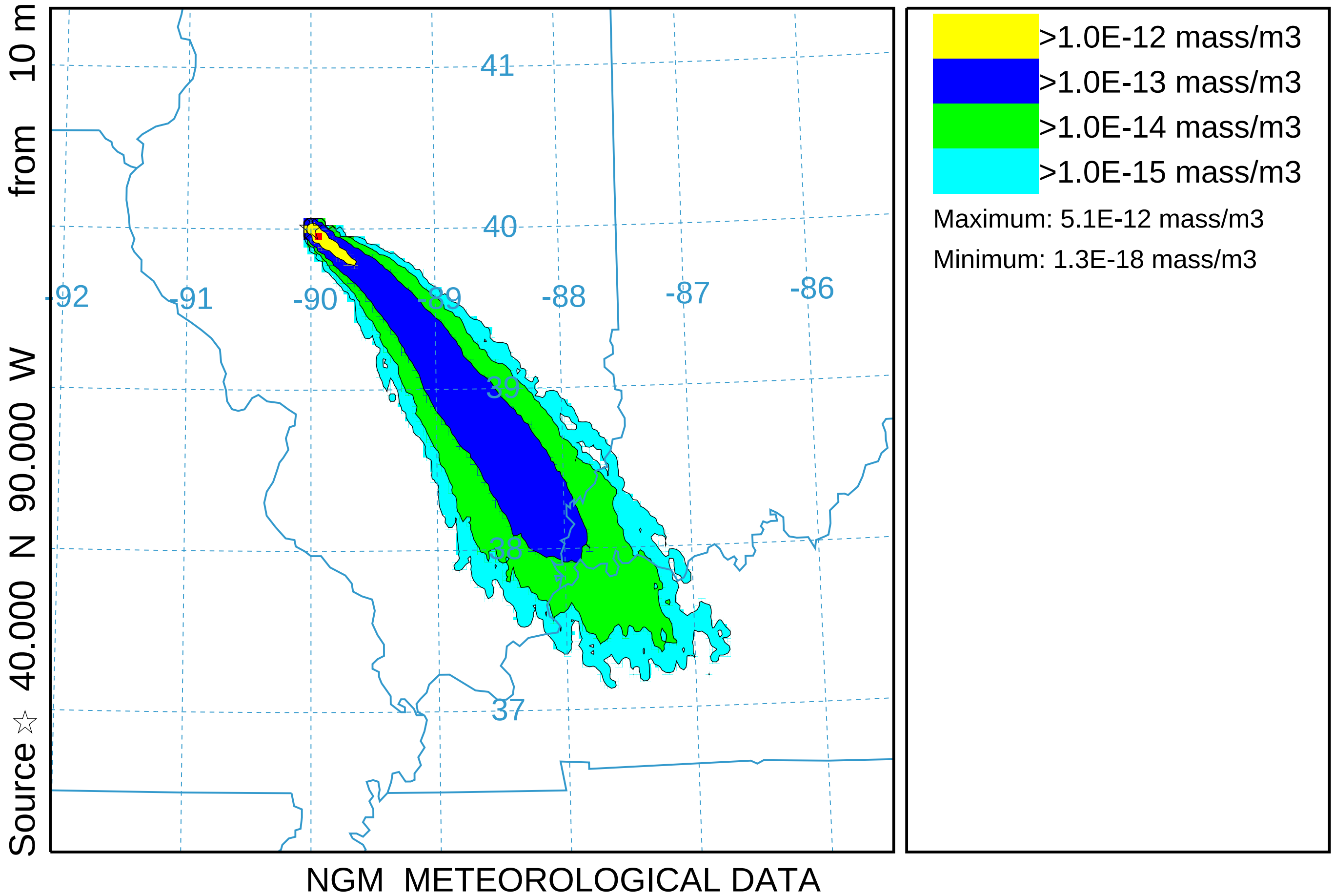
Multiple Meteorology Trajectory (109)

Forward trajectories starting at 0000 UTC 16 Oct 95

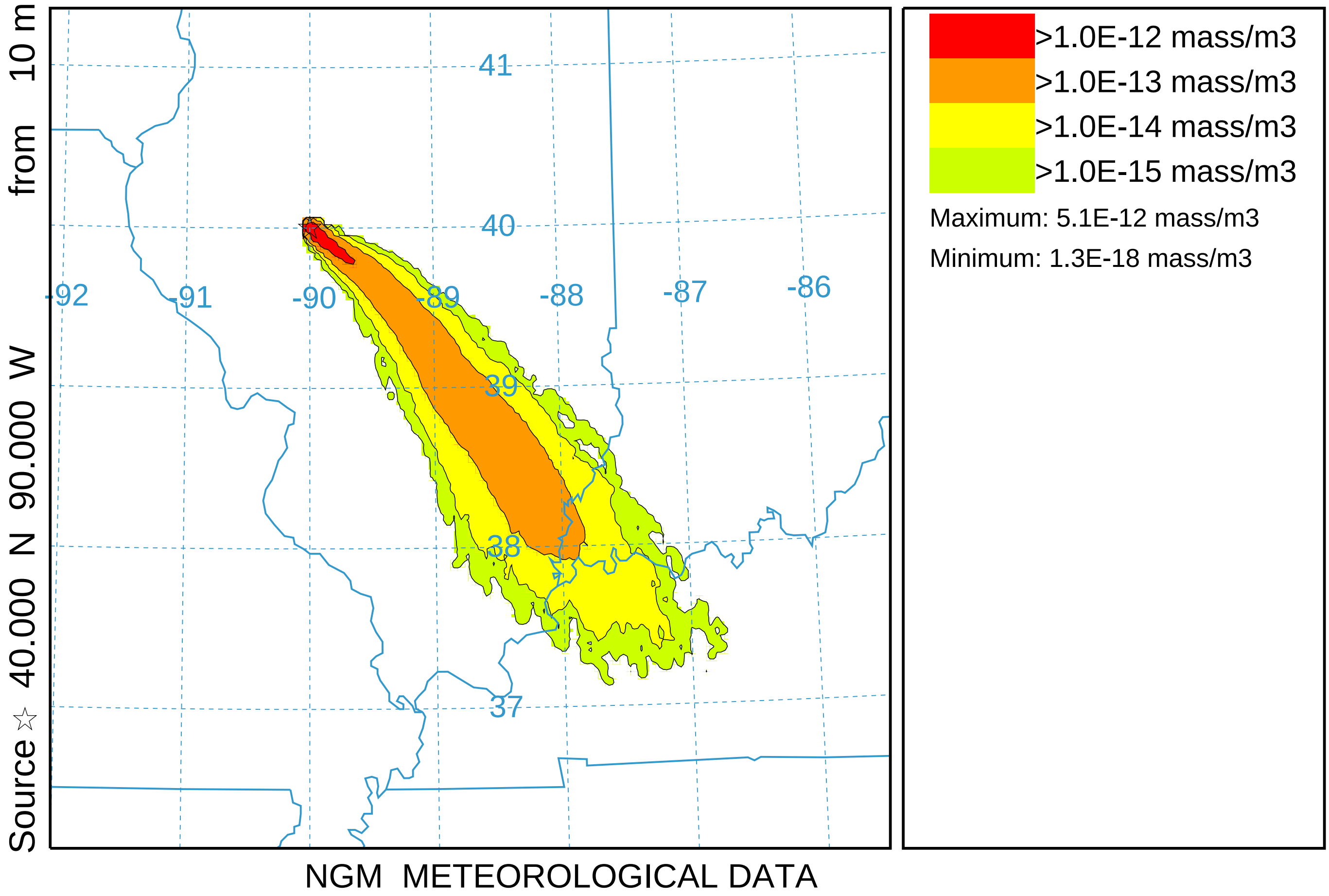
NGM Meteorological Data

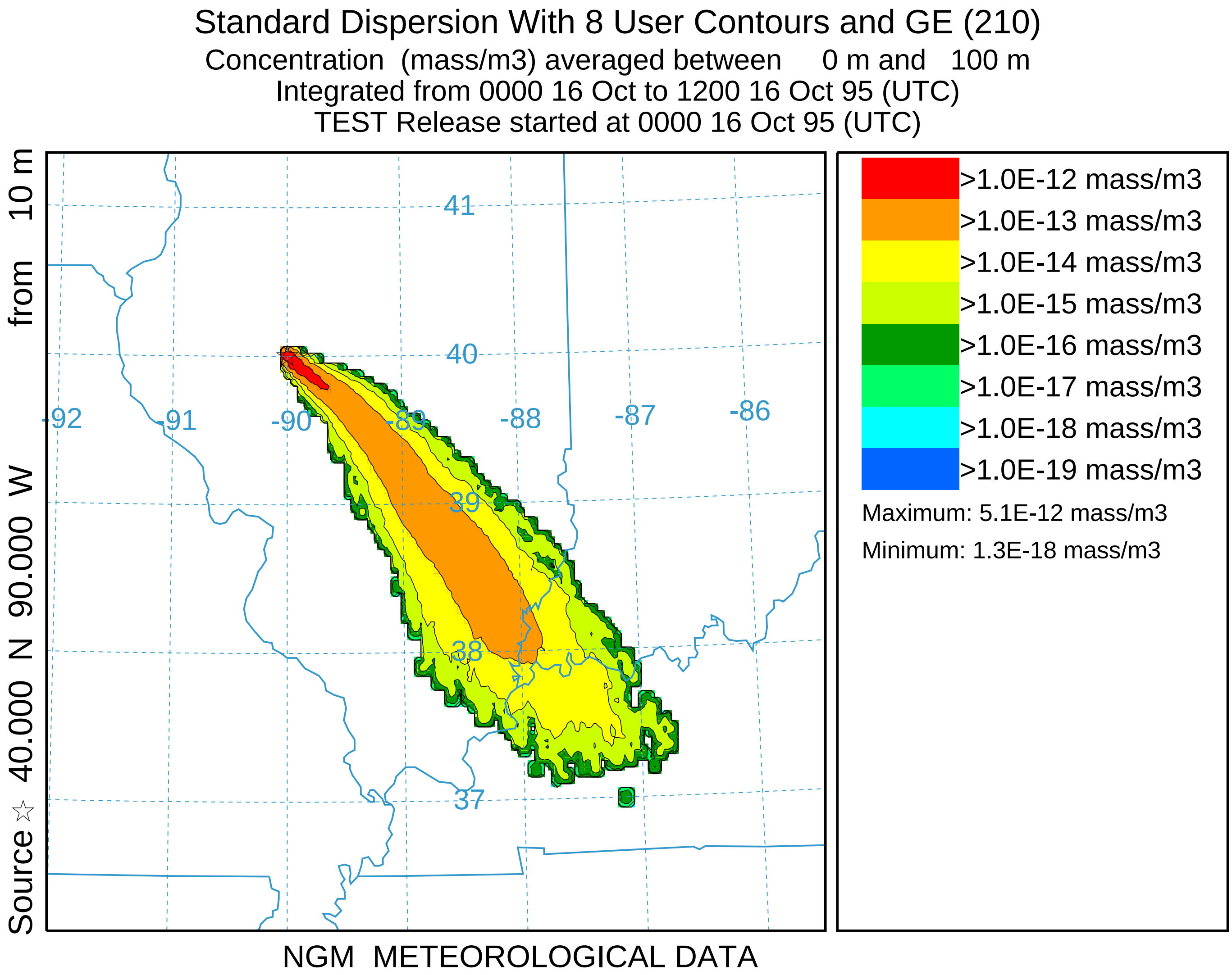


Standard Dispersion Simulation (010)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
TEST Release started at 0000 16 Oct 95 (UTC)



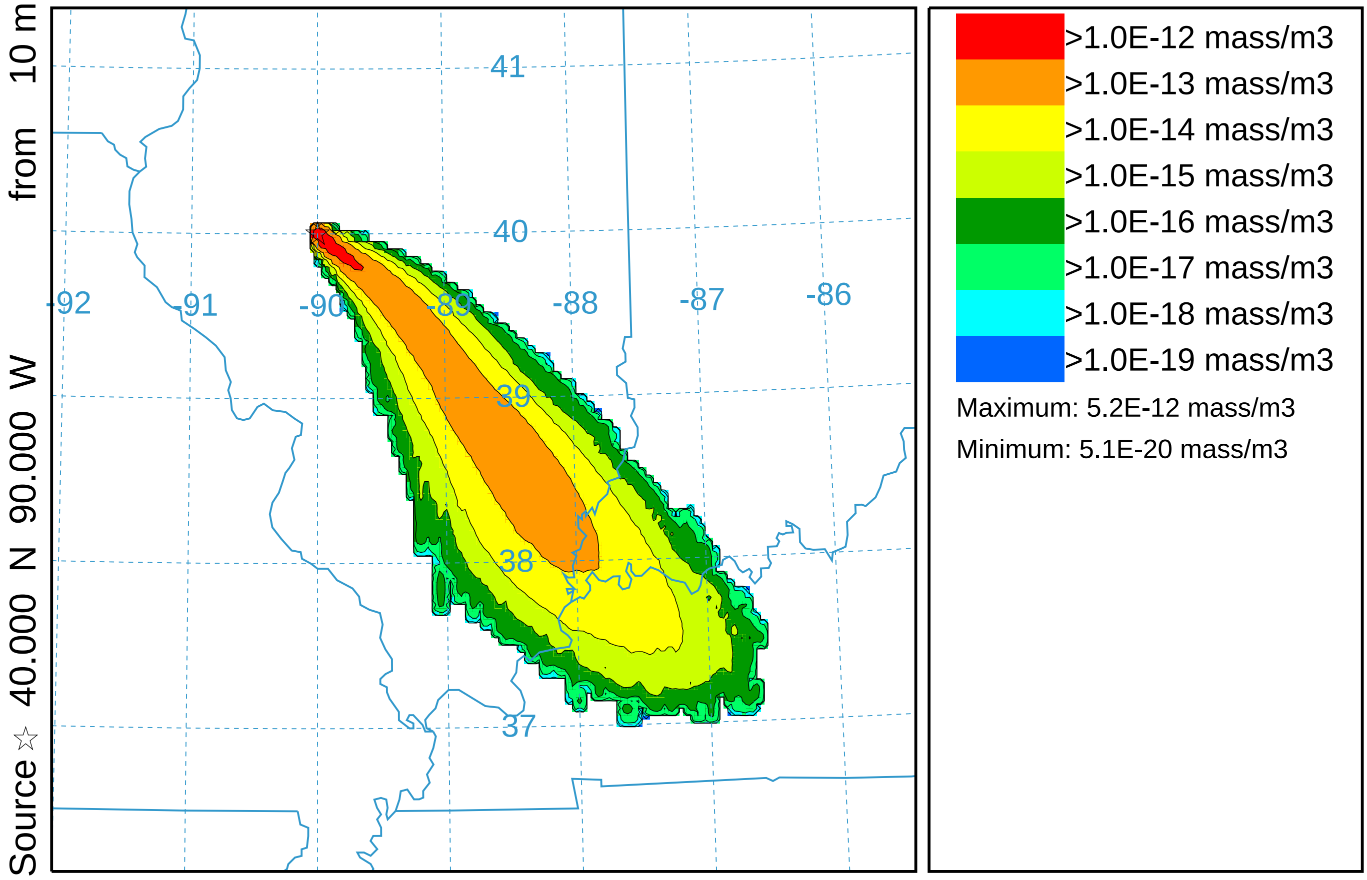
Standard Dispersion With 4 User Set Contours (110)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
TEST Release started at 0000 16 Oct 95 (UTC)



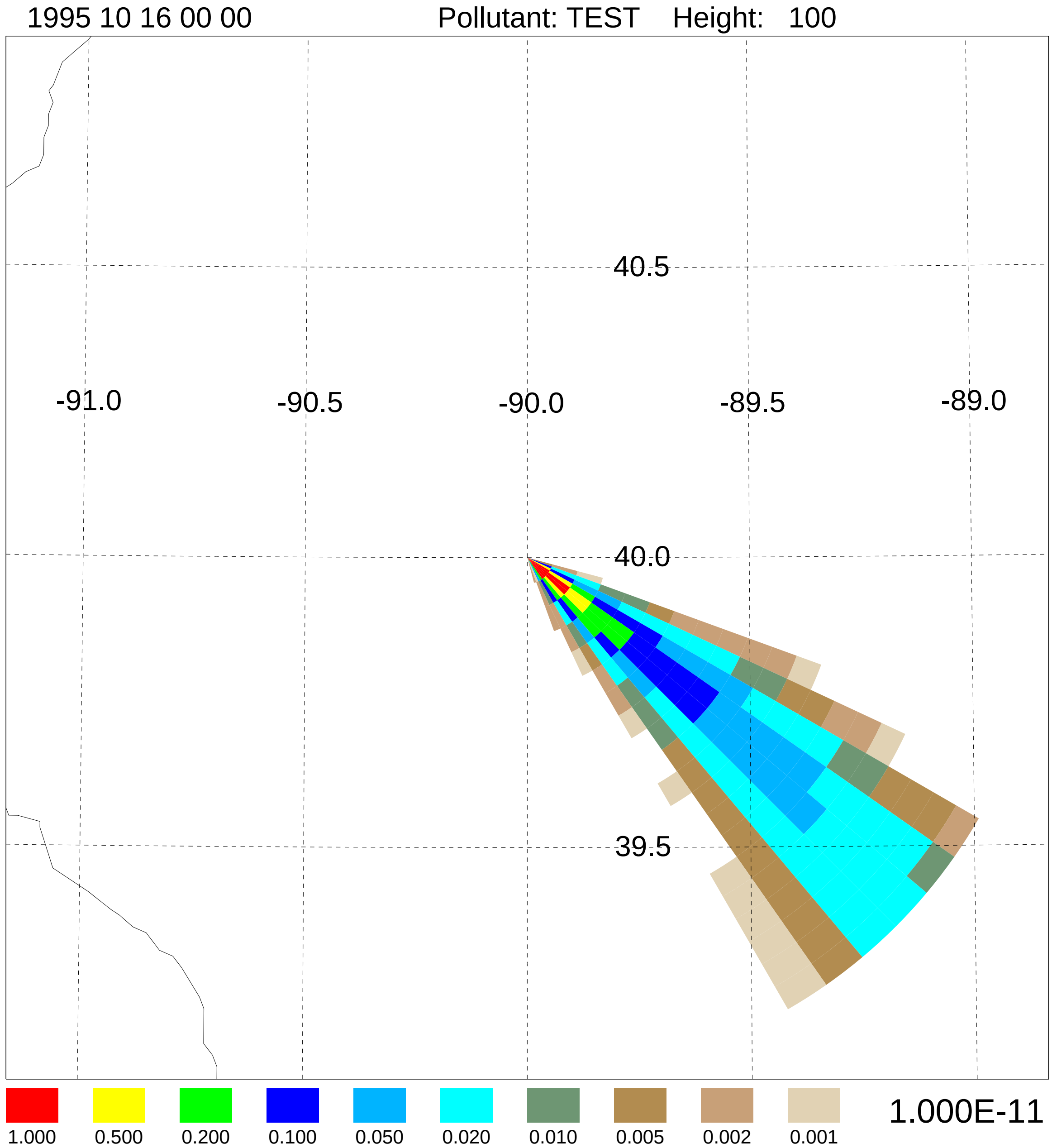


Standard Dispersion With 8 User Contours and GE (210)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
TEST Release started at 0000 16 Oct 95 (UTC)
Source ☆ 40.000 N 90.000 W from 10 m
41
40
39
38
37
-92
-91
-90
-89
-88
-87
-86
>1.0E-12 mass/m3
>1.0E-13 mass/m3
>1.0E-14 mass/m3
>1.0E-15 mass/m3
>1.0E-16 mass/m3
>1.0E-17 mass/m3
>1.0E-18 mass/m3
>1.0E-19 mass/m3
Maximum: 5.1E-12 mass/m3
Minimum: 1.3E-18 mass/m3
NGM METEOROLOGICAL DATA

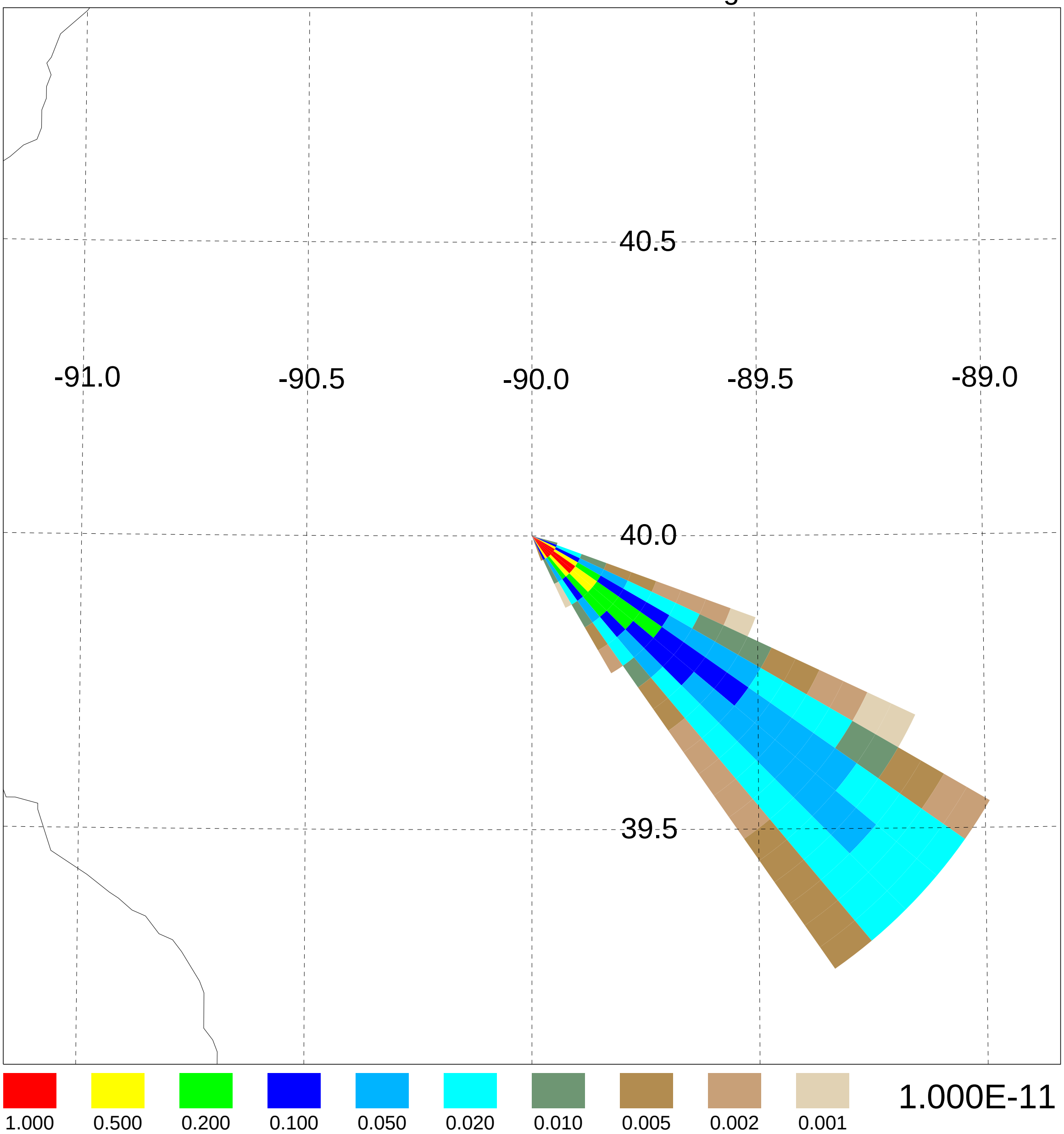
MPI Dispersion using 4 processors (310)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
TEST Release started at 0000 16 Oct 95 (UTC)



NGM METEOROLOGICAL DATA


1995 10 16 00 00
Pollutant: TEST Height: 100
40.5
-91.0
-90.5
-90.0
-89.5
-89.0
40.0
39.5
1.000
0.500
0.200
0.100
0.050
0.020
0.010
0.005
0.002
0.001
1.000E-11

1995 10 16 00 00 Pollutant: TEST Height: 100

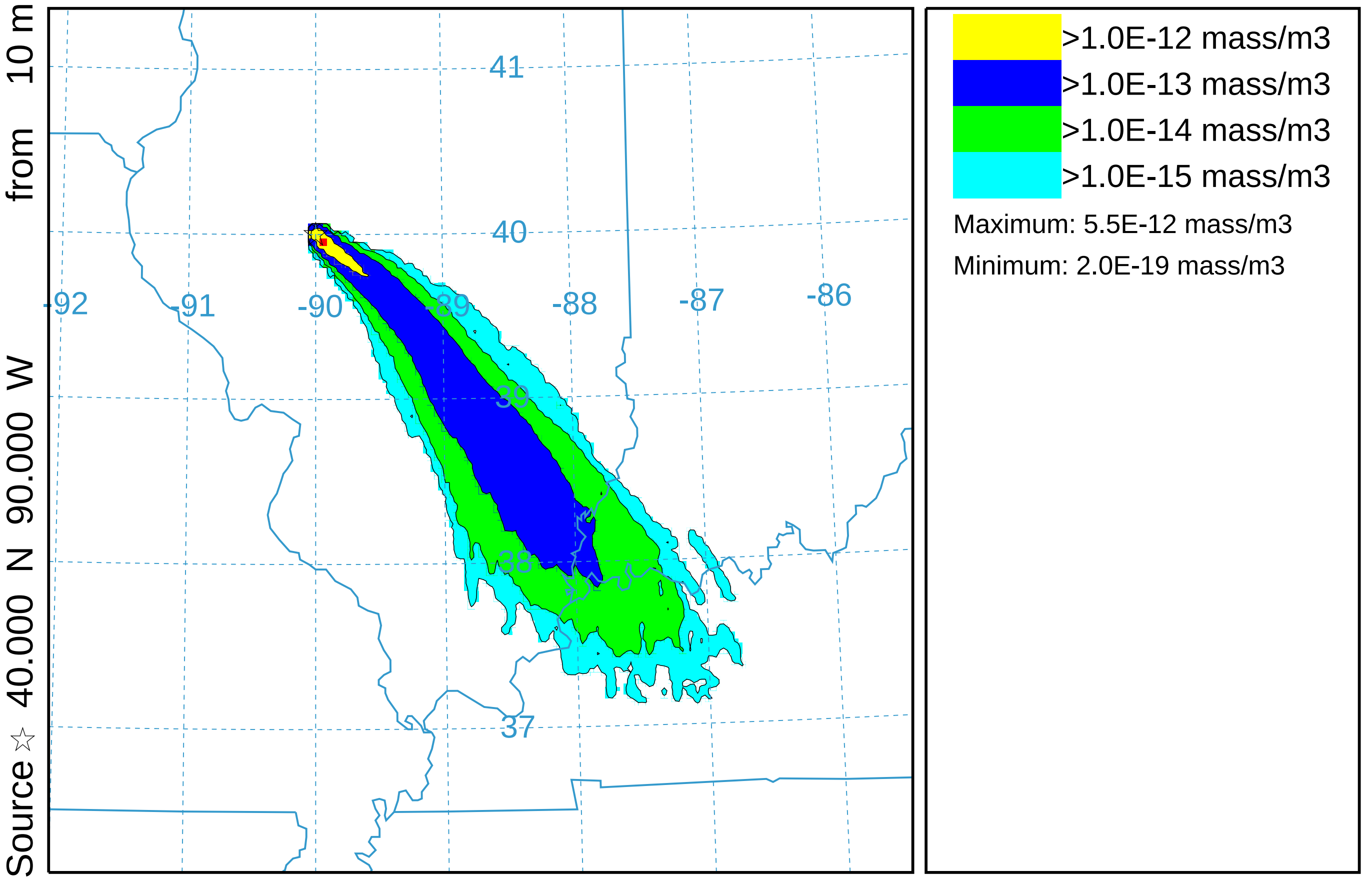


Polar+Rectangular Grid (510)

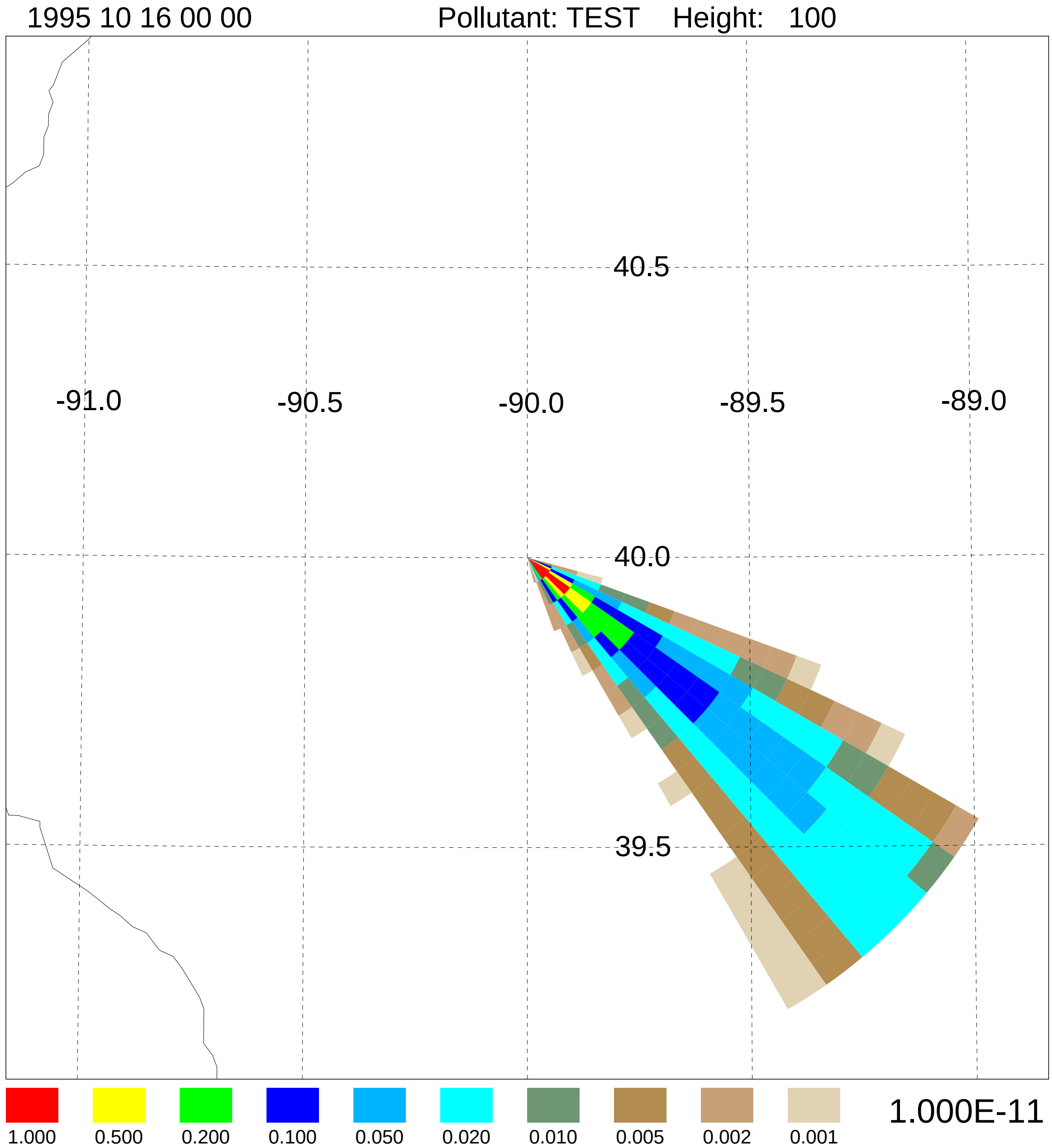
Concentration (mass/m3) averaged between 0 m and 100 m

Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)

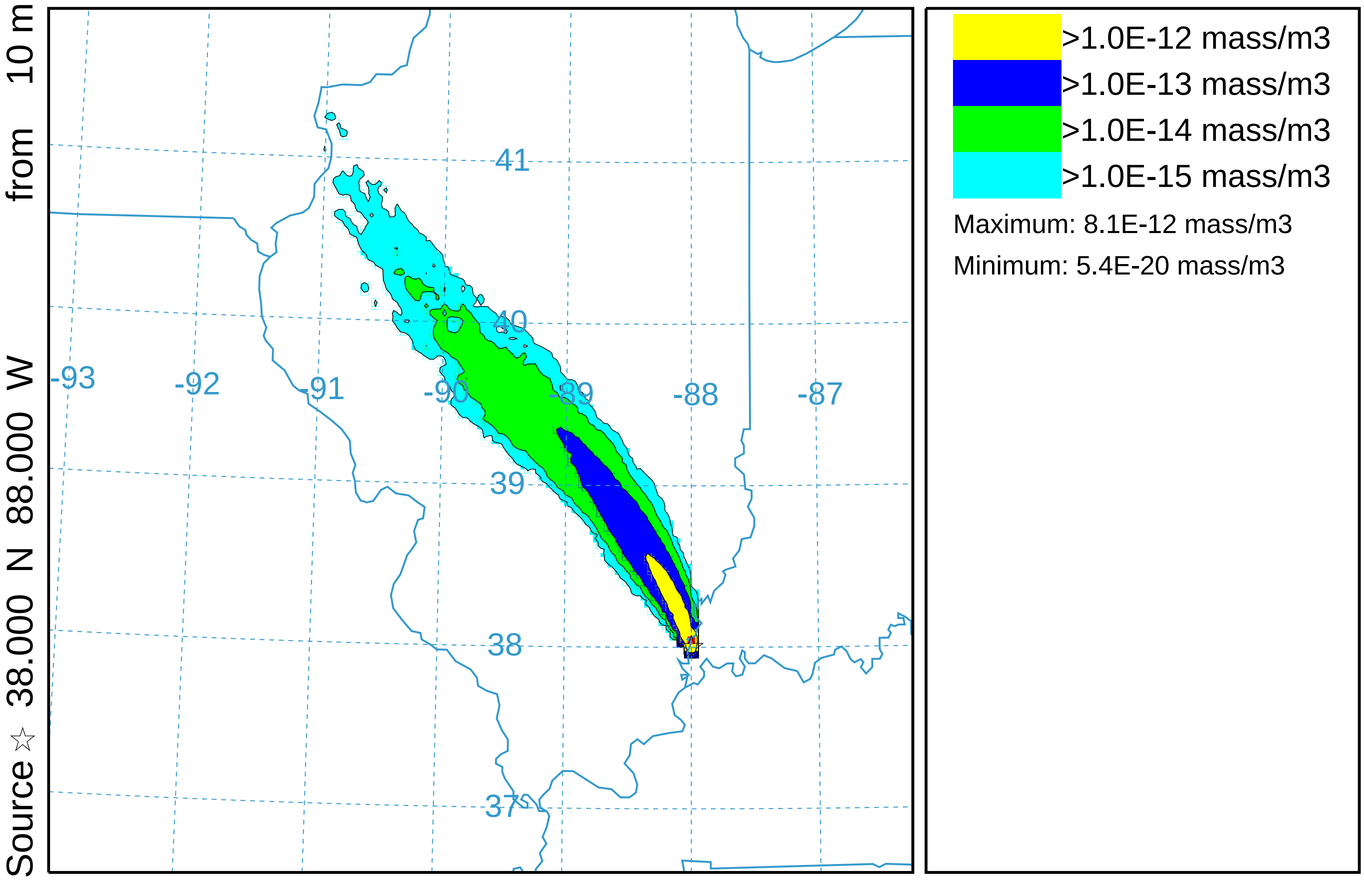
TEST Release started at 0000 16 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

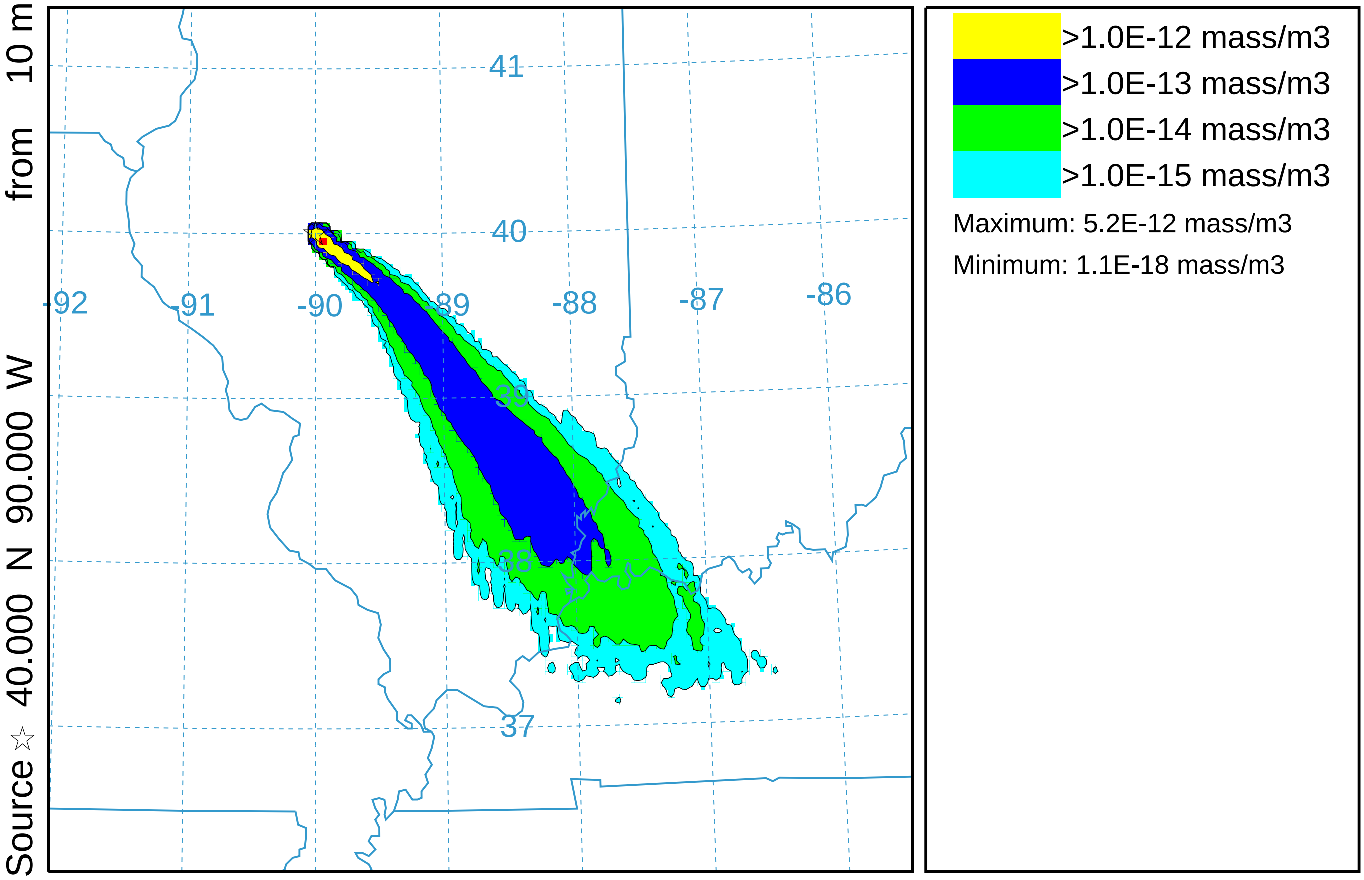

1995 10 16 00 00
Pollutant: TEST Height: 100
40.5
-91.0
-90.5
-90.0
-89.5
-89.0
40.0
39.5
1.000
0.500
0.200
0.100
0.050
0.020
0.010
0.005
0.002
0.001
1.000E-11

Backward Dispersion for Attribution (011)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 1200 16 Oct to 0000 16 Oct 95 (UTC) [backward]
TEST Calculation started at 1200 16 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

Dispersion using Deformation (012)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
TEST Release started at 0000 16 Oct 95 (UTC)



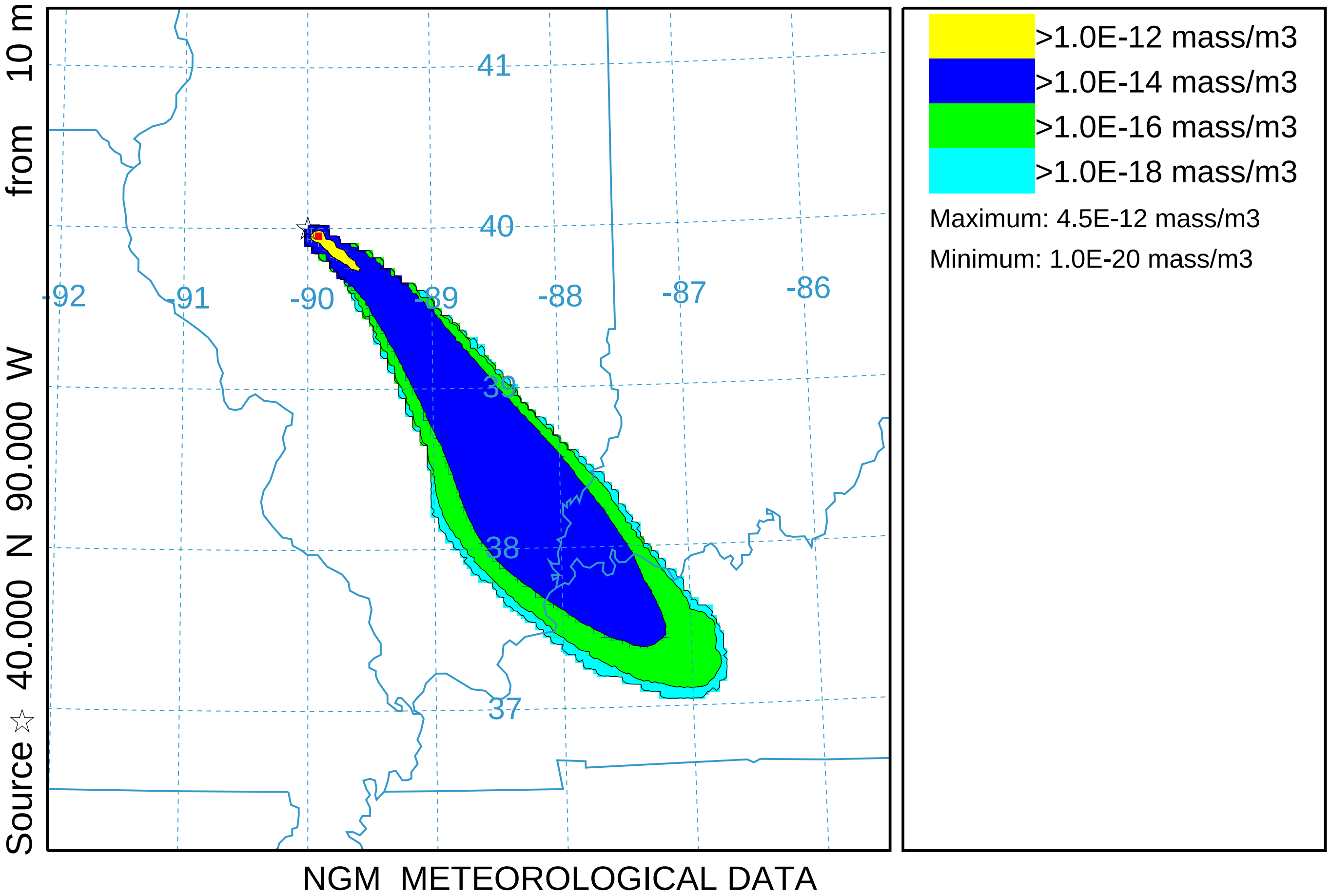
NGM METEOROLOGICAL DATA

Top-Hat Linear Puff Dispersion (013)

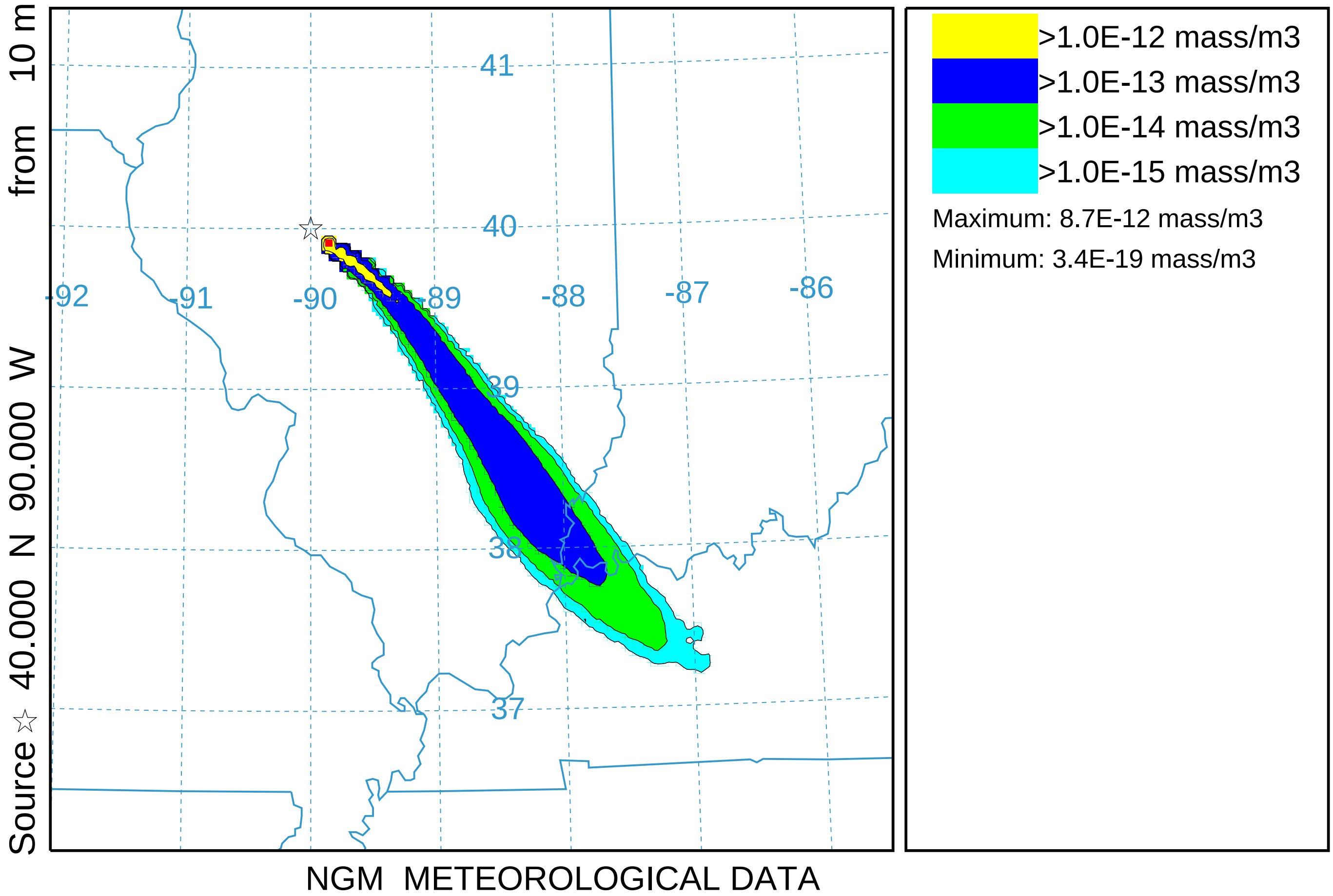
Concentration (mass/m3) averaged between 0 m and 100 m

Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)

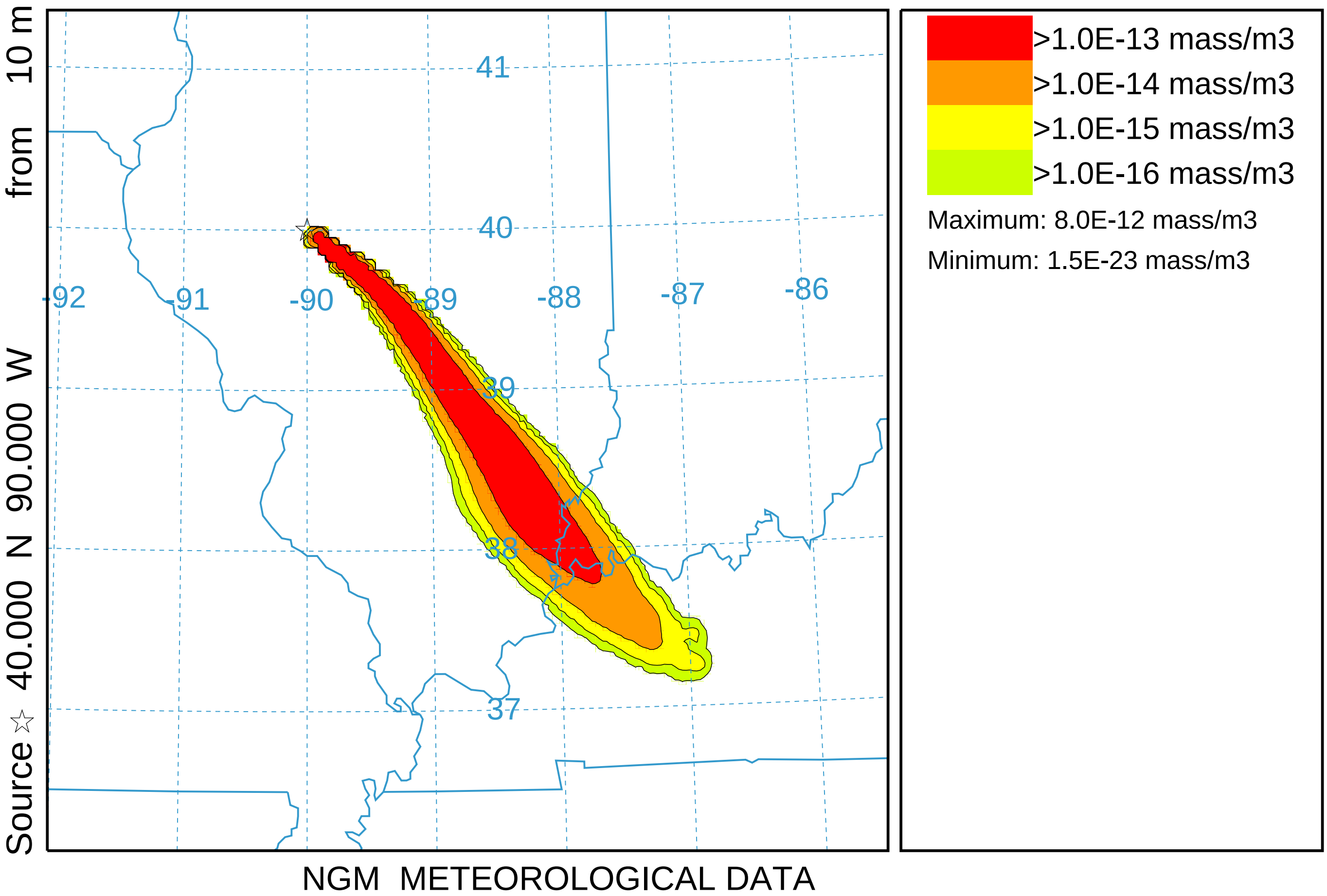
TEST Release started at 0000 16 Oct 95 (UTC)



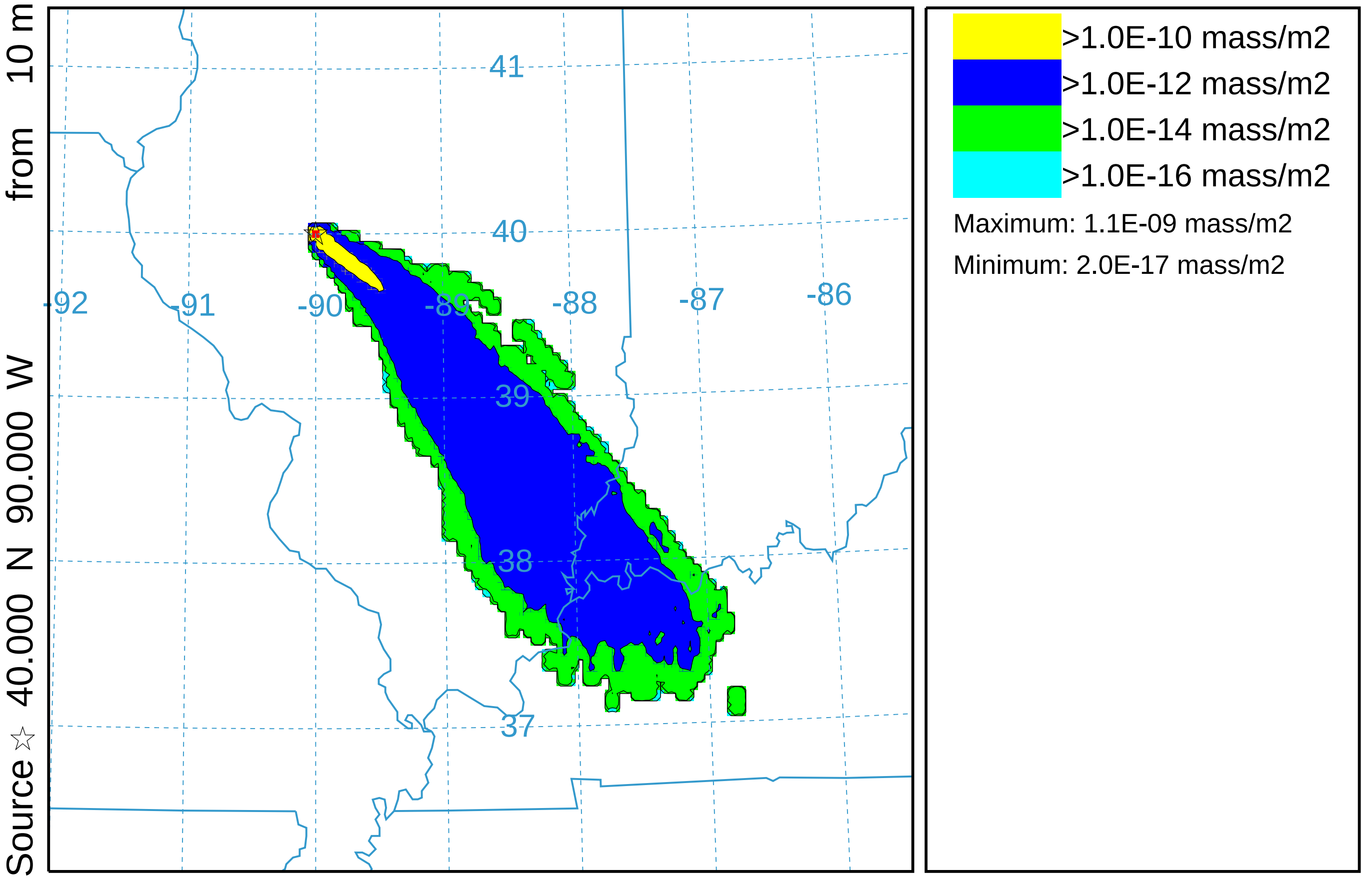
Top-Hat Empirical Puff Dispersion (014)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
TEST Release started at 0000 16 Oct 95 (UTC)



Gaussian Empirical Puff Dispersion (015)
Concentration (mass/m3) averaged between   0 m and 100 m
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
TEST Release started at 0000 16 Oct 95 (UTC)

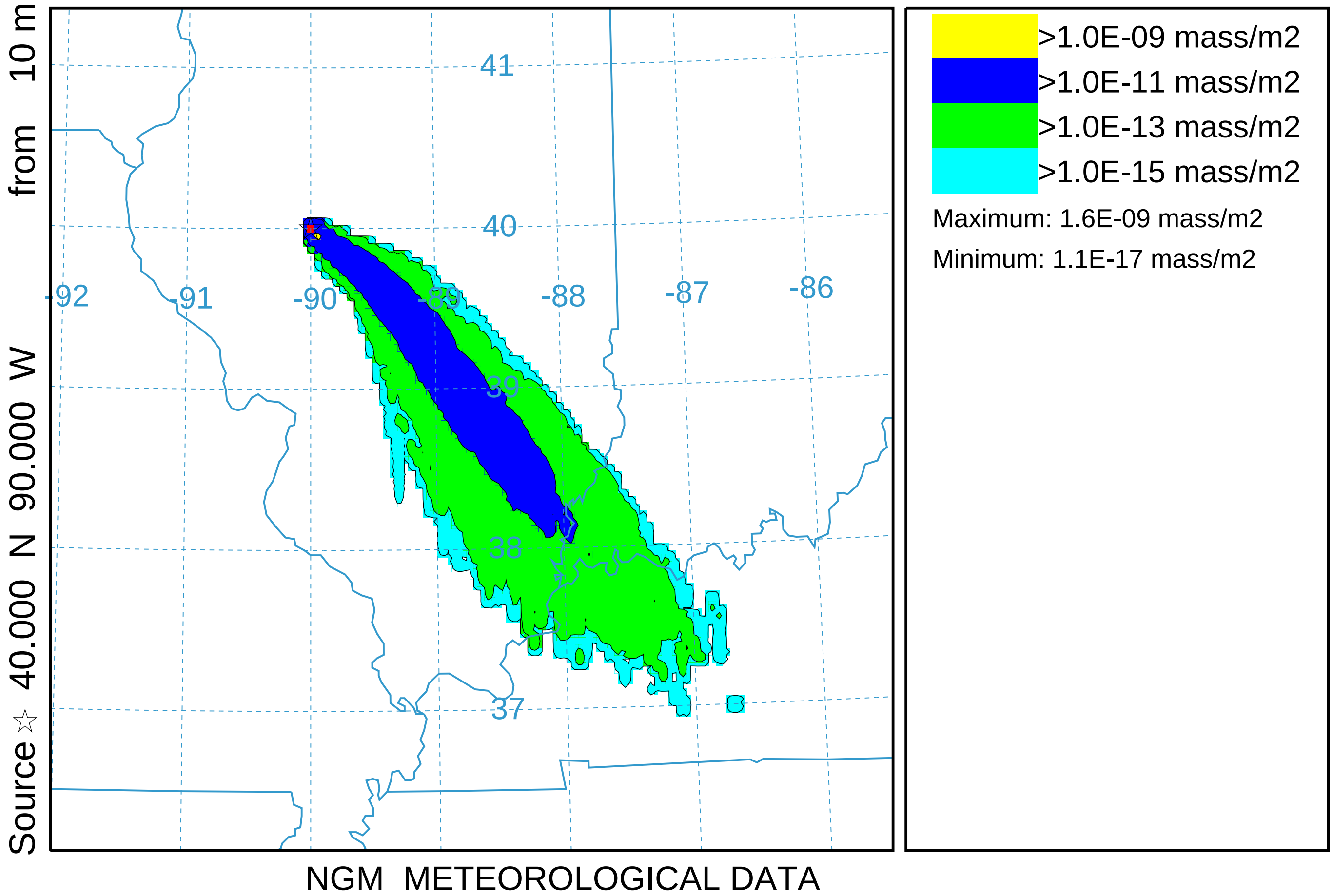


Deposition using Size and Density (016)
Deposition (mass/m2) at ground-level
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
TEST Release started at 0000 16 Oct 95 (UTC)

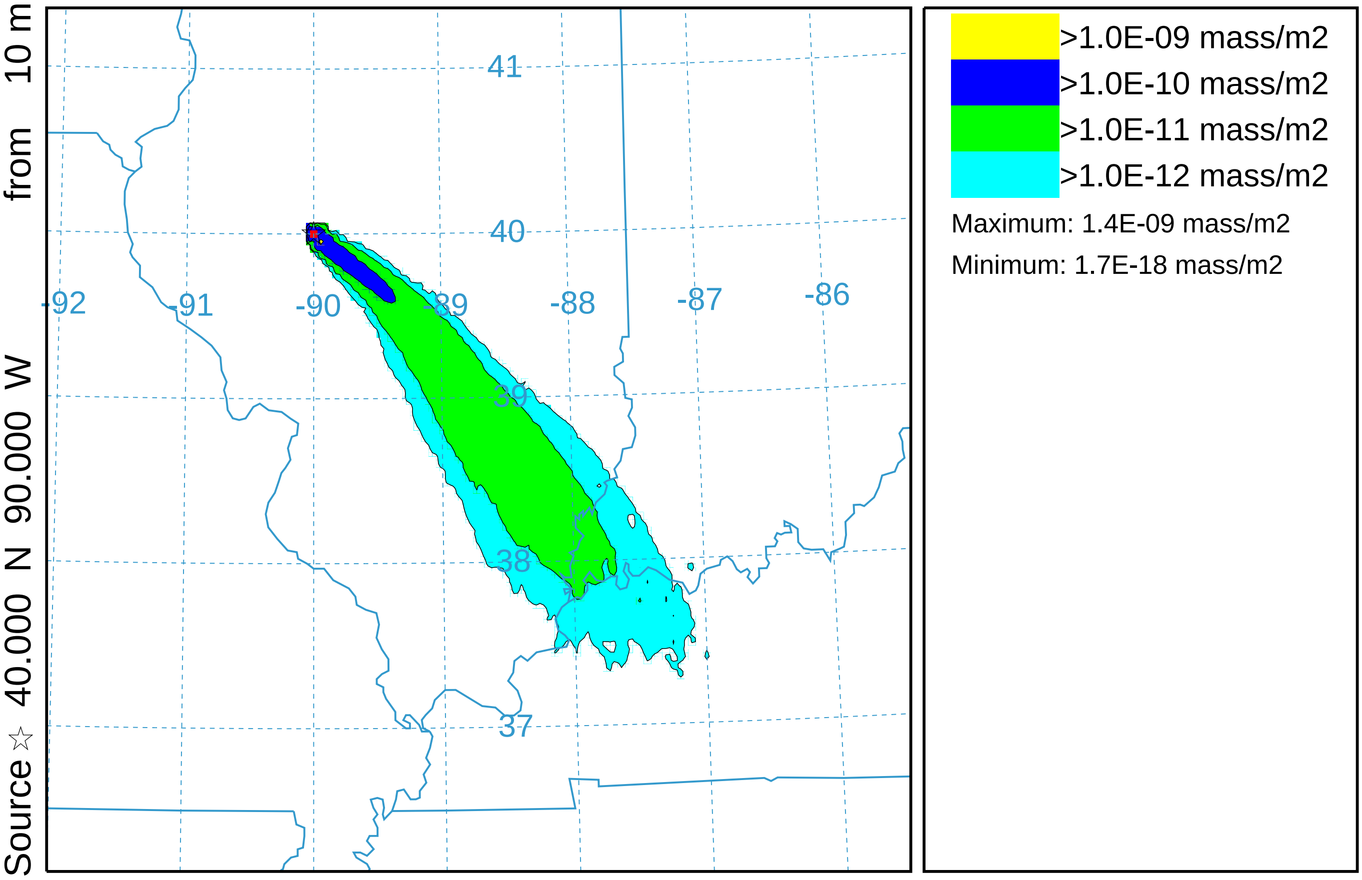


NGM METEOROLOGICAL DATA

Deposition using Multiple Size Bins (116)
Deposition (mass/m2) at ground-level
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
SUM Release started at 0000 16 Oct 95 (UTC)

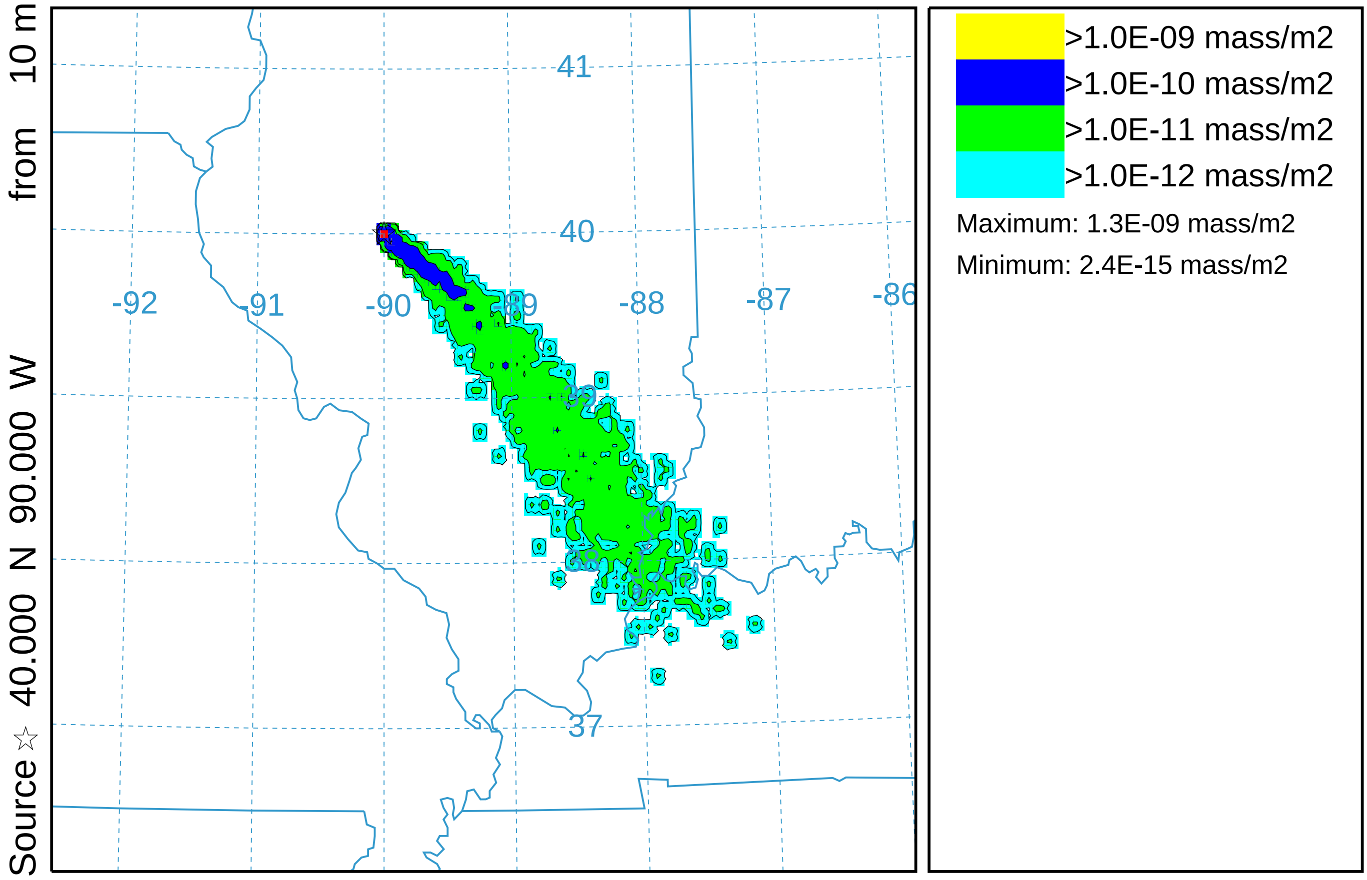


Deposition using specified velocity (017)
Deposition (mass/m2) at ground-level
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
TEST Release started at 0000 16 Oct 95 (UTC)



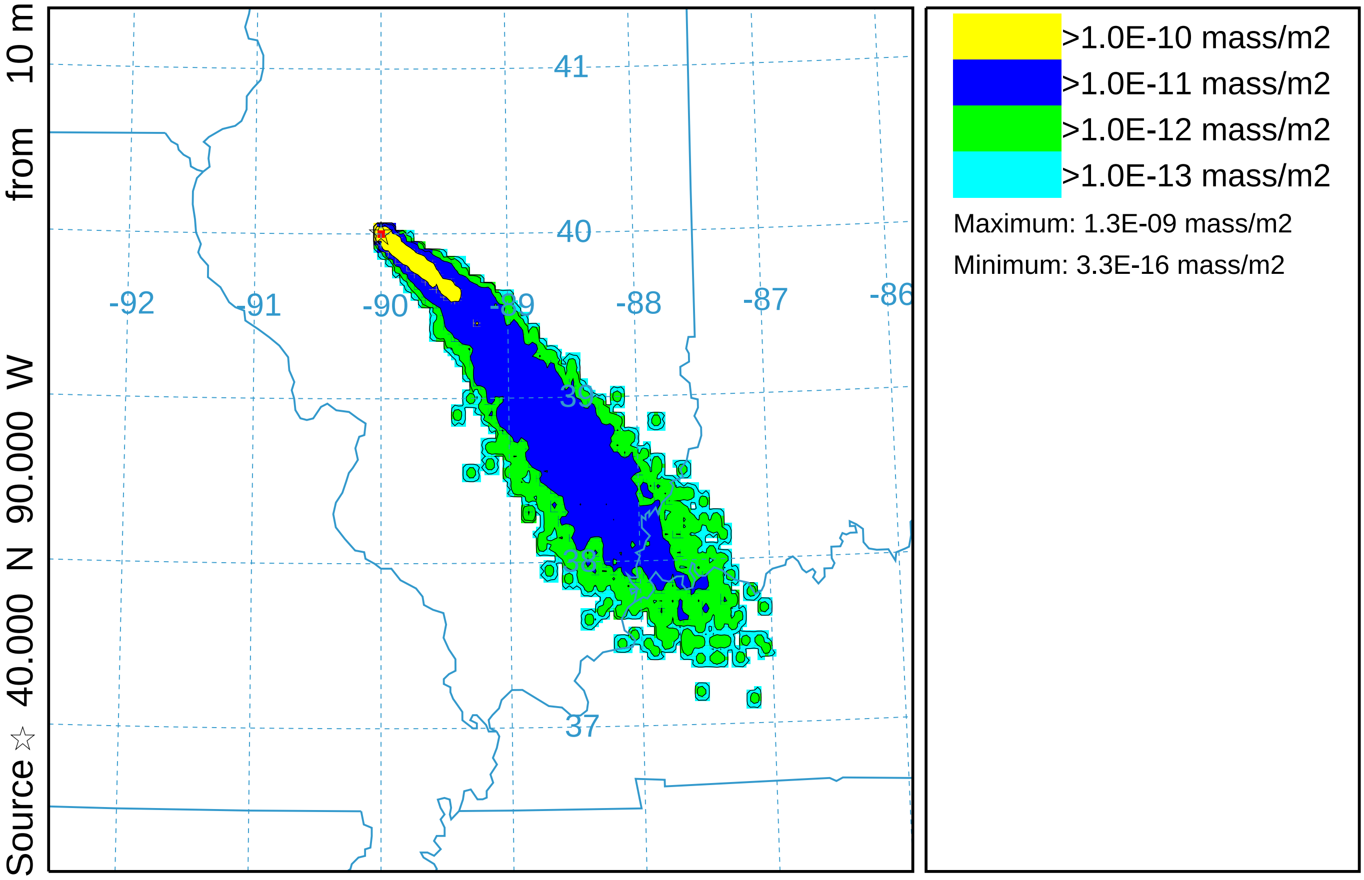
NGM METEOROLOGICAL DATA

Probability Deposition using 1000 particles (018)
Deposition (mass/m2) at ground-level
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
TEST Release started at 0000 16 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

Probability Deposition using 10000 particles (019)
Deposition (mass/m2) at ground-level
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
TEST Release started at 0000 16 Oct 95 (UTC)



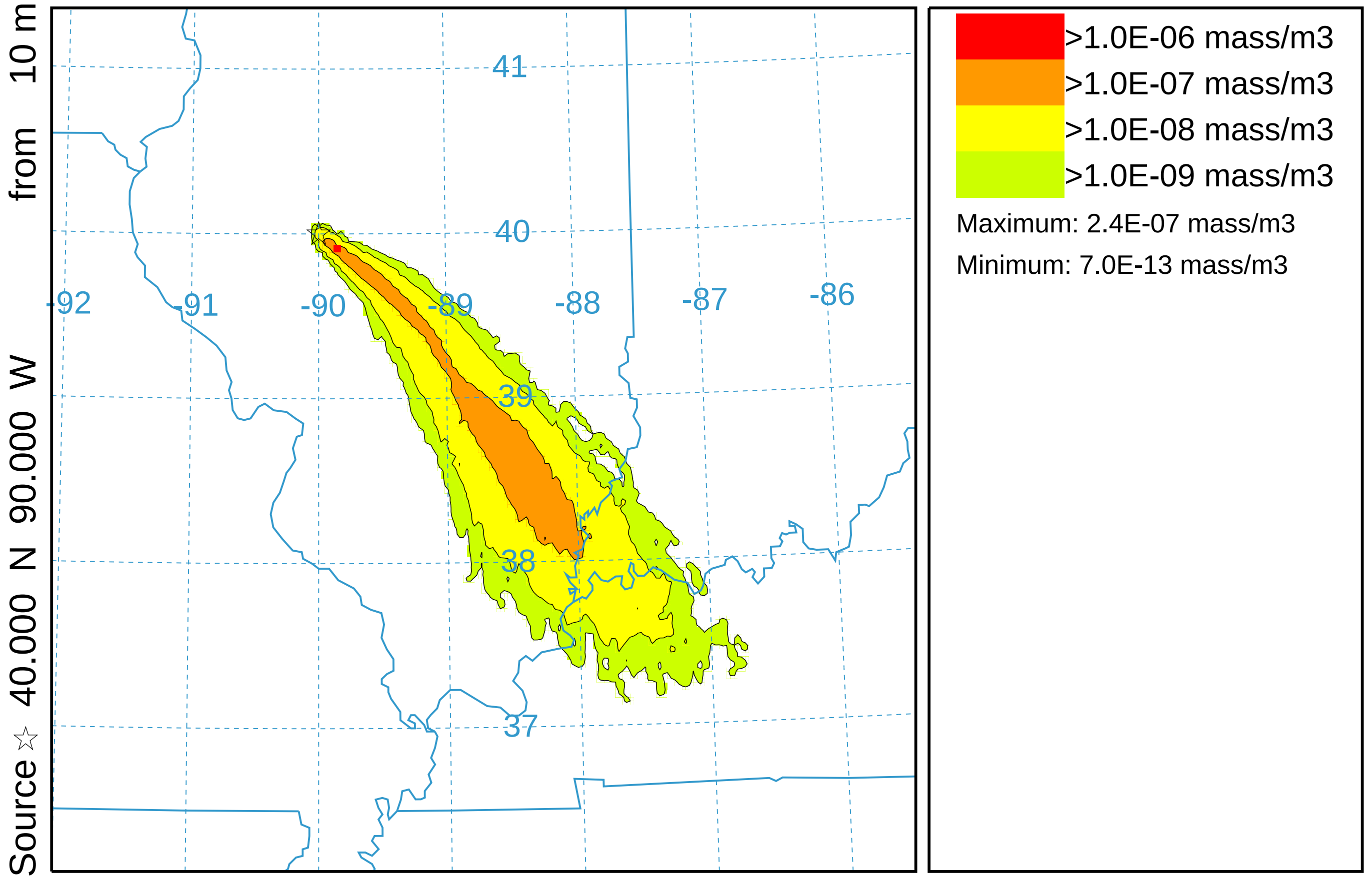
NGM METEOROLOGICAL DATA

Species conversion at 10% per hour (020)

Concentration (mass/m3) averaged between 0 m and 100 m

Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)

POL2 Release started at 0000 16 Oct 95 (UTC)



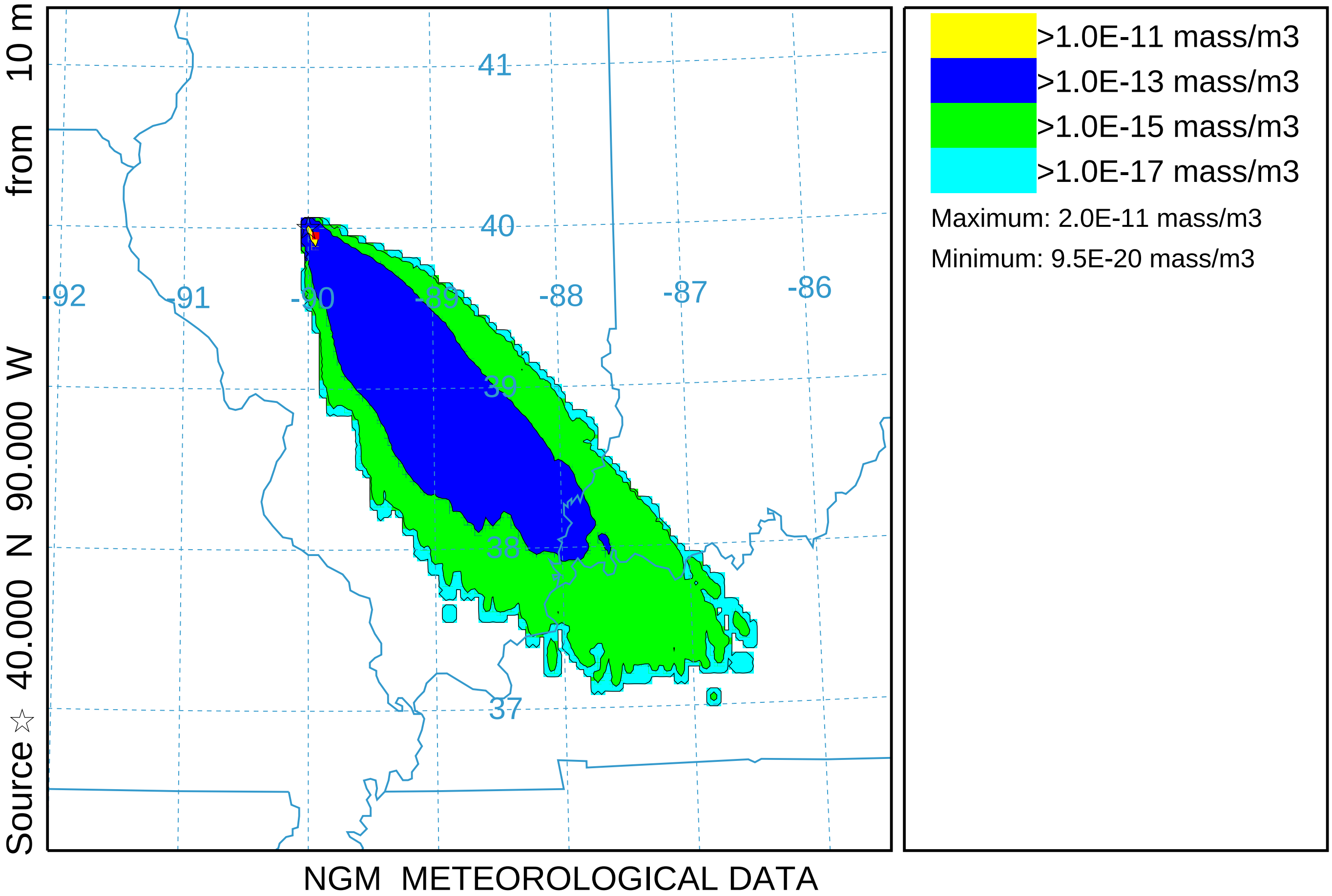
NGM METEOROLOGICAL DATA

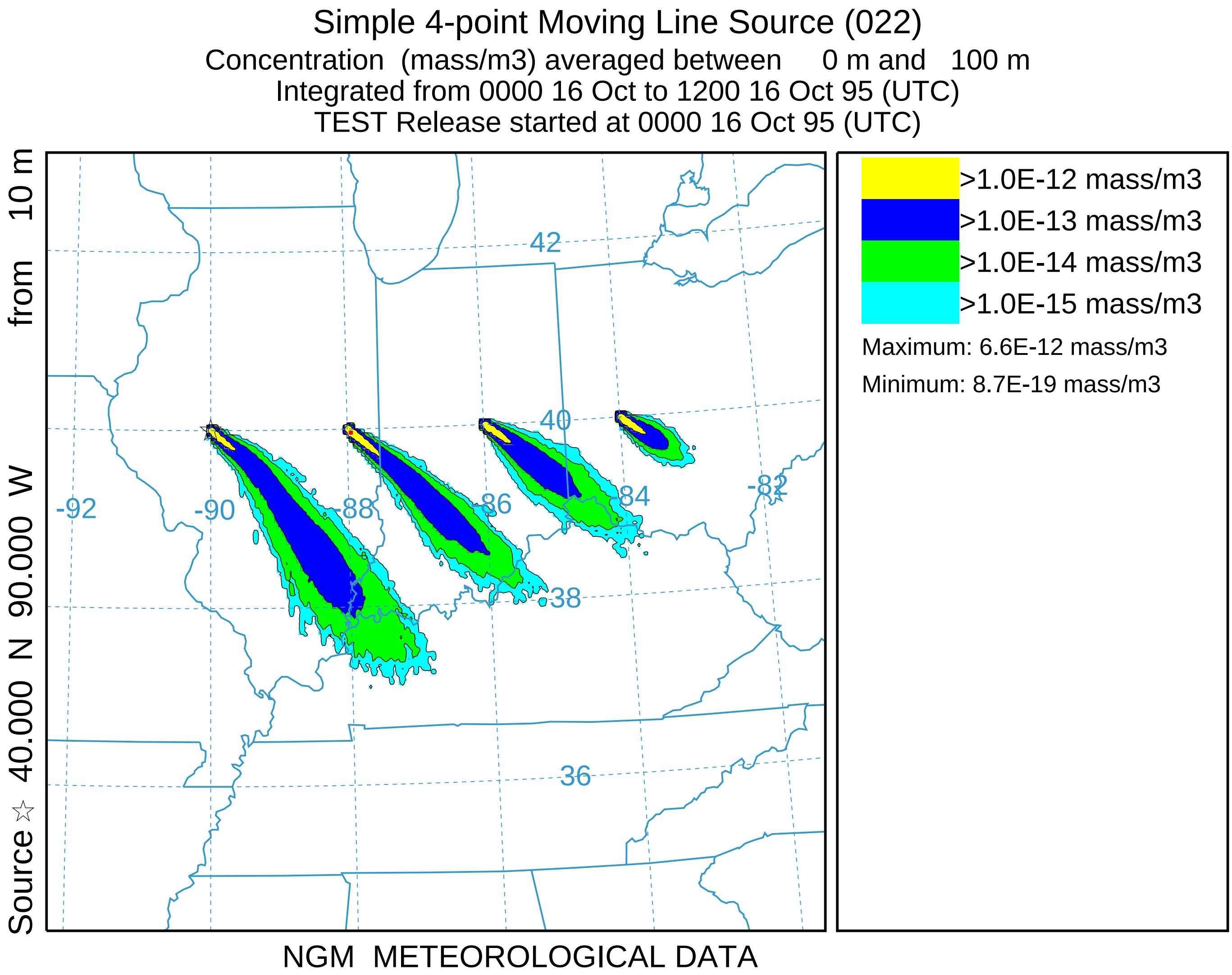
Cycle Short Emissions Every 3 hrs (021)

Concentration (mass/m3) averaged between 0 m and 100 m

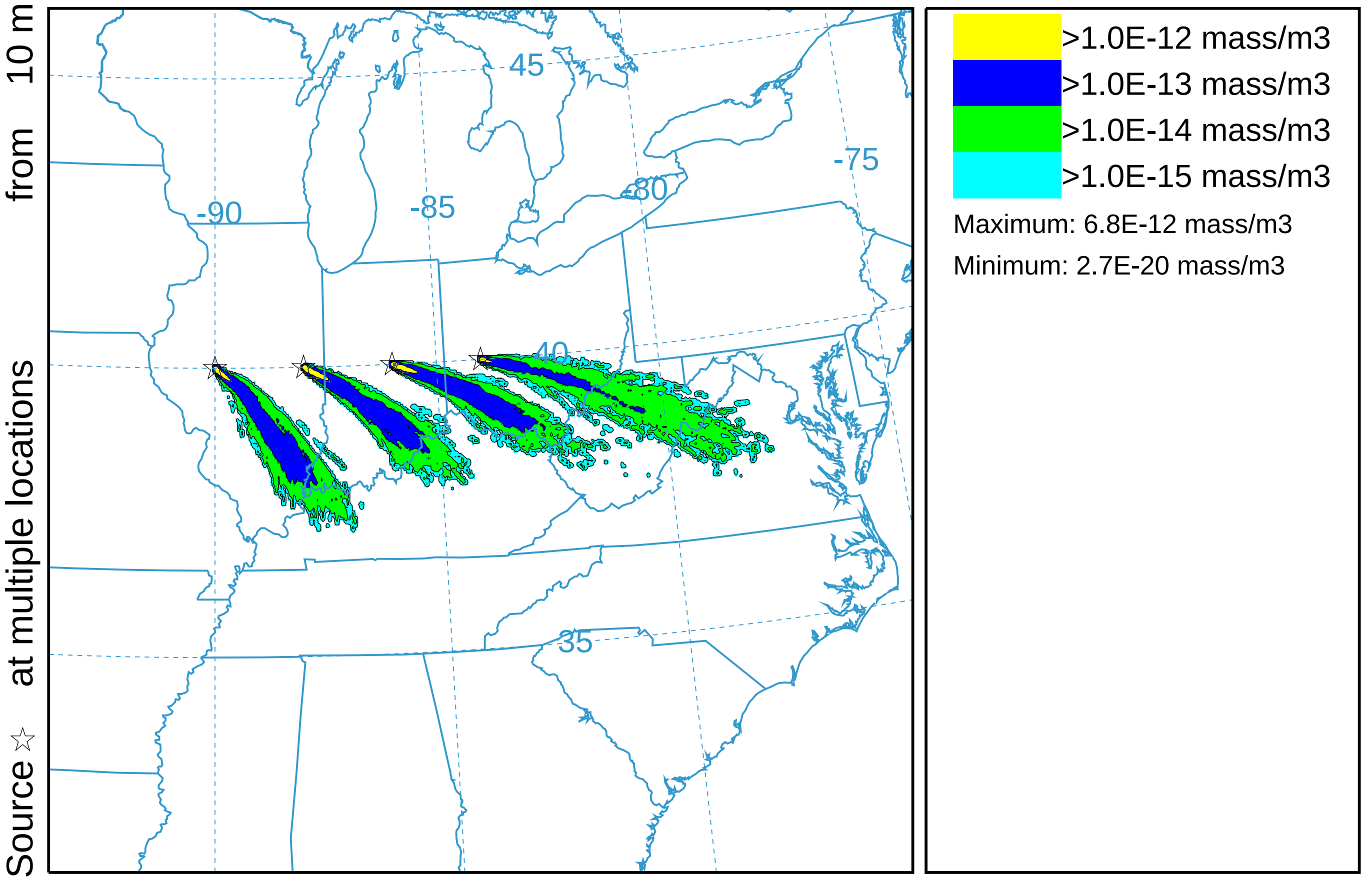
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)

TEST Release started at 0000 16 Oct 95 (UTC)



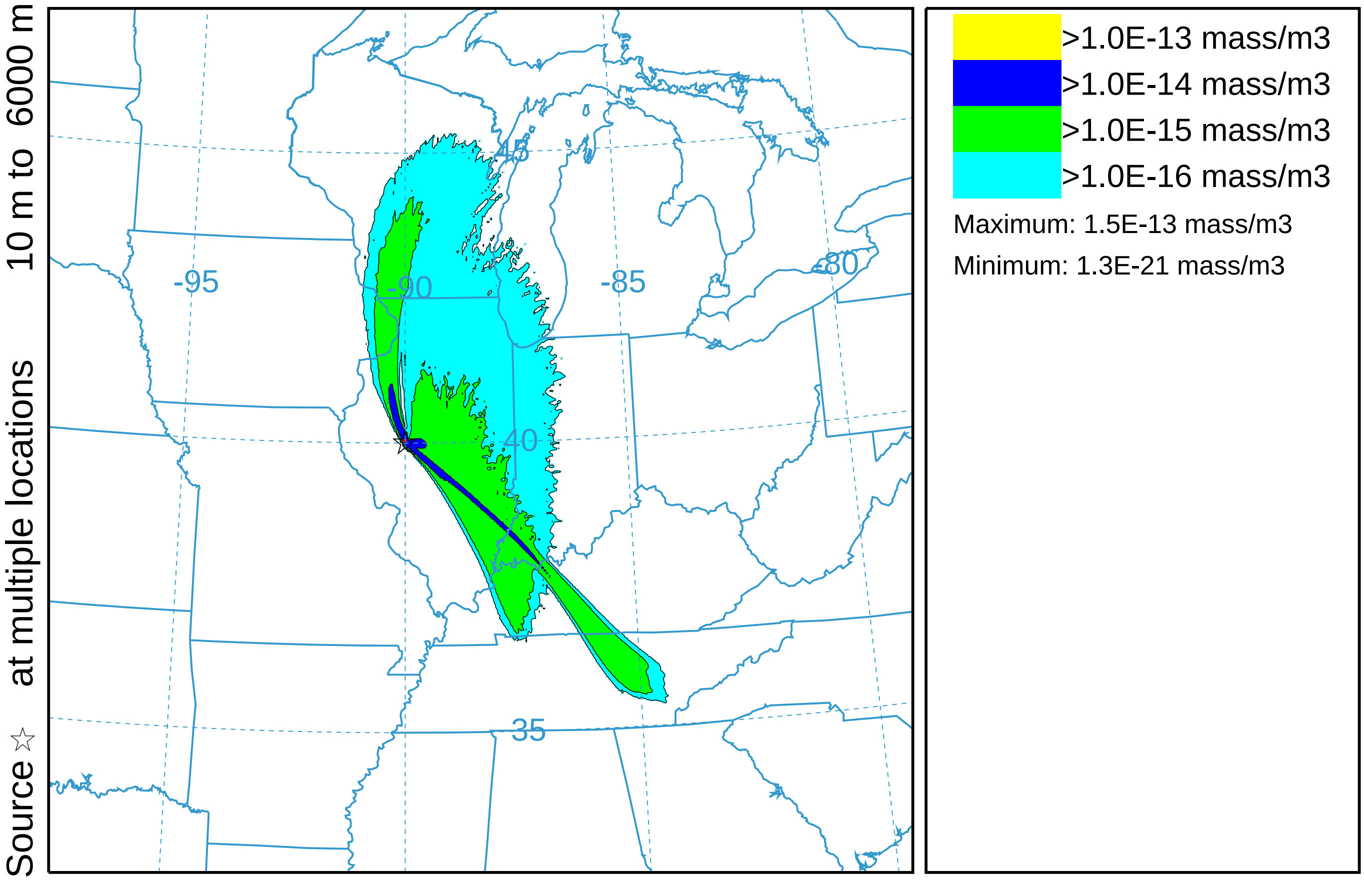

Simple 4-point Moving Line Source (022)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
TEST Release started at 0000 16 Oct 95 (UTC)
Source ☆ 40.000 N 90.000 W from 10 m
42
40
38
36
-92
-90
-88
-86
-84
-82
>1.0E-12 mass/m3
>1.0E-13 mass/m3
>1.0E-14 mass/m3
>1.0E-15 mass/m3
Maximum: 6.6E-12 mass/m3
Minimum: 8.7E-19 mass/m3
NGM METEOROLOGICAL DATA

Simultaneous Multiple Sources in Space (023)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
TEST Release started at 0000 16 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

Simultaneous Multiple Sources in Height (024)
Concentration (mass/m3) averaged between 0 m and 10000 m
Integrated from 0000 17 Oct to 1200 17 Oct 95 (UTC)
TEST Release started at 0000 17 Oct 95 (UTC)



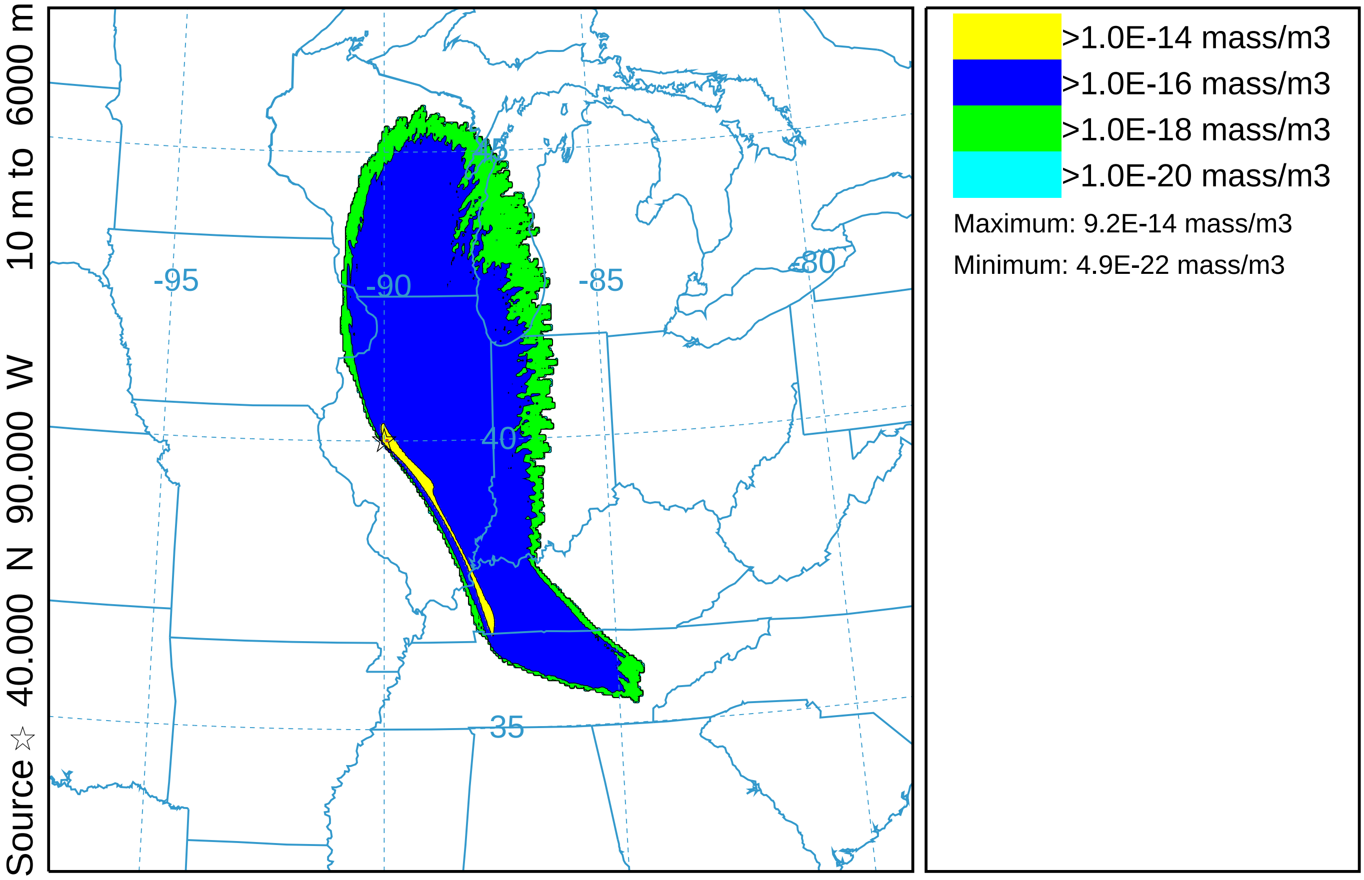
NGM METEOROLOGICAL DATA

Vertical Line Source (025)

Concentration (mass/m3) averaged between 0 m and 10000 m

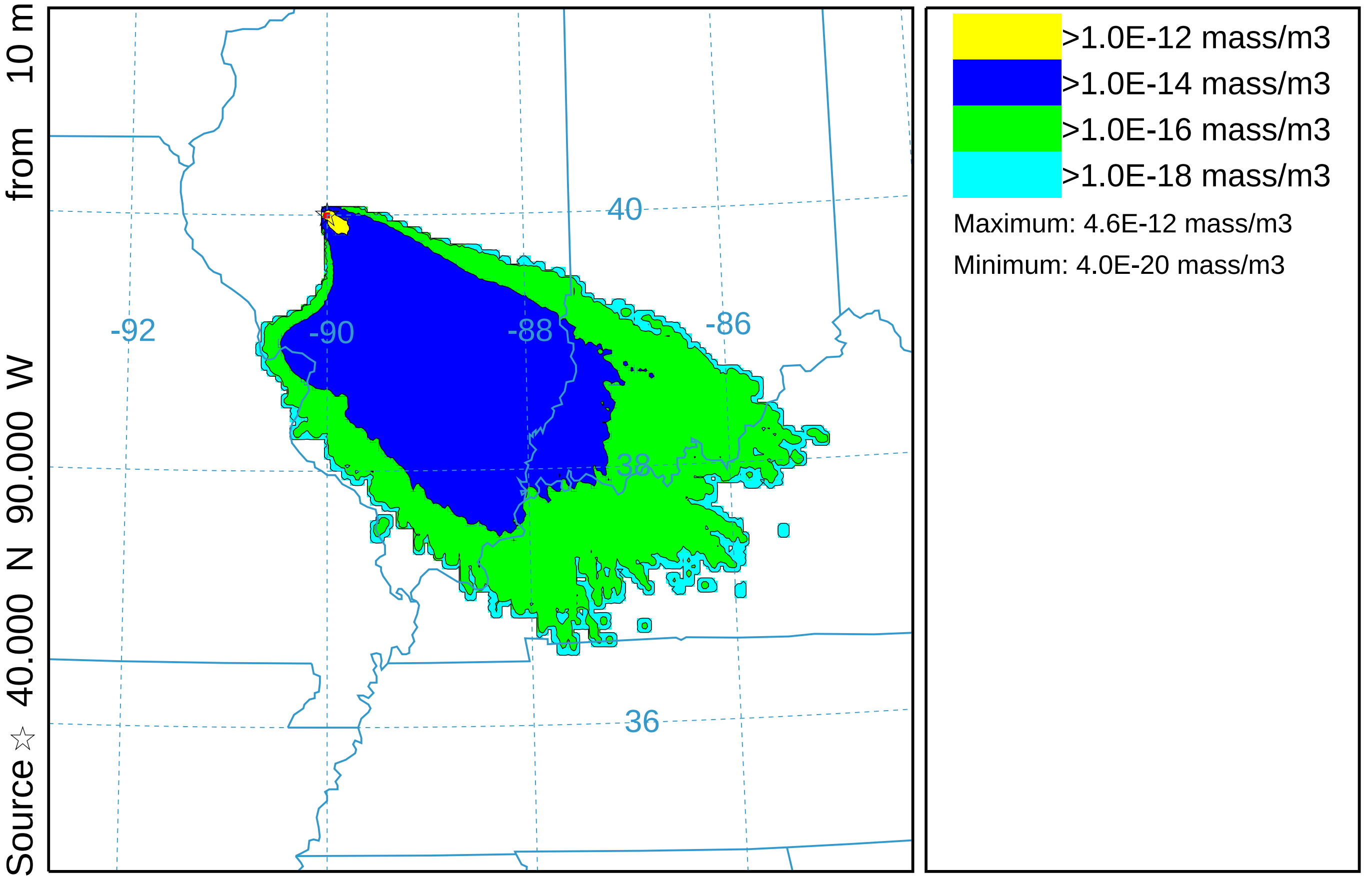
Integrated from 0000 17 Oct to 1200 17 Oct 95 (UTC)

TEST Release started at 0000 17 Oct 95 (UTC)



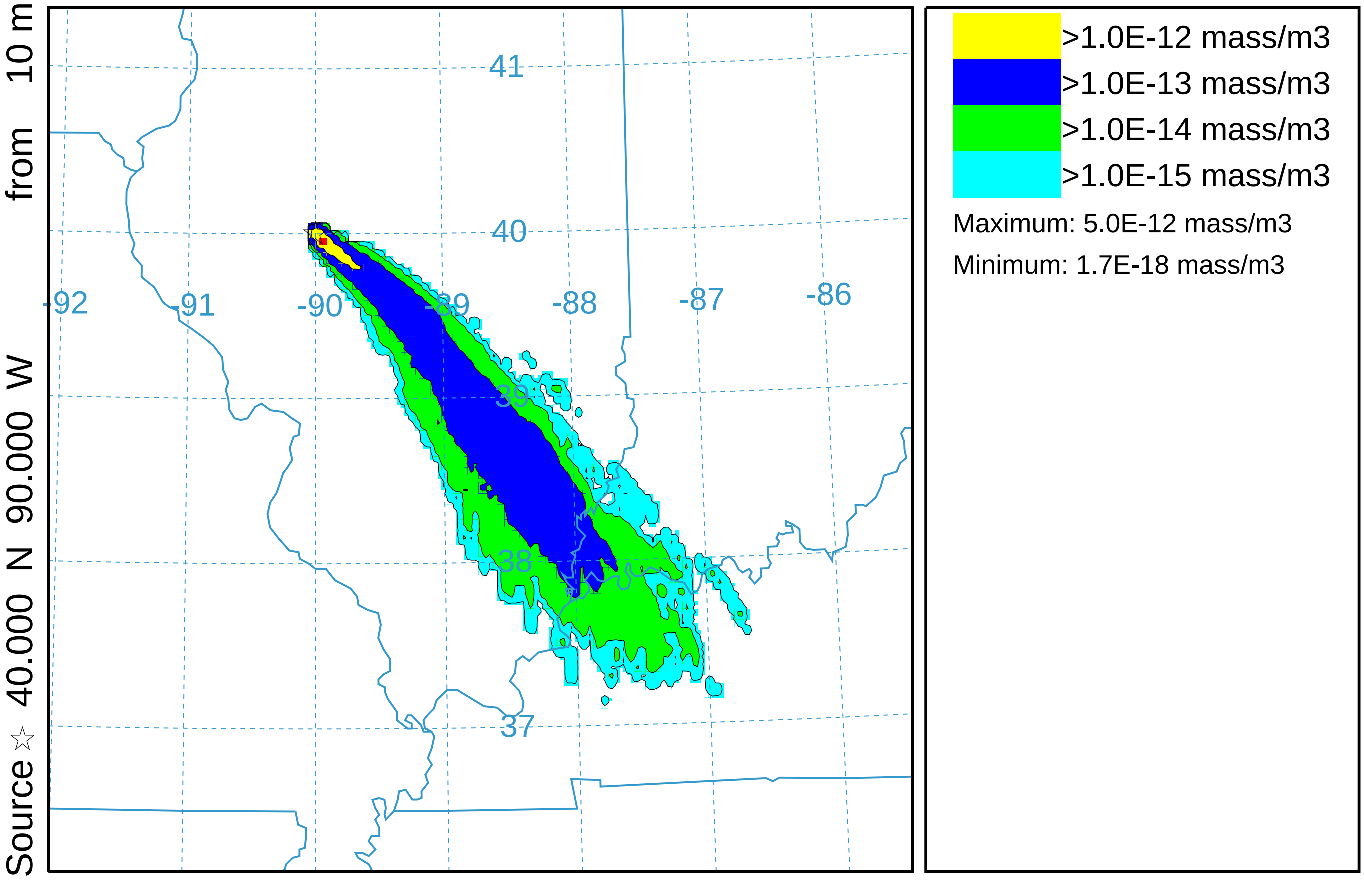
NGM METEOROLOGICAL DATA

9-Member Meteorological Ensemble Means (026)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
MEAN Release started at 0000 16 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

9-Member Variance Ensemble Means (027)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
MEAN Release started at 0000 16 Oct 95 (UTC)



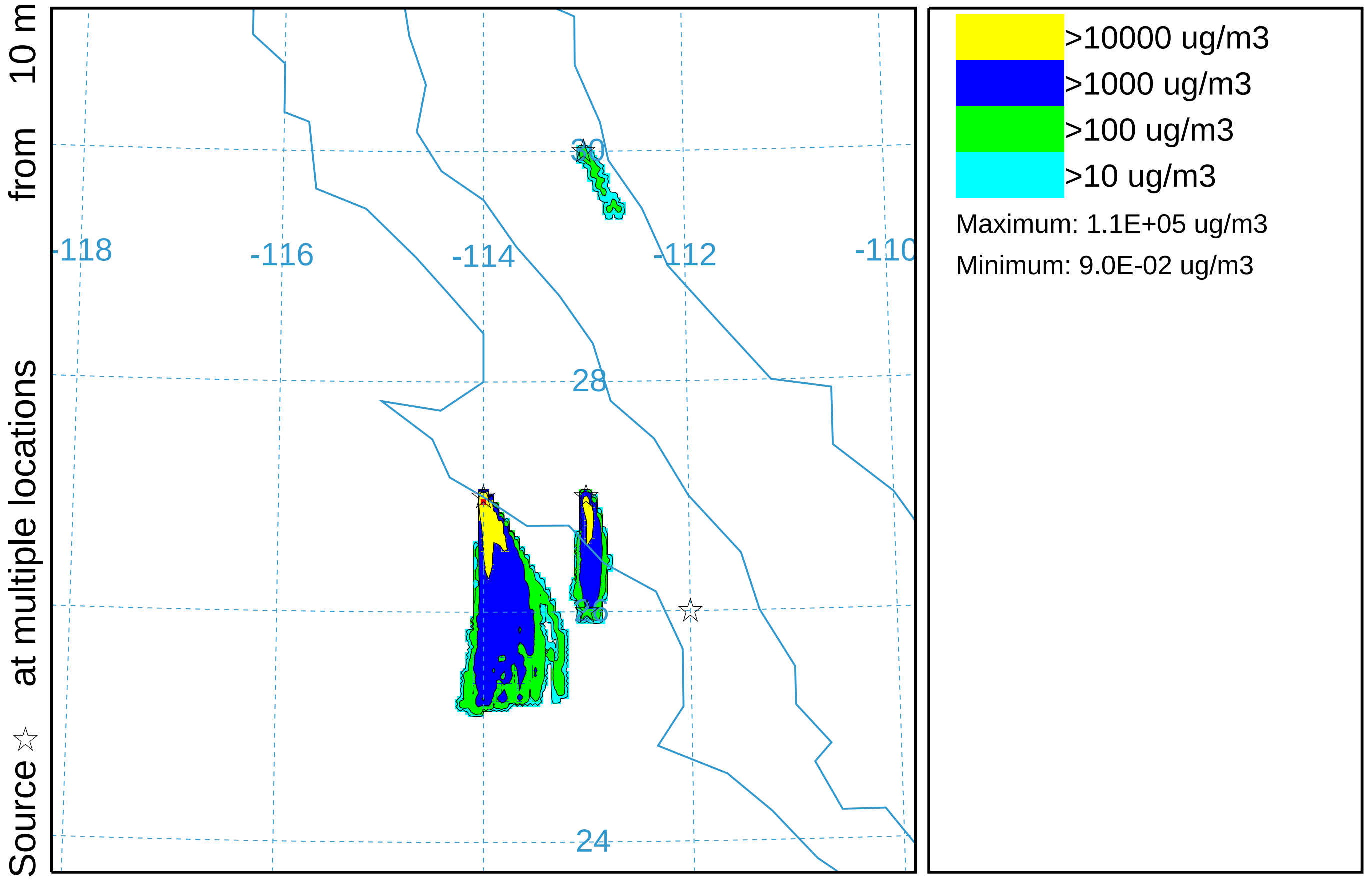
NGM METEOROLOGICAL DATA

Dust Storm Emission Algorithm (028)

Concentration (ug/m3) averaged between 0 m and 100 m

Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)

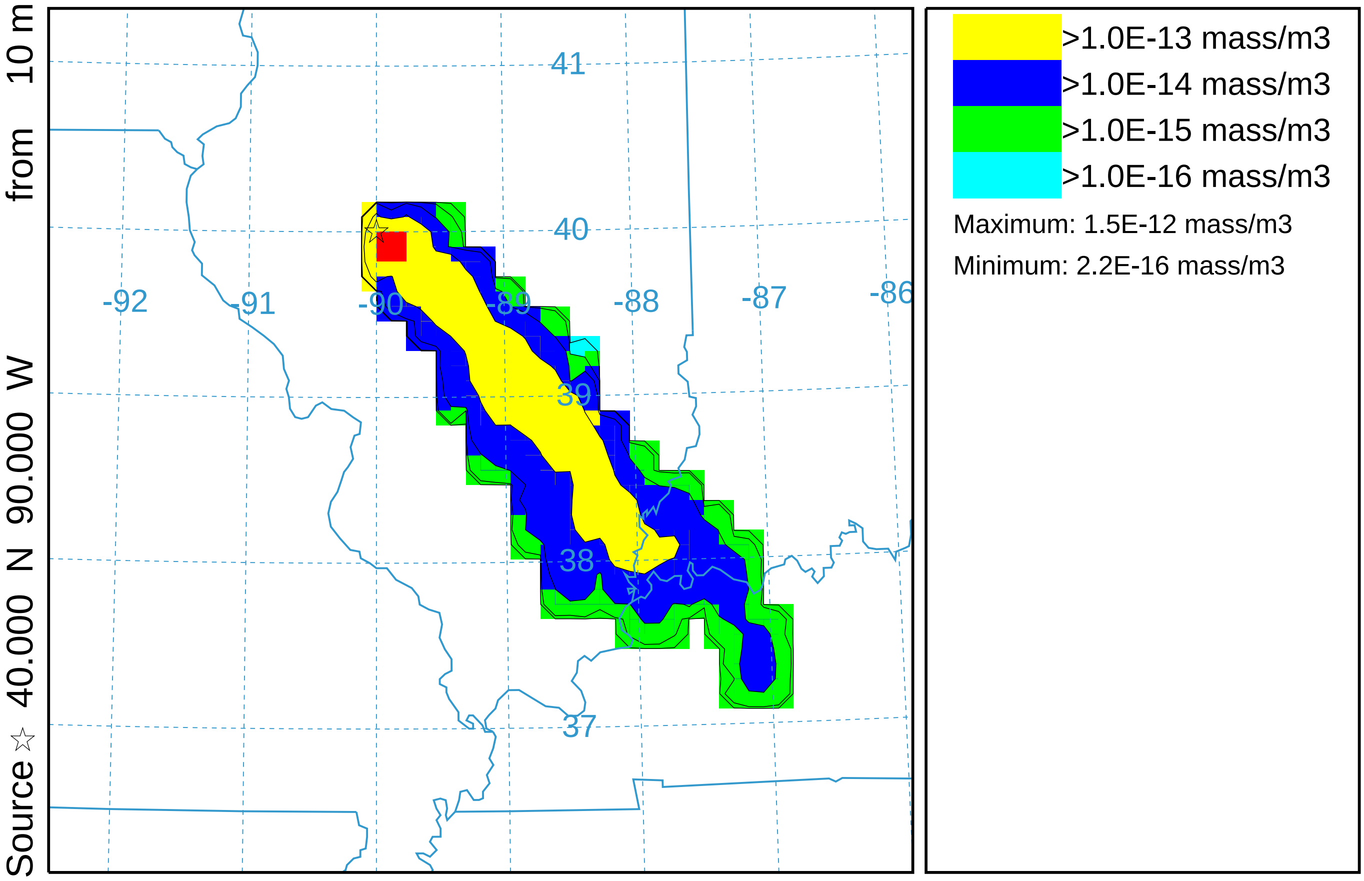
PM10 Release started at 0000 16 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

Matrix - Downwind Receptors (029)

Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)
C(R) Release started at 0000 16 Oct 95 (UTC)



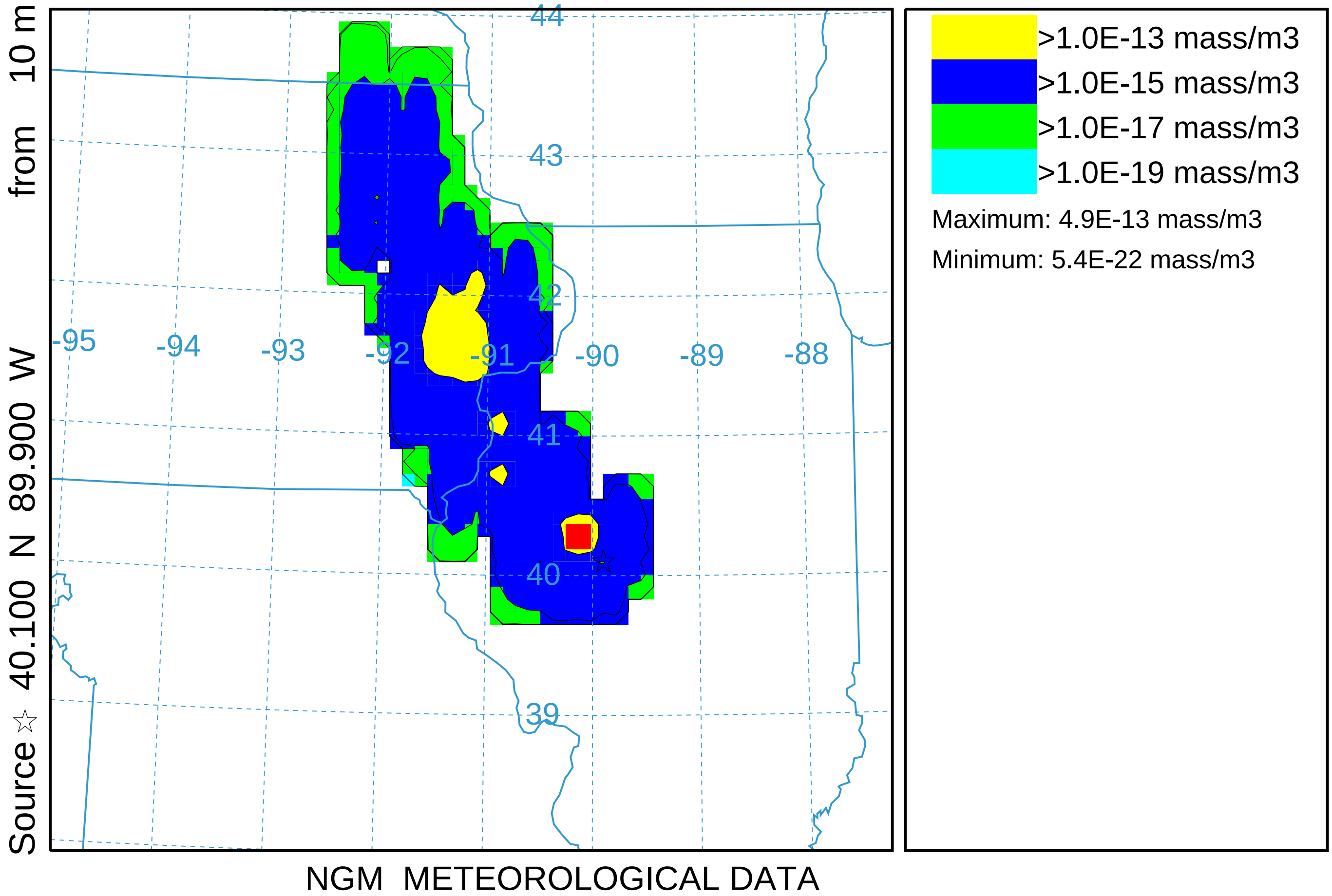
NGM METEOROLOGICAL DATA

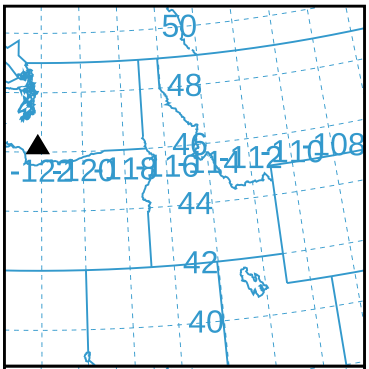
Matrix - Upwind Sources (030)

Concentration (mass/m3) averaged between 0 m and 100 m

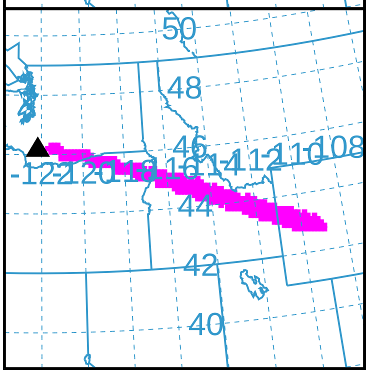
Integrated from 0000 16 Oct to 1200 16 Oct 95 (UTC)

C(S) Release started at 0000 16 Oct 95 (UTC)

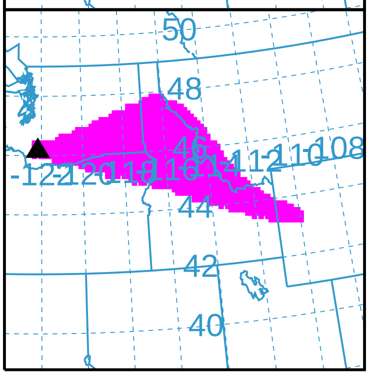




FL550
FL350

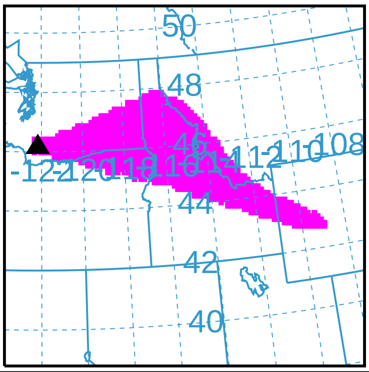


FL350
FL200



FL200
SURFACE

VALID 1200Z 17 OCT 95 (ERUPTION+012H00M)



FL550
SURFACE
(COMPOSITE)

NOAA HYSPLIT GUIDANCE

Created: 0333 GMT 20211208

▲ TEST-131 N4612W12211

SUMMIT 8363 FT

ERUPTION 0000Z 17 OCT 95

DURATION 1 HR

ASH COLUMN FL200

■ ASH CLOUD

****

ALERT UPDATE

****

****

NGM CYCLE

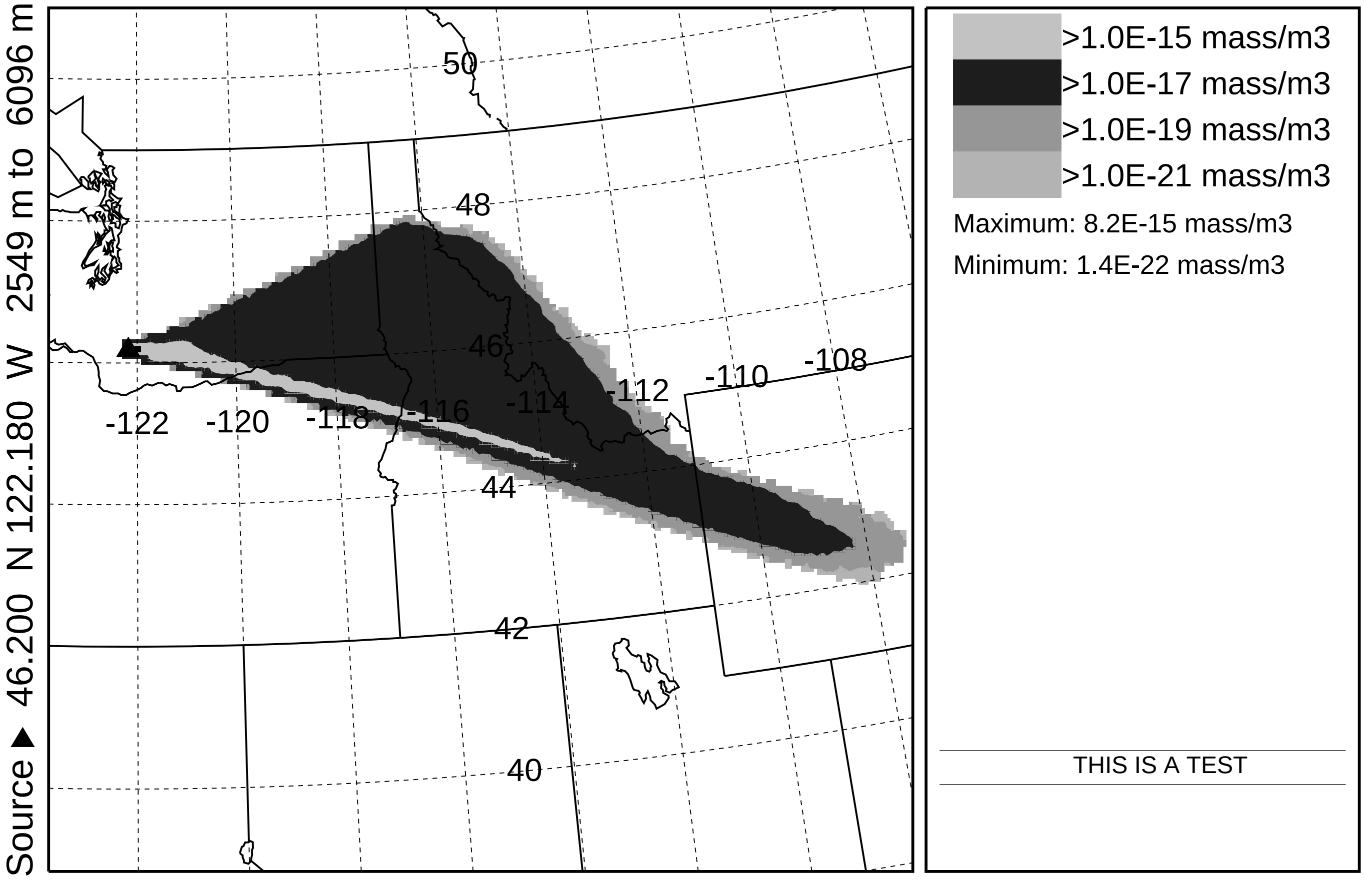
12Z 17 OCT 95

SEE VOLCANIC ASH SIGMETS AND ADVISORIES

Wet deposition: Yes

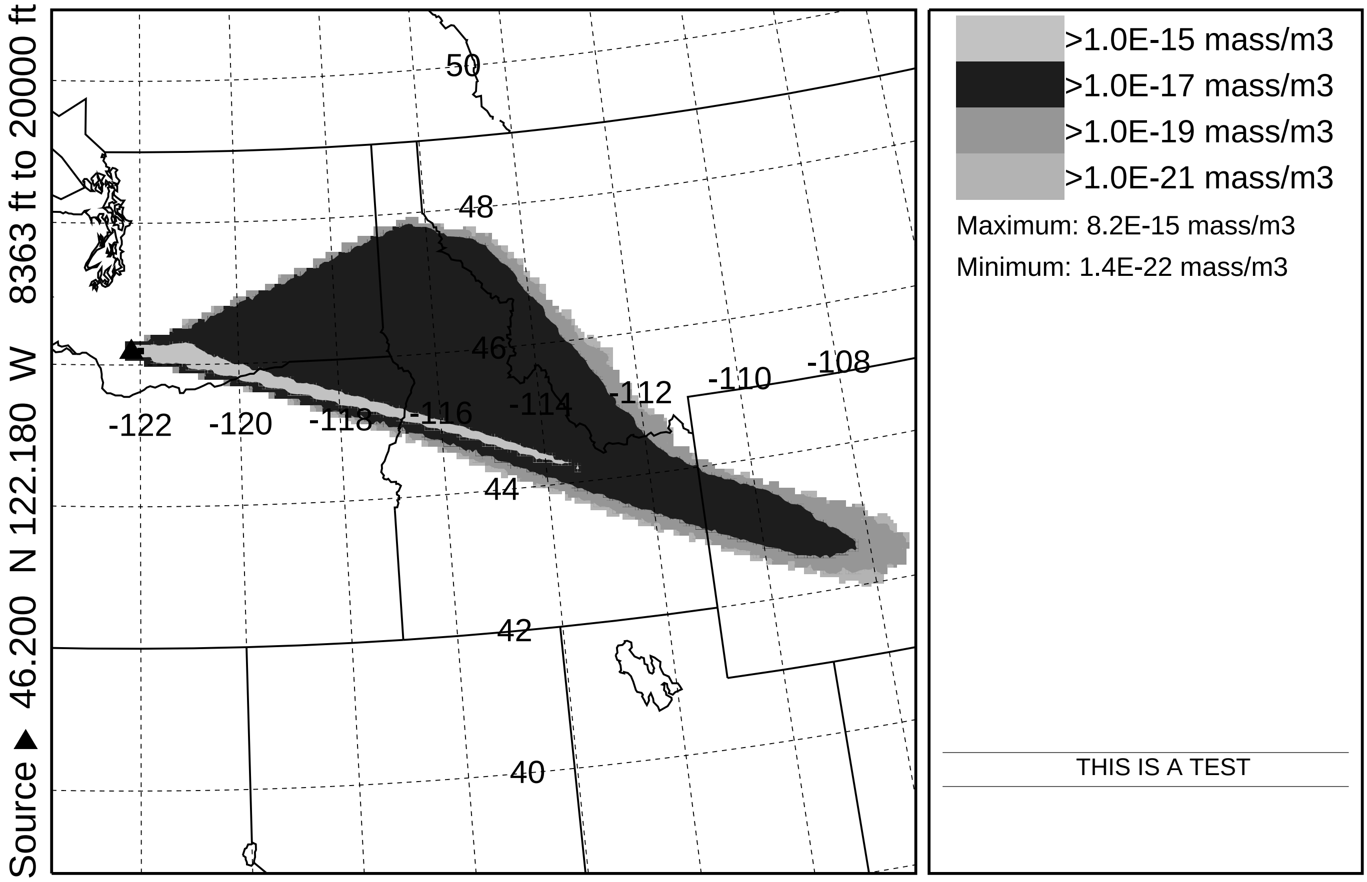
Z-time information (use lower-case):
WWW.SRH.NOAA.GOV/JETSTREAM/SYNOPTIC/TIME.HTM

Volcanic ash (231)
Concentration (mass/m3) averaged between 0 m and 16764 m
Integrated from 0000 17 Oct to 1200 17 Oct 95 (UTC)
SUM Release started at 0000 17 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

Volcanic ash (altitude in feet) (331)
Concentration (mass/m3) averaged between 0 ft and 55000 ft
Integrated from 0000 17 Oct to 1200 17 Oct 95 (UTC)
SUM Release started at 0000 17 Oct 95 (UTC)

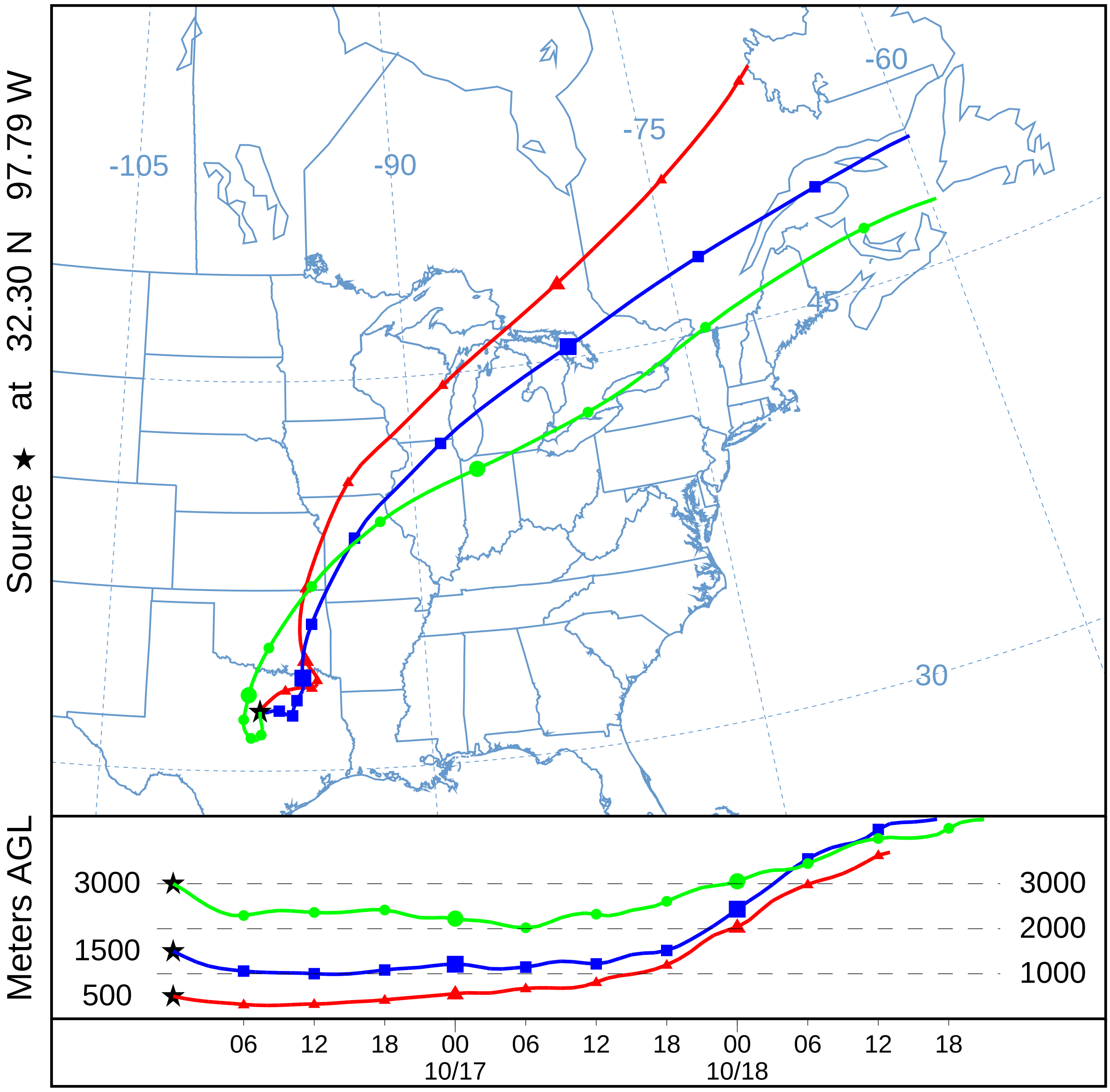


NGM METEOROLOGICAL DATA

RSMC Trajectory (032)

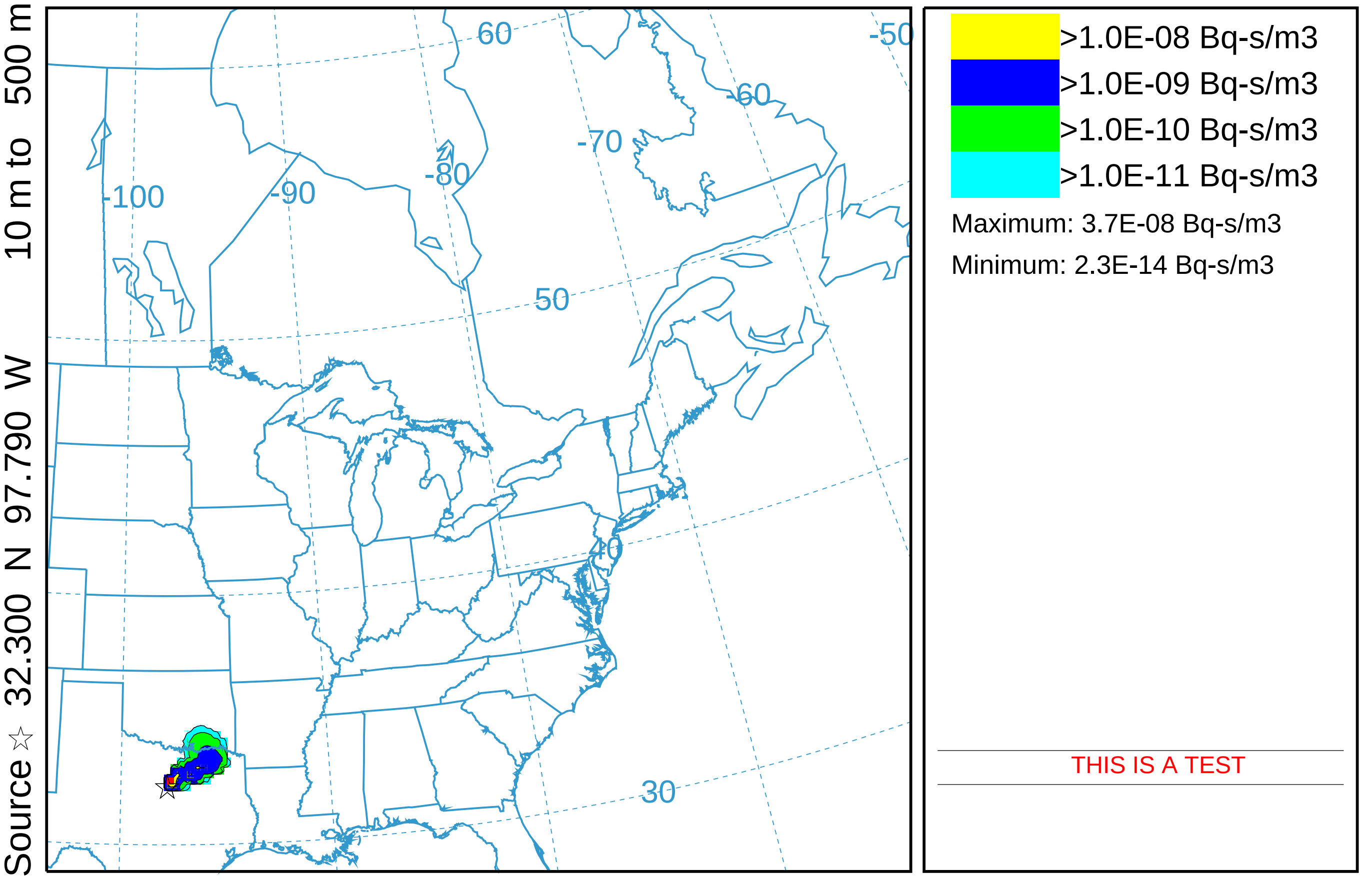
Forward trajectories starting at 0000 UTC 16 Oct 95

NGM Meteorological Data



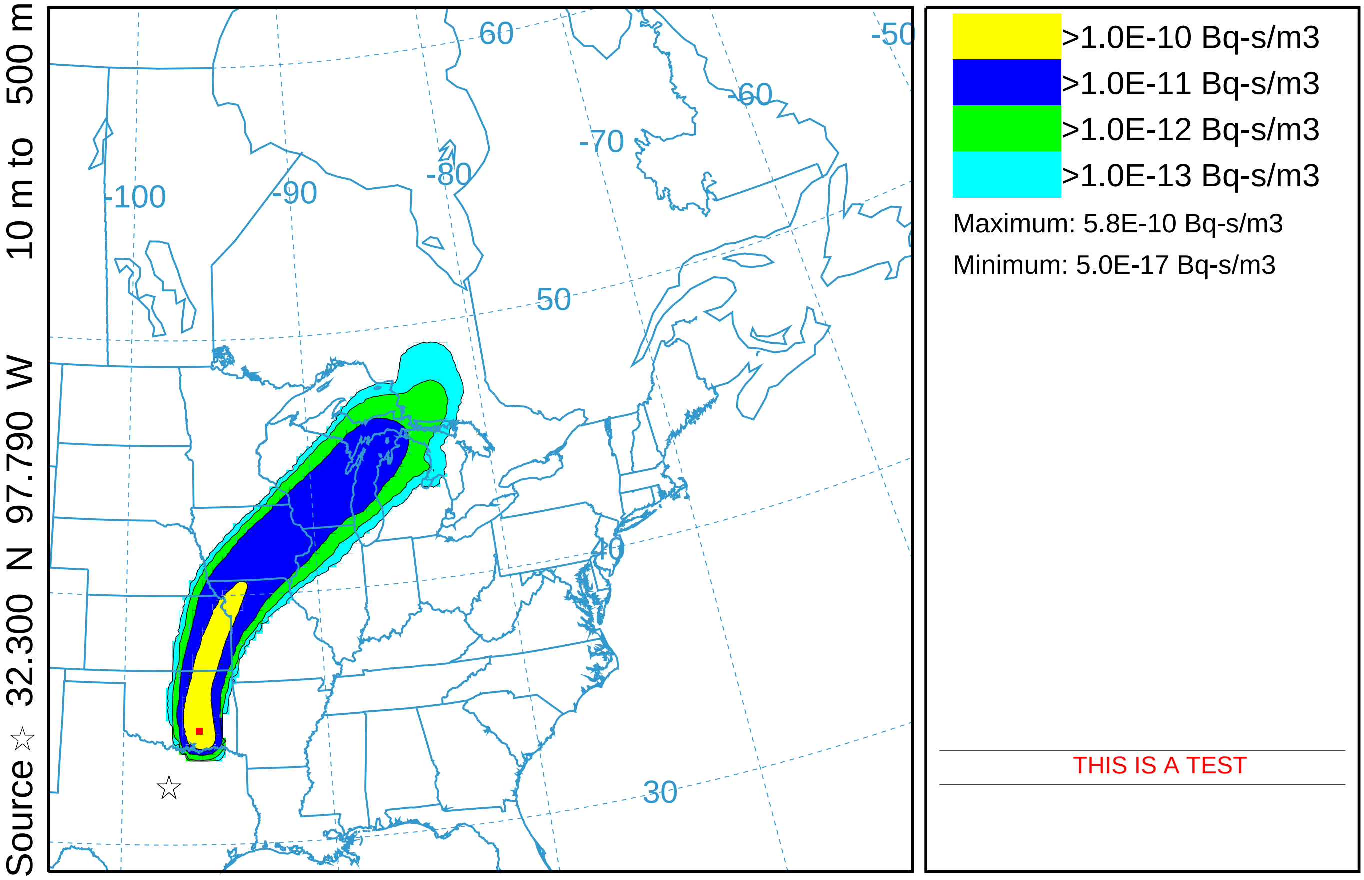
RSMC Dispersion (033)

Exposure (Bq-s/m3) averaged between 0 m and 500 m
Integrated from 0000 16 Oct to 0000 17 Oct 95 (UTC)
C137 Release started at 0000 16 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

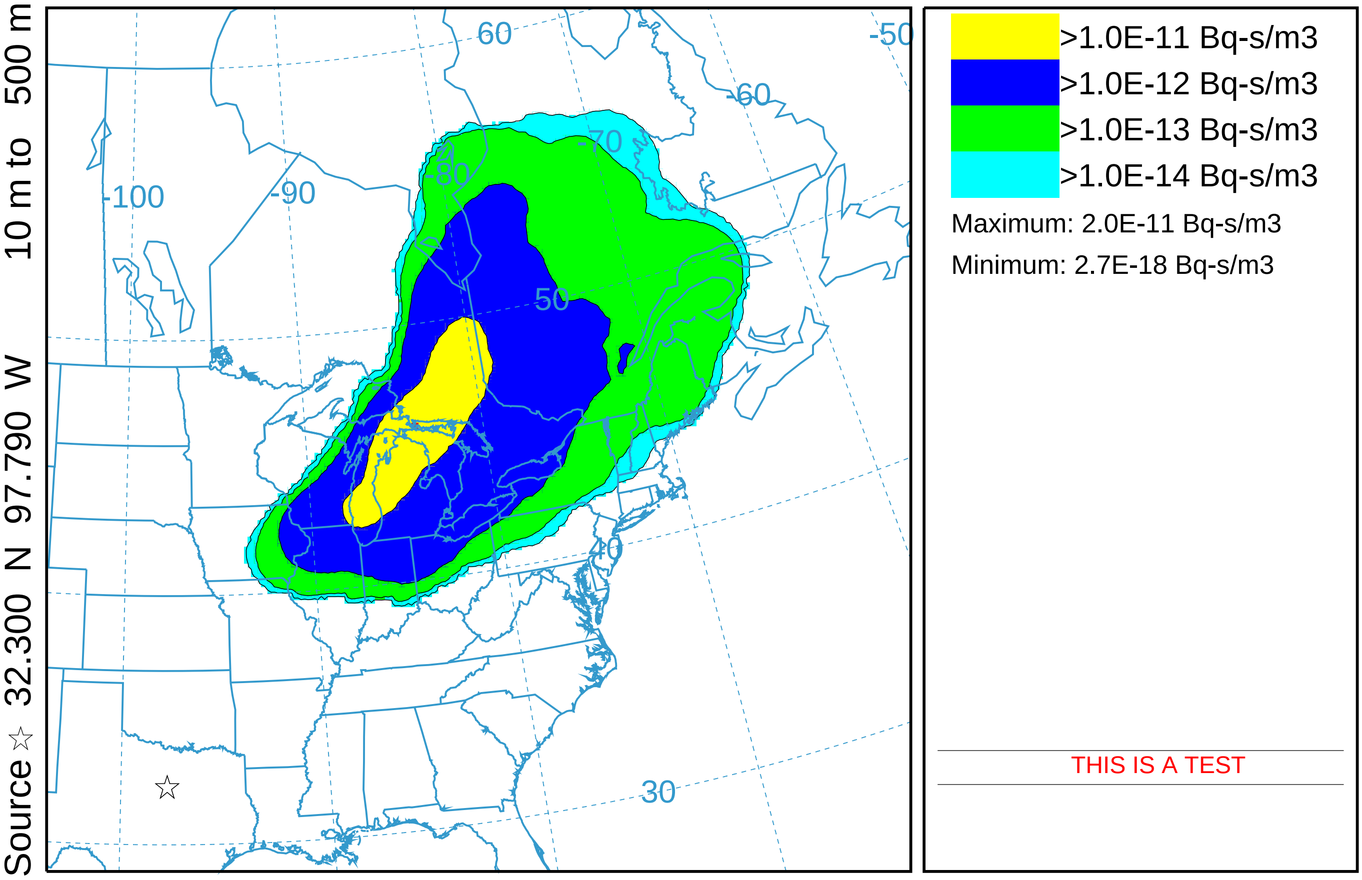
RSMC Dispersion (033)
Exposure (Bq-s/m3) averaged between 0 m and 500 m
Integrated from 0000 17 Oct to 0000 18 Oct 95 (UTC)
C137 Release started at 0000 16 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

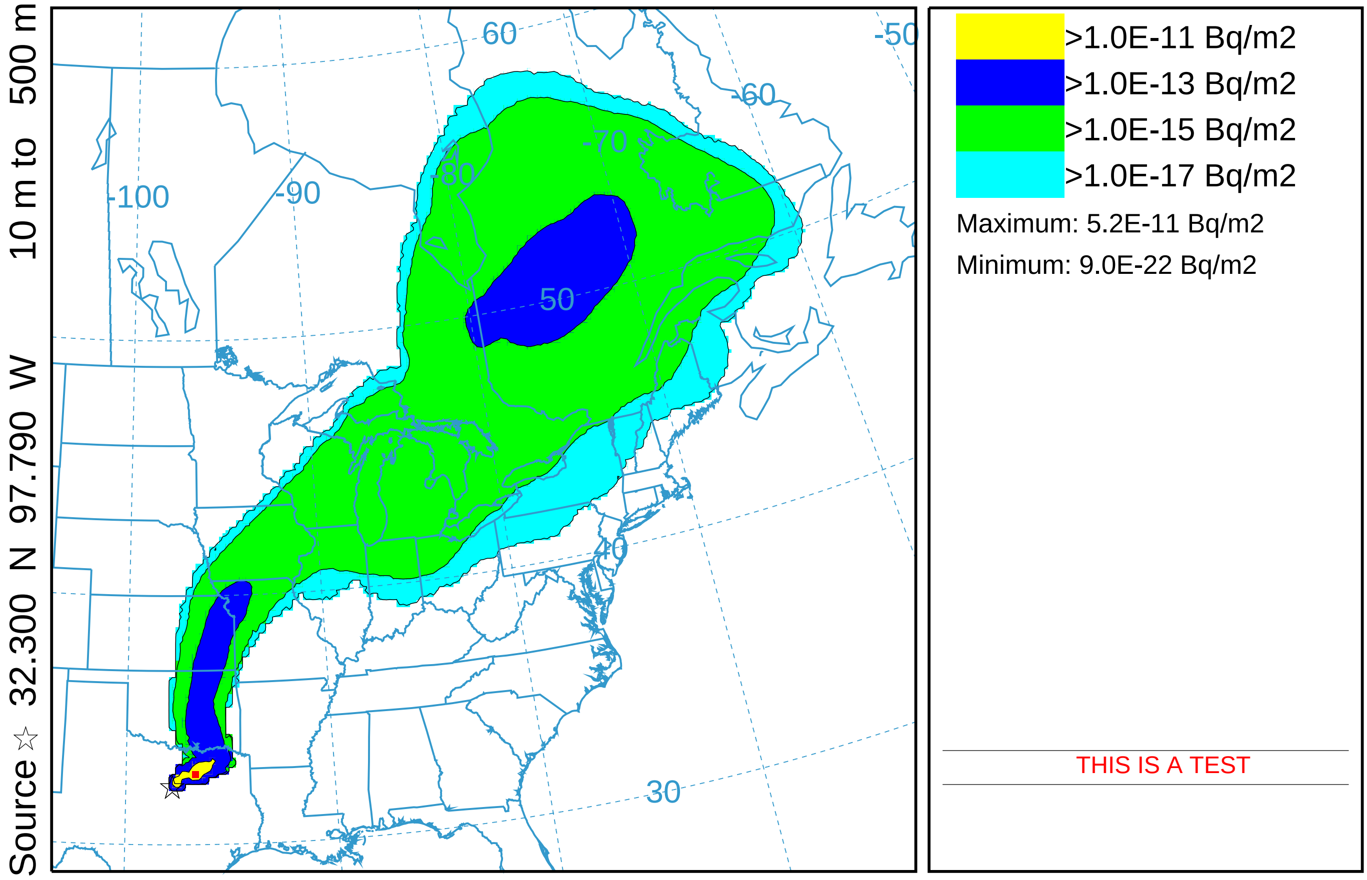
RSMC Dispersion (033)

Exposure (Bq-s/m3) averaged between 0 m and 500 m
Integrated from 0000 18 Oct to 0000 19 Oct 95 (UTC)
C137 Release started at 0000 16 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

RSMC Dispersion (033)
Deposition (Bq/m2) at ground-level
Integrated from 0000 16 Oct to 0000 19 Oct 95 (UTC)
C137 Release started at 0000 16 Oct 95 (UTC)



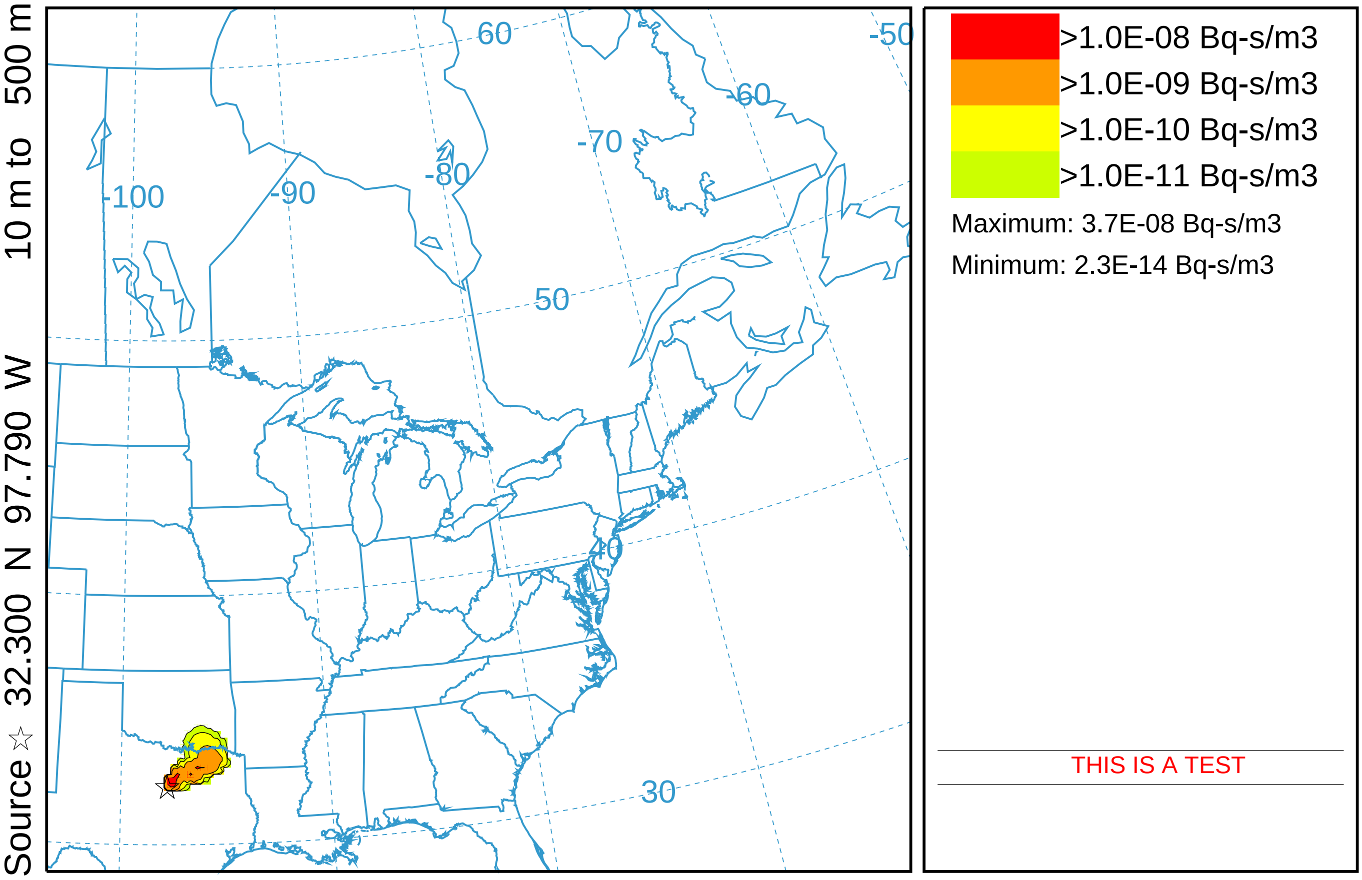
NGM METEOROLOGICAL DATA

RSMC Dispersion With Dynamic Colors (133)

Exposure (Bq-s/m3) averaged between 0 m and 500 m

Integrated from 0000 16 Oct to 0000 17 Oct 95 (UTC)

C137 Release started at 0000 16 Oct 95 (UTC)



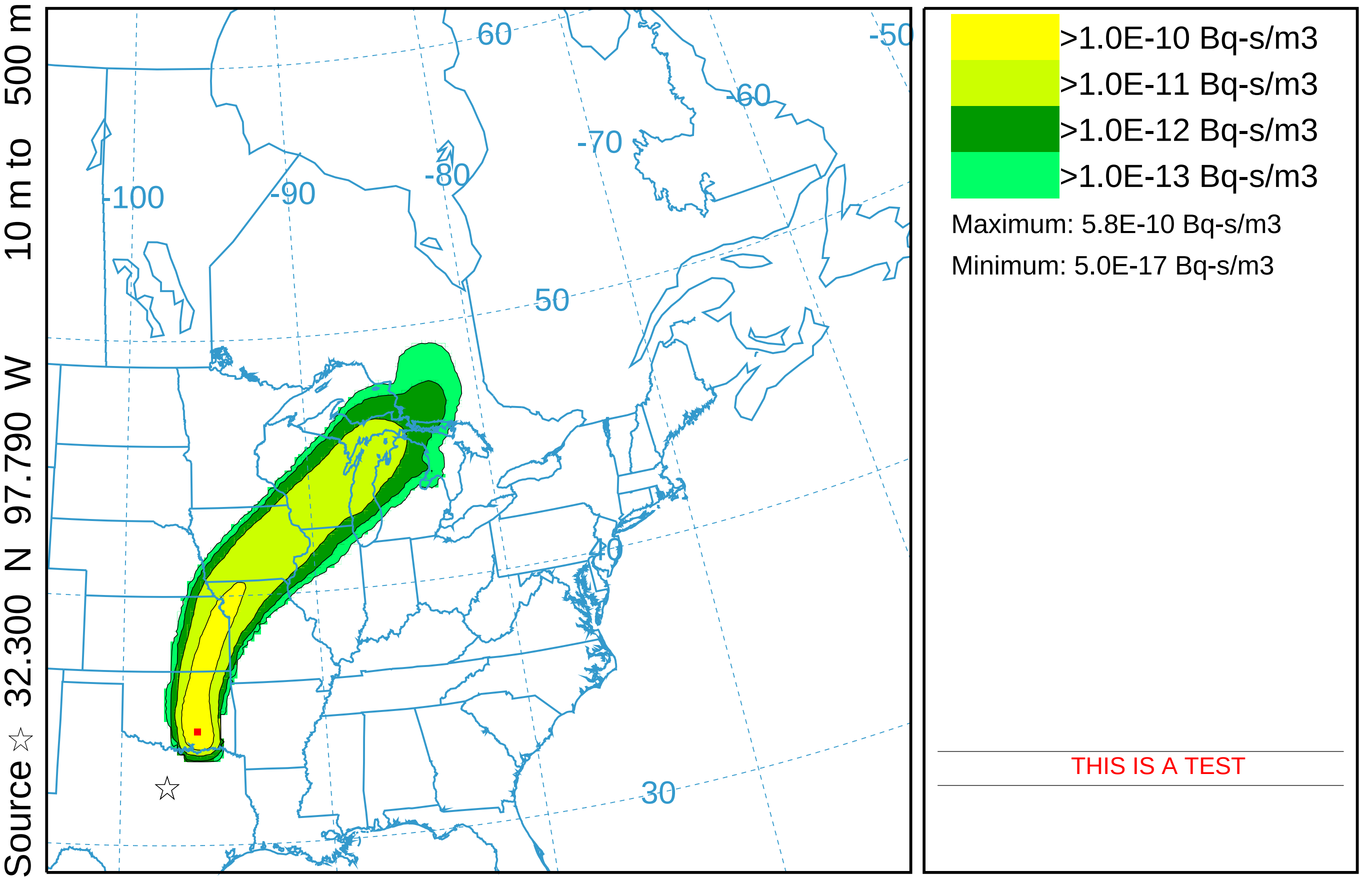
NGM METEOROLOGICAL DATA

RSMC Dispersion With Dynamic Colors (133)

Exposure (Bq-s/m3) averaged between 0 m and 500 m

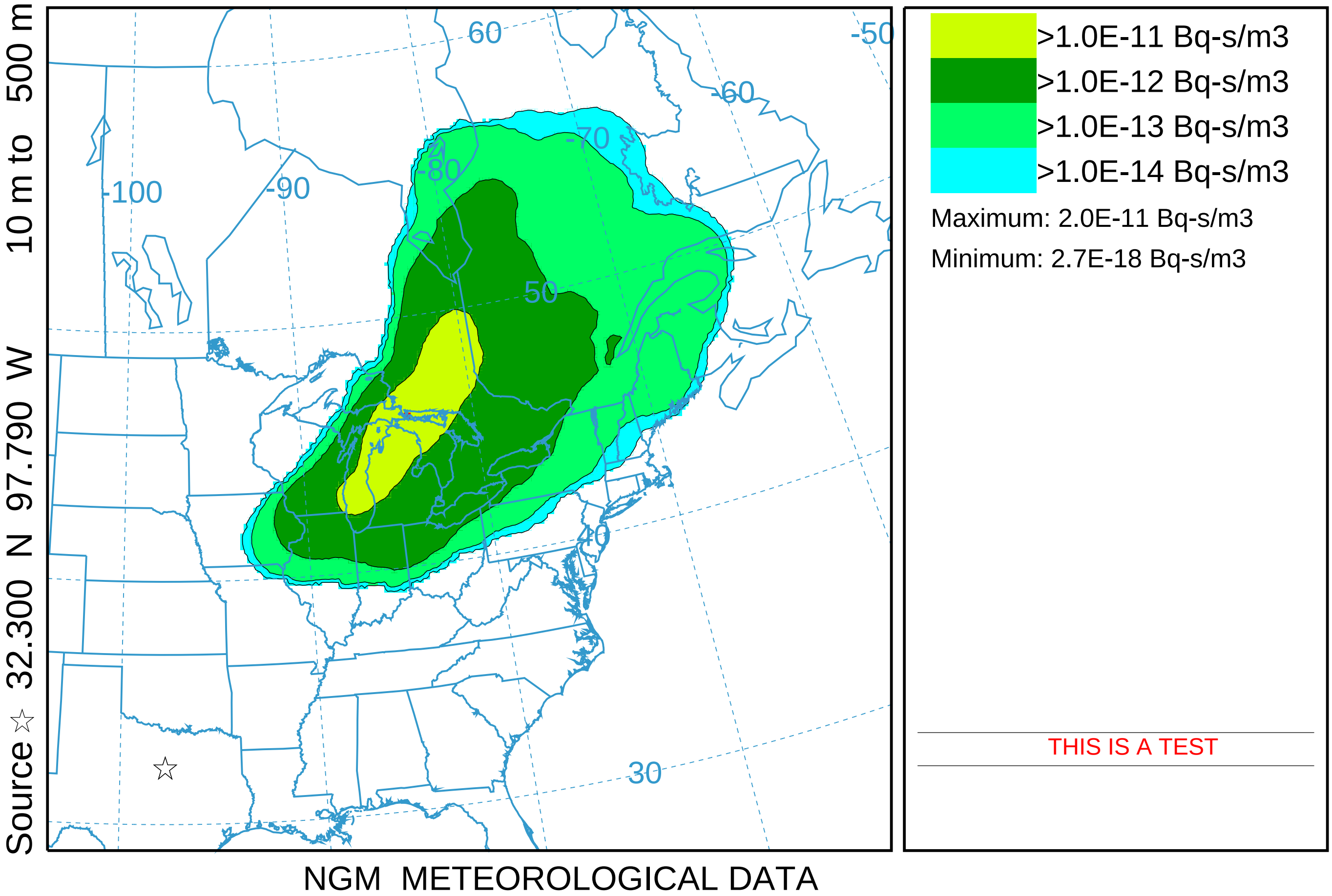
Integrated from 0000 17 Oct to 0000 18 Oct 95 (UTC)

C137 Release started at 0000 16 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

RSMC Dispersion With Dynamic Colors (133)
Exposure (Bq-s/m3) averaged between 0 m and 500 m
Integrated from 0000 18 Oct to 0000 19 Oct 95 (UTC)
C137 Release started at 0000 16 Oct 95 (UTC)

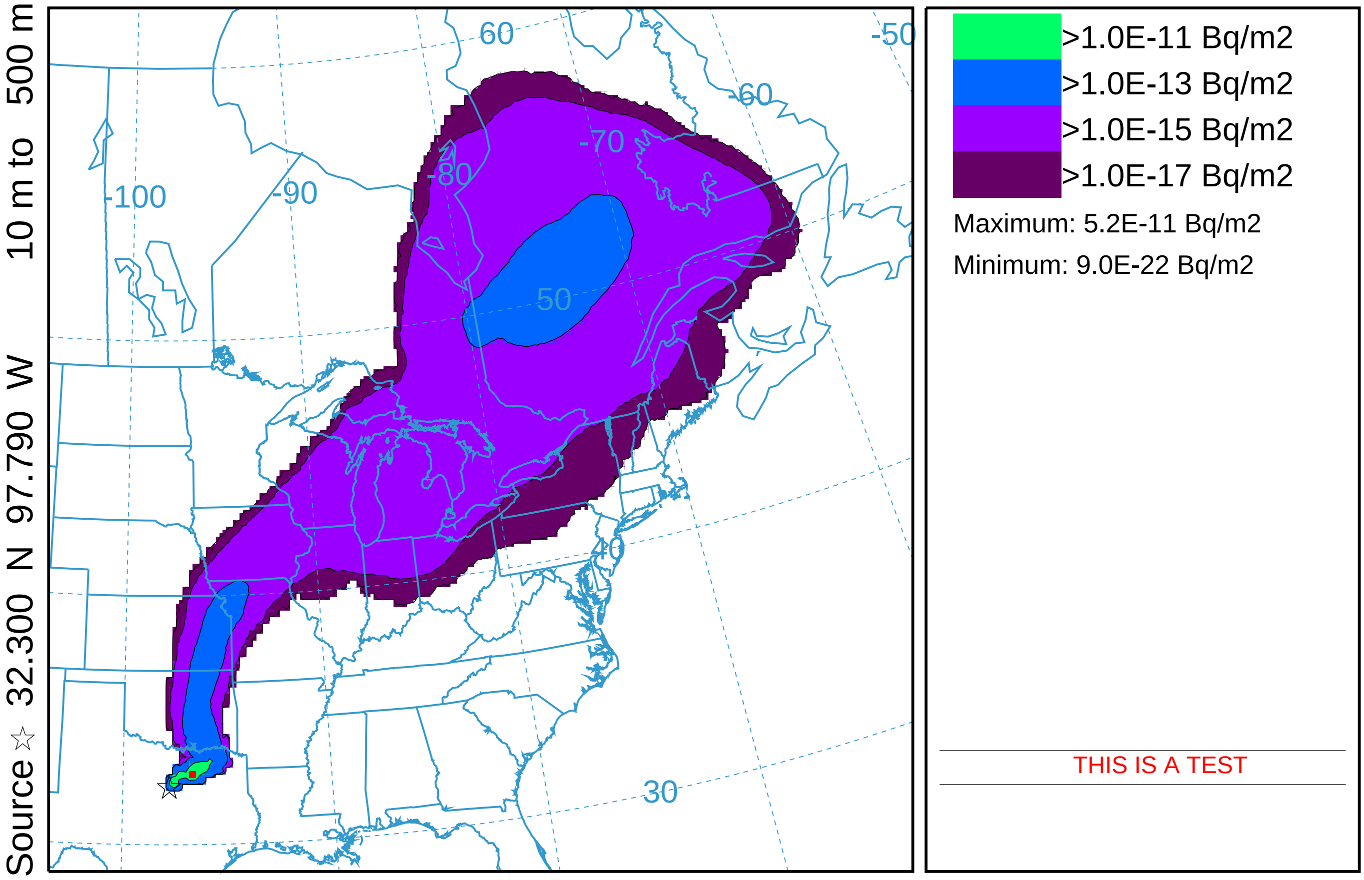


RSMC Dispersion With Dynamic Colors (133)

Deposition (Bq/m2) at ground-level

Integrated from 0000 16 Oct to 0000 19 Oct 95 (UTC)

C137 Release started at 0000 16 Oct 95 (UTC)



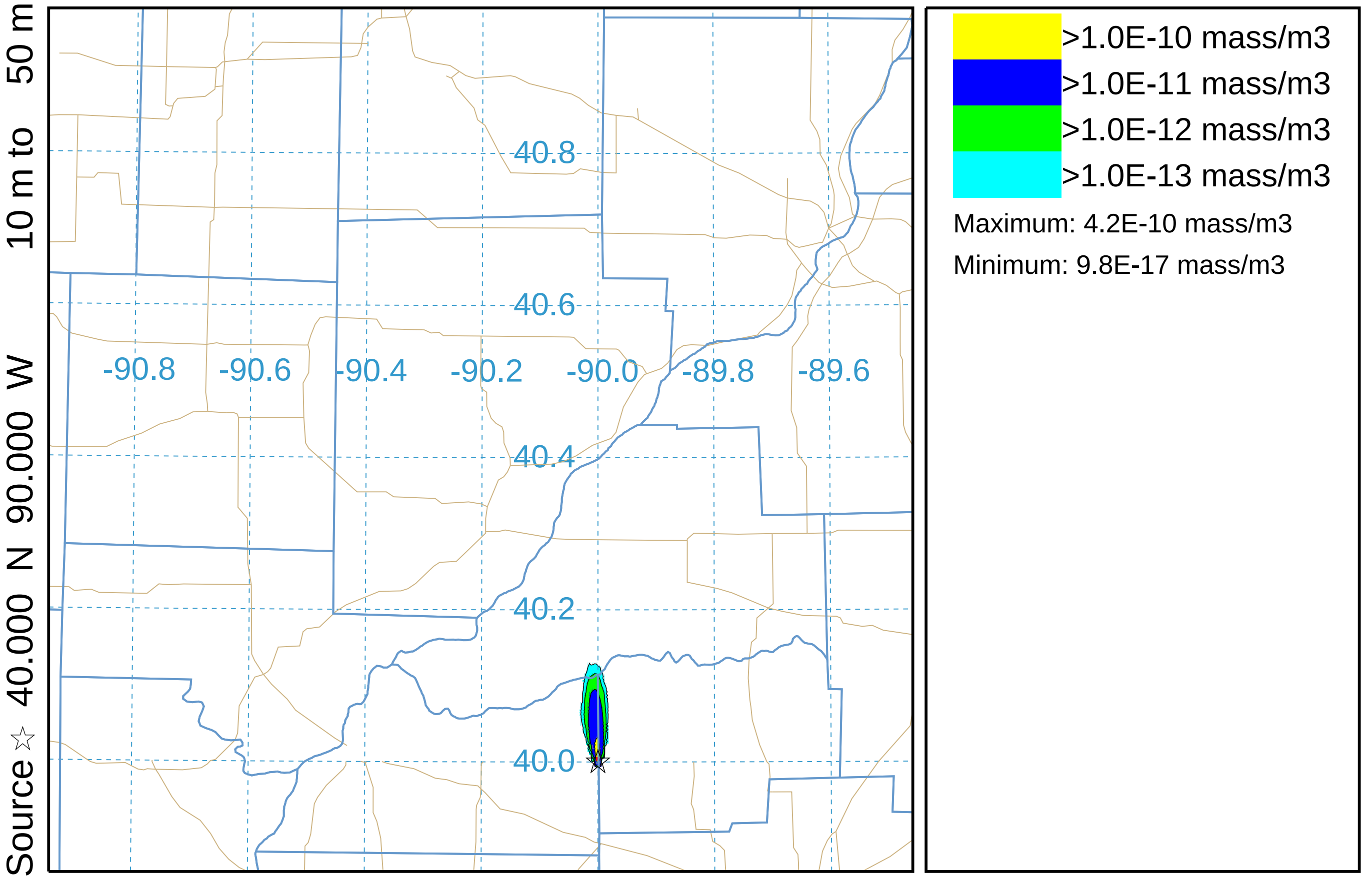
NGM METEOROLOGICAL DATA

HLS fine (134)

Concentration (mass/m3) averaged between 0 m and 500 m

Integrated from 0000 17 Oct to 0100 17 Oct 95 (UTC)

Release started at 0000 17 Oct 95 (UTC)



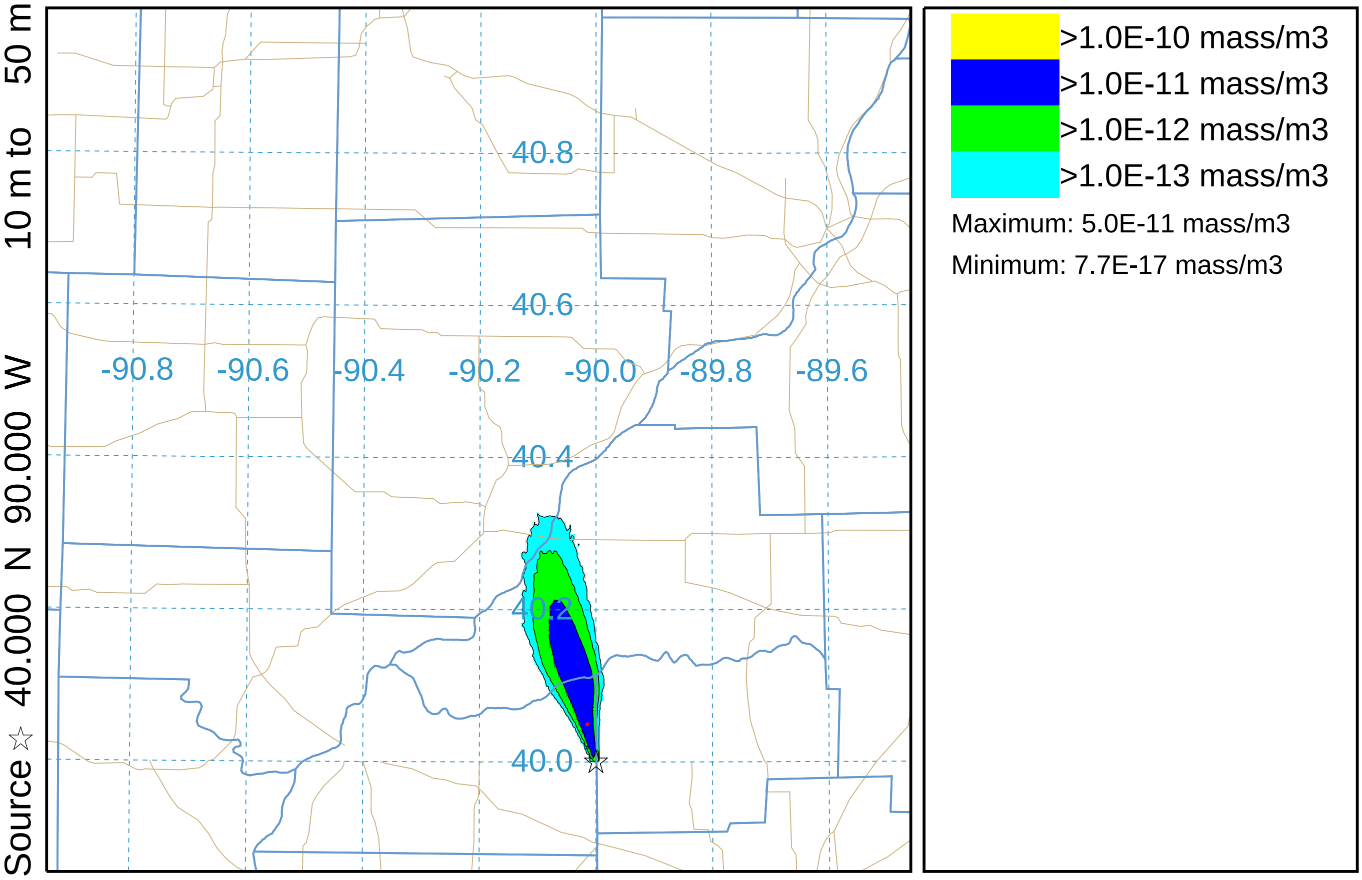
NGM METEOROLOGICAL DATA

HLS fine (134)

Concentration (mass/m3) averaged between 0 m and 500 m

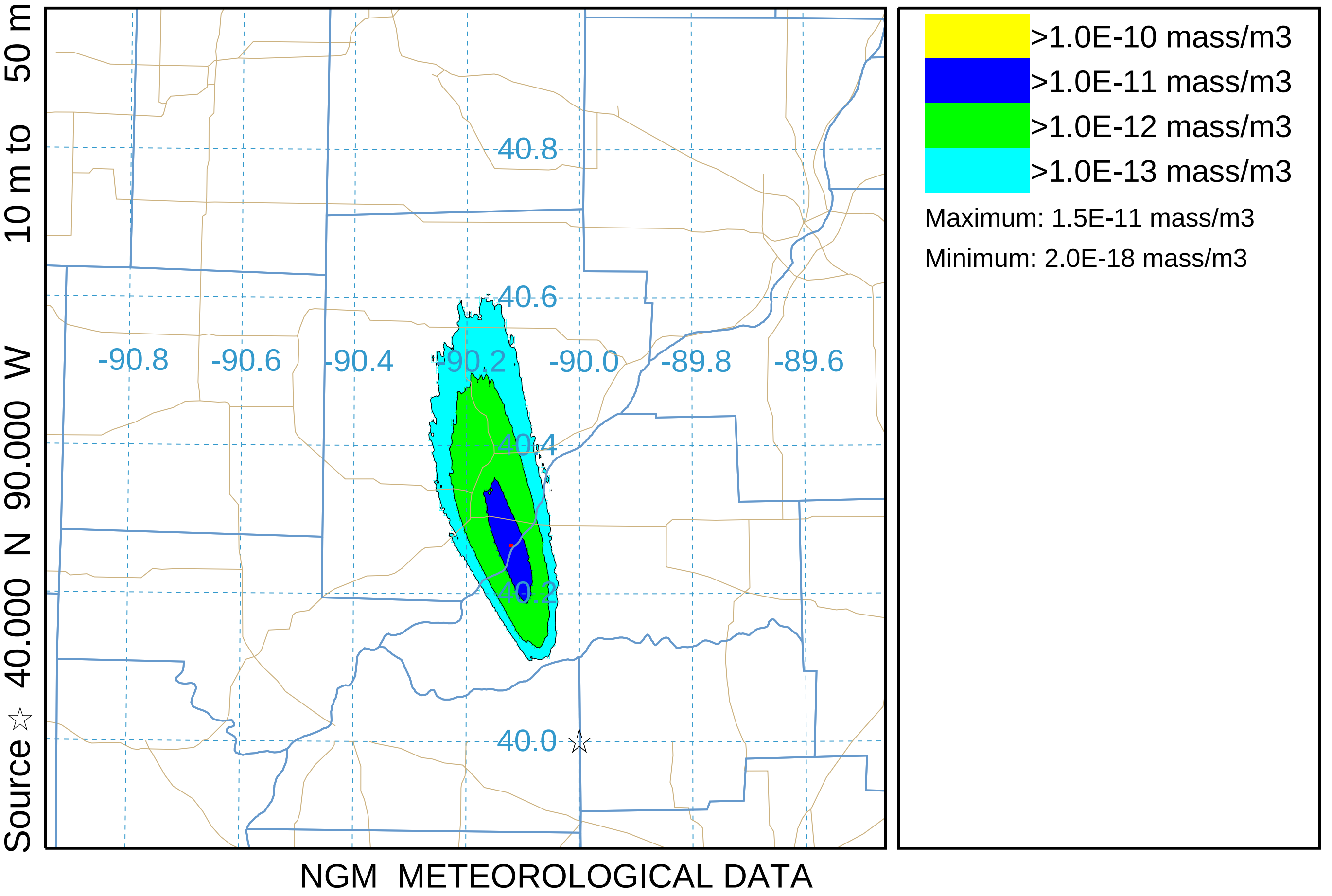
Integrated from 0100 17 Oct to 0200 17 Oct 95 (UTC)

Release started at 0000 17 Oct 95 (UTC)

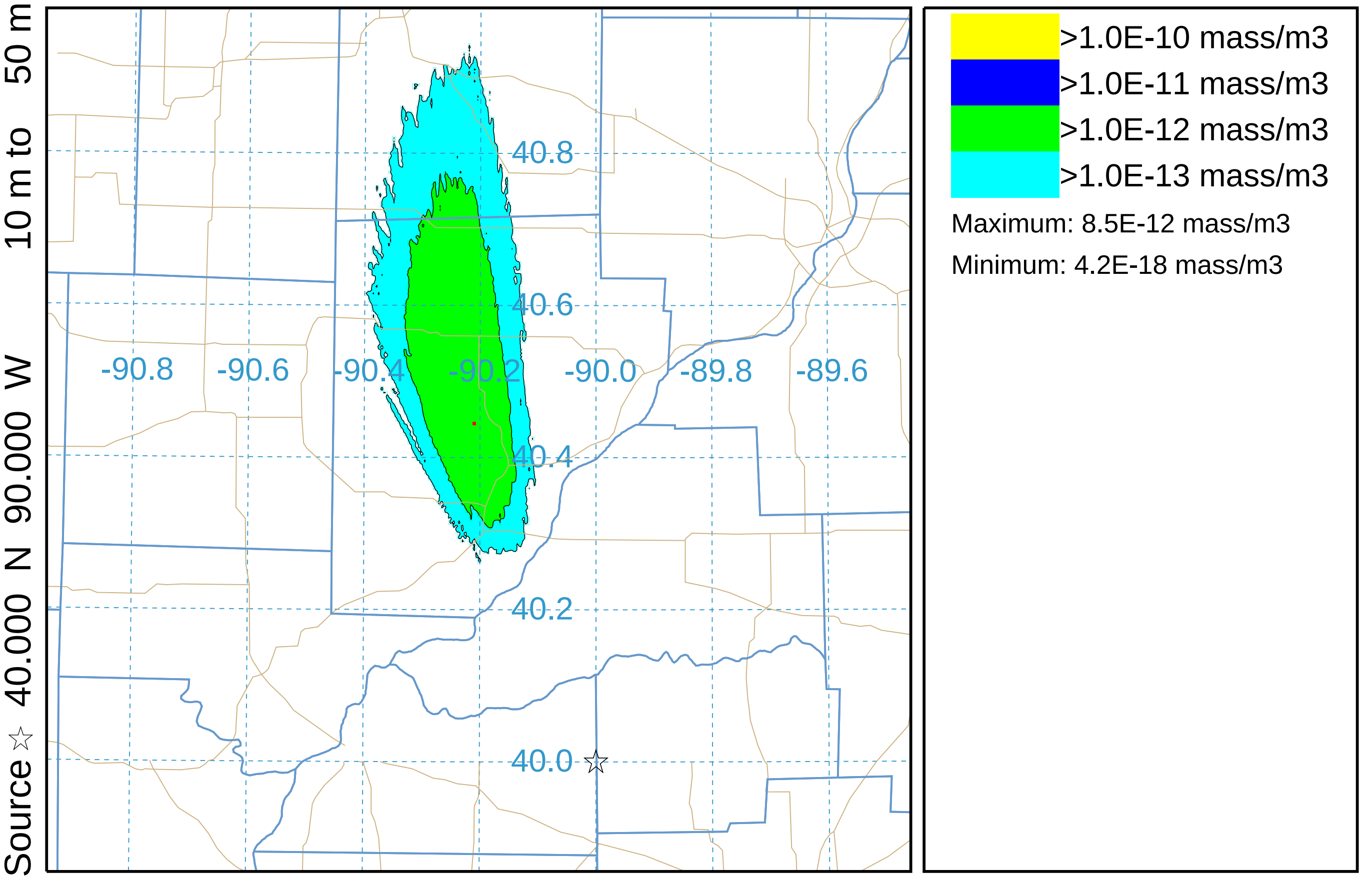


NGM METEOROLOGICAL DATA

HLS fine (134)
Concentration (mass/m3) averaged between 0 m and 500 m
Integrated from 0200 17 Oct to 0300 17 Oct 95 (UTC)
Release started at 0000 17 Oct 95 (UTC)



HLS fine (134)
Concentration (mass/m3) averaged between 0 m and 500 m
Integrated from 0300 17 Oct to 0400 17 Oct 95 (UTC)
Release started at 0000 17 Oct 95 (UTC)



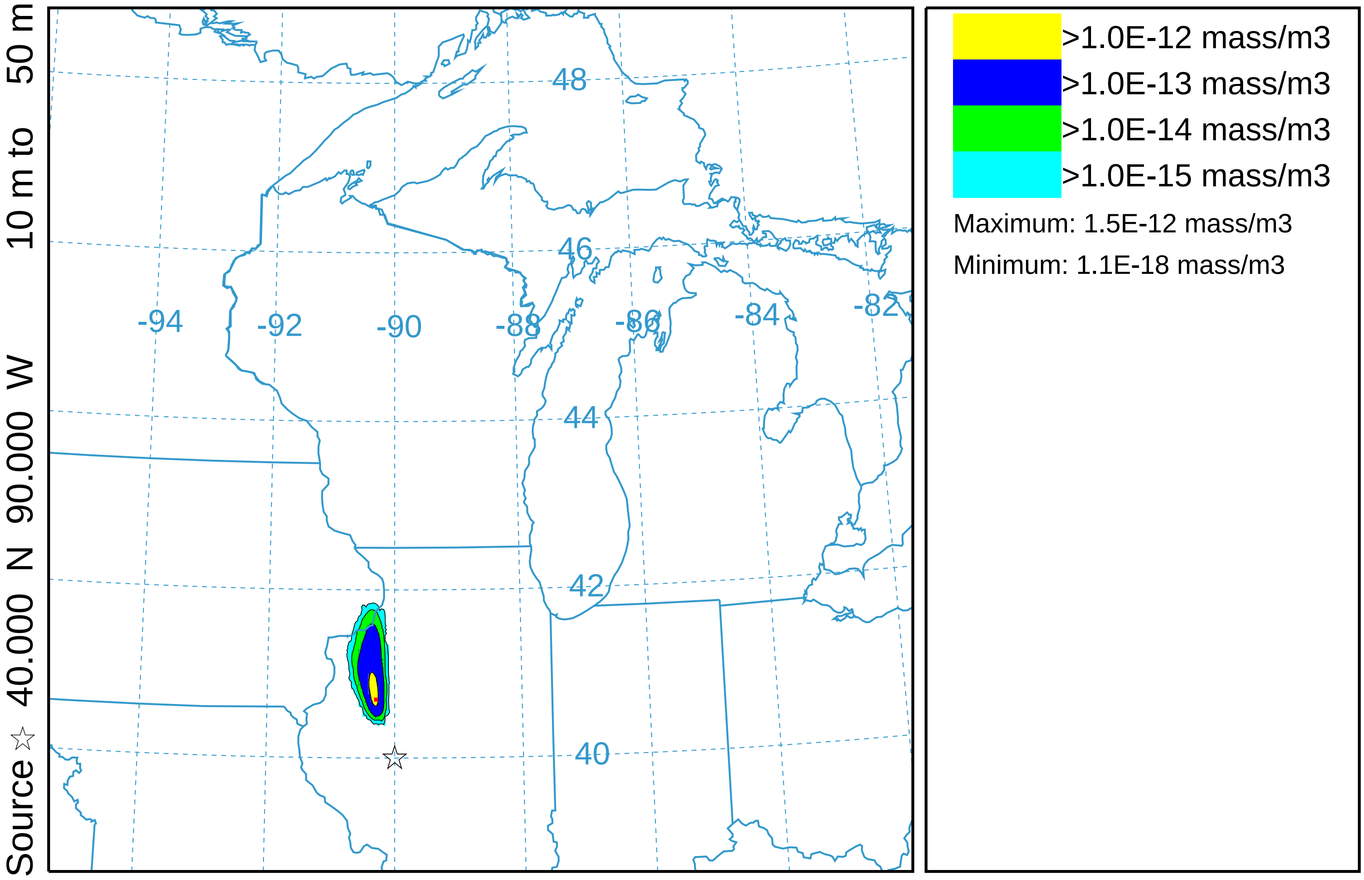
NGM METEOROLOGICAL DATA

HLS coarse (234)

Concentration (mass/m3) averaged between 0 m and 500 m

Integrated from 0400 17 Oct to 0600 17 Oct 95 (UTC)

Release started at 0000 17 Oct 95 (UTC)



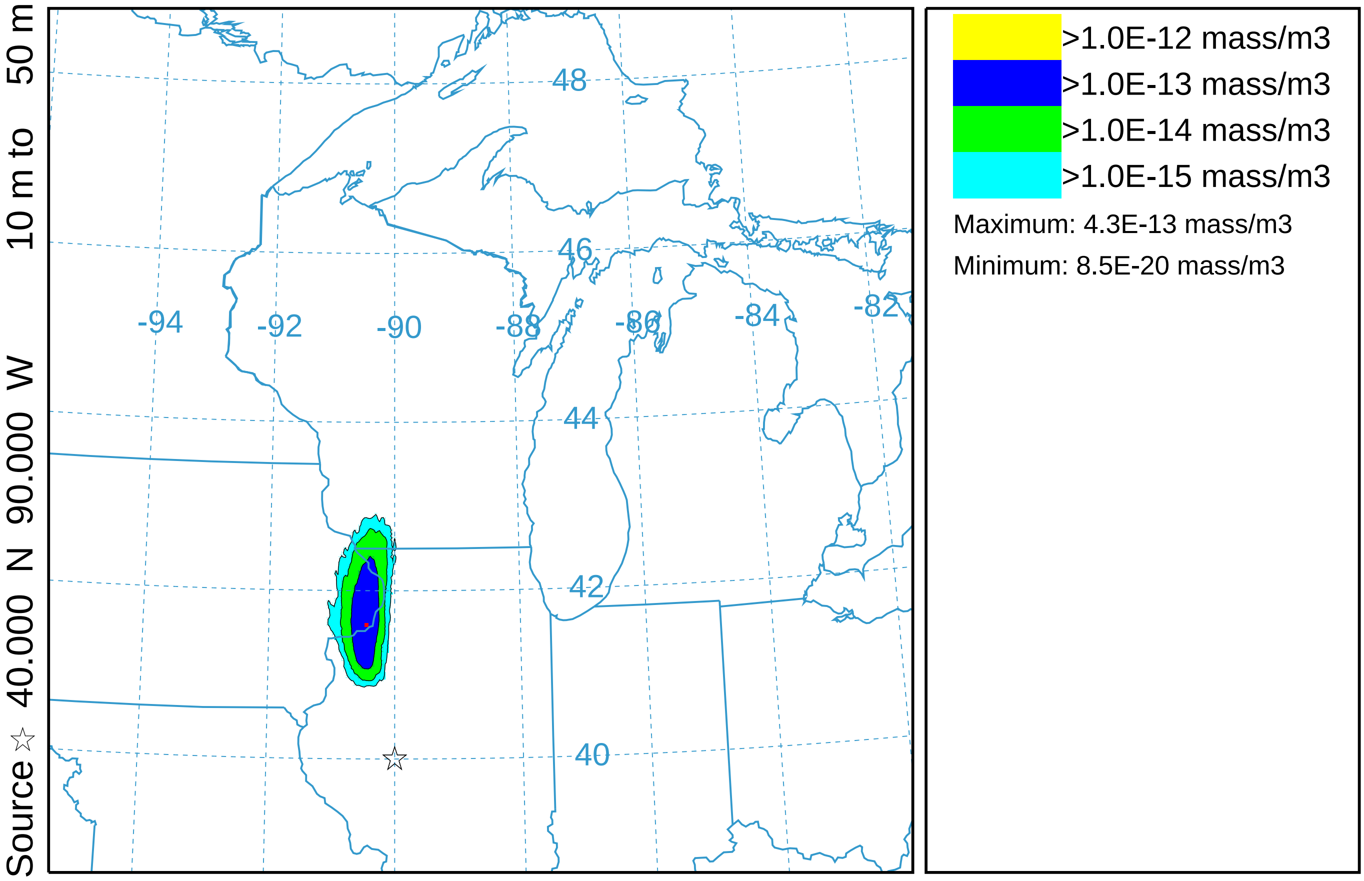
NGM METEOROLOGICAL DATA

HLS coarse (234)

Concentration (mass/m3) averaged between 0 m and 500 m

Integrated from 0600 17 Oct to 0800 17 Oct 95 (UTC)

Release started at 0000 17 Oct 95 (UTC)



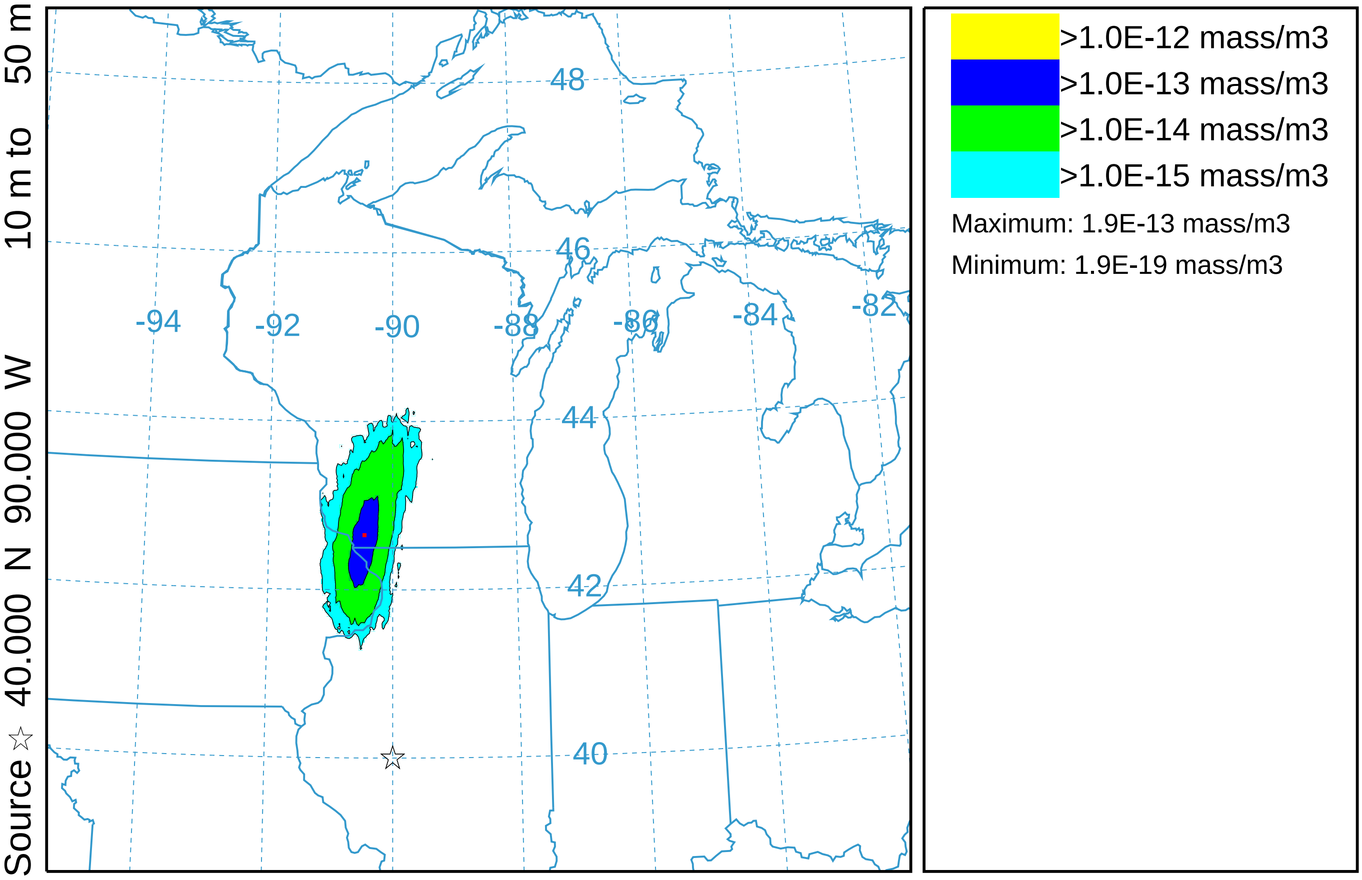
NGM METEOROLOGICAL DATA

HLS coarse (234)

Concentration (mass/m3) averaged between 0 m and 500 m

Integrated from 0800 17 Oct to 1000 17 Oct 95 (UTC)

Release started at 0000 17 Oct 95 (UTC)



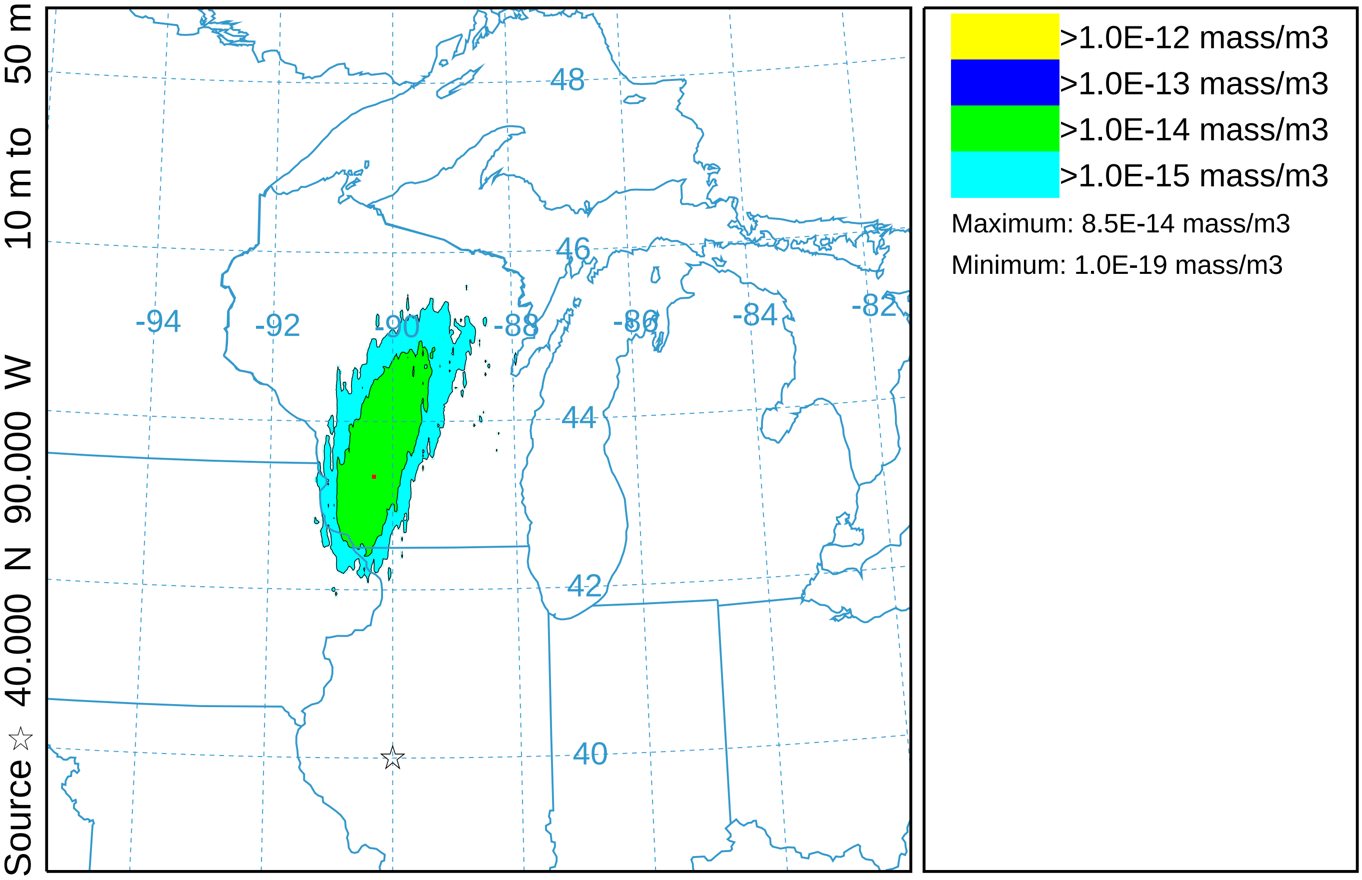
NGM METEOROLOGICAL DATA

HLS coarse (234)

Concentration (mass/m3) averaged between 0 m and 500 m

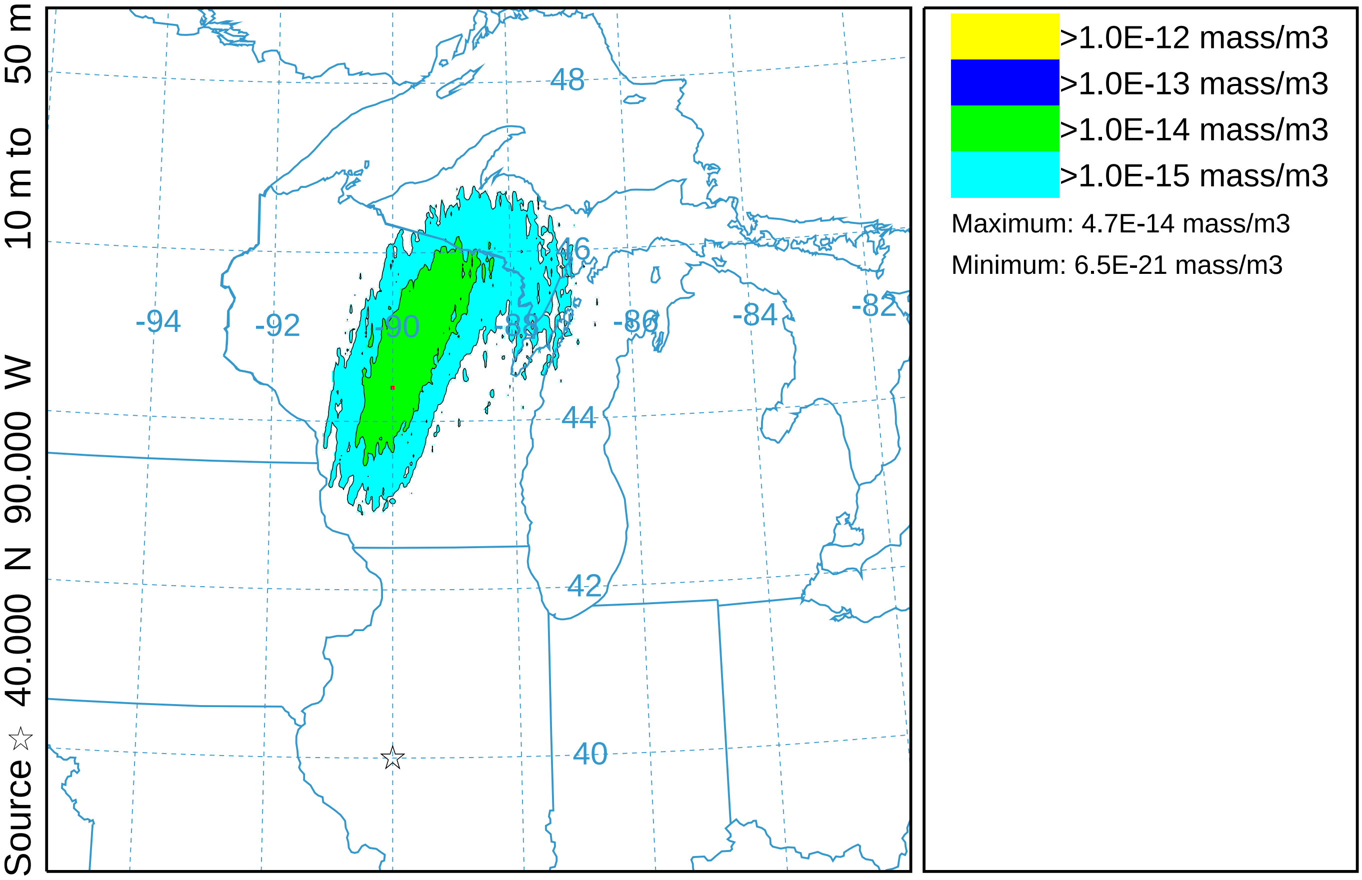
Integrated from 1000 17 Oct to 1200 17 Oct 95 (UTC)

Release started at 0000 17 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

HLS coarse (234)
Concentration (mass/m3) averaged between 0 m and 500 m
Integrated from 1200 17 Oct to 1400 17 Oct 95 (UTC)
Release started at 0000 17 Oct 95 (UTC)



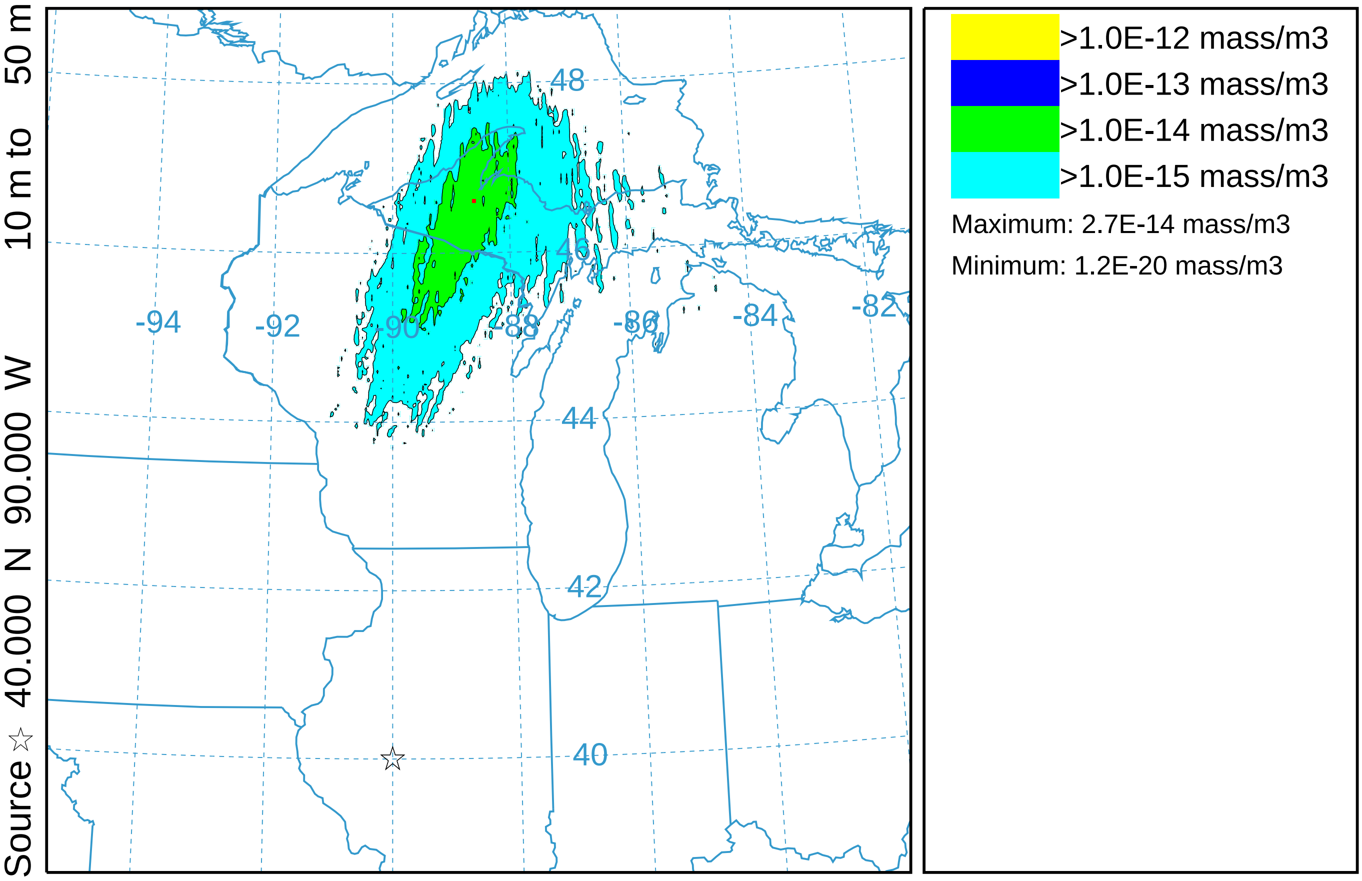
NGM METEOROLOGICAL DATA

HLS coarse (234)

Concentration (mass/m3) averaged between 0 m and 500 m

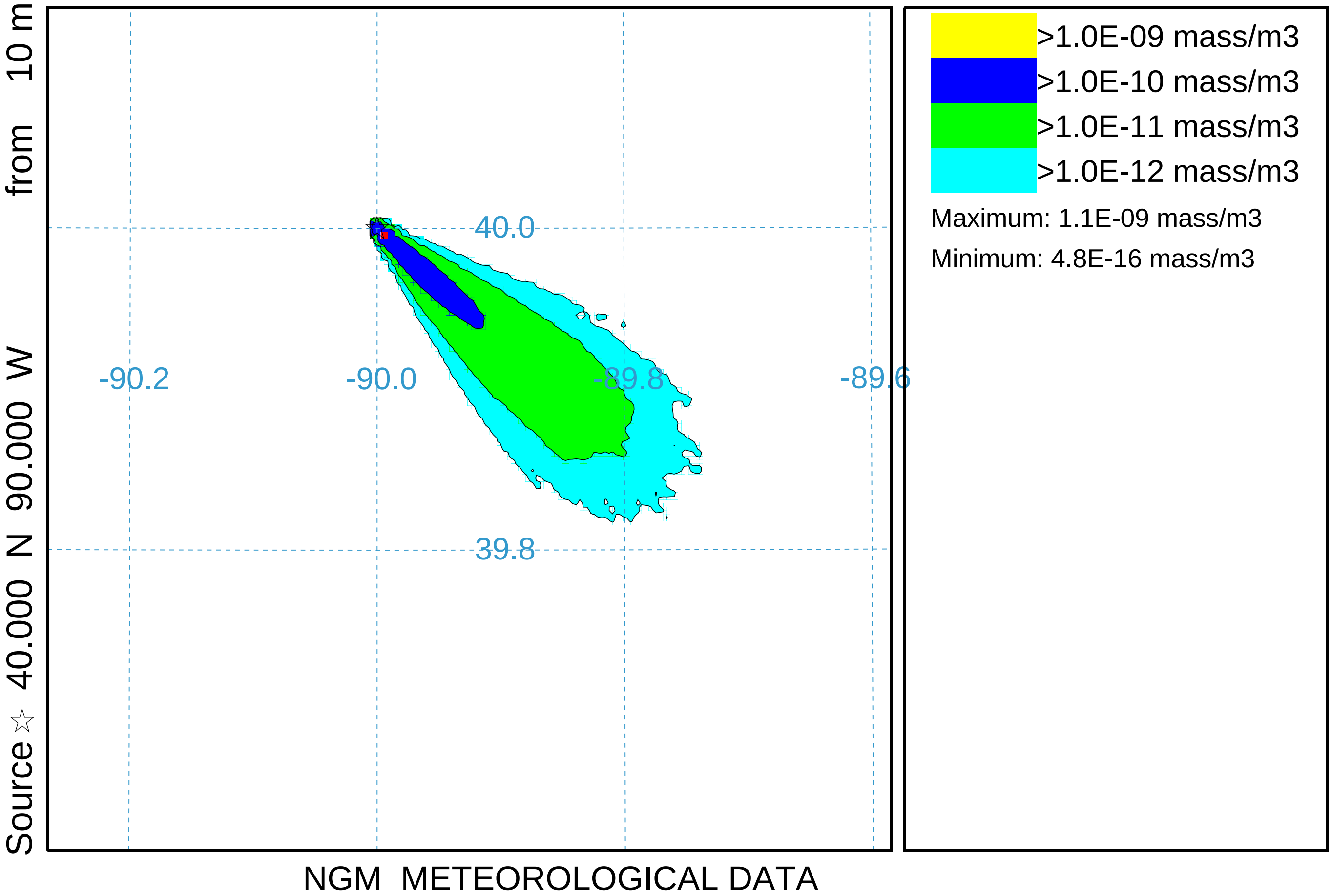
Integrated from 1400 17 Oct to 1600 17 Oct 95 (UTC)

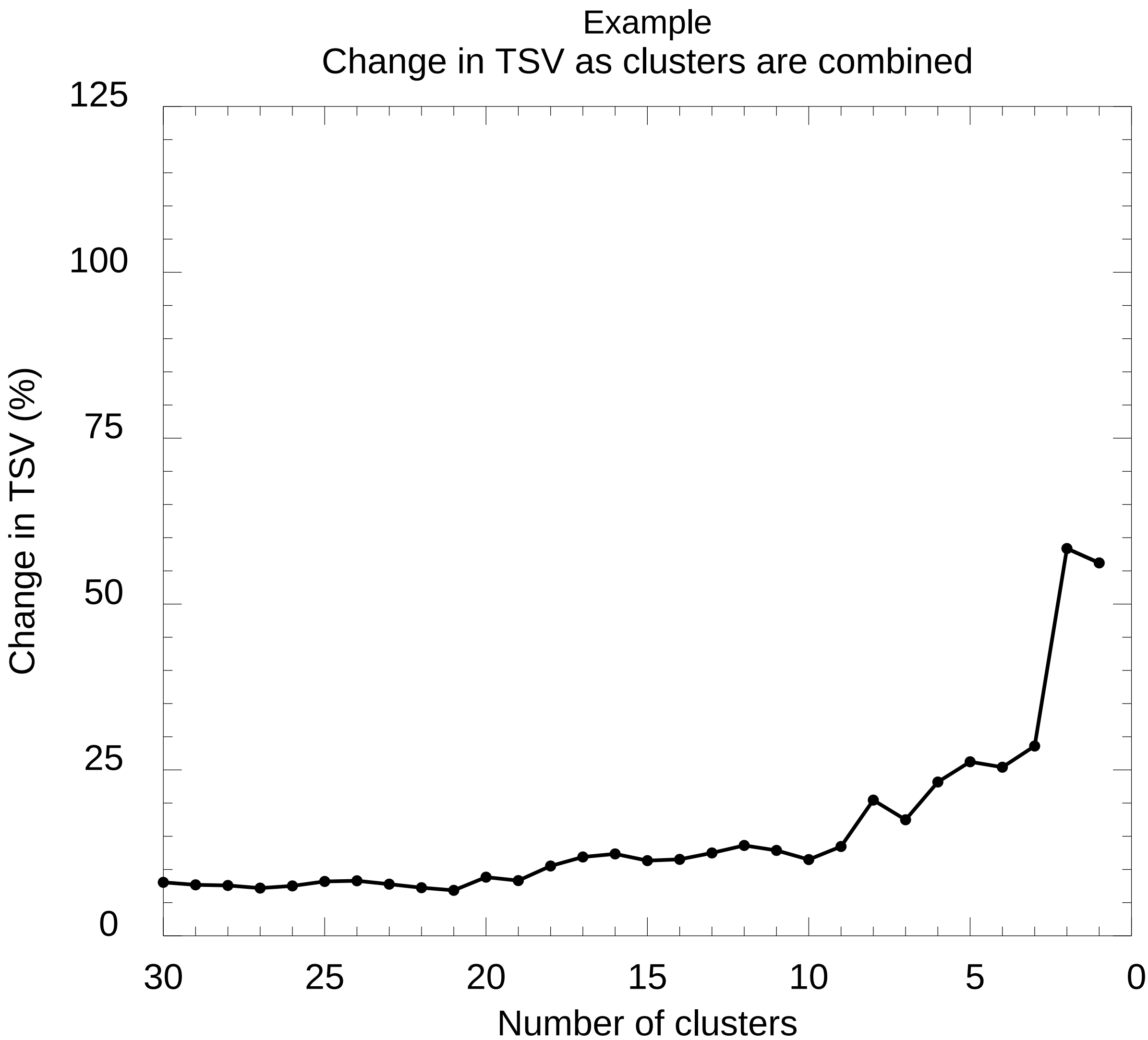
Release started at 0000 17 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

Short-Range Dispersion Simulation (035)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0000 16 Oct to 0100 16 Oct 95 (UTC)
TEST Release started at 0000 16 Oct 95 (UTC)

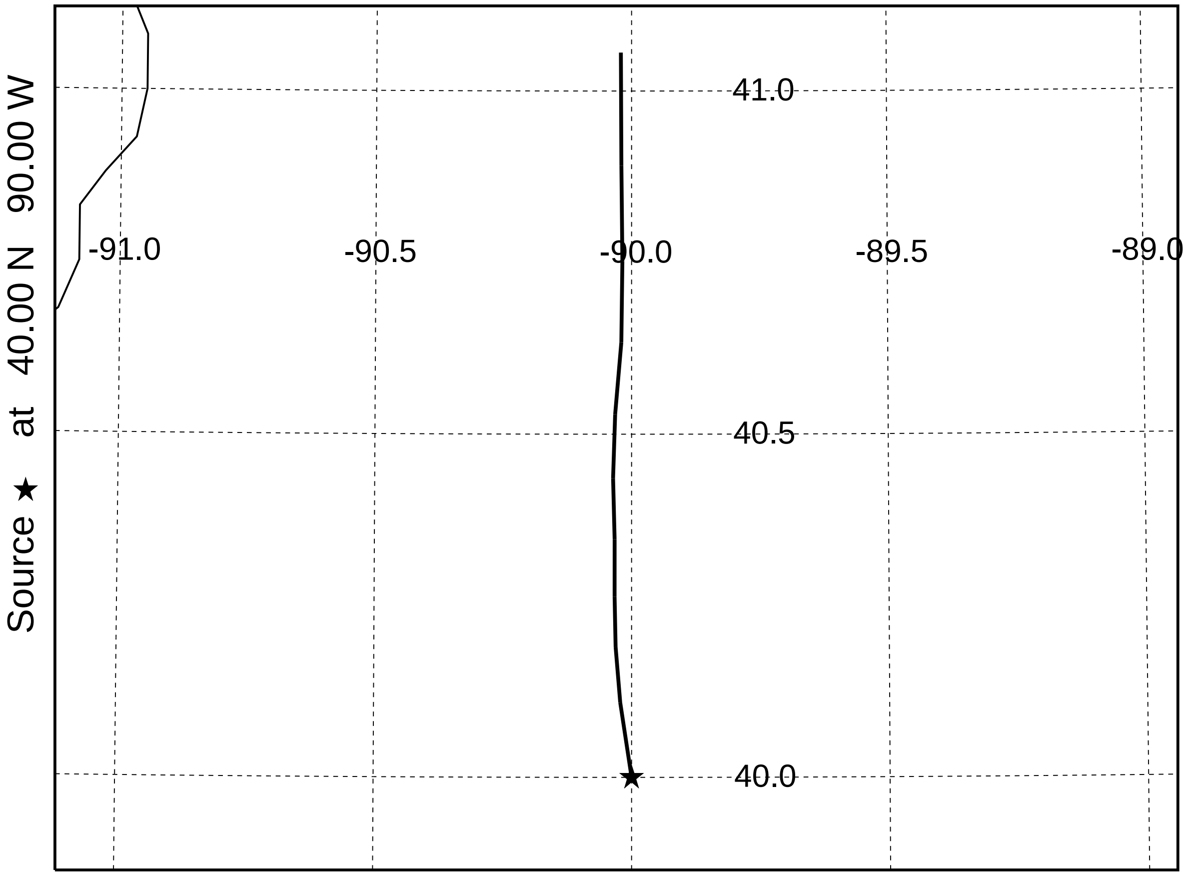



Example
Change in TSV as clusters are combined
Change in TSV (%)
125
100
75
50
25
0
30
25
20
15
10
5
0
Number of clusters

TRAJECTORIES IN CLUSTER 0 of 3 Example

1 forward trajectories starting at various times

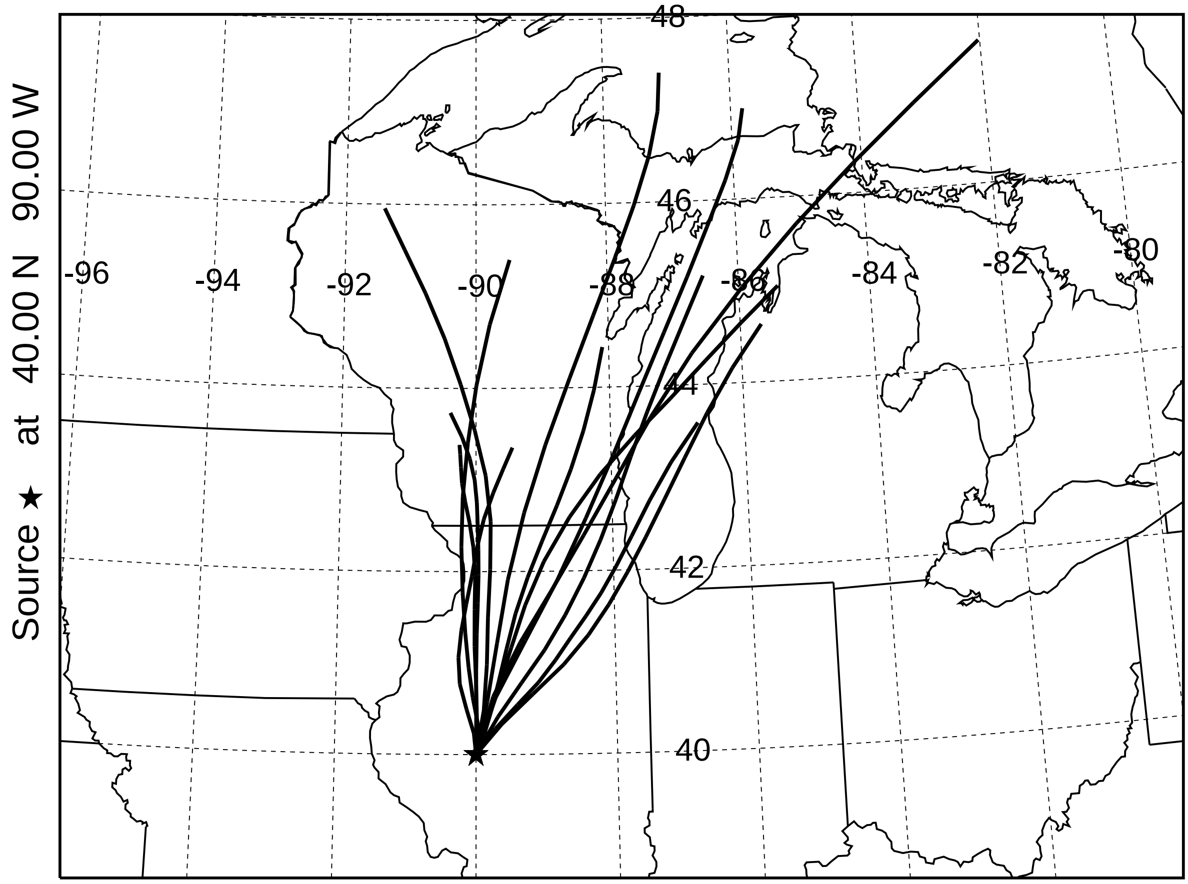
NGM Meteorological Data



TRAJECTORIES IN CLUSTER 1 of 3 Example

13 forward trajectories starting at various times

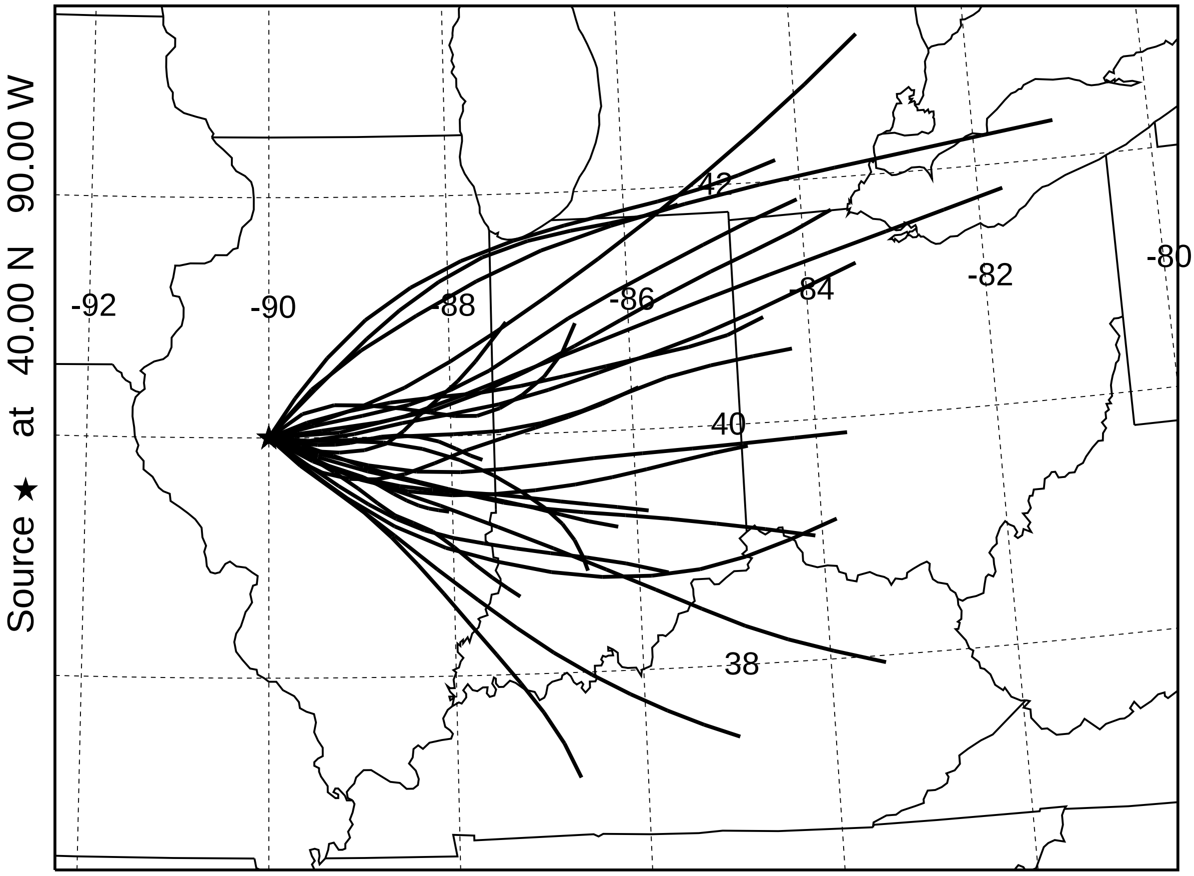
NGM Meteorological Data



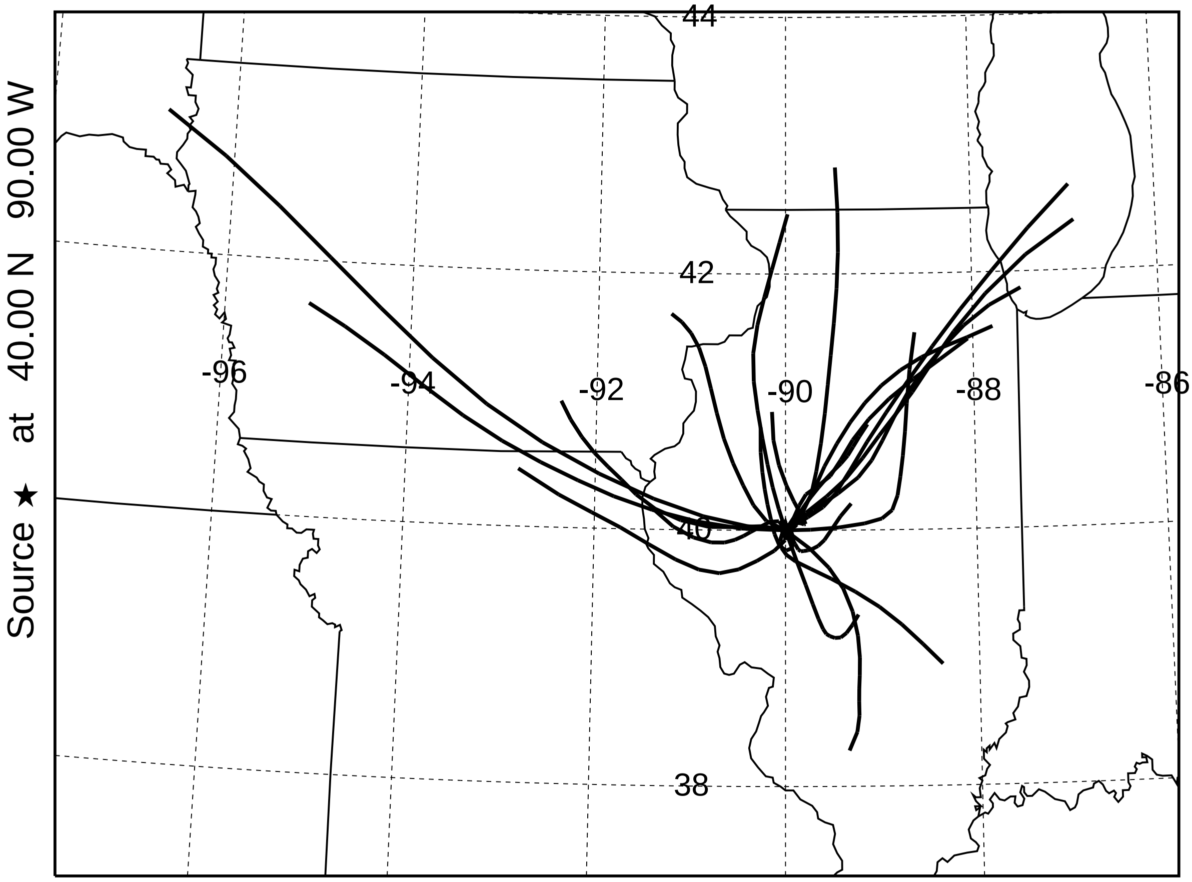
TRAJECTORIES IN CLUSTER 2 of 3 Example

27 forward trajectories starting at various times

NGM Meteorological Data



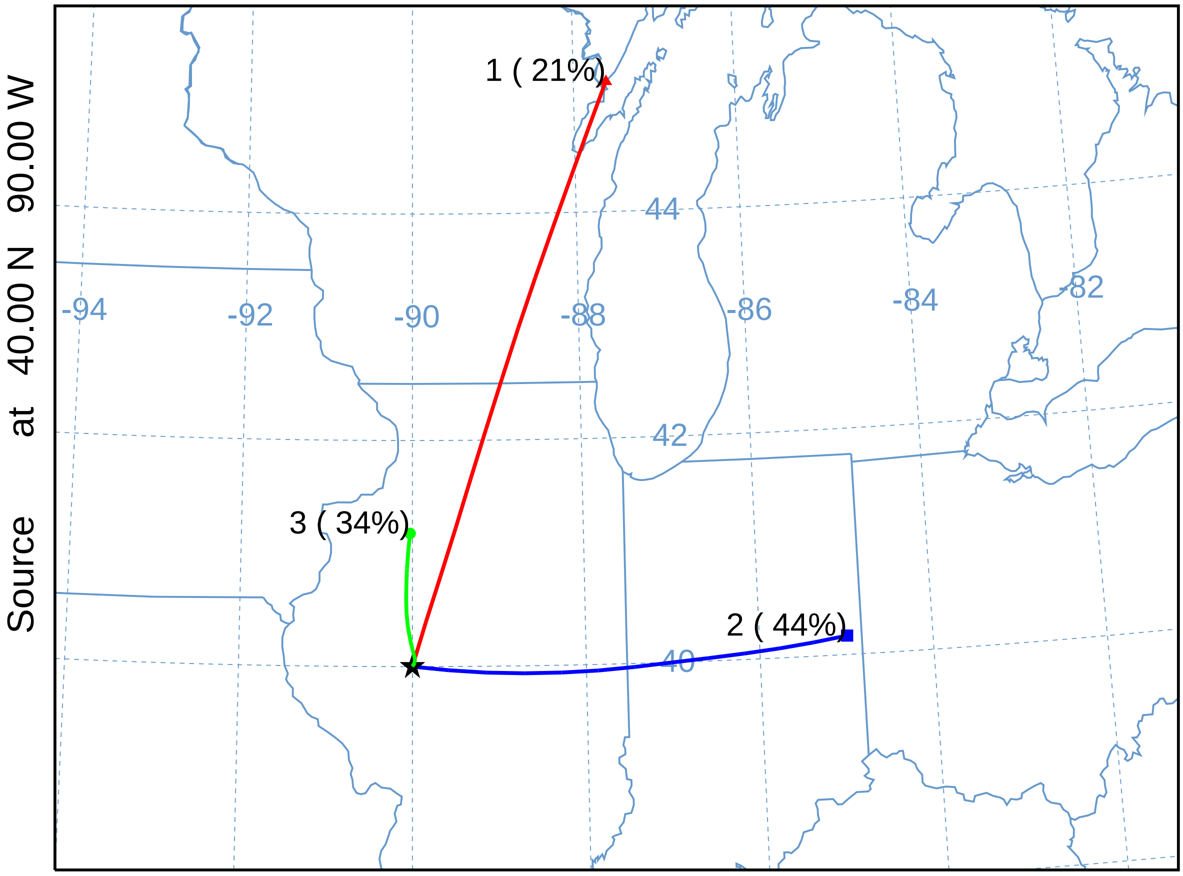
TRAJECTORIES IN CLUSTER 3 of 3 Example
21 forward trajectories starting at various times
NGM Meteorological Data



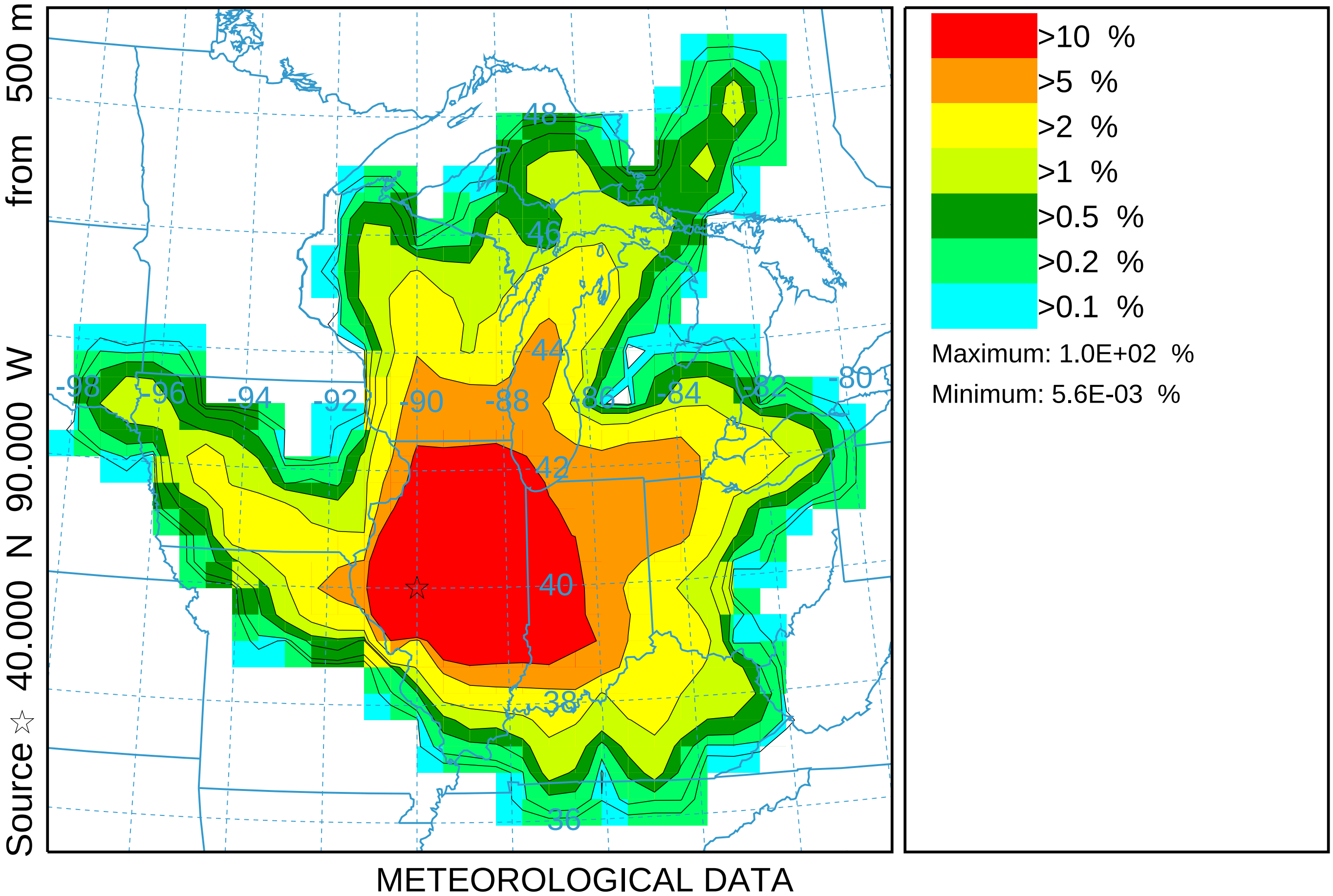
CLUSTER MEAN TRAJECTORIES Example

61 forward trajectories

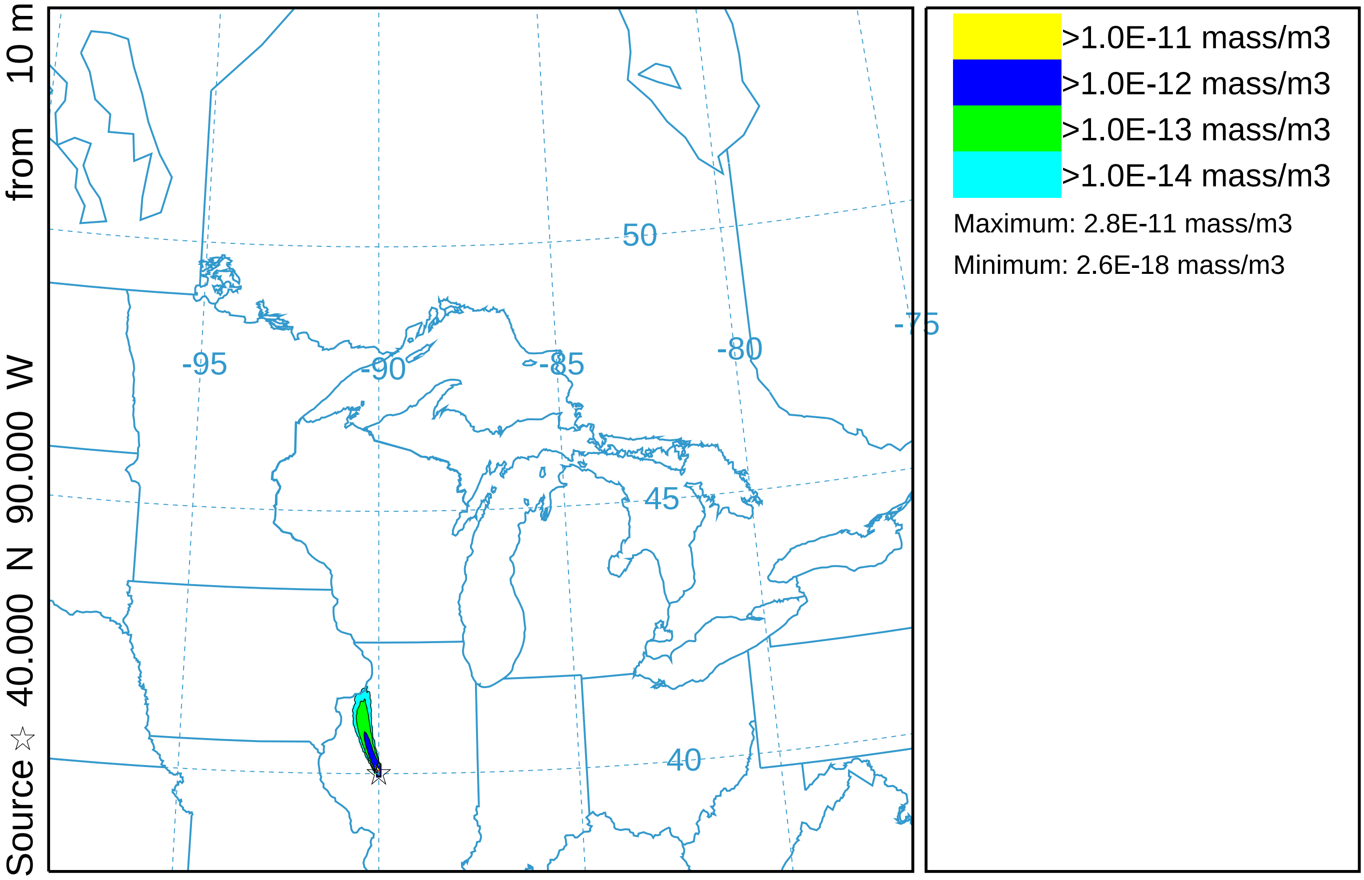
NGM Meteorological Data



Trajectory Frequency Plot
Values (%) averaged between 0 m and 99999 m
Integrated from 0000 01 Oct to 0000 31 Oct 95 (UTC)
Freq Release started at 0000 00 00 (UTC)

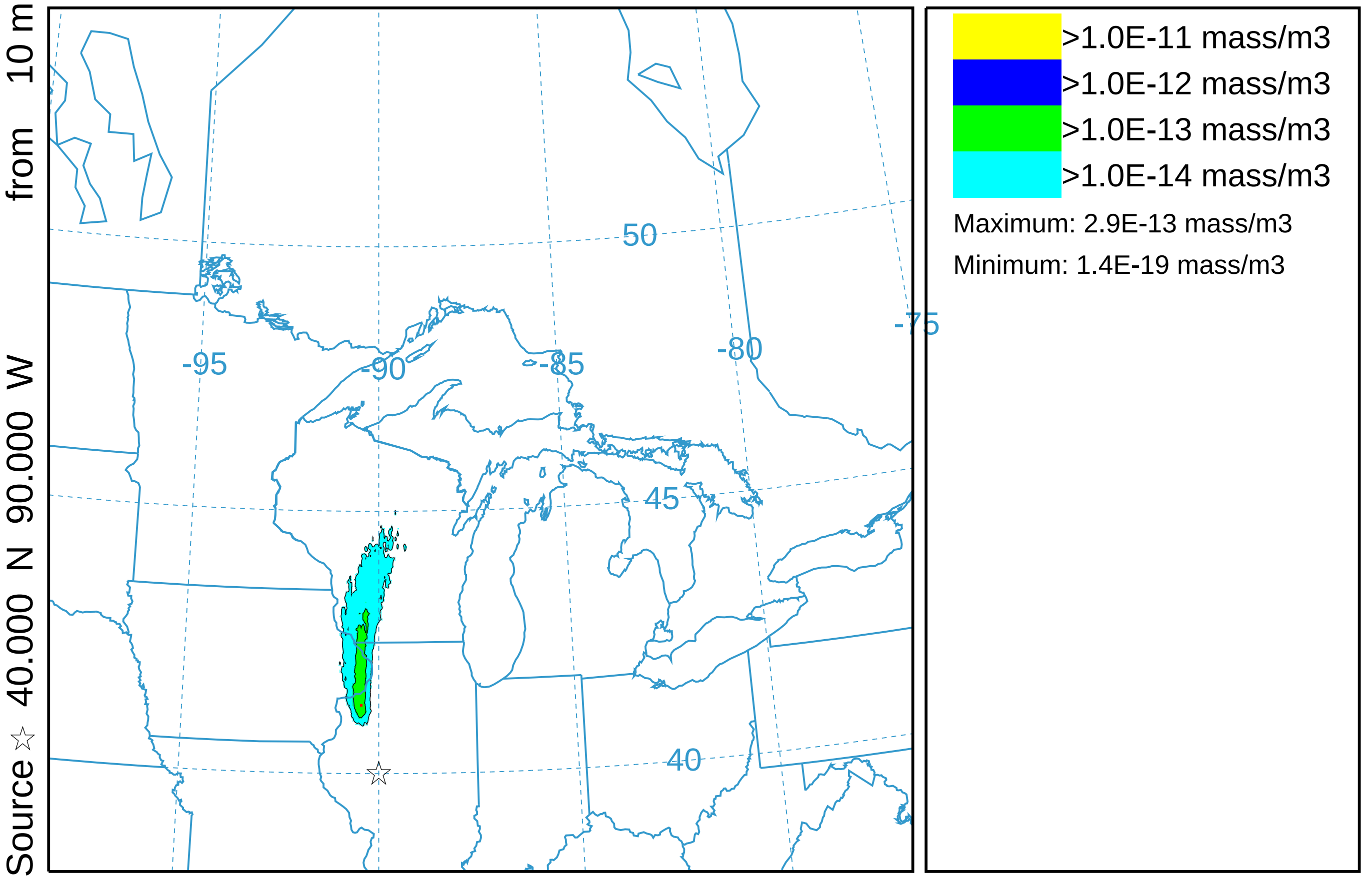


Transport to 24 hours (036)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0000 17 Oct to 0600 17 Oct 95 (UTC)
TEST Release started at 0000 17 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

Transport to 24 hours (036)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0600 17 Oct to 1200 17 Oct 95 (UTC)
TEST Release started at 0000 17 Oct 95 (UTC)



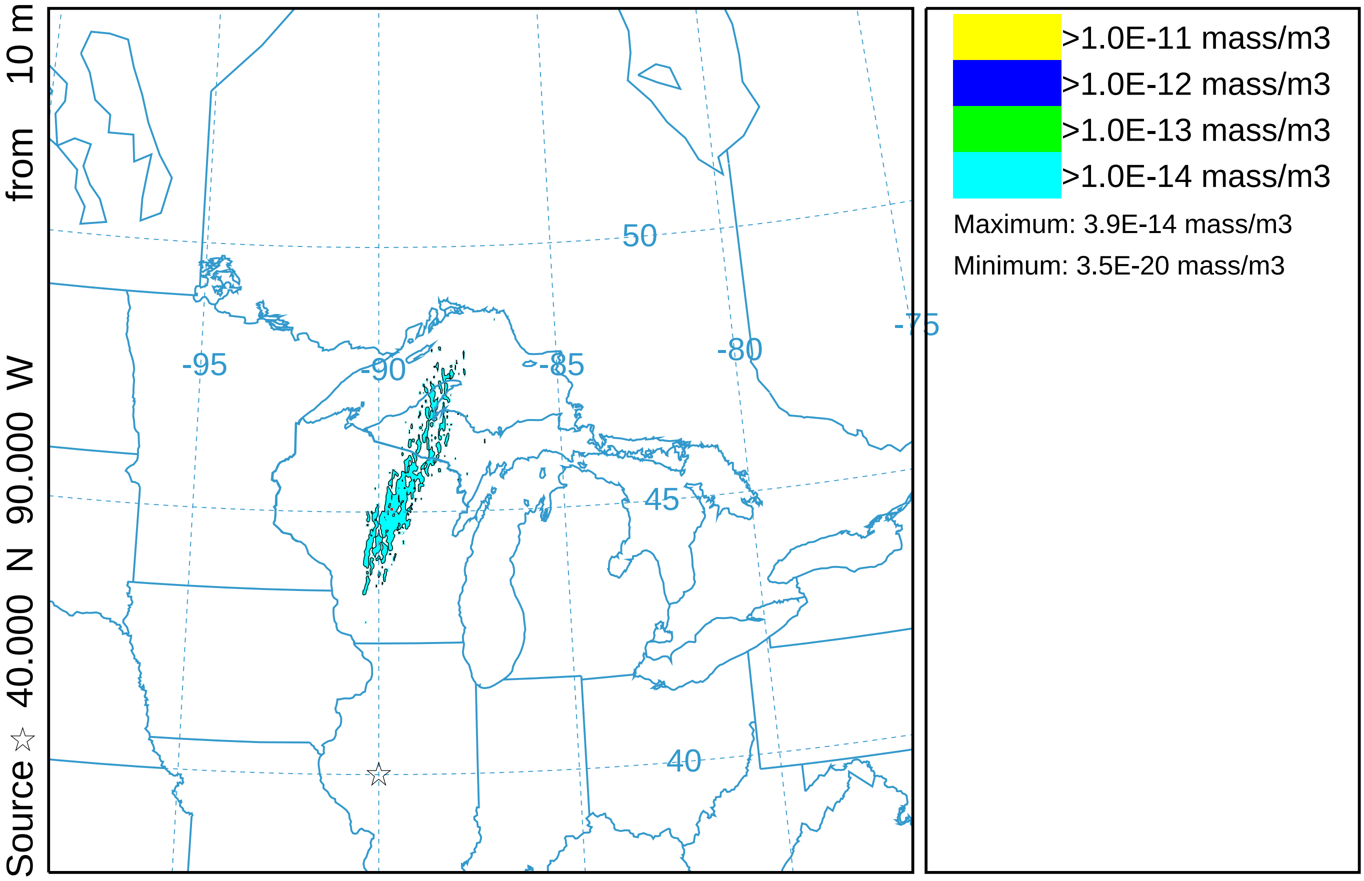
NGM METEOROLOGICAL DATA

Transport to 24 hours (036)

Concentration (mass/m3) averaged between 0 m and 100 m

Integrated from 1200 17 Oct to 1800 17 Oct 95 (UTC)

TEST Release started at 0000 17 Oct 95 (UTC)



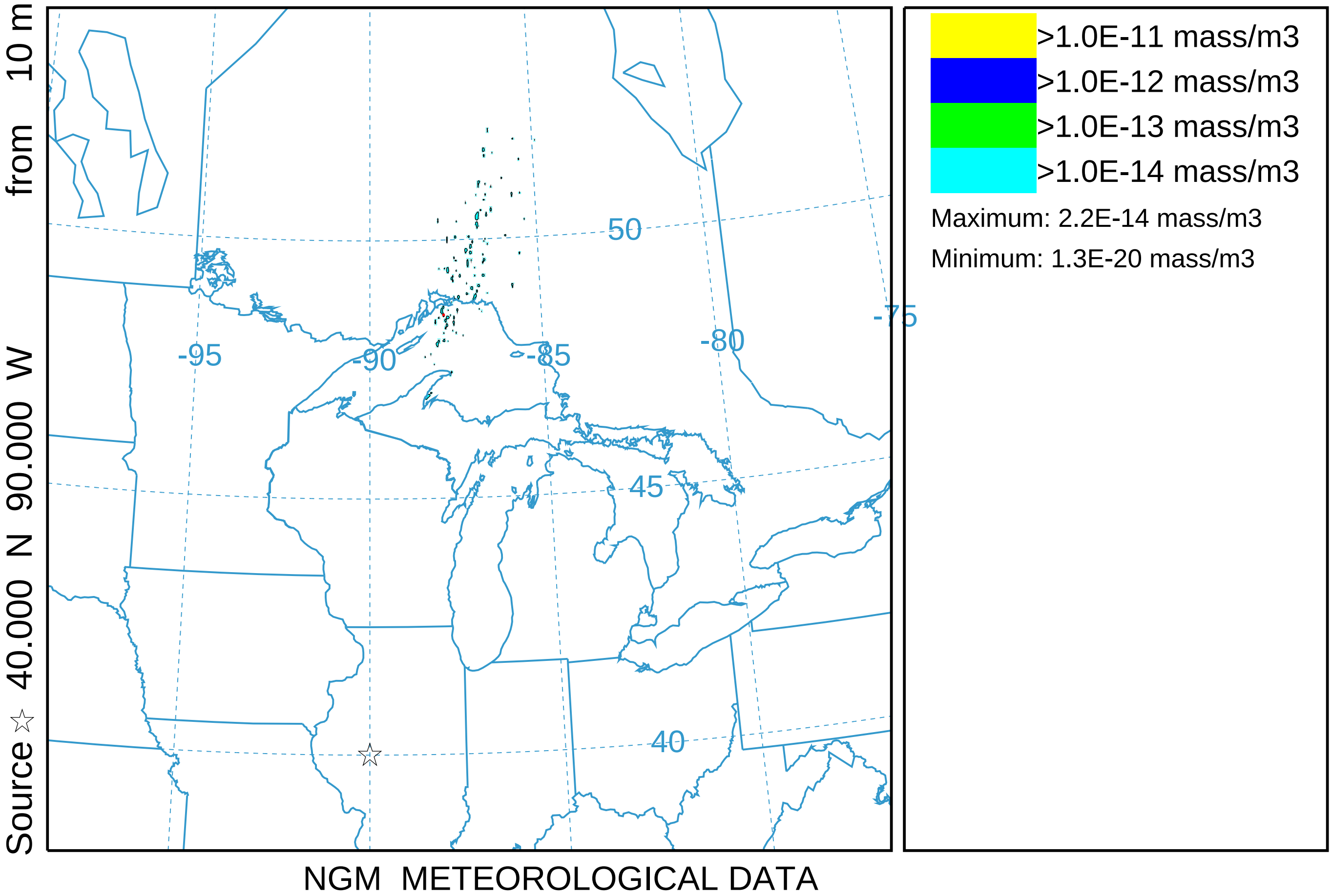
NGM METEOROLOGICAL DATA

Transport to 24 hours (036)

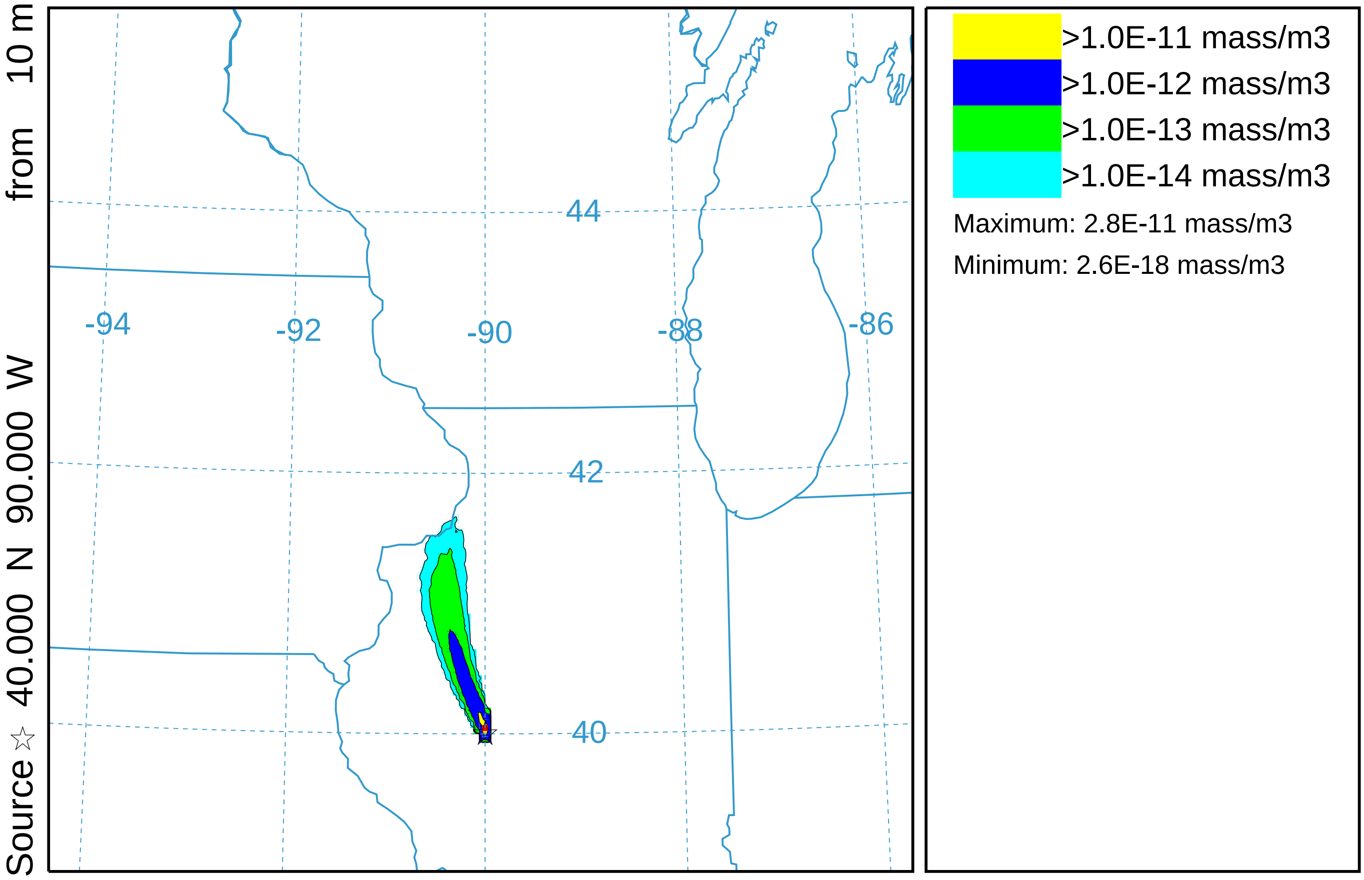
Concentration (mass/m3) averaged between 0 m and 100 m

Integrated from 1800 17 Oct to 0000 18 Oct 95 (UTC)

TEST Release started at 0000 17 Oct 95 (UTC)

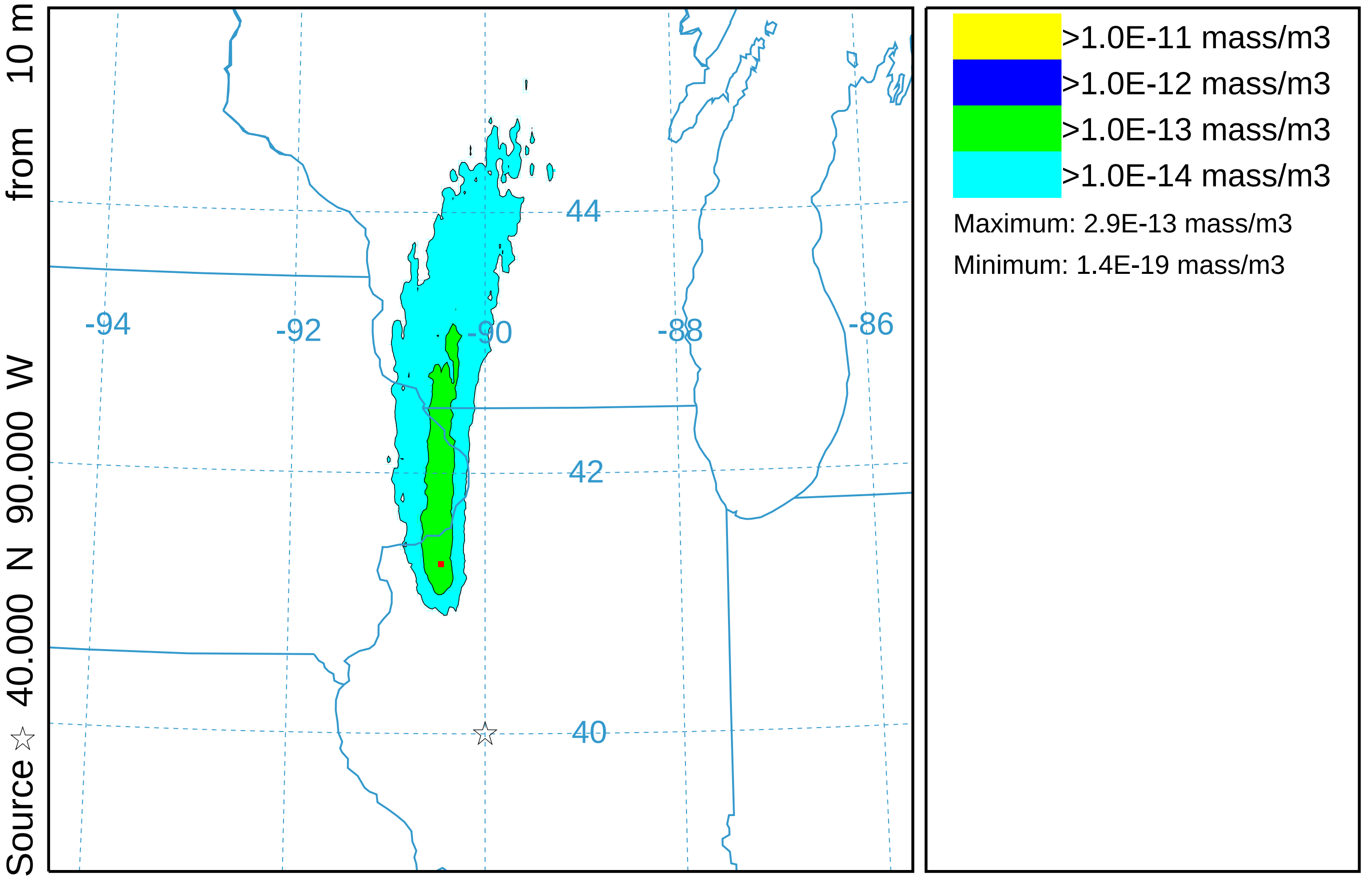


GEM@12h - Lagrangian grid (037)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0000 17 Oct to 0600 17 Oct 95 (UTC)
TEST Release started at 0000 17 Oct 95 (UTC)



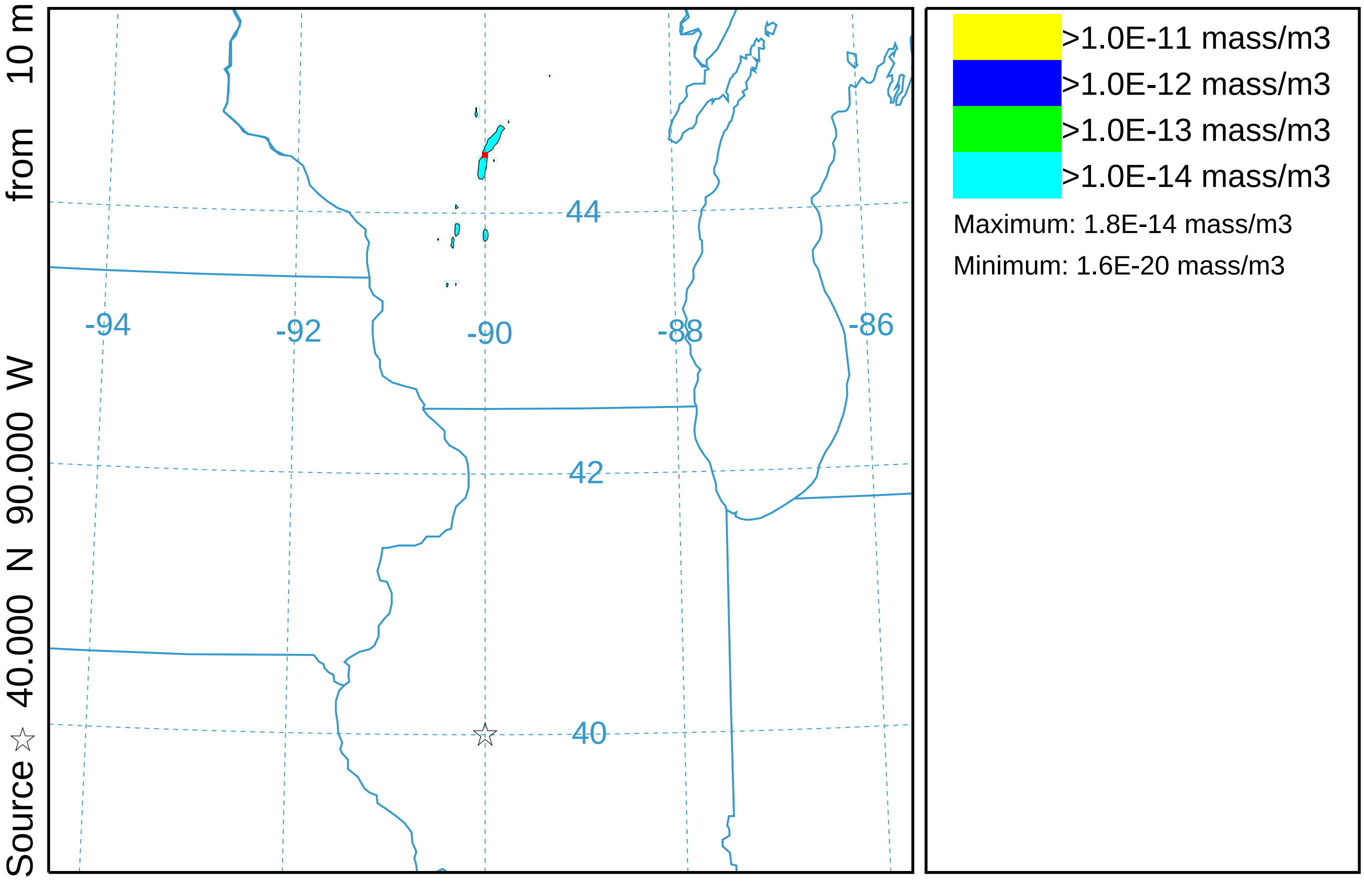
NGM METEOROLOGICAL DATA

GEM@12h - Lagrangian grid (037)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 0600 17 Oct to 1200 17 Oct 95 (UTC)
TEST Release started at 0000 17 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

GEM@12h - Lagrangian grid (037)
Concentration (mass/m3) averaged between 0 m and 100 m
Integrated from 1200 17 Oct to 1800 17 Oct 95 (UTC)
TEST Release started at 0000 17 Oct 95 (UTC)



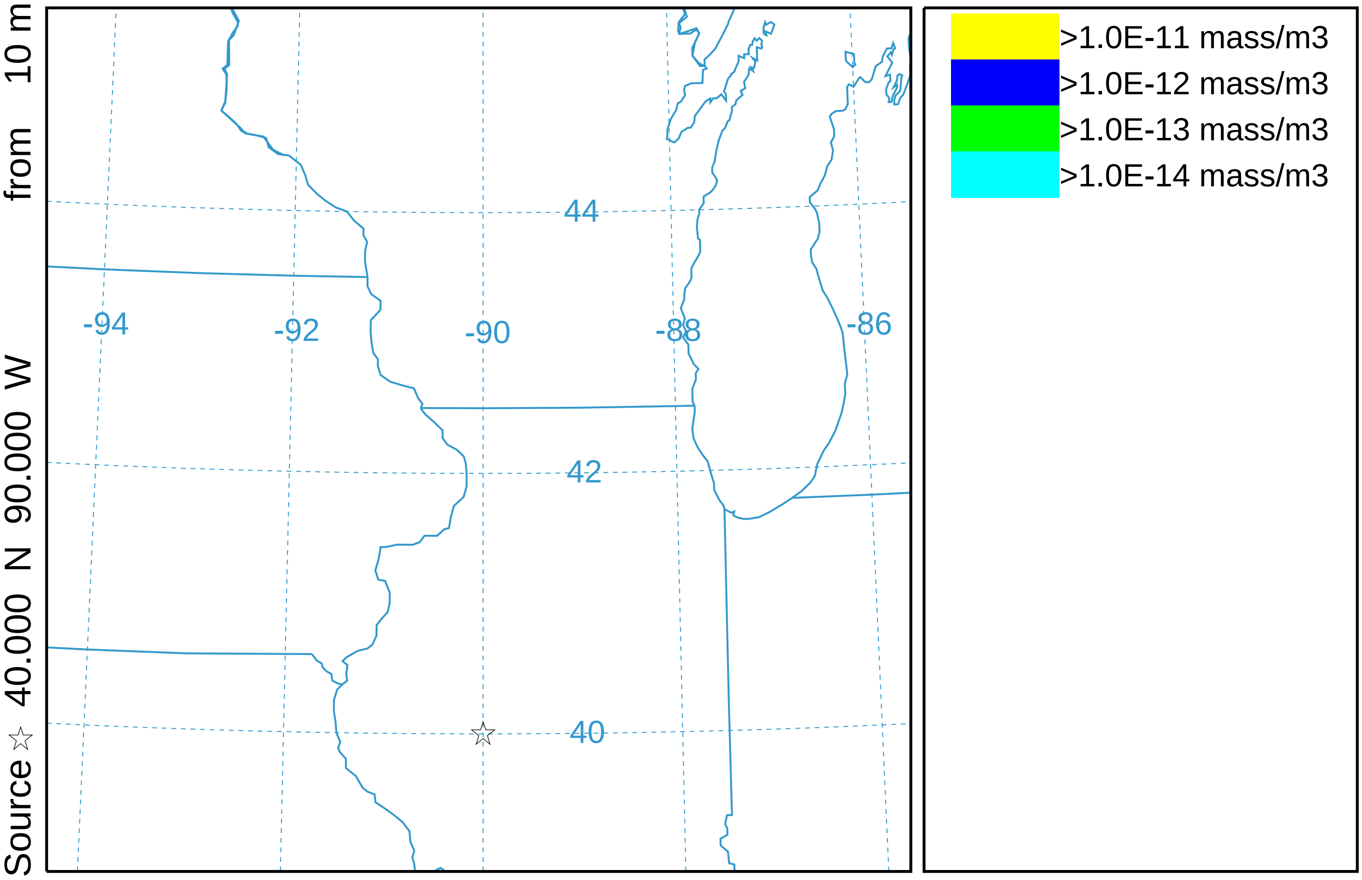
NGM METEOROLOGICAL DATA

GEM@12h - Lagrangian grid (037)

Concentration (mass/m3) averaged between 0 m and 100 m

Integrated from 1800 17 Oct to 0000 18 Oct 95 (UTC)

TEST Release started at 0000 17 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

GEM@12h - Eulerian grid (137)

Concentration (/m3) at level 232 m

Integrated from 0600 17 Oct to 0600 17 Oct 95 (UTC)

P001 Release started at 0000 17 Oct 95 (UTC)

LEGEND

>1.0E-17 /m3
>1.0E-18 /m3
>1.0E-19 /m3
>1.0E-20 /m3
>1.0E-21 /m3
>1.0E-22 /m3
>1.0E-23 /m3
>1.0E-24 /m3
>1.0E-25 /m3
>1.0E-26 /m3
>1.0E-27 /m3
>1.0E-28 /m3

Source ☆ 40.000 N 90.000 W from 10 m

GEMx METEOROLOGICAL DATA

GEM@12h - Eulerian grid (137)
Concentration (/m3) at level 232 m
Integrated from 1200 17 Oct to 1200 17 Oct 95 (UTC)
P001 Release started at 0000 17 Oct 95 (UTC)

Source ☆ 40.000 N 90.000 W from 10 m

LEGEND

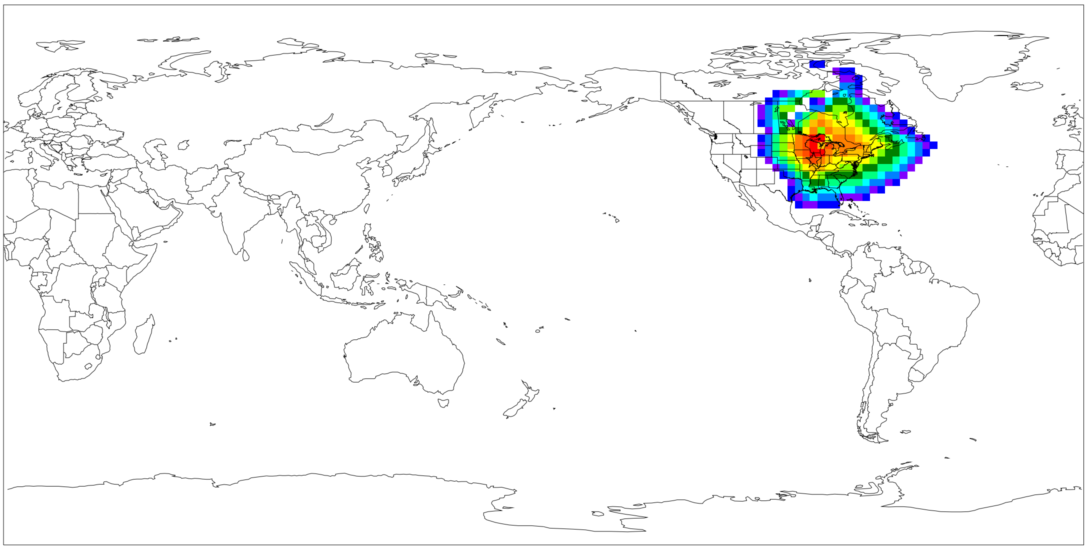
>1.0E-17 /m3
>1.0E-18 /m3
>1.0E-19 /m3
>1.0E-20 /m3
>1.0E-21 /m3
>1.0E-22 /m3
>1.0E-23 /m3
>1.0E-24 /m3
>1.0E-25 /m3
>1.0E-26 /m3
>1.0E-27 /m3
>1.0E-28 /m3

Max: 3.3E-17 /m3
Min: 3.8E-21 /m3

GEMx METEOROLOGICAL DATA

GEM@12h - Eulerian grid (137)
Concentration (/m3) at level 232 m
Integrated from 1800 17 Oct to 1800 17 Oct 95 (UTC)
P001 Release started at 0000 17 Oct 95 (UTC)

Source ☆ 40.000 N 90.000 W from 10 m



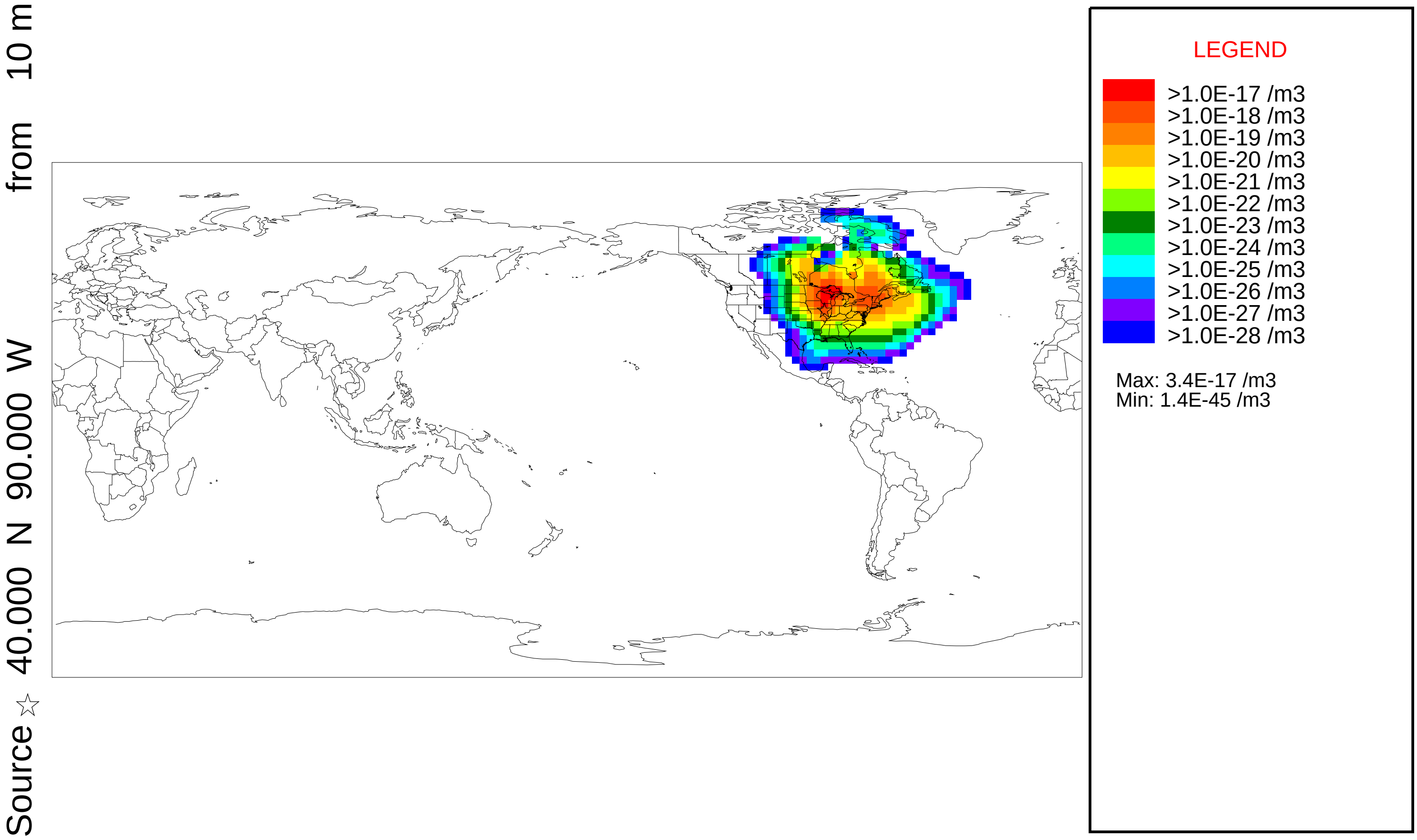
LEGEND

- >1.0E-17 /m3
- >1.0E-18 /m3
- >1.0E-19 /m3
- >1.0E-20 /m3
- >1.0E-21 /m3
- >1.0E-22 /m3
- >1.0E-23 /m3
- >1.0E-24 /m3
- >1.0E-25 /m3
- >1.0E-26 /m3
- >1.0E-27 /m3
- >1.0E-28 /m3

Max: 4.9E-17 /m3
Min: 1.4E-45 /m3

GEMx METEOROLOGICAL DATA

GEM@12h - Eulerian grid (137)
Concentration (/m3) at level 232 m
Integrated from 0000 18 Oct to 0000 18 Oct 95 (UTC)
P001 Release started at 0000 17 Oct 95 (UTC)



GEMx METEOROLOGICAL DATA

GEM@12h - Lagrangian + Eulerian (237)

Concentration (/m3) at level 232 m

Integrated from 0600 17 Oct to 0600 17 Oct 95 (UTC)

P001 Release started at 0000 17 Oct 95 (UTC)

Source ☆ 40.000 N 90.000 W from 10 m

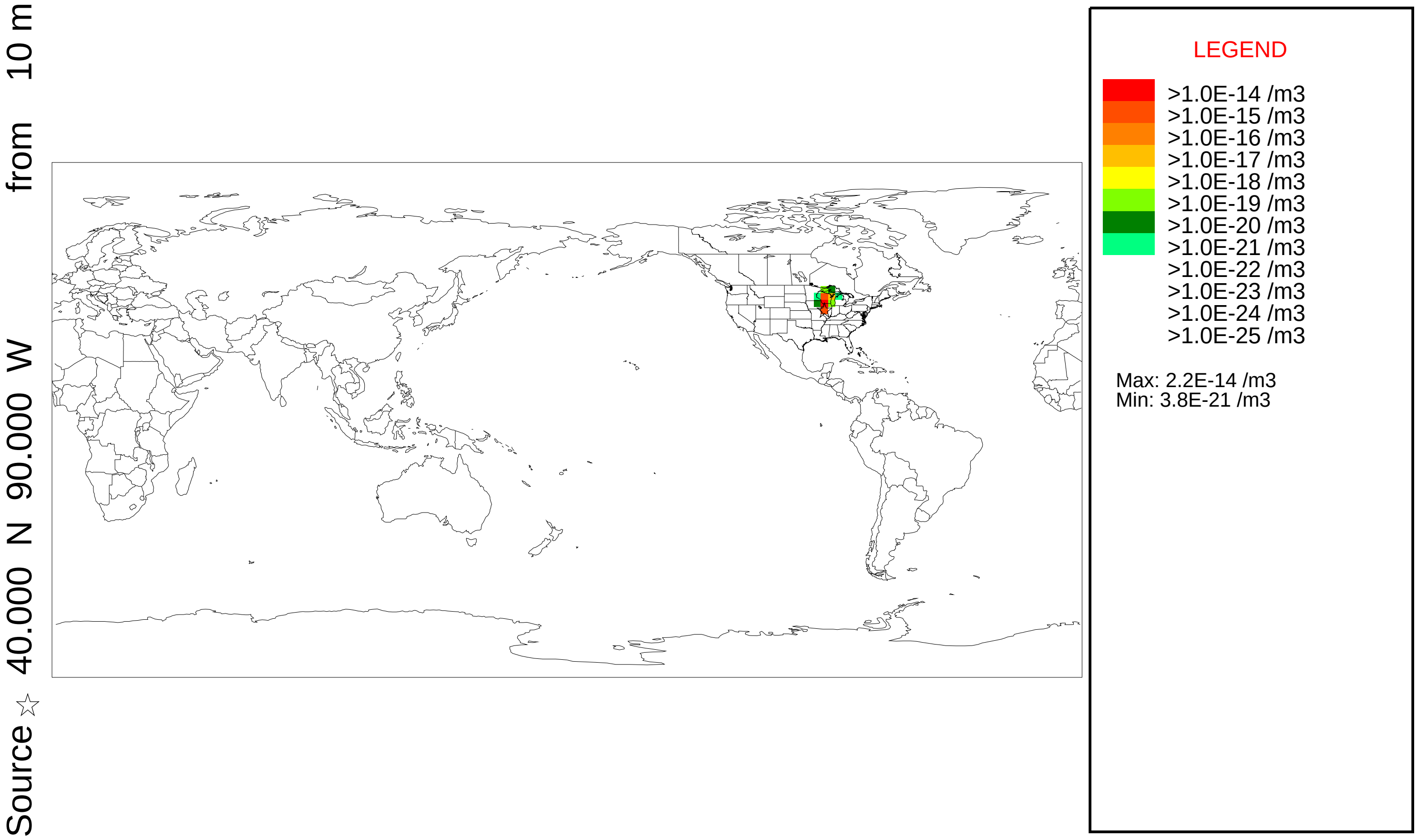
LEGEND

>1.0E-14 /m3
>1.0E-15 /m3
>1.0E-16 /m3
>1.0E-17 /m3
>1.0E-18 /m3
>1.0E-19 /m3
>1.0E-20 /m3
>1.0E-21 /m3
>1.0E-22 /m3
>1.0E-23 /m3
>1.0E-24 /m3
>1.0E-25 /m3

Max: 8.2E-14 /m3
Min: 1.4E-15 /m3

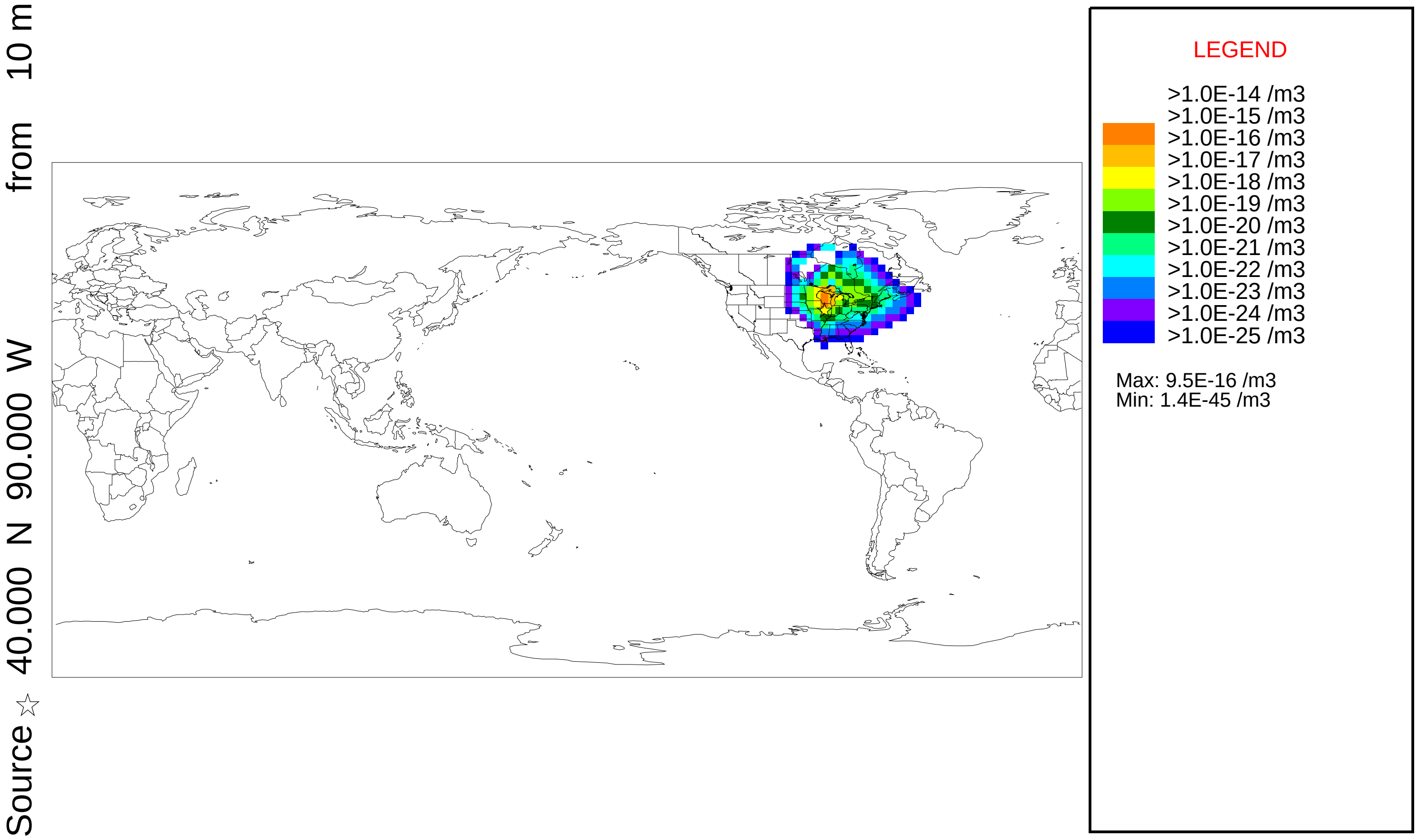
GEMx METEOROLOGICAL DATA

GEM@12h - Lagrangian + Eulerian (237)
Concentration (/m3) at level 232 m
Integrated from 1200 17 Oct to 1200 17 Oct 95 (UTC)
P001 Release started at 0000 17 Oct 95 (UTC)



GEMx METEOROLOGICAL DATA

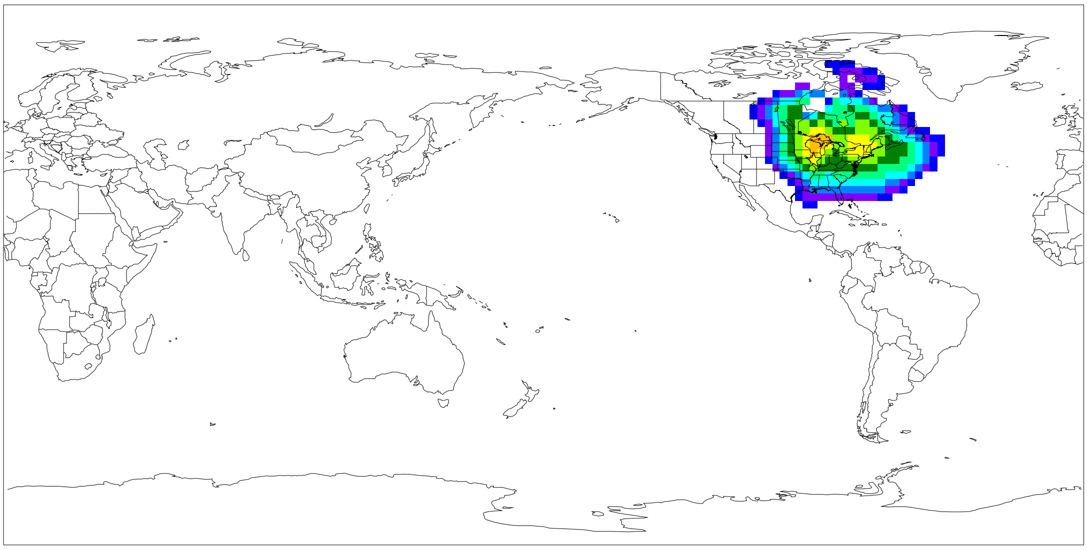
GEM@12h - Lagrangian + Eulerian (237)
Concentration (/m3) at level 232 m
Integrated from 1800 17 Oct to 1800 17 Oct 95 (UTC)
P001 Release started at 0000 17 Oct 95 (UTC)



GEMx METEOROLOGICAL DATA

GEM@12h - Lagrangian + Eulerian (237)
Concentration (/m3) at level 232 m
Integrated from 0000 18 Oct to 0000 18 Oct 95 (UTC)
P001 Release started at 0000 17 Oct 95 (UTC)

Source ☆ 40.000 N 90.000 W from 10 m



LEGEND

>1.0E-14 /m3
>1.0E-15 /m3
>1.0E-16 /m3
>1.0E-17 /m3
>1.0E-18 /m3
>1.0E-19 /m3
>1.0E-20 /m3
>1.0E-21 /m3
>1.0E-22 /m3
>1.0E-23 /m3
>1.0E-24 /m3
>1.0E-25 /m3

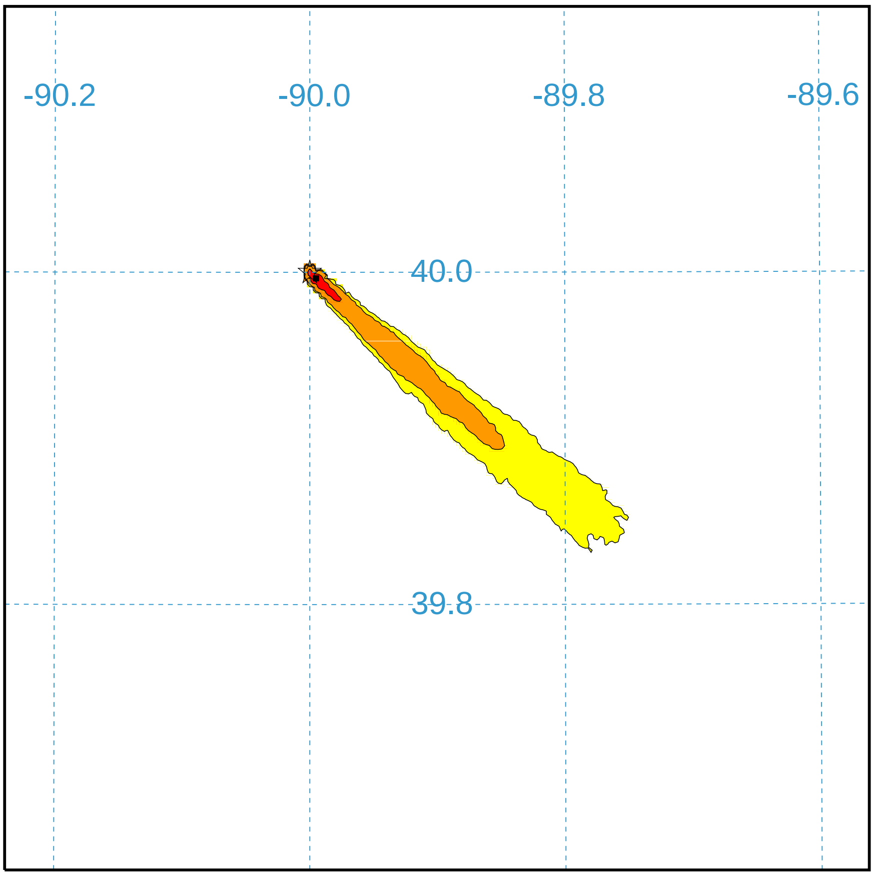
Max: 3.4E-17 /m3
Min: 1.4E-45 /m3

GEMx METEOROLOGICAL DATA

Chemical Dispersion Simulation (038)

Concentration (ppm) averaged between 0 m and 100 m
Integrated from 0000 16 Oct to 0100 16 Oct 95 (UTC)
CHEM Release started at 0000 16 Oct 95 (UTC)

Source ☆ 40.000 N 90.000 W from 50 m



NGM METEOROLOGICAL DATA

Not for Public Dissemination

AEGL-3	>20.0 ppm
AEGL-2	>2.0 ppm
AEGL-1	>0.5 ppm

Maximum: 1.0E+02 ppm
Minimum: 4.1E-06 ppm

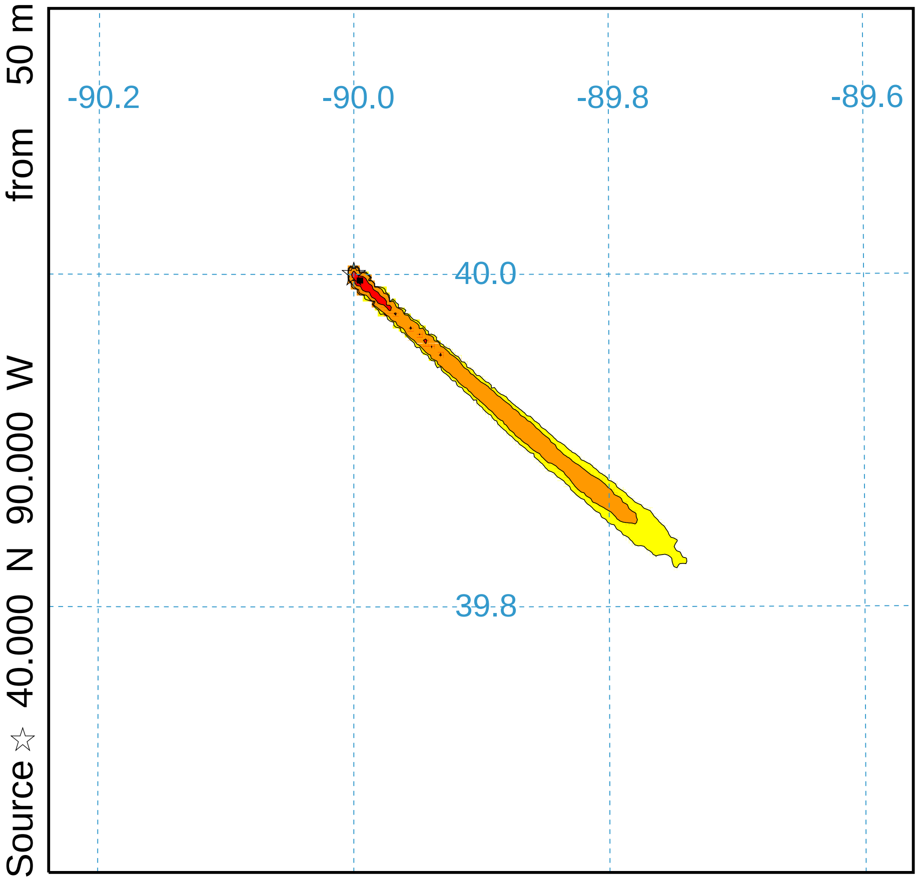
ACUTE (SHORT-TERM) EFFECTS

AEGL-3 : Life-threatening health effects possible.

AEGL-2 : Irreversible or other serious health effects that could impair the ability to take protective action.

AEGL-1 : Mild, transient health effects.

Chemical Dispersion Simulation (138)
Concentration (ppm) averaged between 0 m and 100 m
Integrated from 0000 16 Oct to 0100 16 Oct 95 (UTC)
CHEM Release started at 0000 16 Oct 95 (UTC)



NGM METEOROLOGICAL DATA

Not for Public Dissemination

AEGL-3	>20.0 ppm
AEGL-2	>2.0 ppm
AEGL-1	>0.5 ppm

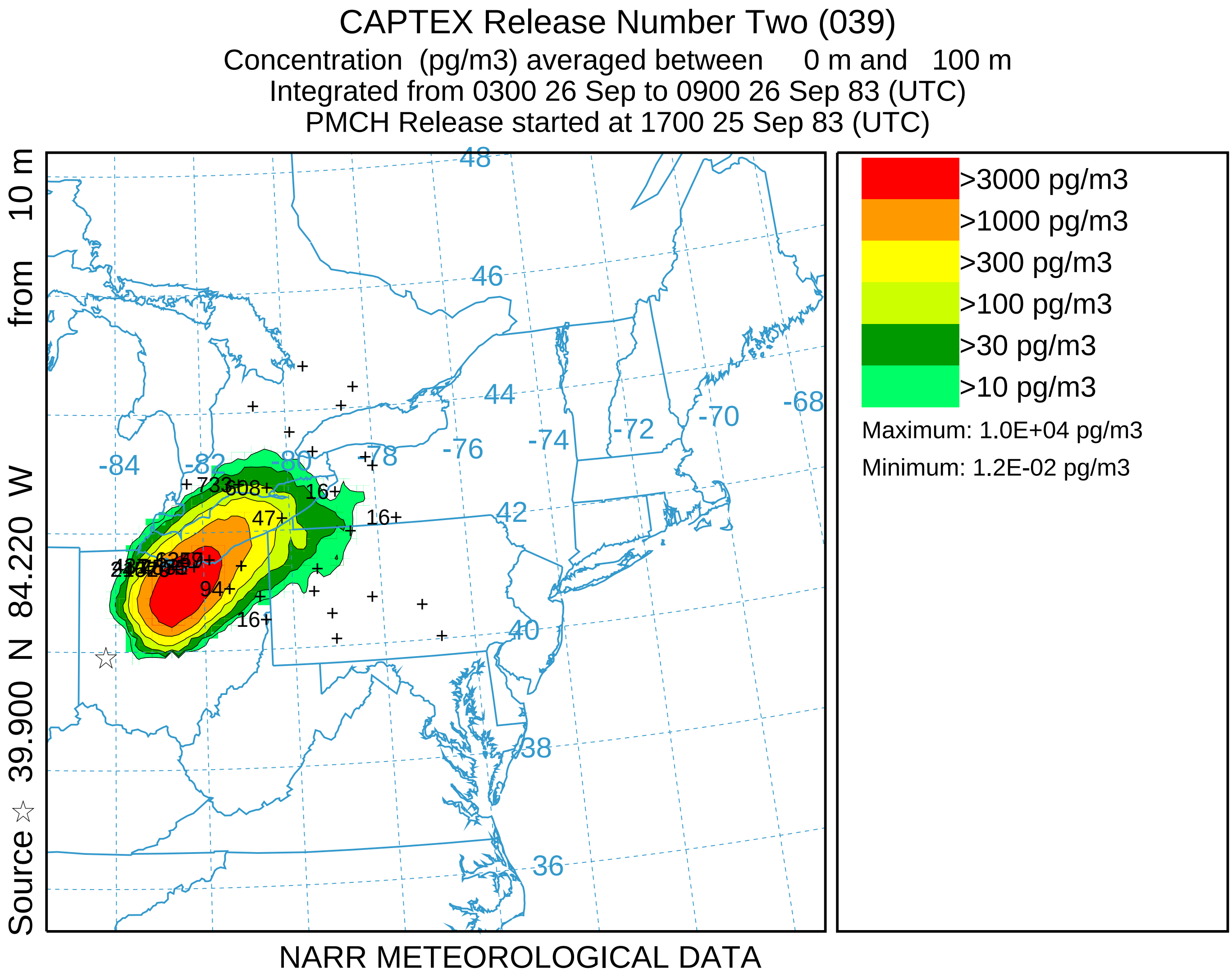
Maximum: 1.0E+02 ppm
Minimum: 5.7E-05 ppm

ACUTE (SHORT-TERM) EFFECTS

AEGL-3 : Life-threatening health effects possible.

AEGL-2 : Irreversible or other serious health effects that could impair the ability to take protective action.

AEGL-1 : Mild, transient health effects.



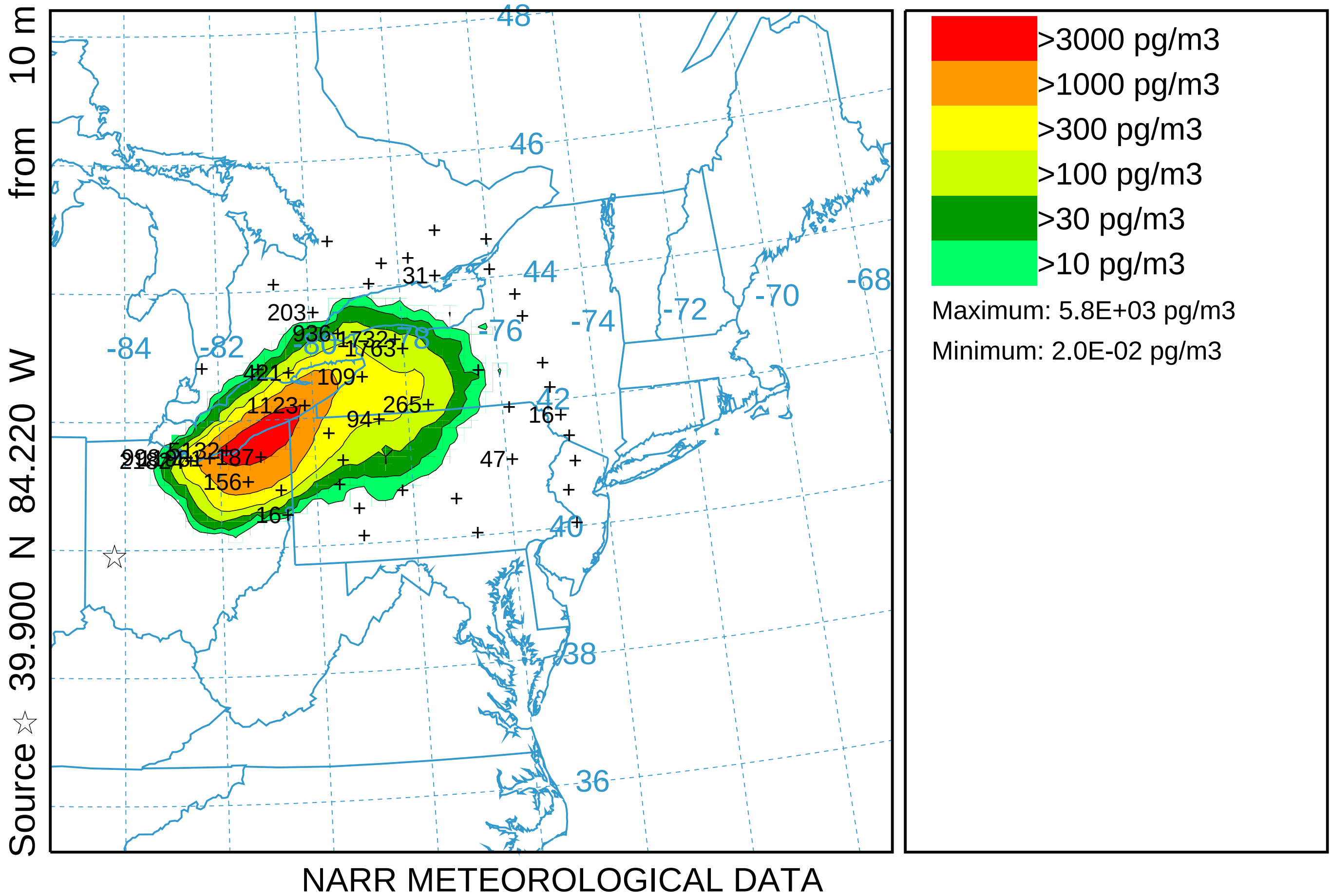
CAPTEX Release Number Two (039)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 0300 26 Sep to 0900 26 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)
>3000 pg/m3
>1000 pg/m3
>300 pg/m3
>100 pg/m3
>30 pg/m3
>10 pg/m3
Maximum: 1.0E+04 pg/m3
Minimum: 1.2E-02 pg/m3
Source ☆ 39.900 N 84.220 W from 10 m
NARR METEOROLOGICAL DATA

CAPTEX Release Number Two (039)

Concentration (pg/m3) averaged between 0 m and 100 m

Integrated from 0900 26 Sep to 1500 26 Sep 83 (UTC)

PMCH Release started at 1700 25 Sep 83 (UTC)

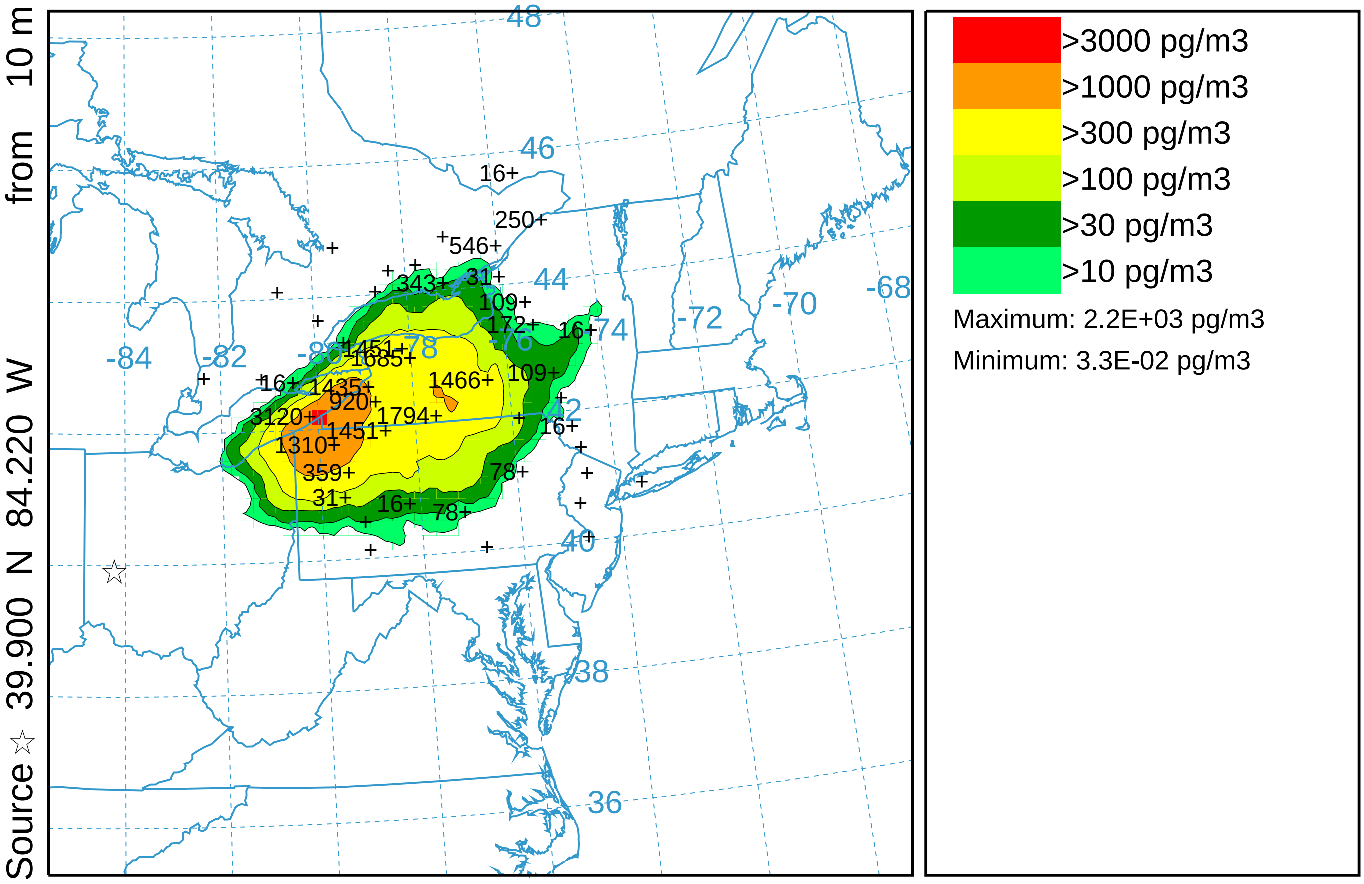


CAPTEX Release Number Two (039)

Concentration (pg/m3) averaged between 0 m and 100 m

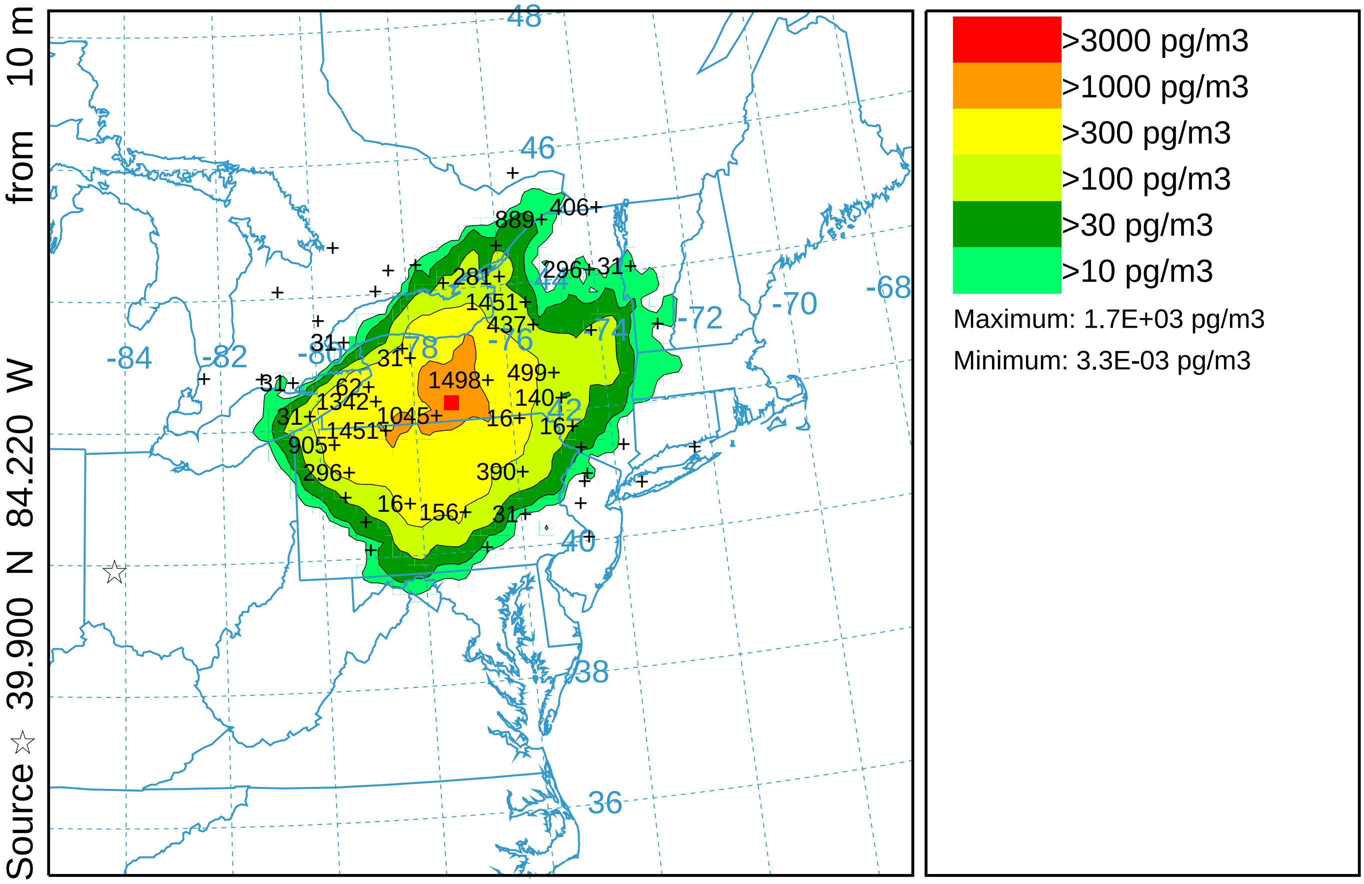
Integrated from 1500 26 Sep to 2100 26 Sep 83 (UTC)

PMCH Release started at 1700 25 Sep 83 (UTC)



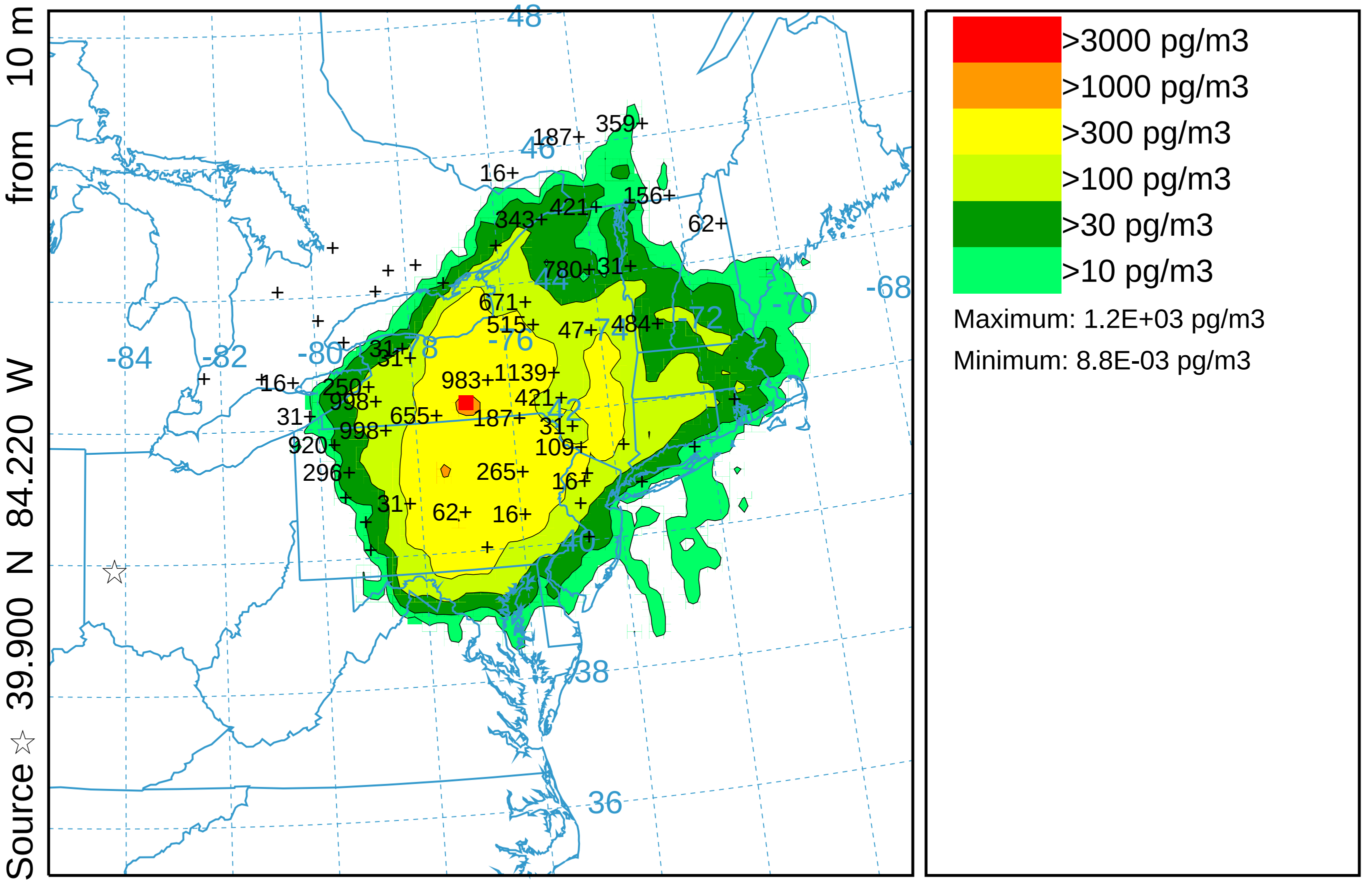
NARR METEOROLOGICAL DATA

CAPTEX Release Number Two (039)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 2100 26 Sep to 0300 27 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



NARR METEOROLOGICAL DATA

CAPTEX Release Number Two (039)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 0300 27 Sep to 0900 27 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



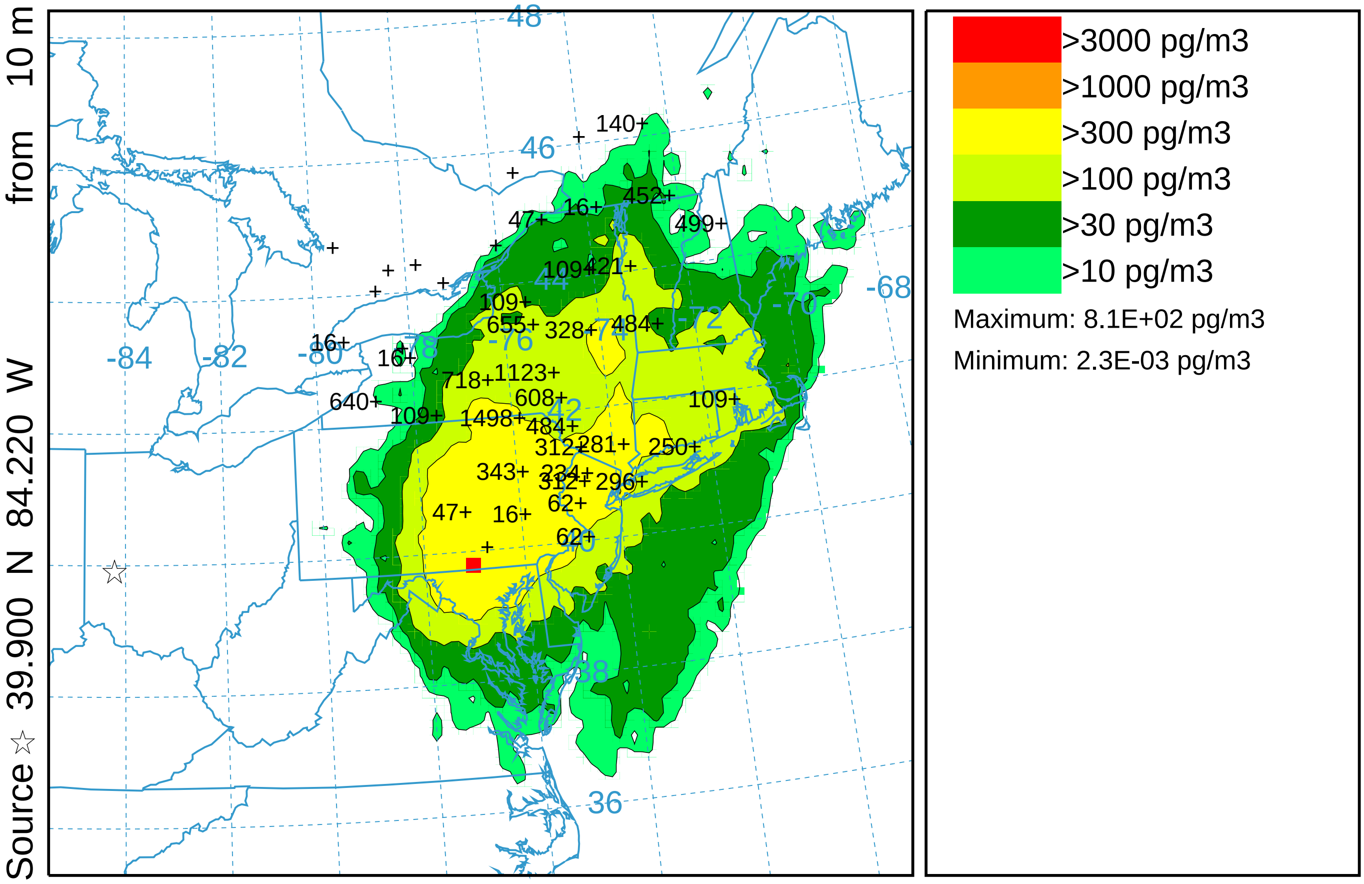
NARR METEOROLOGICAL DATA

CAPTEX Release Number Two (039)

Concentration (pg/m3) averaged between 0 m and 100 m

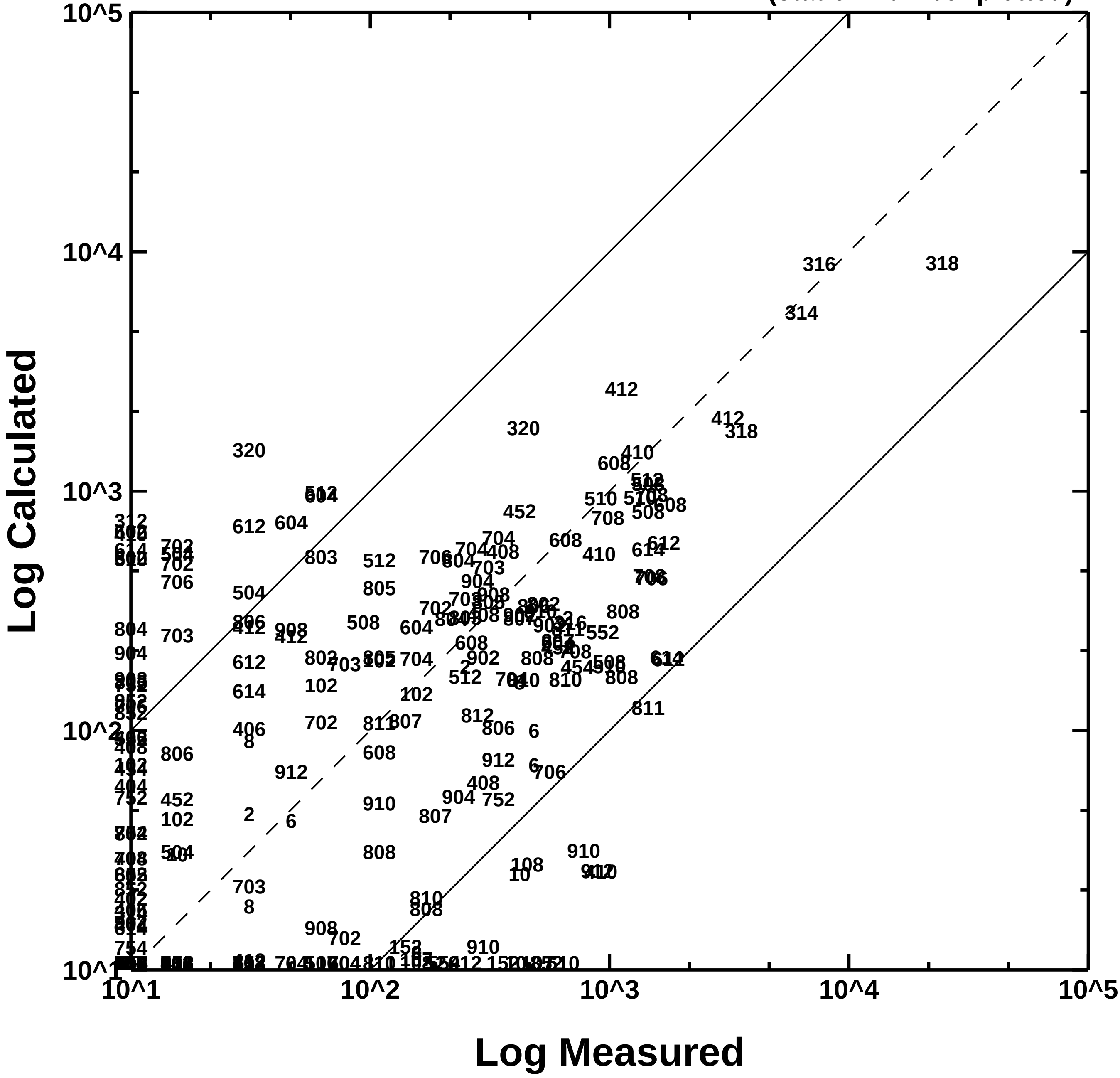
Integrated from 0900 27 Sep to 1500 27 Sep 83 (UTC)

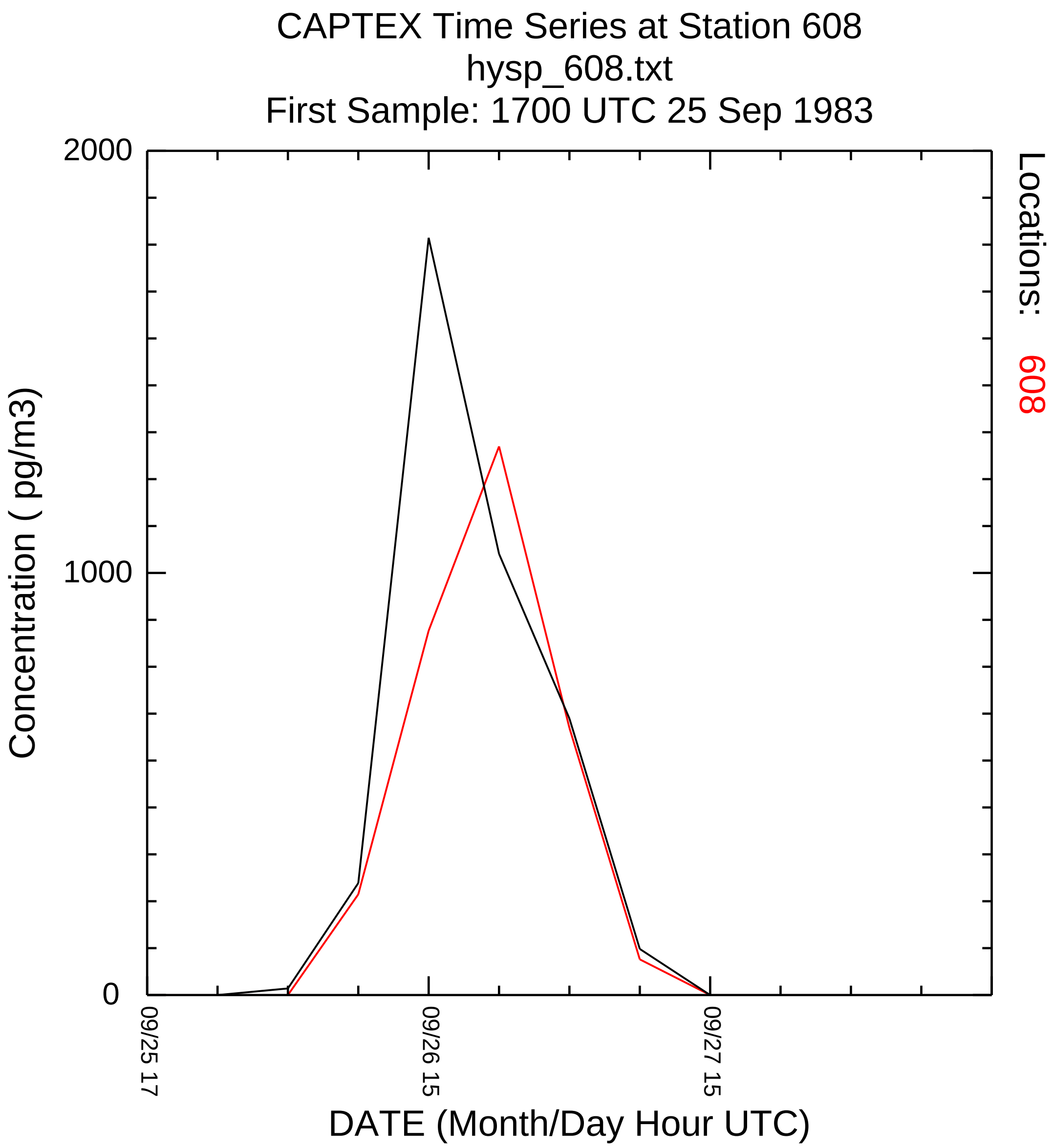
PMCH Release started at 1700 25 Sep 83 (UTC)



NARR METEOROLOGICAL DATA

CAPTEX
Release Number Two
Calculated vs Measured Concentrations
(station number plotted)





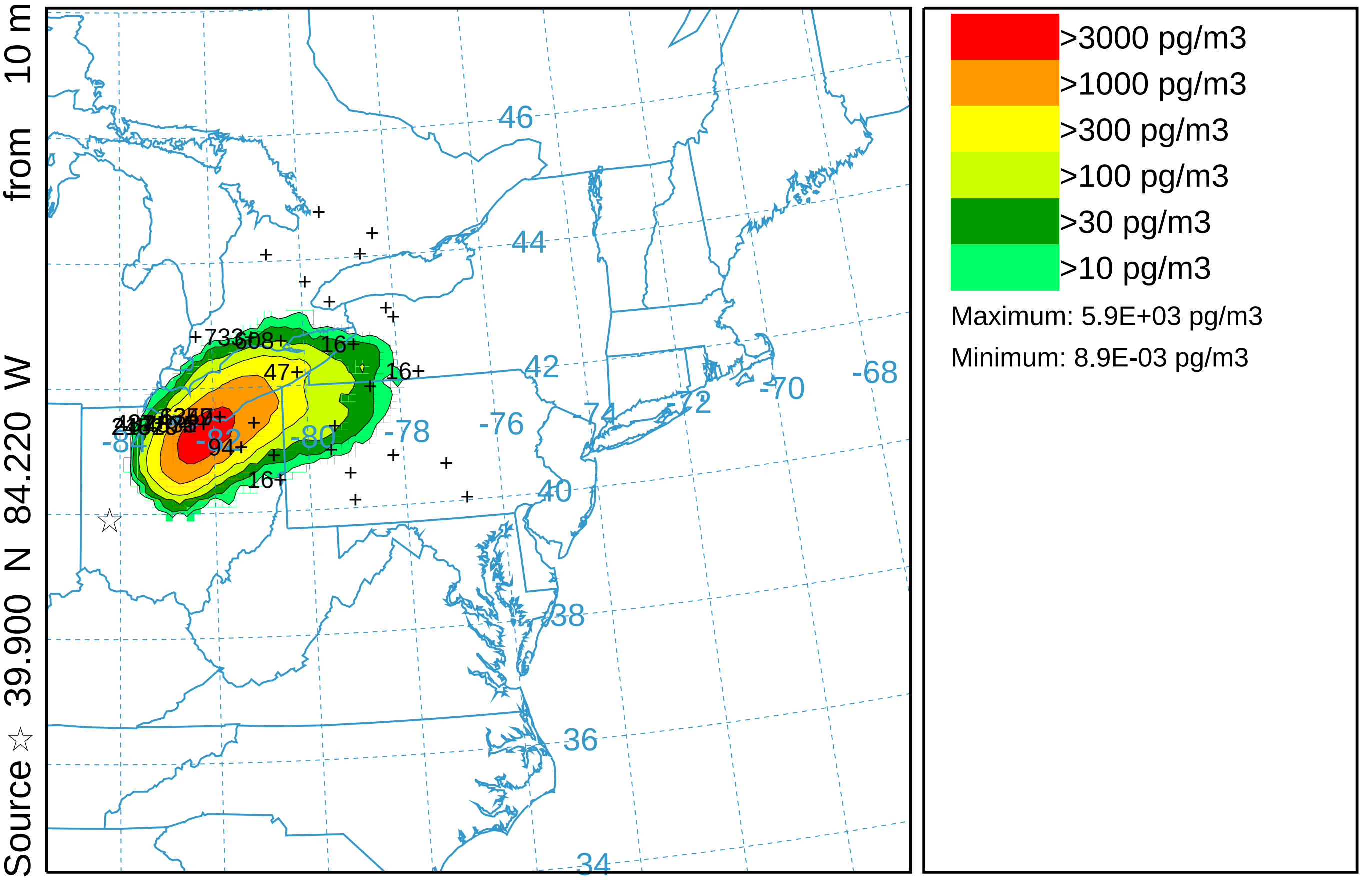
CAPTEX Time Series at Station 608
hysp_608.txt
First Sample: 1700 UTC 25 Sep 1983
2000
1000
0
Concentration (pg/m3)
09/25 17
09/26 15
09/27 15
DATE (Month/Day Hour UTC)
Locations:
608

CAPTEX Release Number Two var VSCALE (040)

Concentration (pg/m3) averaged between 0 m and 100 m

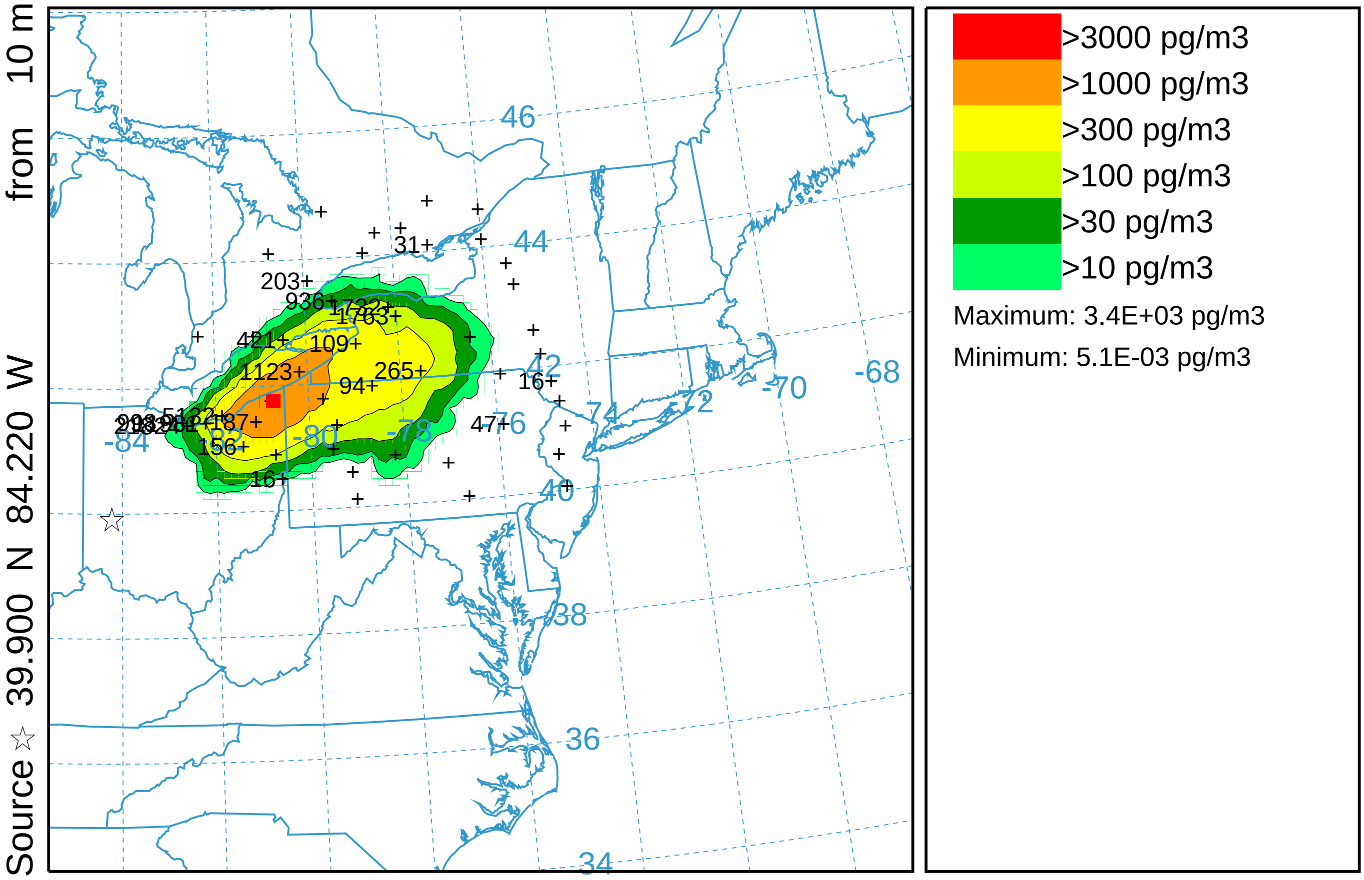
Integrated from 0300 26 Sep to 0900 26 Sep 83 (UTC)

PMCH Release started at 1700 25 Sep 83 (UTC)



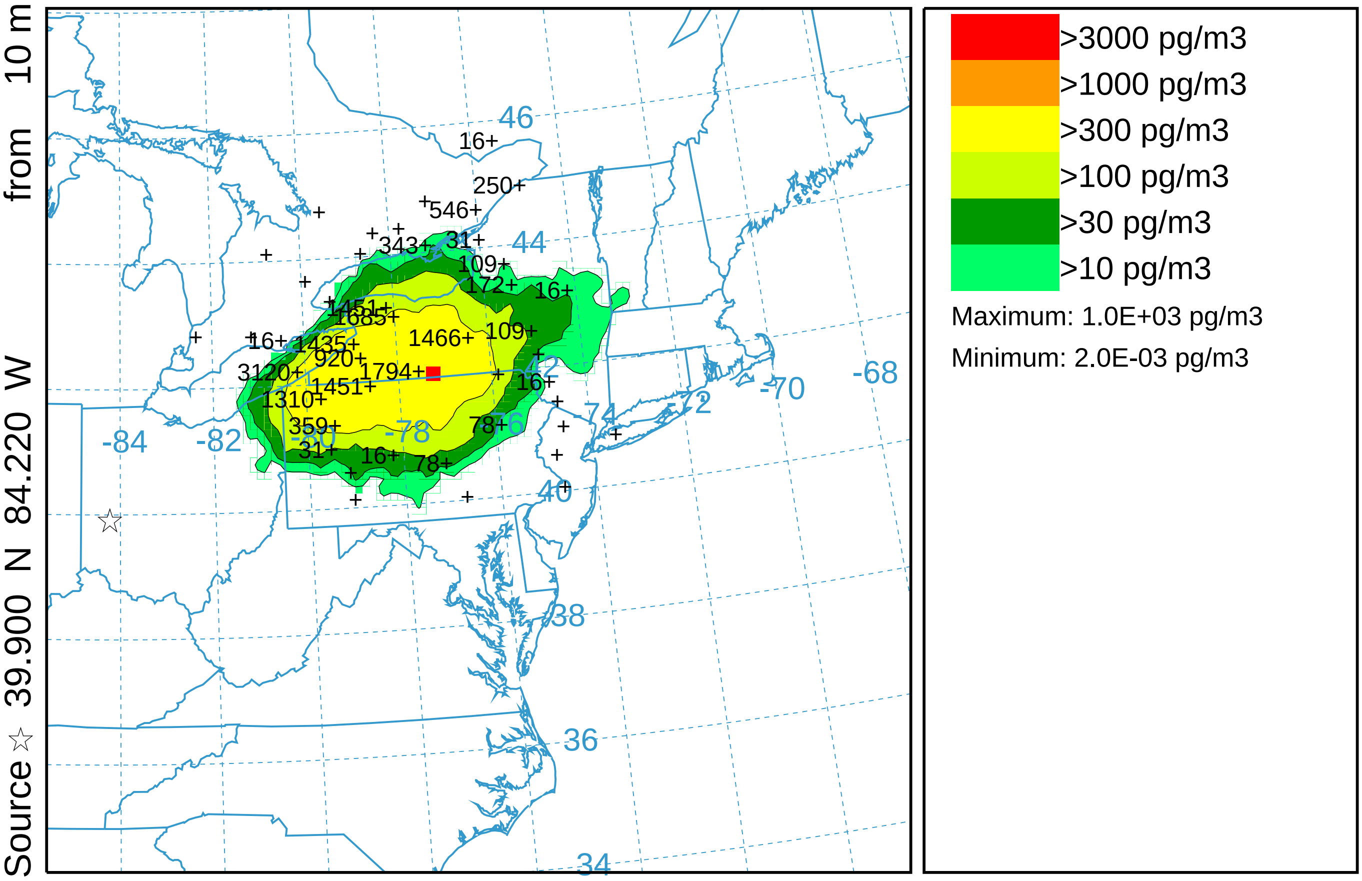
NARR METEOROLOGICAL DATA

CAPTEX Release Number Two var VSCALE (040)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 0900 26 Sep to 1500 26 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



NARR METEOROLOGICAL DATA

CAPTEX Release Number Two var VSCALE (040)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 1500 26 Sep to 2100 26 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



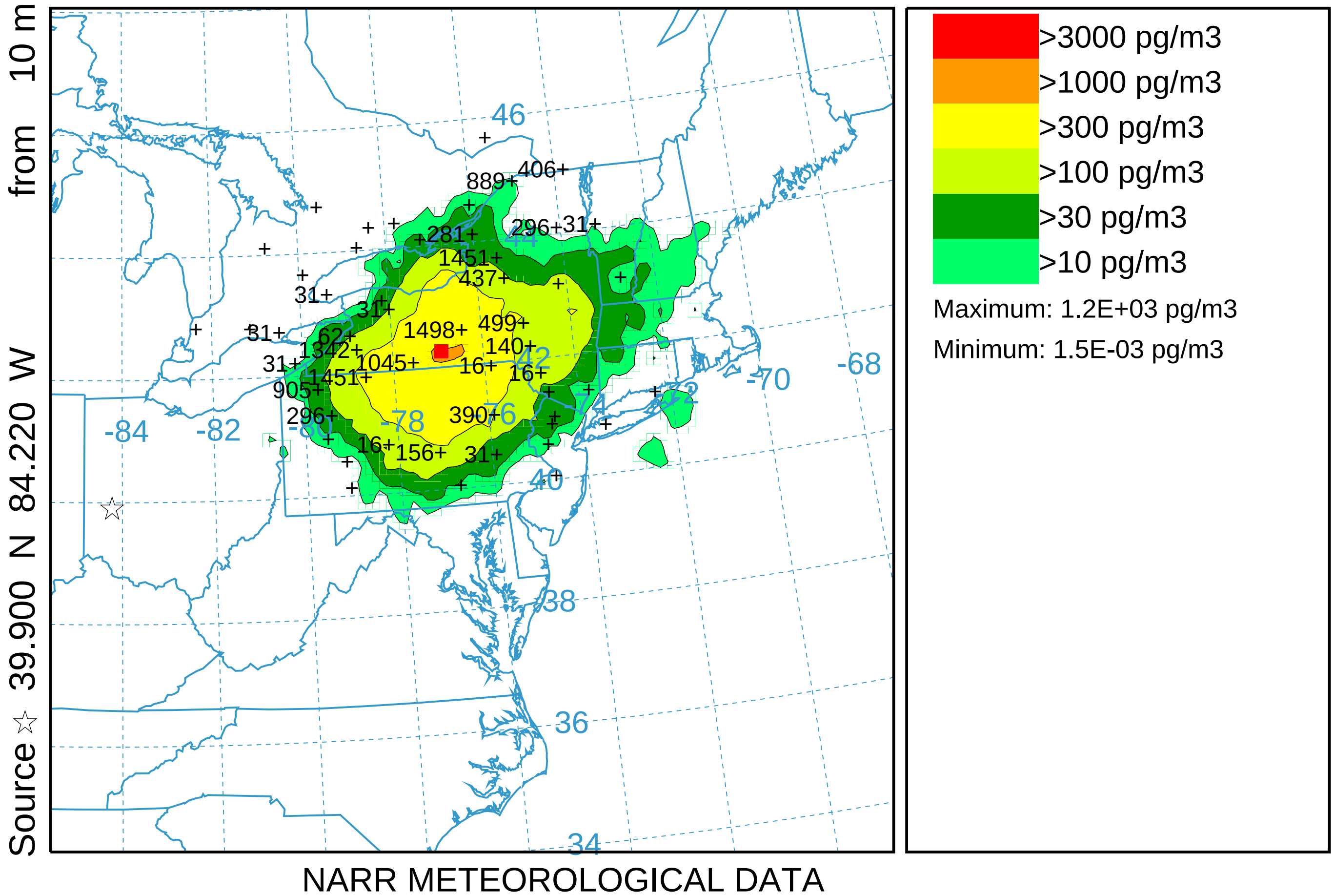
NARR METEOROLOGICAL DATA

CAPTEX Release Number Two var VSCALE (040)

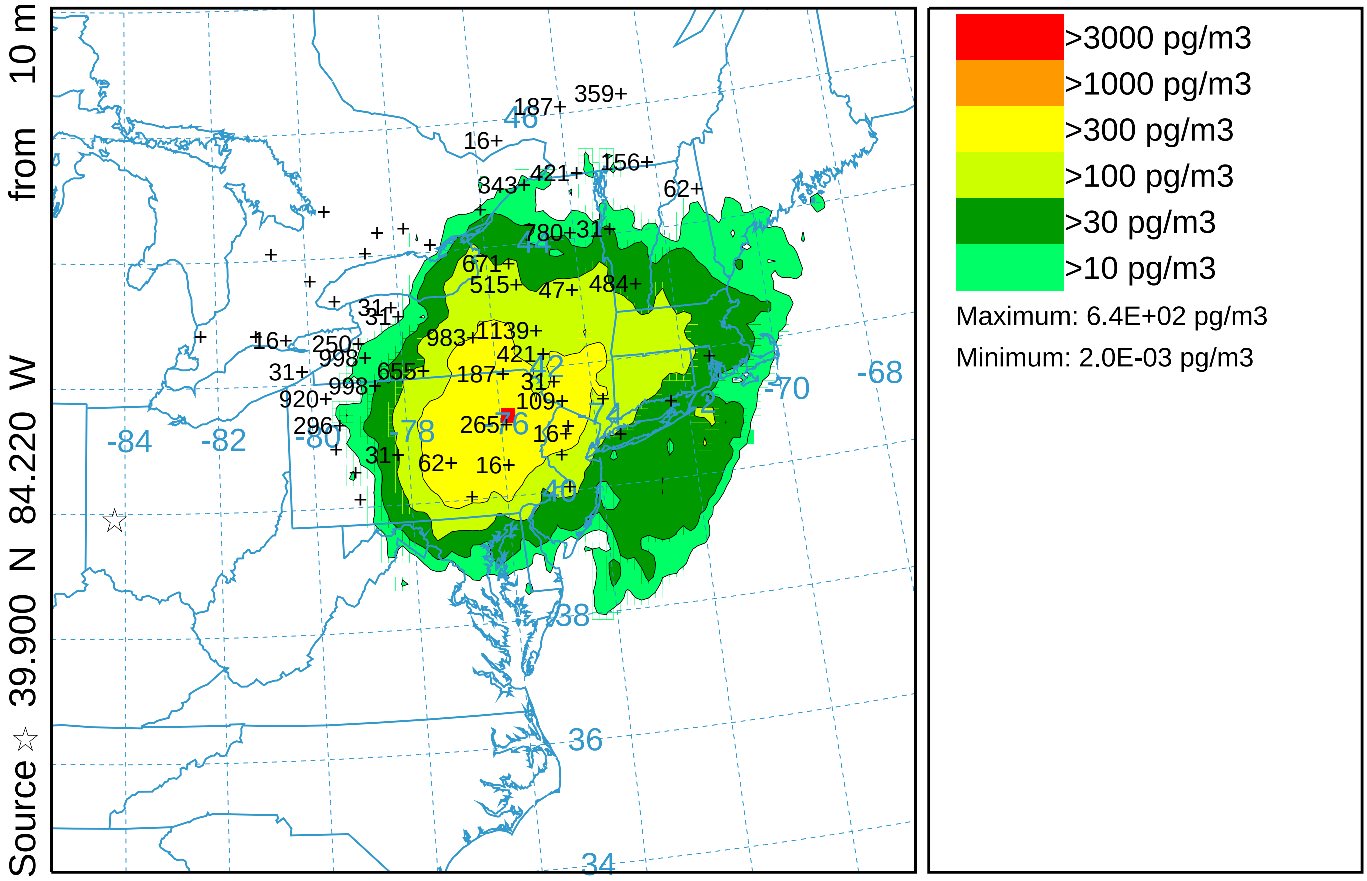
Concentration (pg/m3) averaged between 0 m and 100 m

Integrated from 2100 26 Sep to 0300 27 Sep 83 (UTC)

PMCH Release started at 1700 25 Sep 83 (UTC)

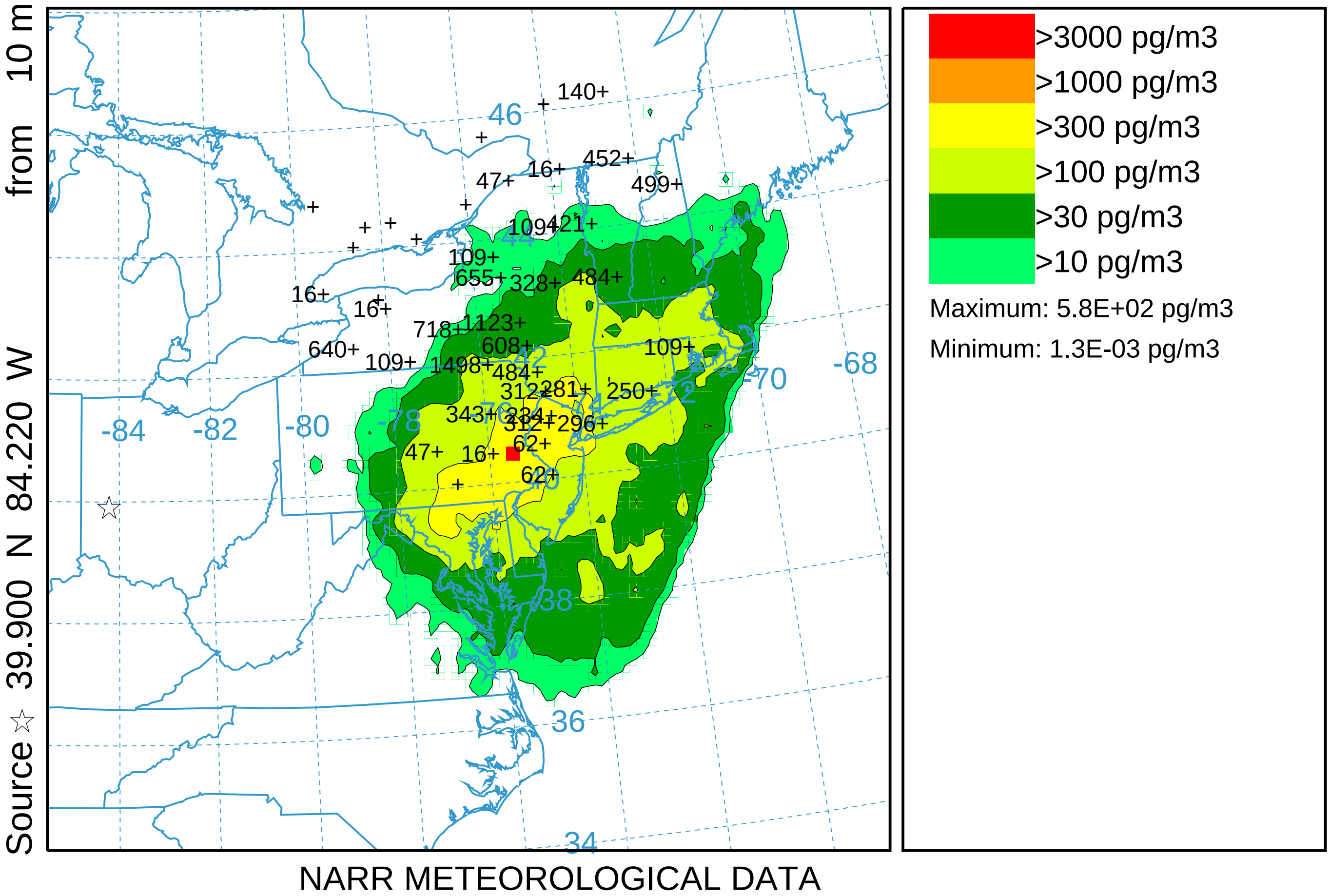


CAPTEX Release Number Two var VSCALE (040)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 0300 27 Sep to 0900 27 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



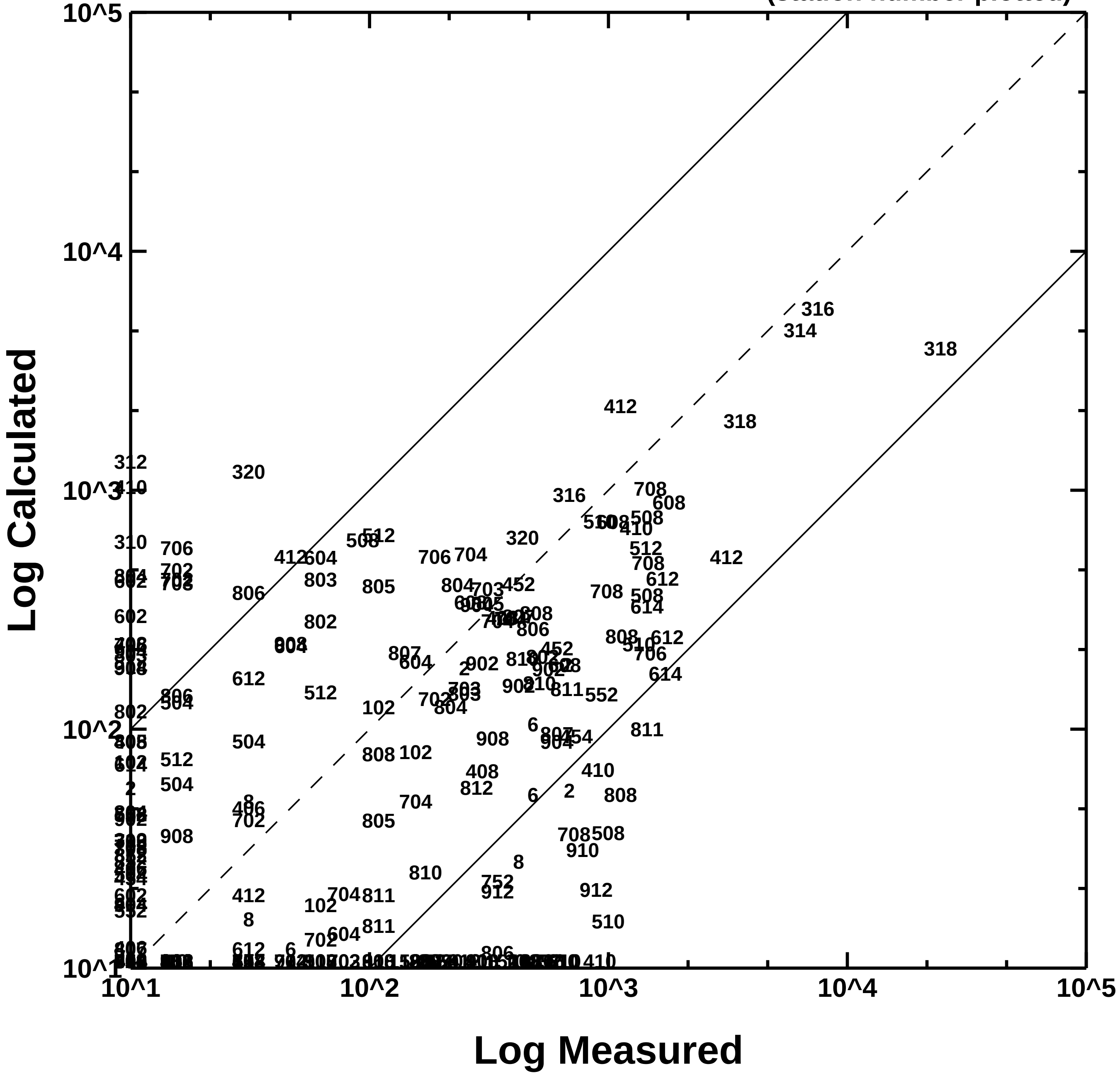
NARR METEOROLOGICAL DATA

CAPTEX Release Number Two var VSCALE (040)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 0900 27 Sep to 1500 27 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)

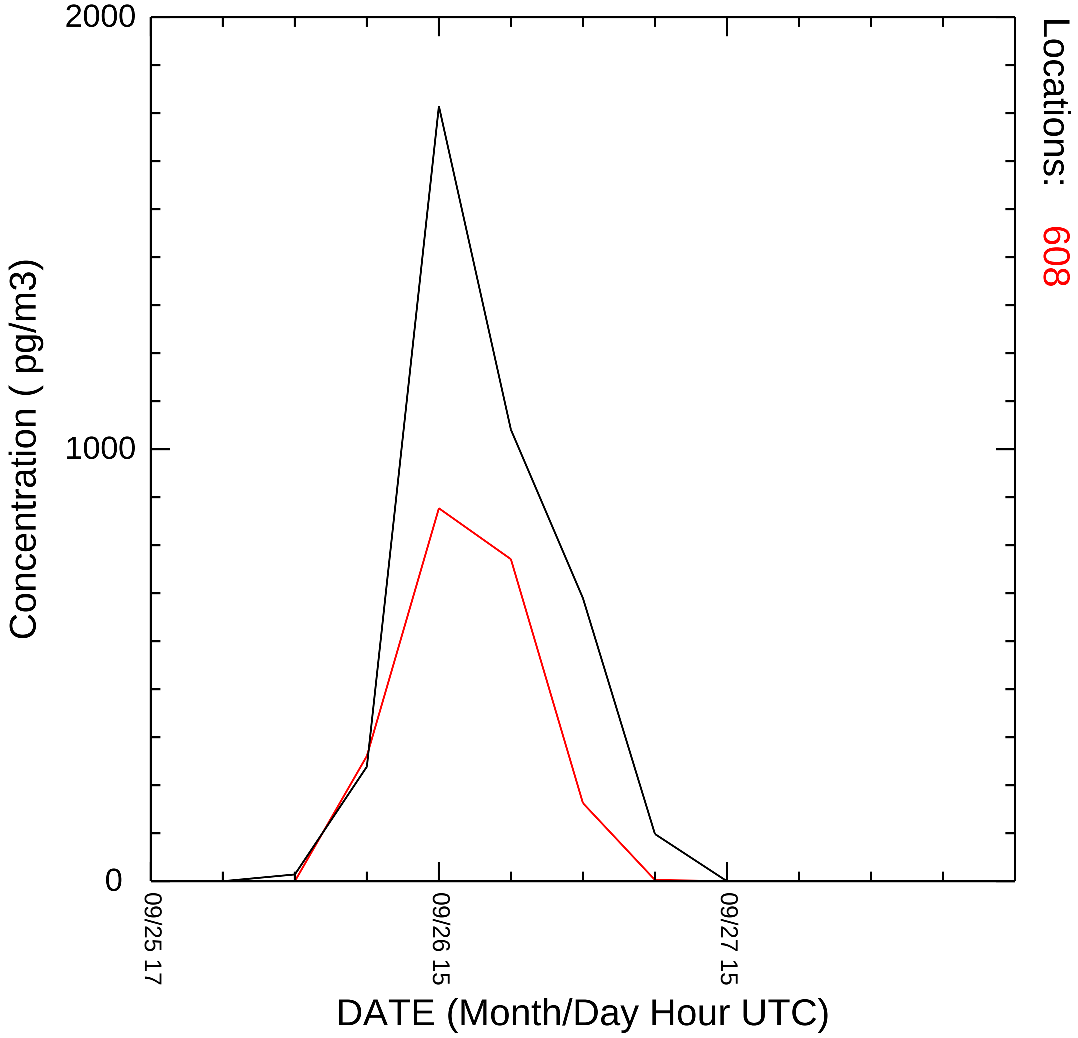


CAPTEX
Release Number Two
Calculated vs Measured Concentrations

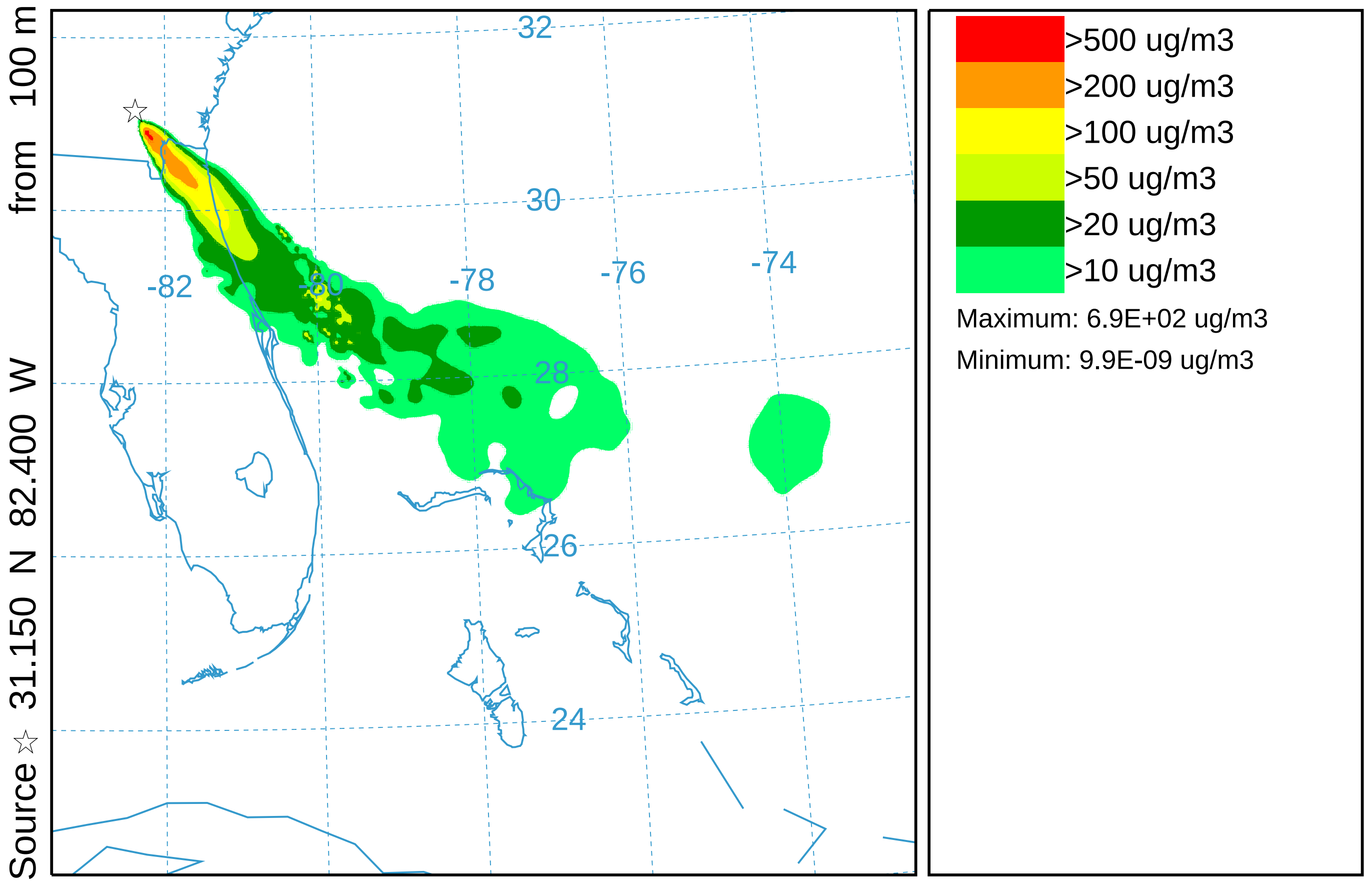
(station number plotted)



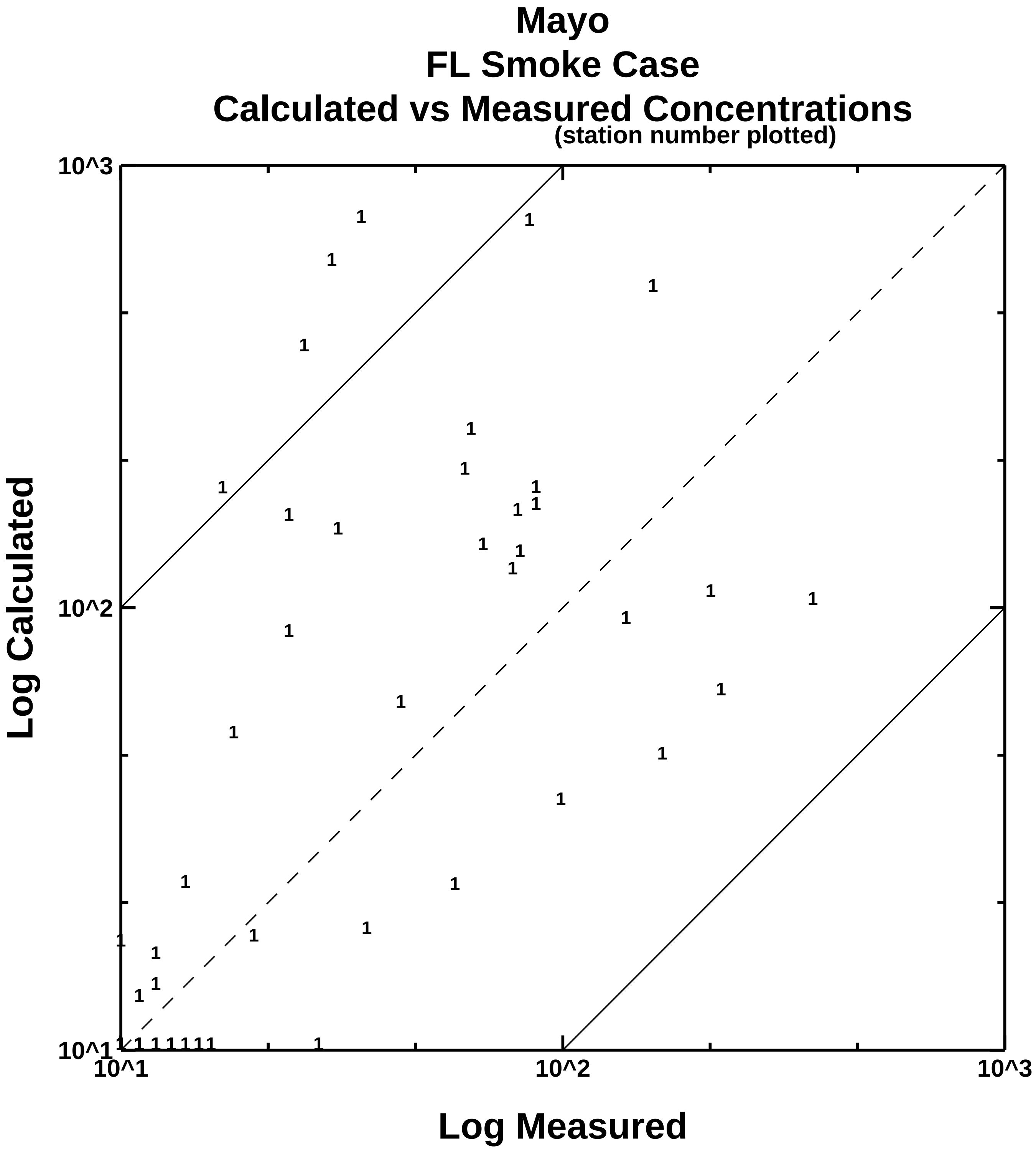
CAPTEX Time Series at Station 608 var VSCALE
hysp_608.txt
First Sample: 1700 UTC 25 Sep 1983



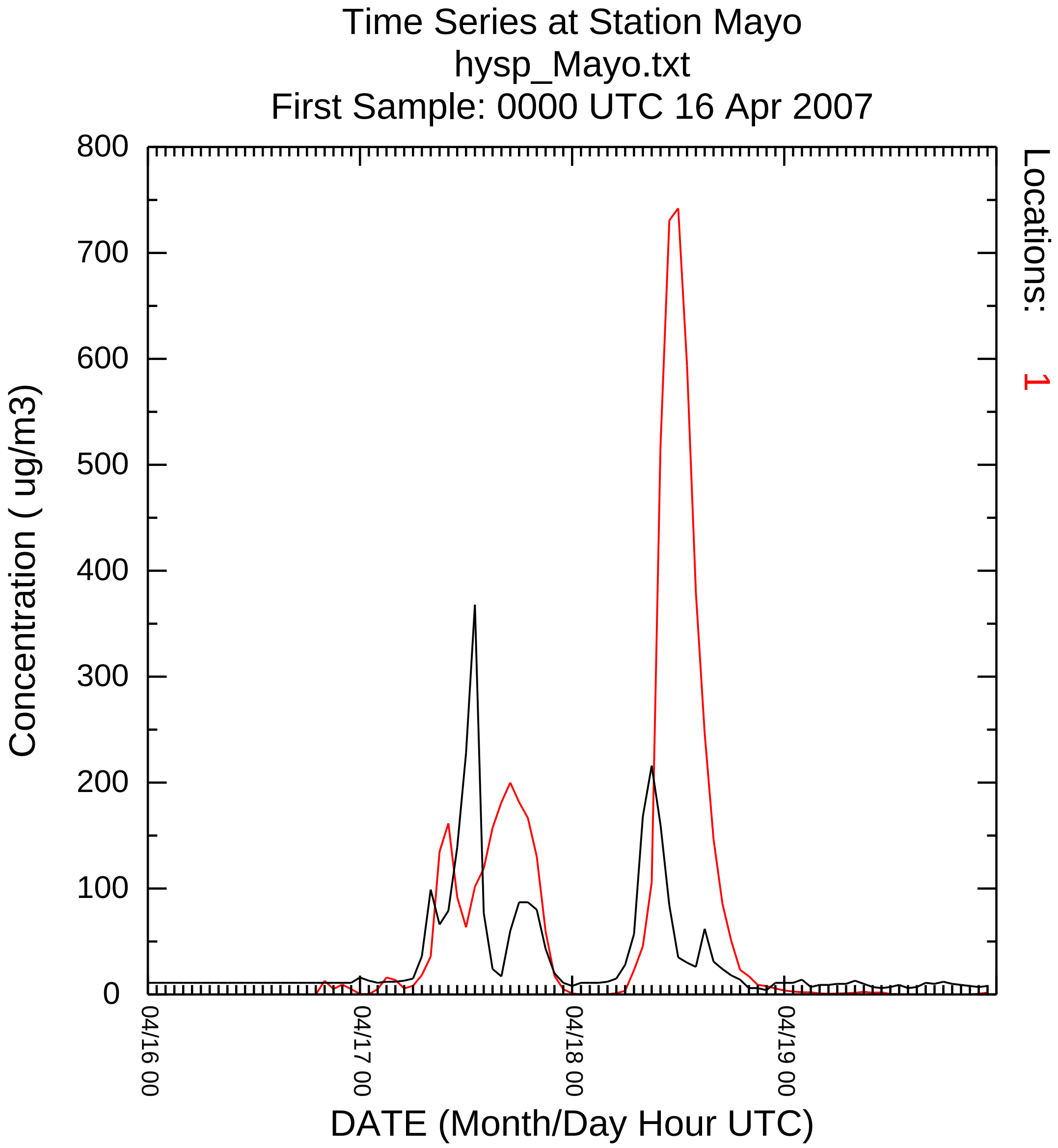
Sweat Farm Road Smoke Simulation (041)
Concentration (ug/m3) averaged between 0 m and 100 m
Integrated from 1500 17 Apr to 1600 17 Apr 07 (UTC)
PM25 Release started at 1800 16 Apr 07 (UTC)



EDAS METEOROLOGICAL DATA

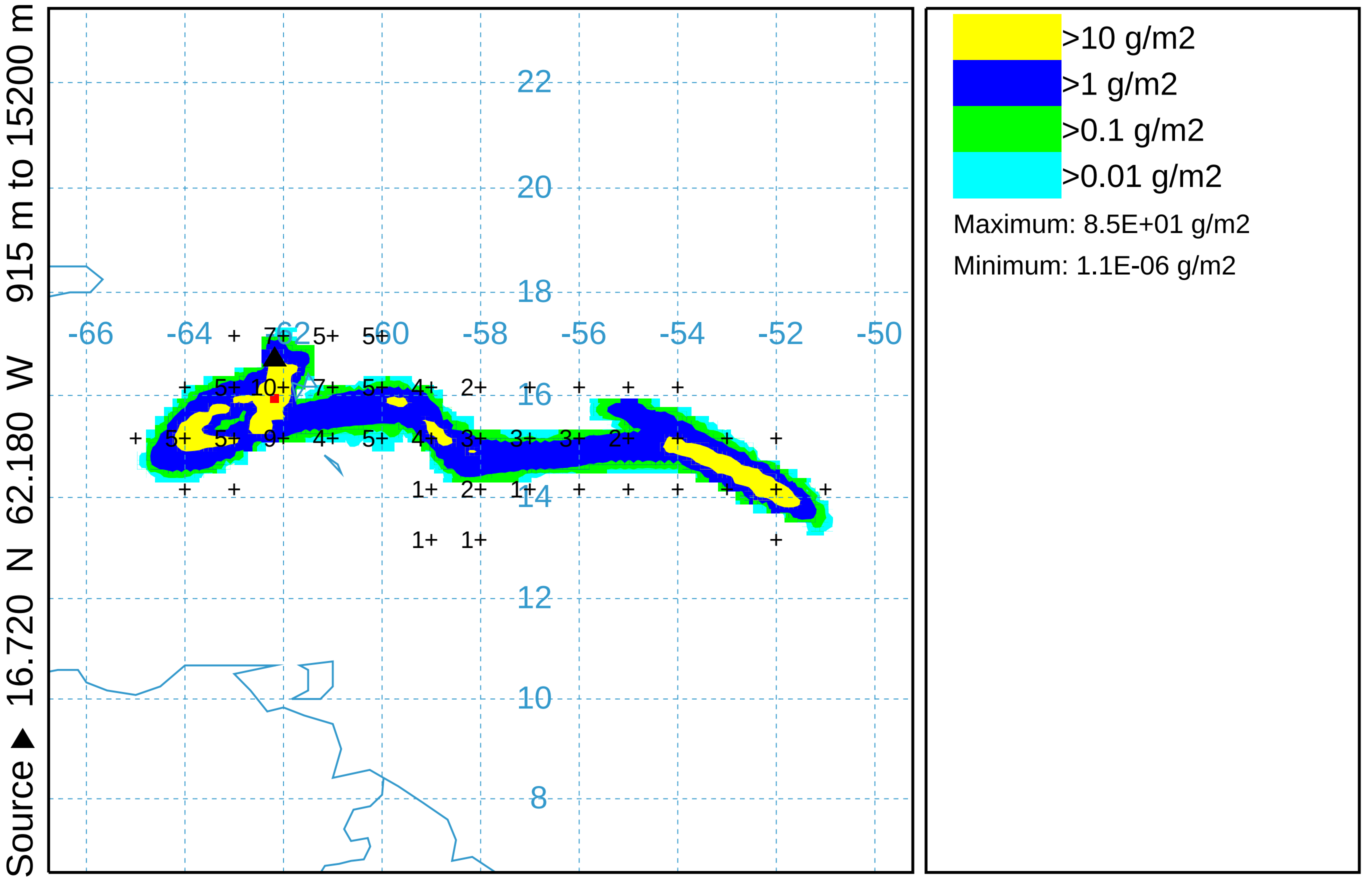


Mayo
FL Smoke Case
Calculated vs Measured Concentrations
(station number plotted)
10^3
10^2
10^1
10^1
10^2
10^3
Log Calculated
Log Measured


Time Series at Station Mayo
hysp_Mayo.txt
First Sample: 0000 UTC 16 Apr 2007
Concentration (ug/m3)
800
700
600
500
400
300
200
100
0
04/16 00
04/17 00
04/18 00
04/19 00
DATE (Month/Day Hour UTC)
Locations:
1

Volcano Soufriere_Hills Simulation (042)

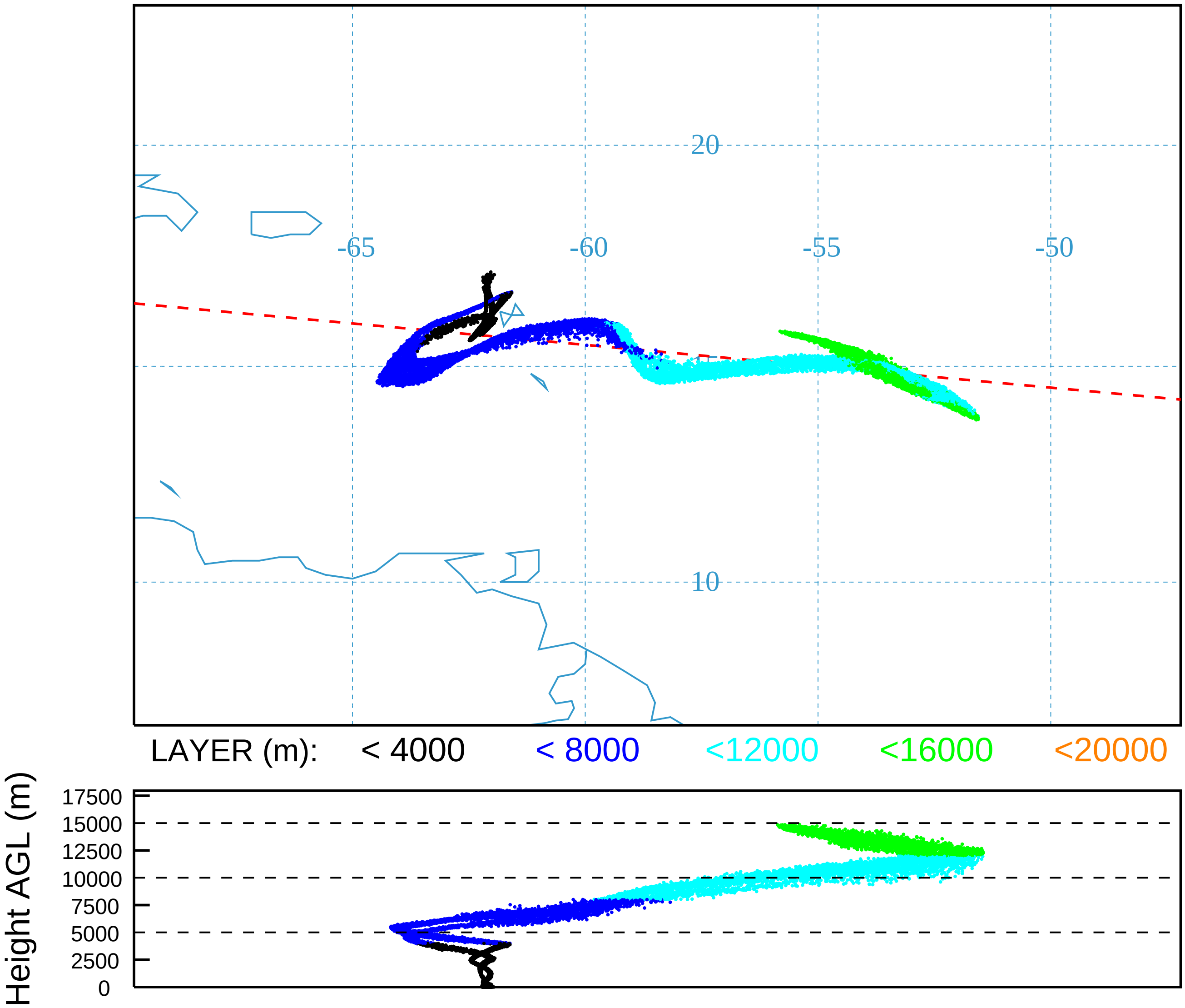
Mass loading (g/m2) between 0 m and 18000 m
Integrated from 0600 12 Feb to 0700 12 Feb 10 (UTC)
SUM Release started at 1635 11 Feb 10 (UTC)



GDAS METEOROLOGICAL DATA

Job Start: Mon Mar 10 17:48:15 UTC 2014
Volcano Soufriere_Hills lat: 16.72 lon: -62.18 Source Hgt: 915 to 15200 m, msl
Release Quantity*: 1.24E+9 kg Start: 10 02 11 16 35 Duration: 1 hrs, 0 min
Vertical distribution: uniform GSD: Default Particles: 32000
Pollutant Averaging/Integration Period: 1 hr
Wet Removal (below/in-cloud): 0.0 / 0.0 Calculated ash mass loading per
Meteorological Data: GDAS1 Mastin et al, doi:10.1016/
Observed ash mass loading from (2013) j.jvolgeores.2009.01.008
Pavolonis et al. doi:10.1002/jgrd.50173 m63=0.03 constant=55 exponent=4.5

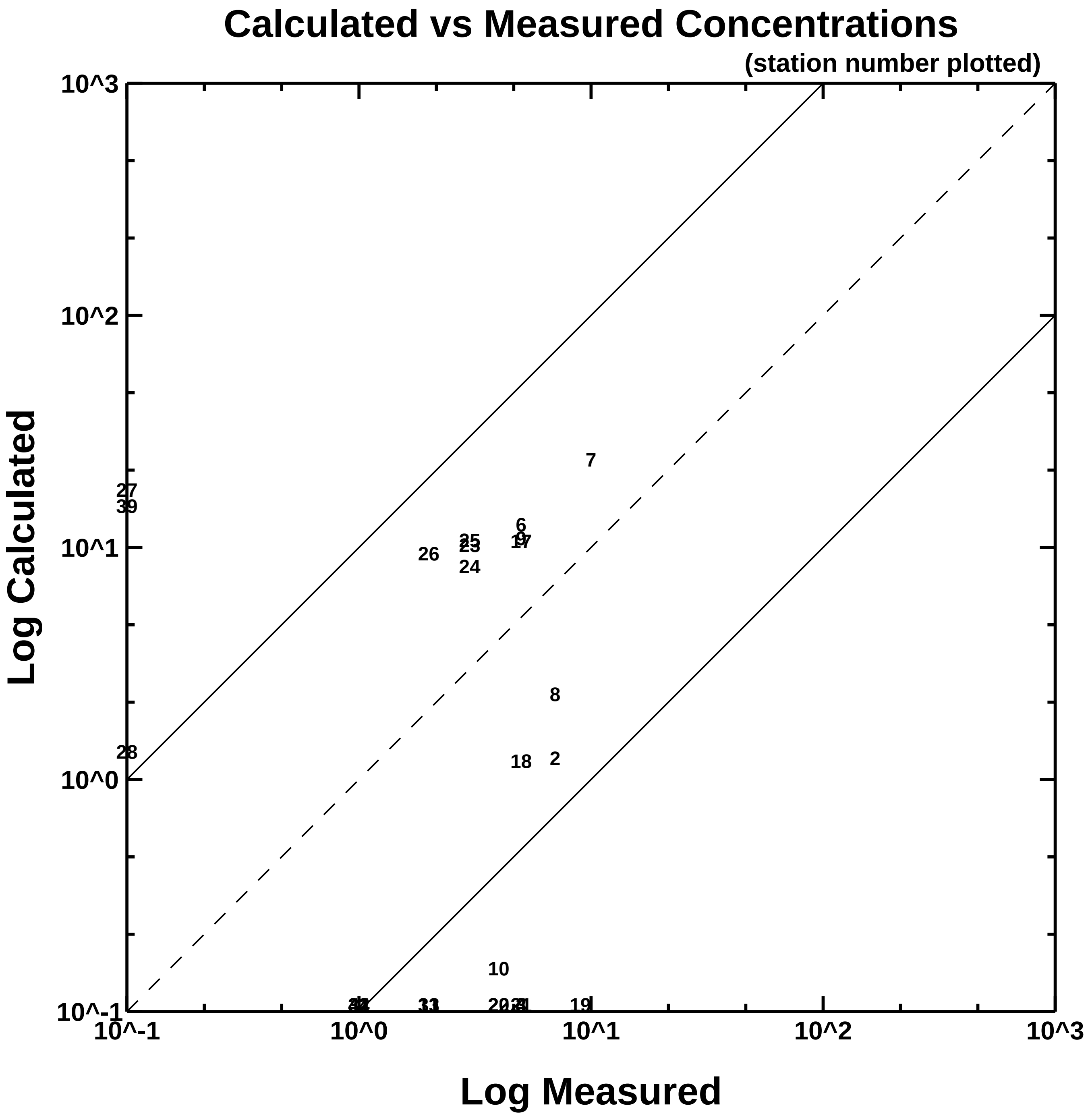
NOAA HYSPLIT MODEL
PARTICLE CROSS-SECTIONS
PARTICLE POSITIONS AT 06 00 12 Feb 10



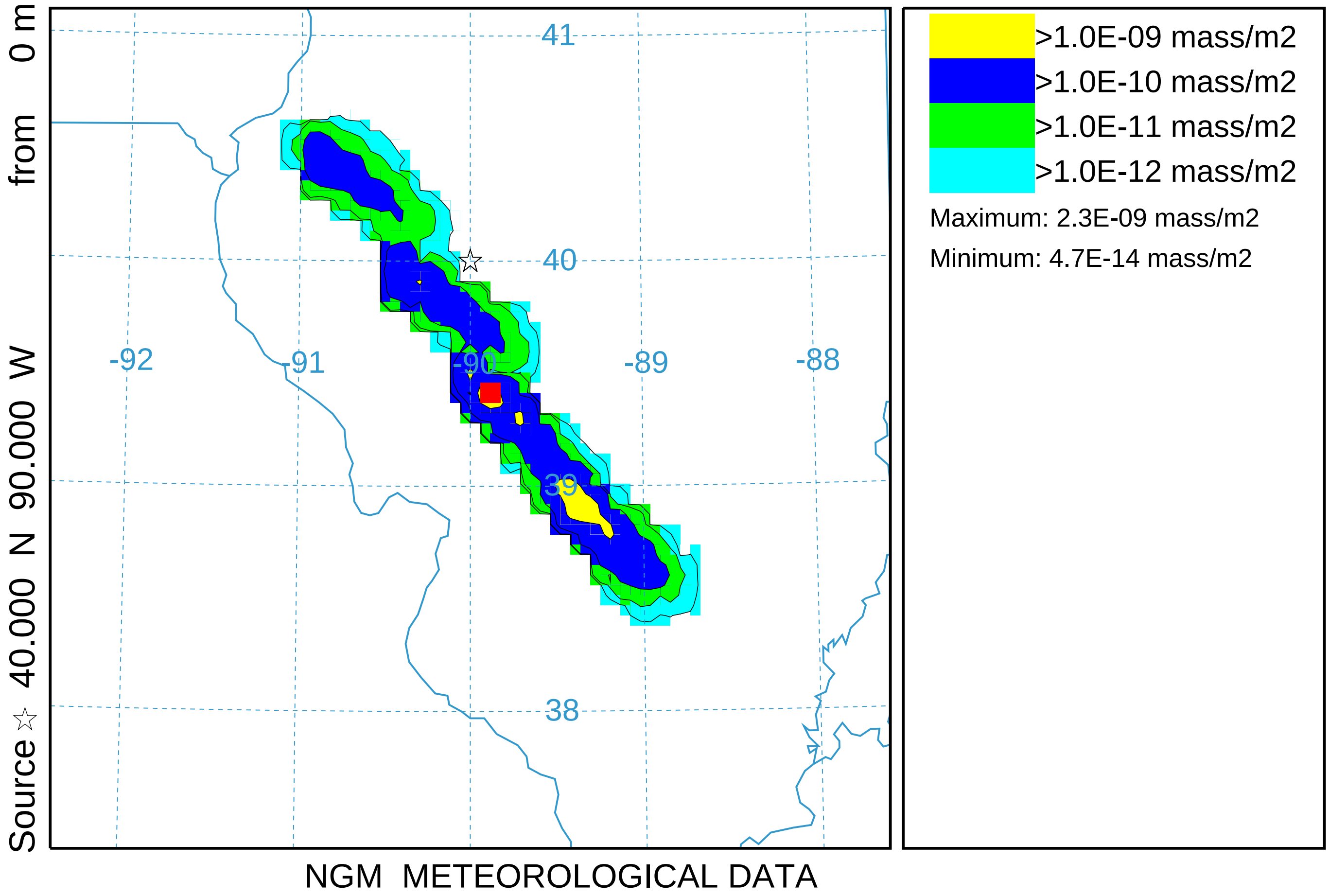
NUMBER OF PARTICLES PLOTTED: 16666 (skip interval 03)

Job Start: Mon Mar 10 17:48:15 UTC 2014
Volcano Soufriere_Hills lat: 16.72 lon: -62.18 Source Hgt: 915 to 15200 m, msl
Release Quantity*: 1.24E+9 kg Start: 10 02 11 16 35 Duration: 1 hrs, 0 min
Vertical distribution: uniform GSD: Default Particles: 32000
Pollutant Averaging/Integration Period: 1 hr
Wet Removal (below/in-cloud): 0.0 / 0.0 Calculated ash mass loading per
Meteorological Data: GDAS1 Mastin et al, doi:10.1016/
Observed ash mass loading from (2013) j.jvolgeores.2009.01.008
Pavolonis et al. doi:10.1002/jgrd.50173 m63=0.03 constant=55 exponent=4.5

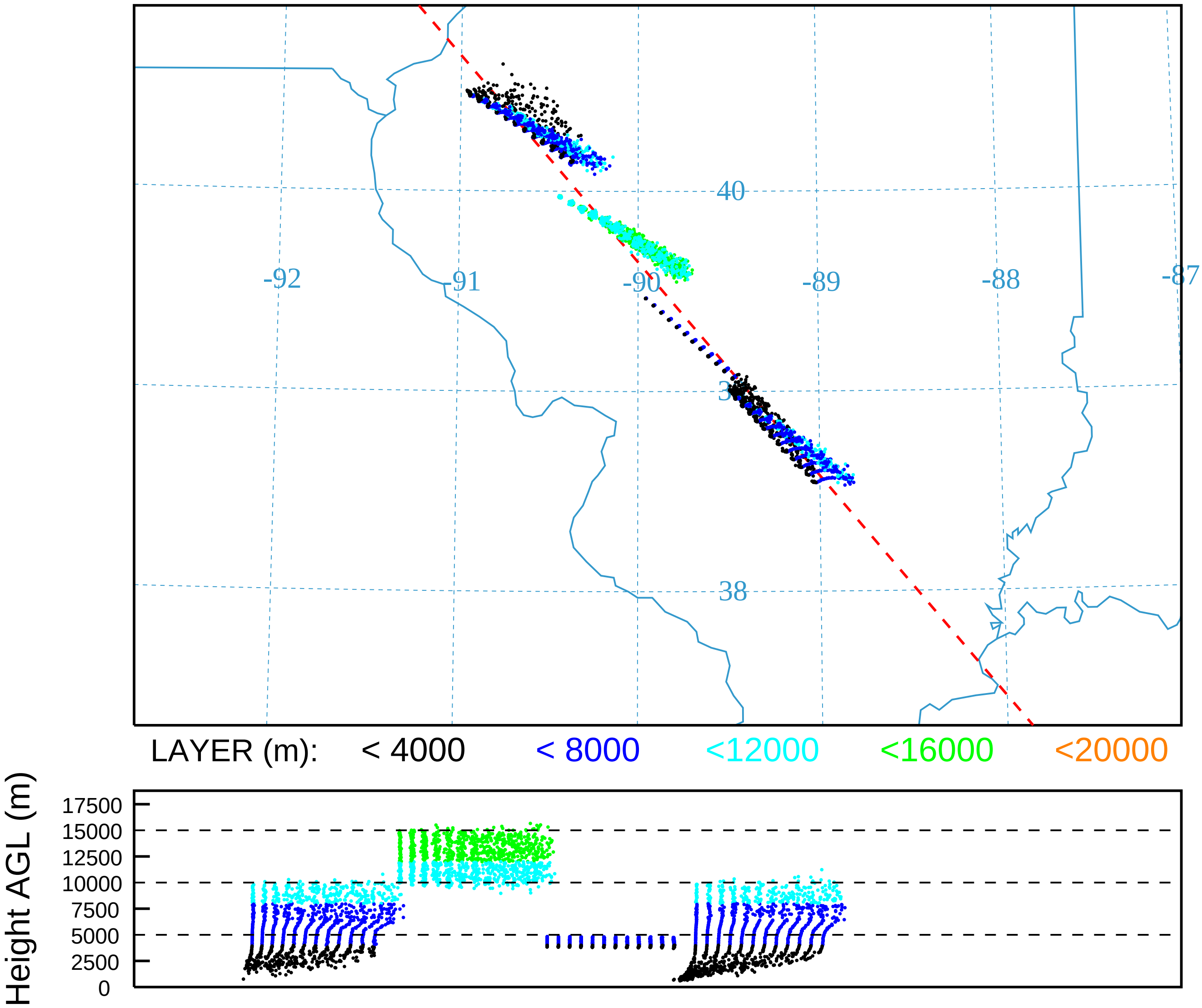
SOUFRIERE_HILLS



Four line sources with an EMITTIMES file (043)
Mass loading (mass/m2) between 0 m and 20000 m
Integrated from 0000 17 Oct to 0100 17 Oct 95 (UTC)
SUM Release started at 0000 17 Oct 95 (UTC)

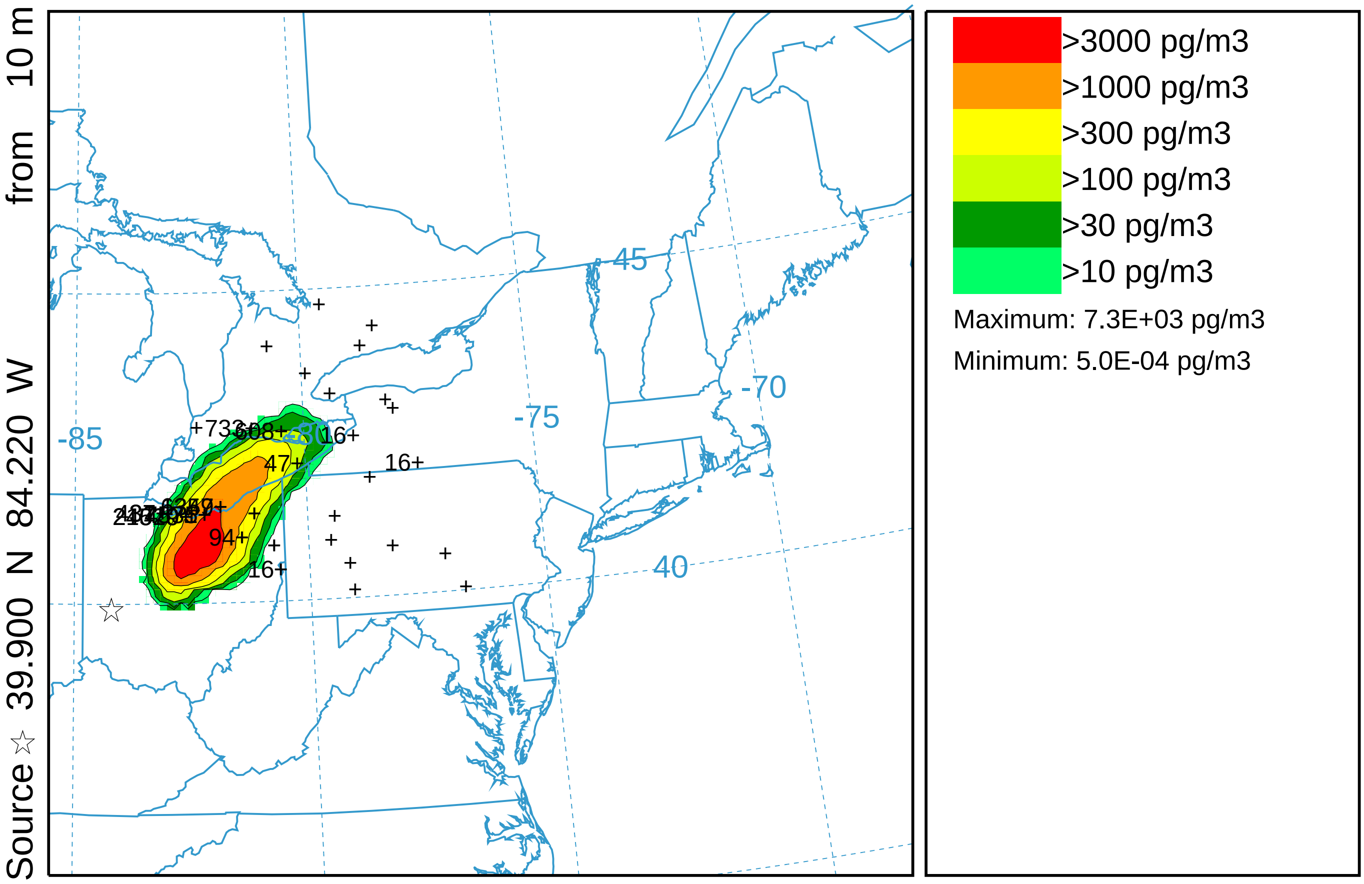


NOAA HYSPLIT MODEL
PARTICLE CROSS-SECTIONS
PARTICLE POSITIONS AT 01 00 17 Oct 95



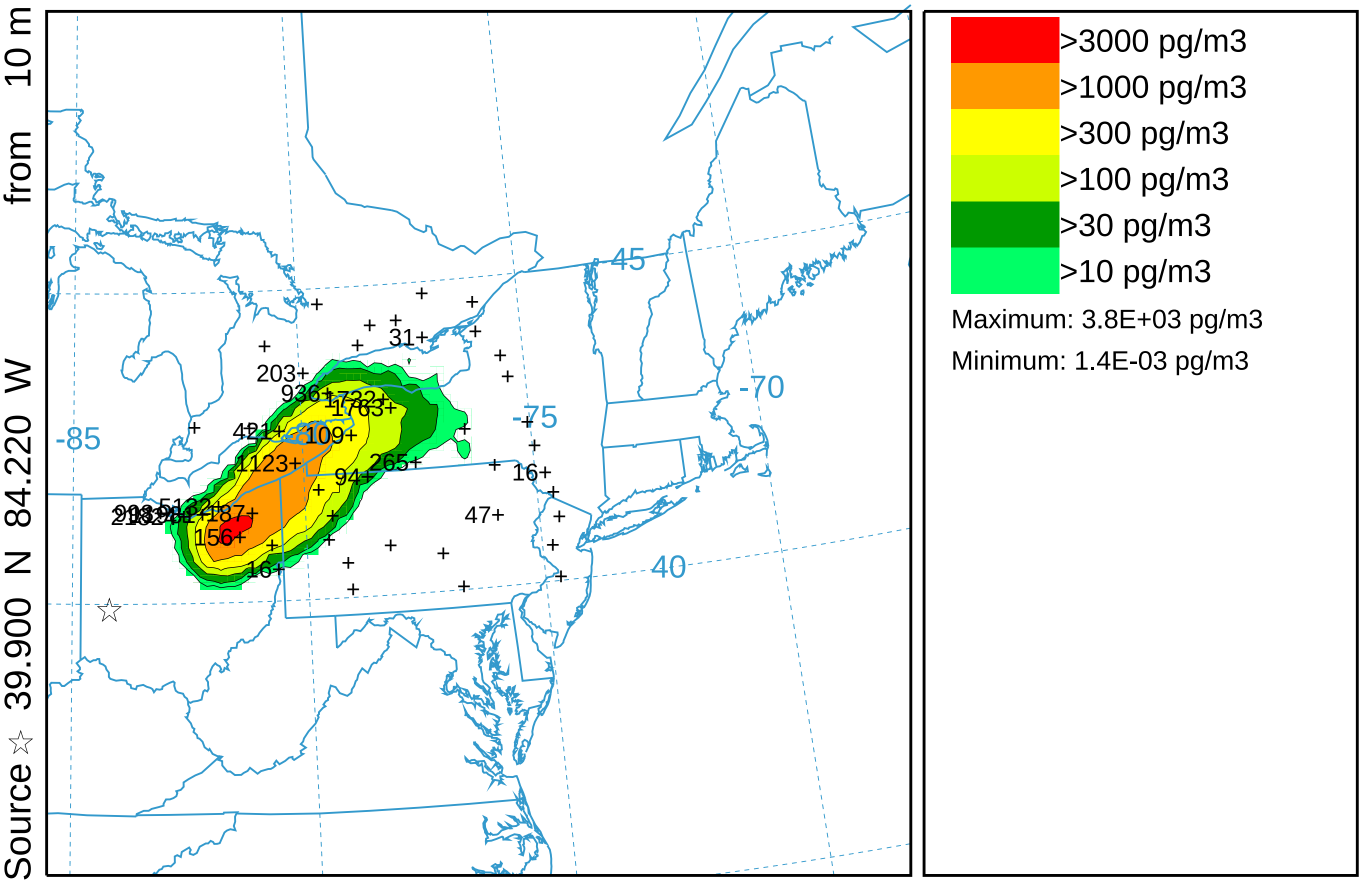
NUMBER OF PARTICLES PLOTTED: 5040

CAPTEX Release Number Two ARW-WRF kblt=2 (050)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 0300 26 Sep to 0900 26 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



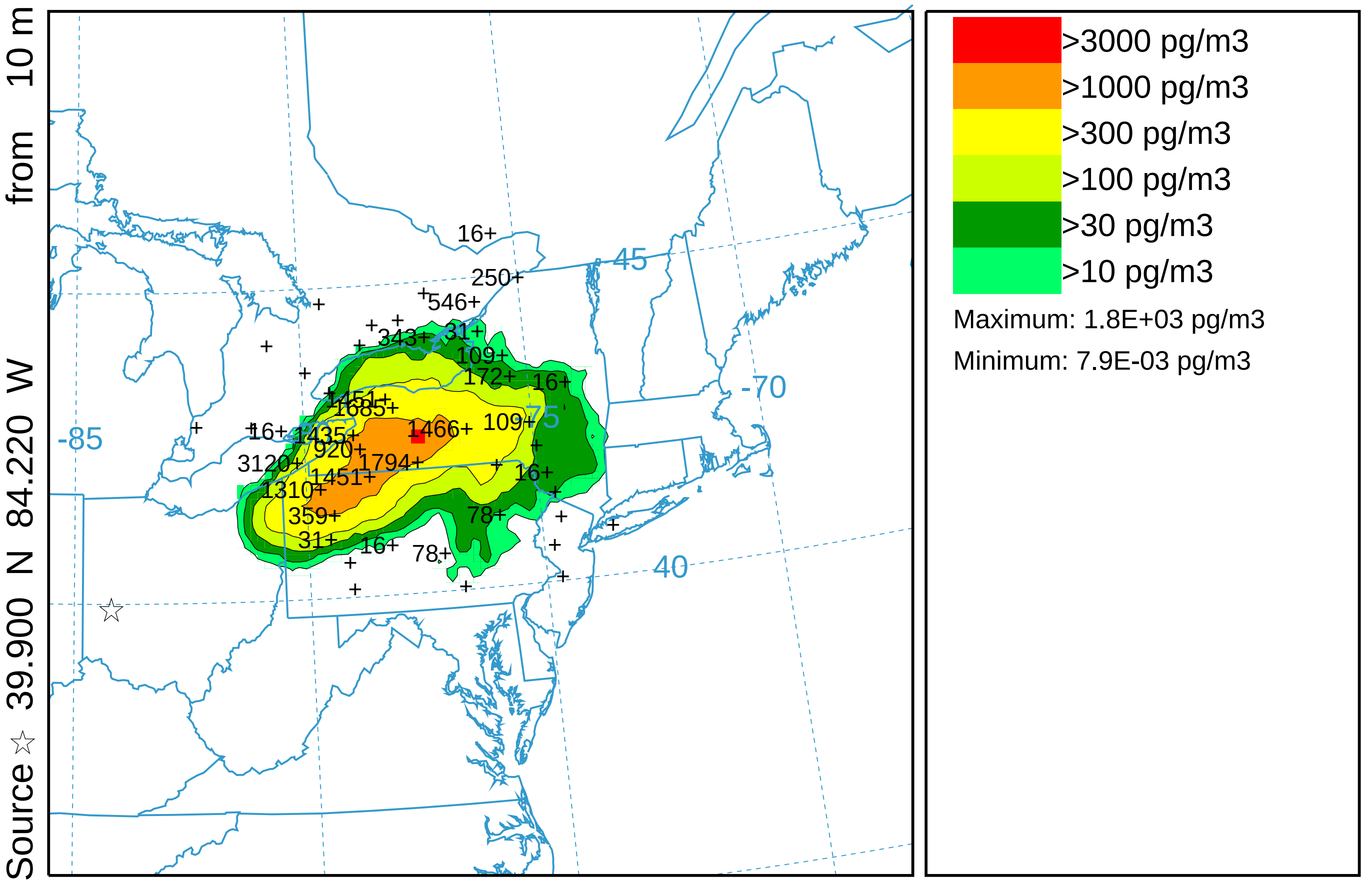
0600 25 Sep 83 AWRF FORECAST INITIALIZATION

CAPTEX Release Number Two ARW-WRF kblt=2 (050)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 0900 26 Sep to 1500 26 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



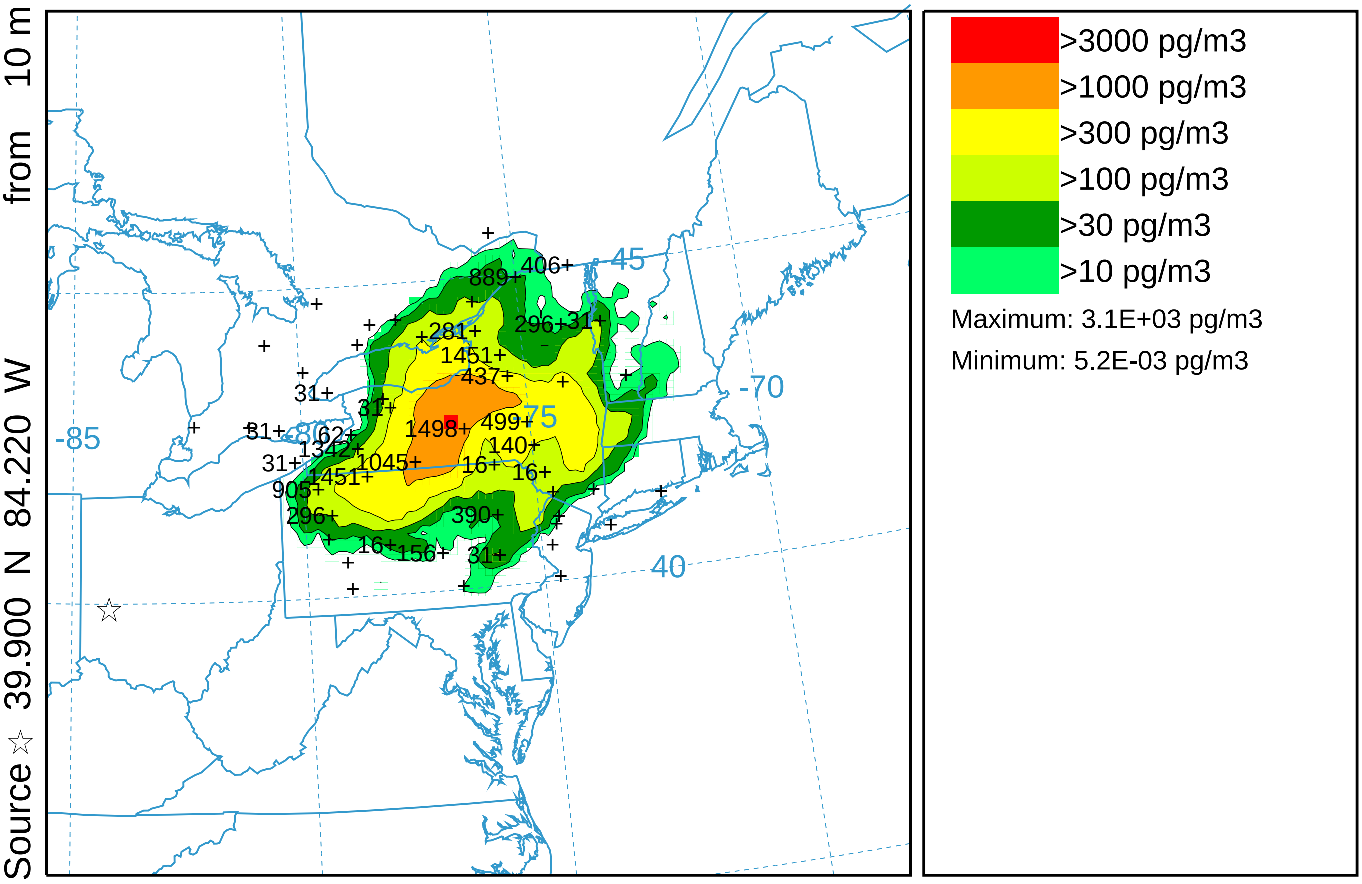
0600 25 Sep 83 AWRF FORECAST INITIALIZATION

CAPTEX Release Number Two ARW-WRF kblt=2 (050)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 1500 26 Sep to 2100 26 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



0600 25 Sep 83 AWRF FORECAST INITIALIZATION

CAPTEX Release Number Two ARW-WRF kblt=2 (050)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 2100 26 Sep to 0300 27 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



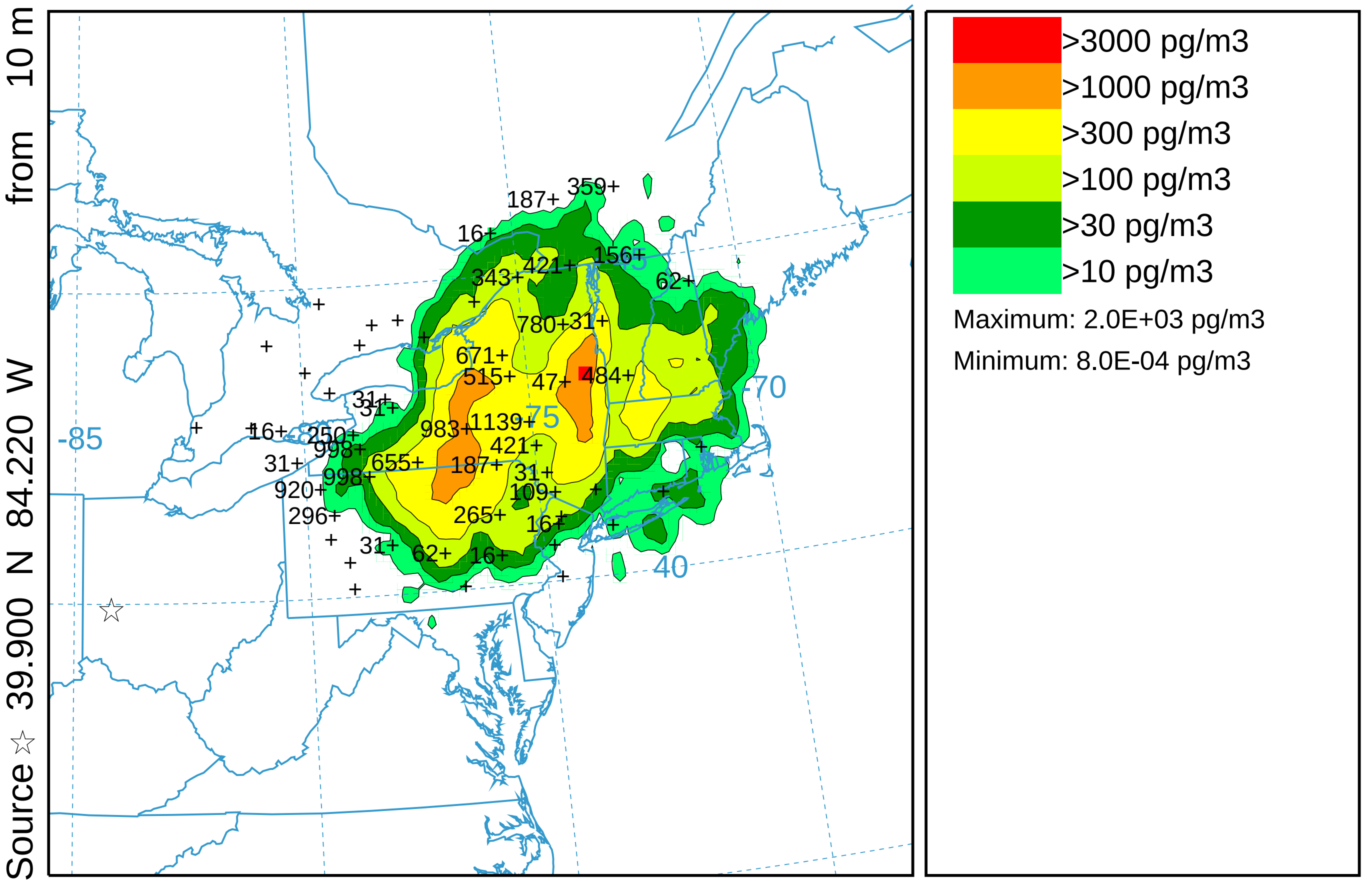
AWRF METEOROLOGICAL DATA

CAPTEX Release Number Two ARW-WRF kblt=2 (050)

Concentration (pg/m3) averaged between 0 m and 100 m

Integrated from 0300 27 Sep to 0900 27 Sep 83 (UTC)

PMCH Release started at 1700 25 Sep 83 (UTC)



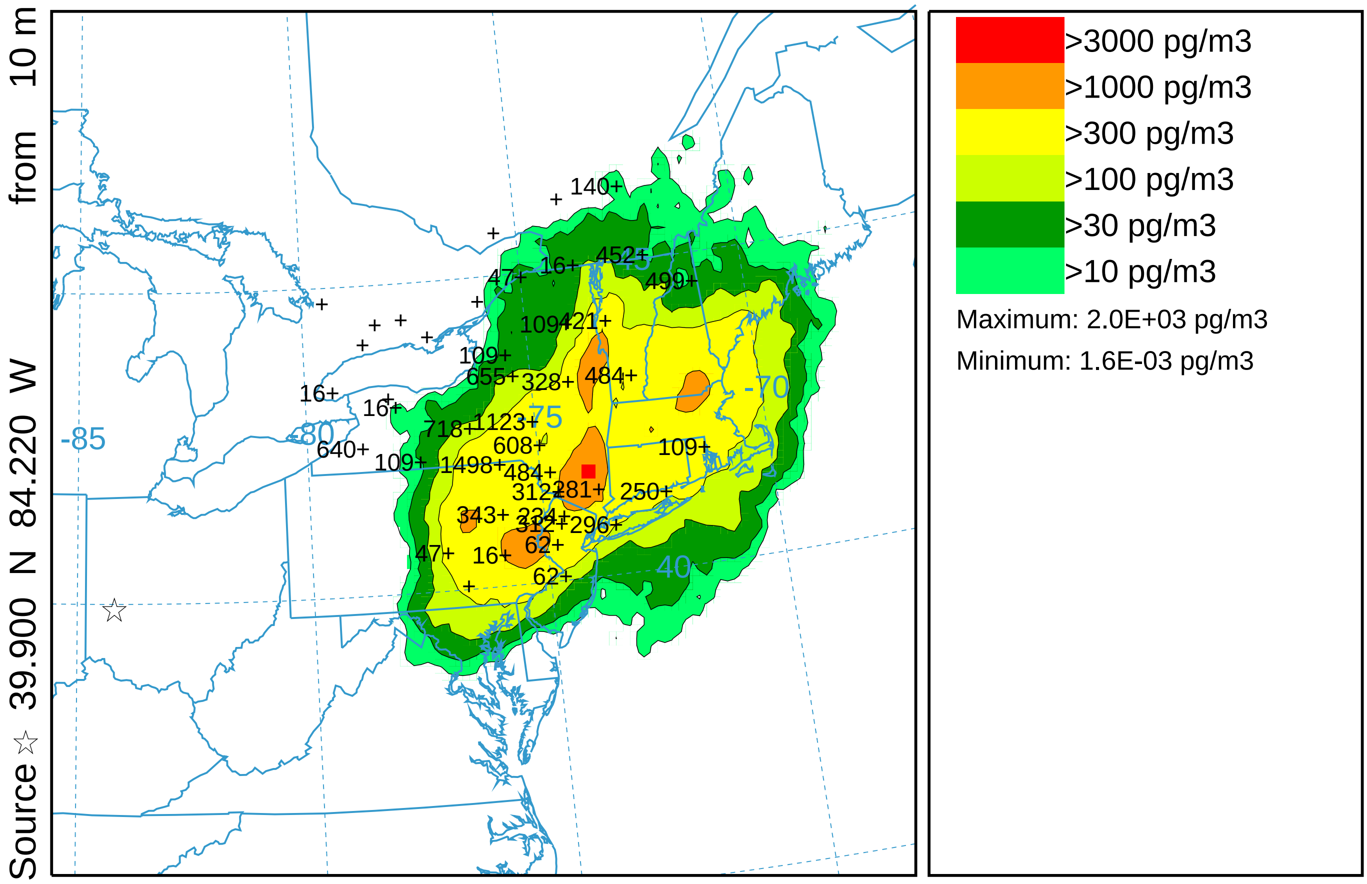
0600 26 Sep 83 AWRF FORECAST INITIALIZATION

CAPTEX Release Number Two ARW-WRF kblt=2 (050)

Concentration (pg/m3) averaged between 0 m and 100 m

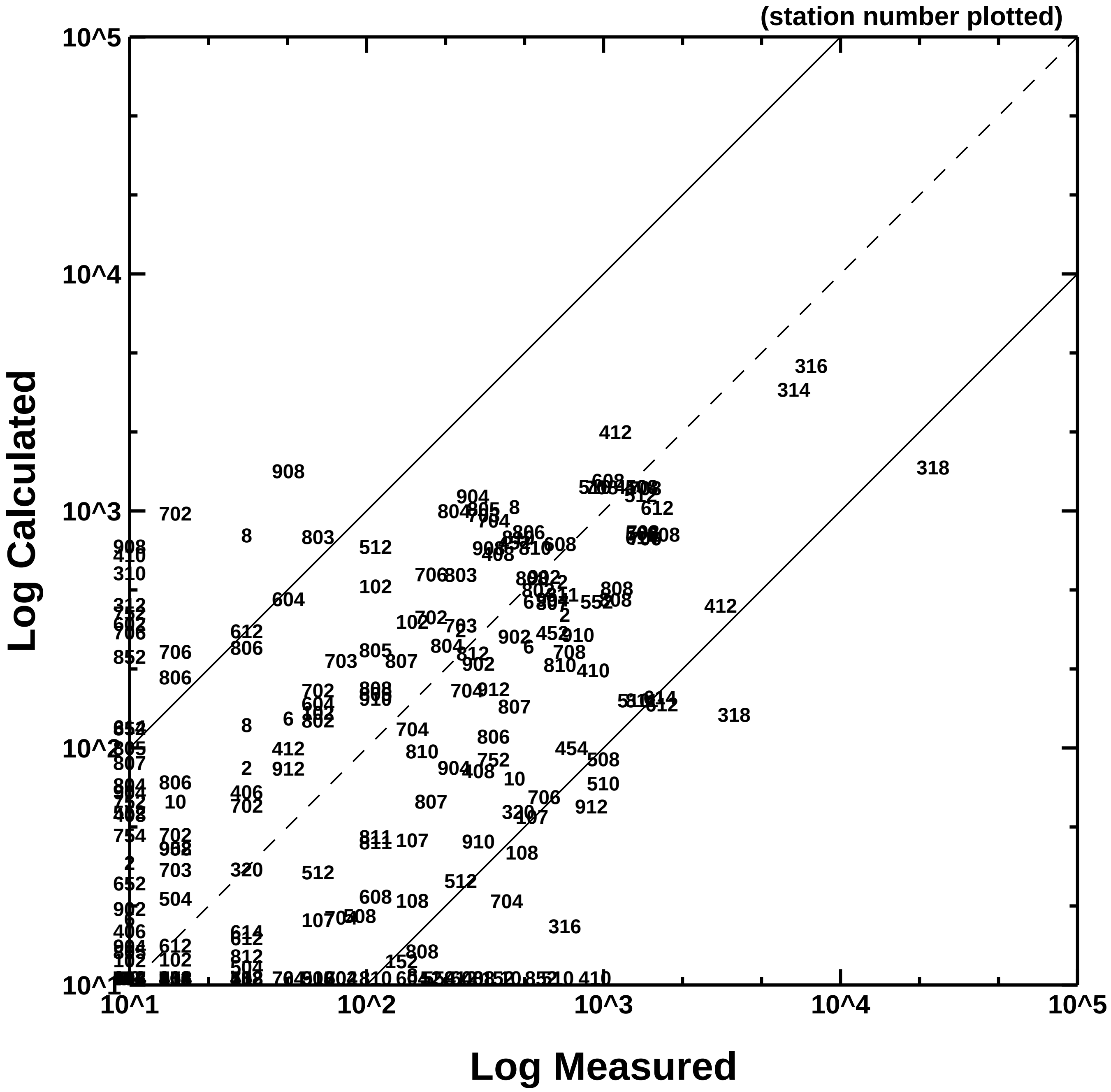
Integrated from 0900 27 Sep to 1500 27 Sep 83 (UTC)

PMCH Release started at 1700 25 Sep 83 (UTC)

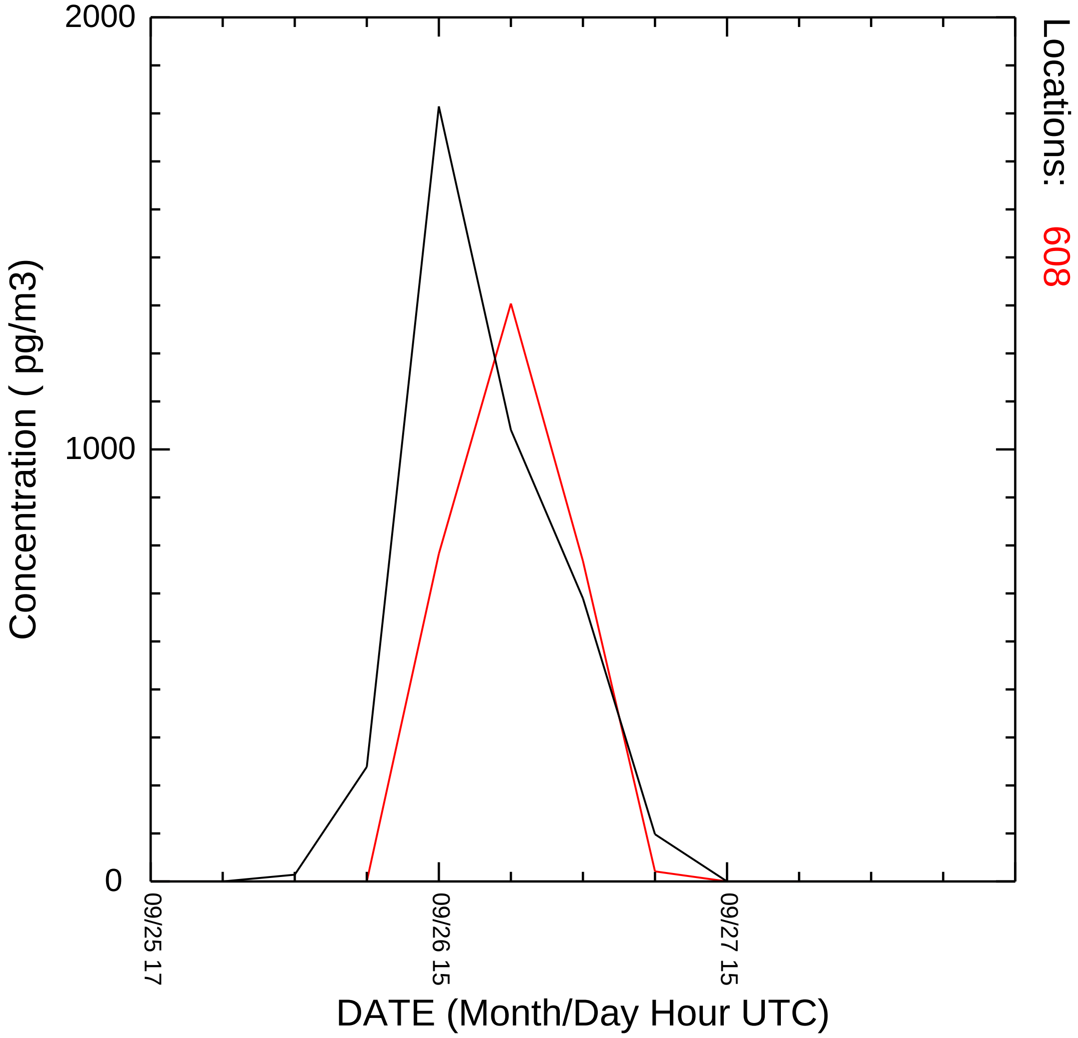


0600 26 Sep 83 AWRF FORECAST INITIALIZATION

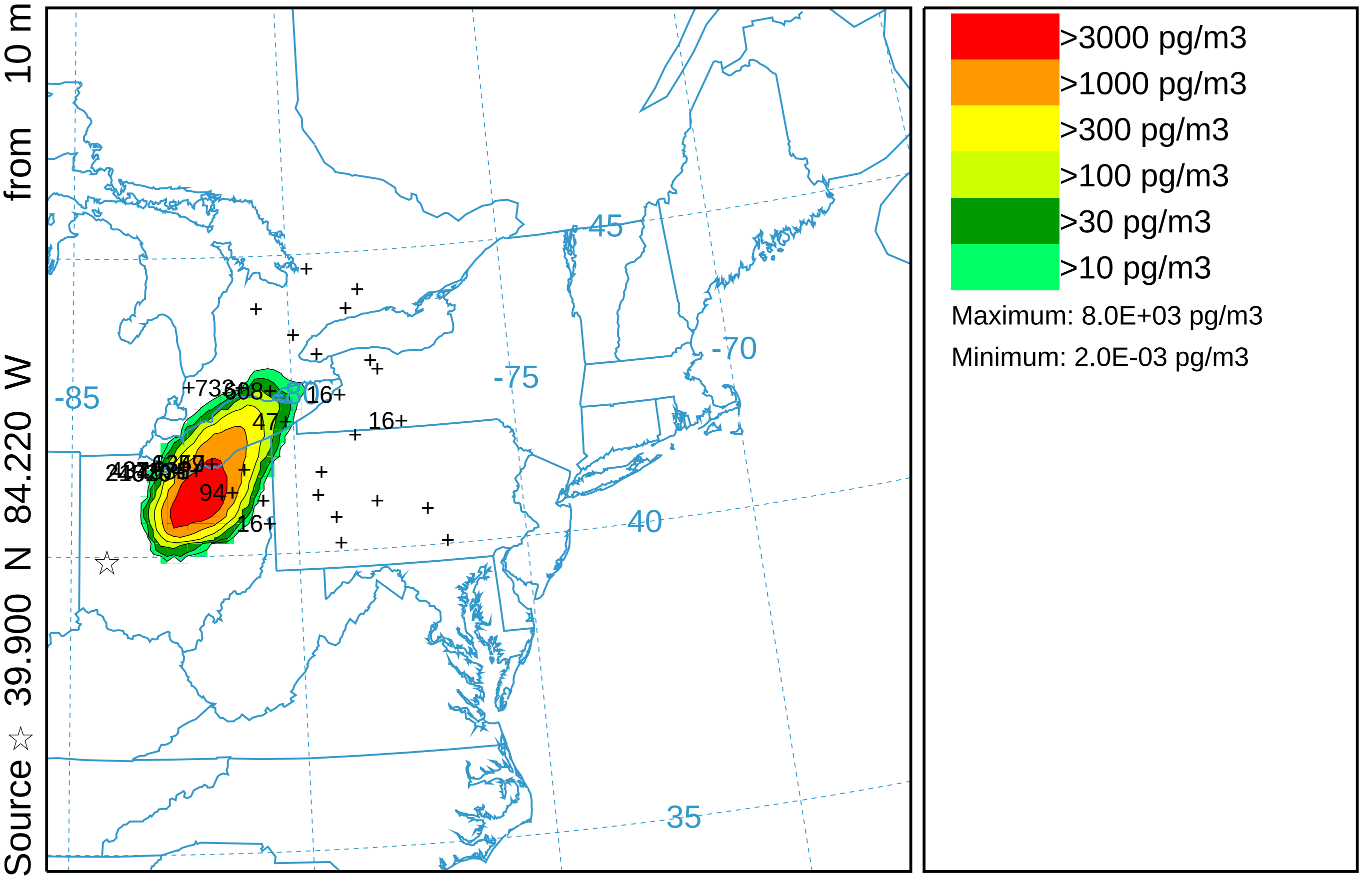
CAPTEX
Release Number Two
Calculated vs Measured Concentrations



CAPTEX Time Series at Station 608 ARW-WRF kblt=2
hysp_608.txt
First Sample: 1700 UTC 25 Sep 1983



CAPTEX Release Number Two ARW-WRF kblt=3 (051)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 0300 26 Sep to 0900 26 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



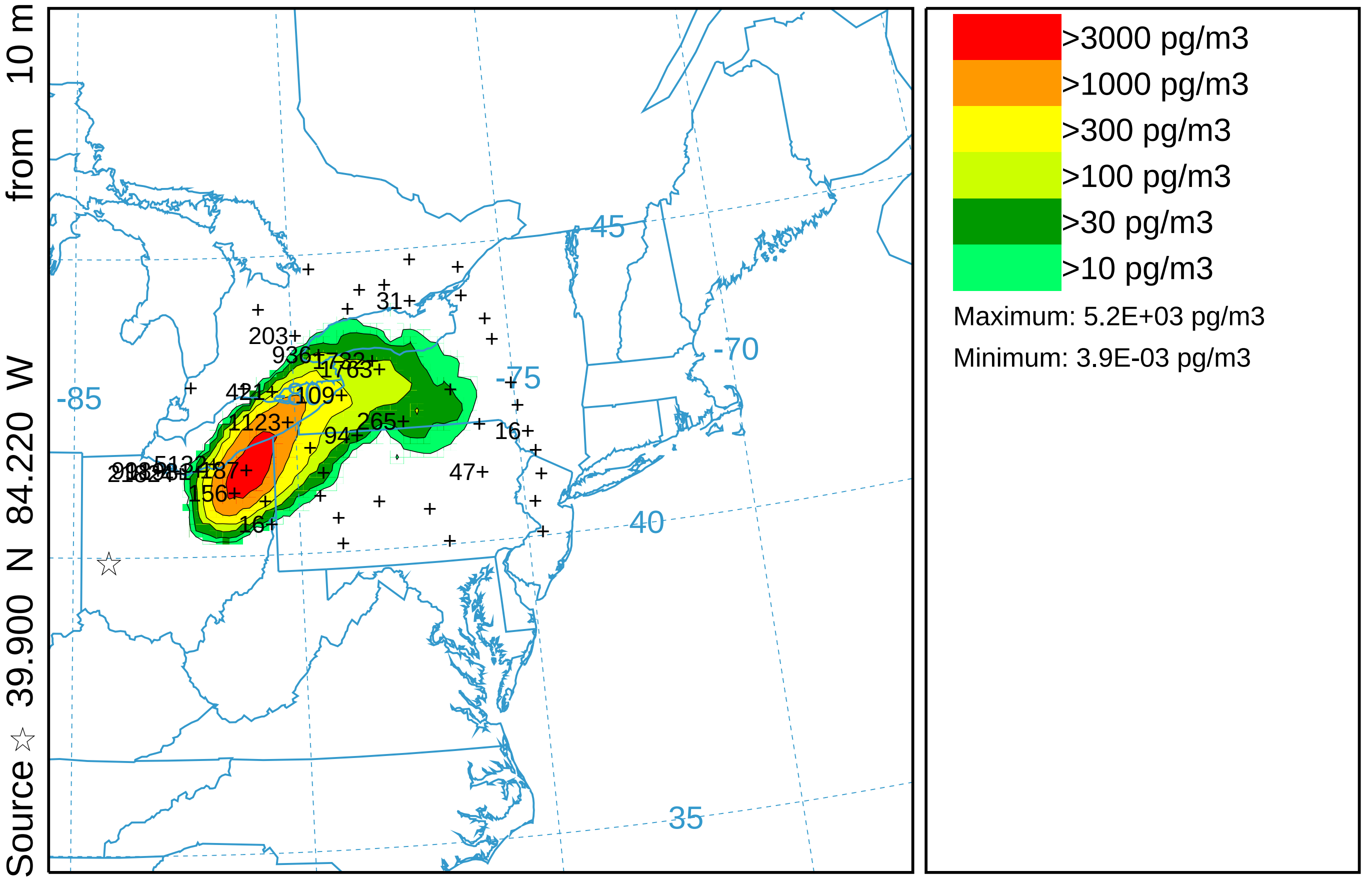
0600 25 Sep 83 AWRF FORECAST INITIALIZATION

CAPTEX Release Number Two ARW-WRF kblt=3 (051)

Concentration (pg/m3) averaged between 0 m and 100 m

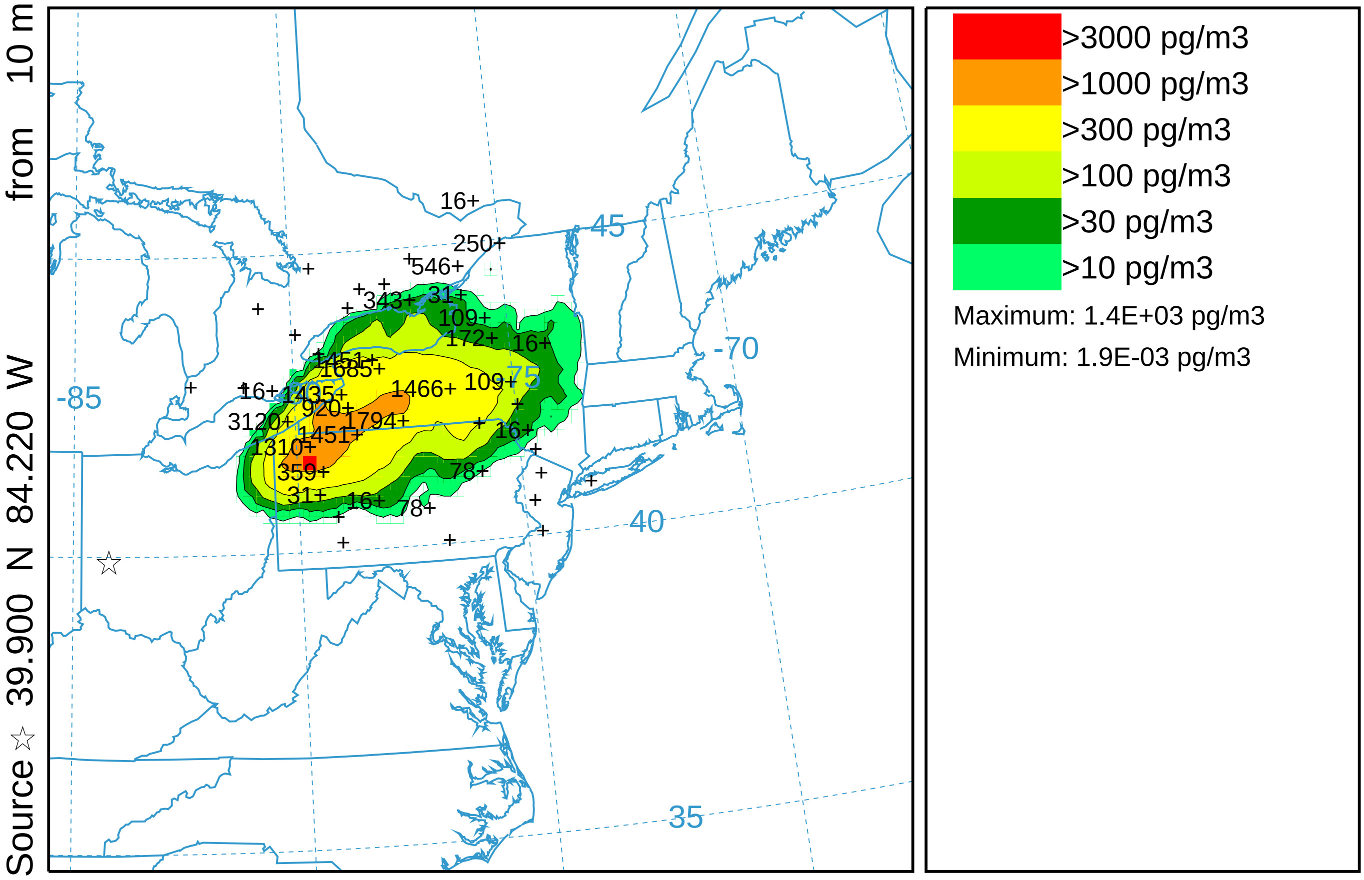
Integrated from 0900 26 Sep to 1500 26 Sep 83 (UTC)

PMCH Release started at 1700 25 Sep 83 (UTC)



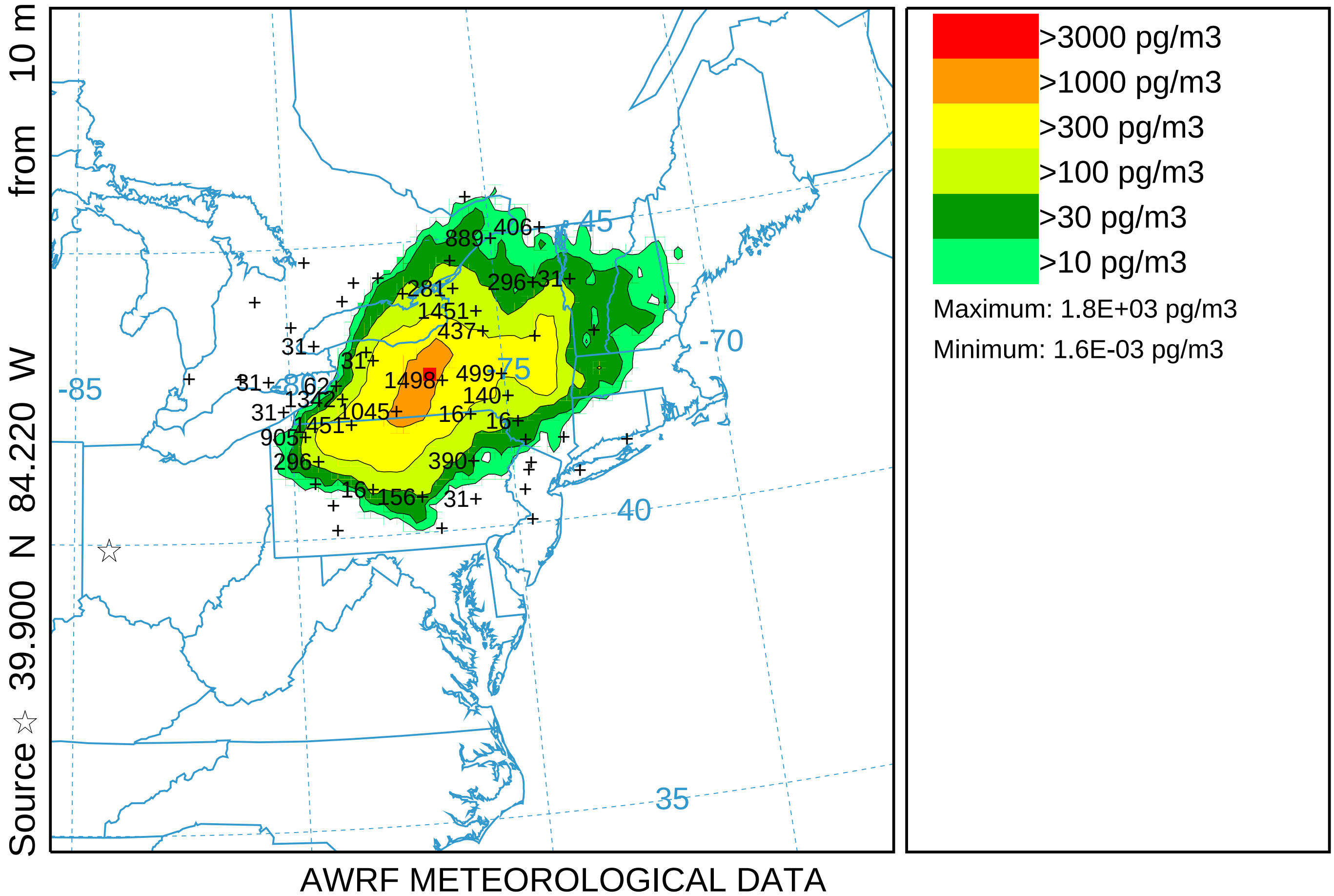
0600 25 Sep 83 AWRF FORECAST INITIALIZATION

CAPTEX Release Number Two ARW-WRF kblt=3 (051)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 1500 26 Sep to 2100 26 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)

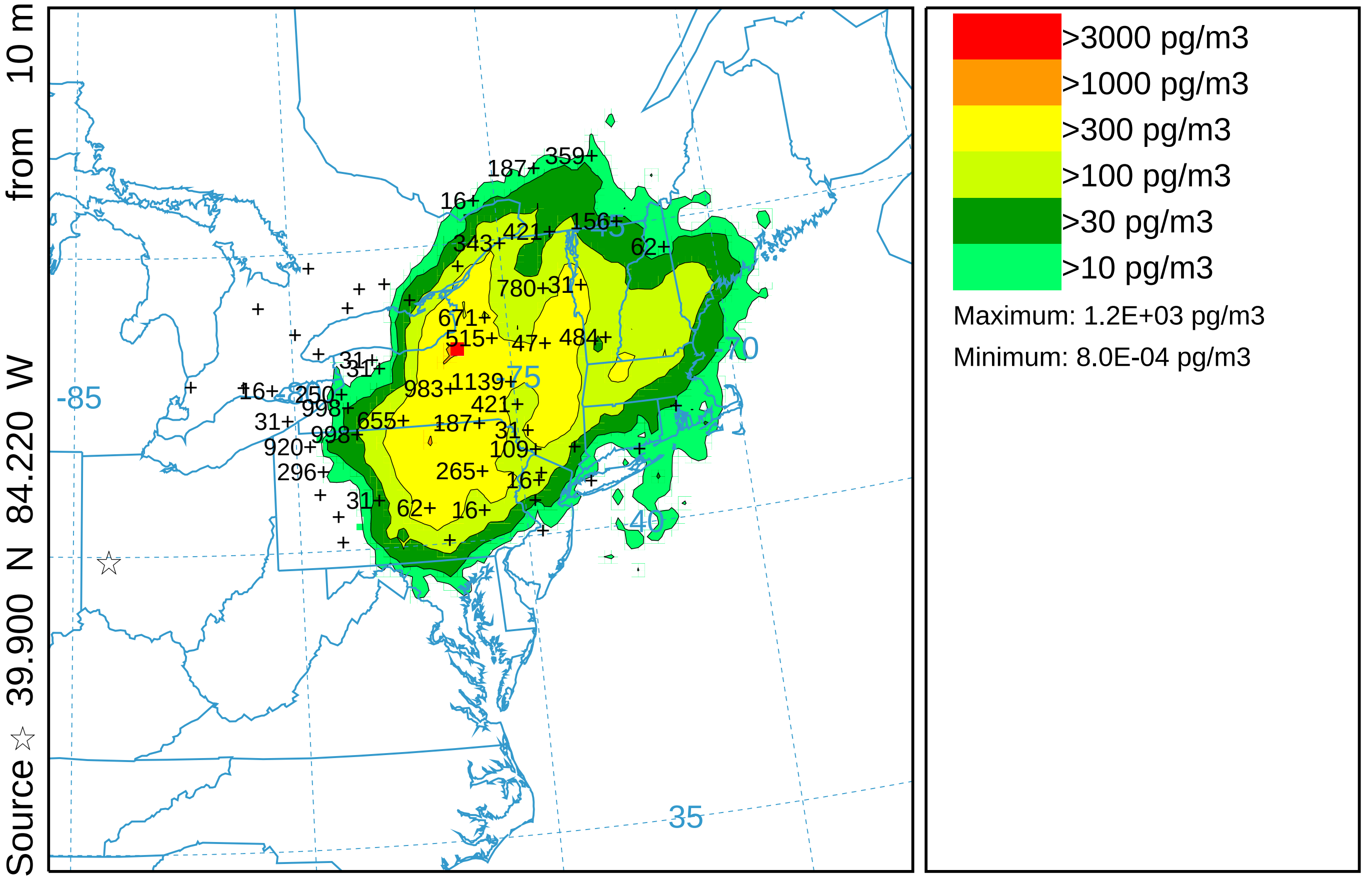


0600 25 Sep 83 AWRF FORECAST INITIALIZATION

CAPTEX Release Number Two ARW-WRF kblt=3 (051)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 2100 26 Sep to 0300 27 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)

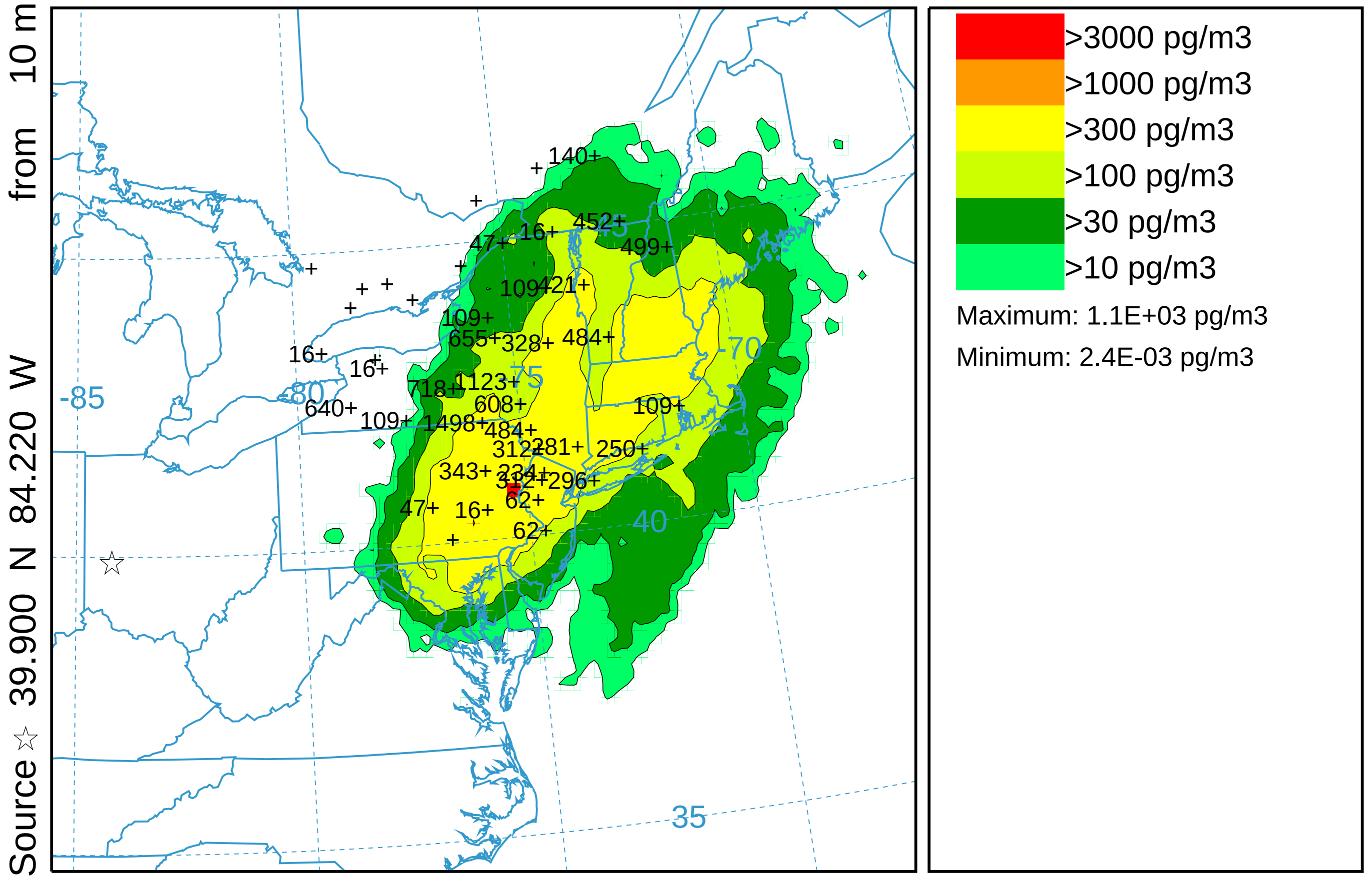


CAPTEX Release Number Two ARW-WRF kblt=3 (051)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 0300 27 Sep to 0900 27 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



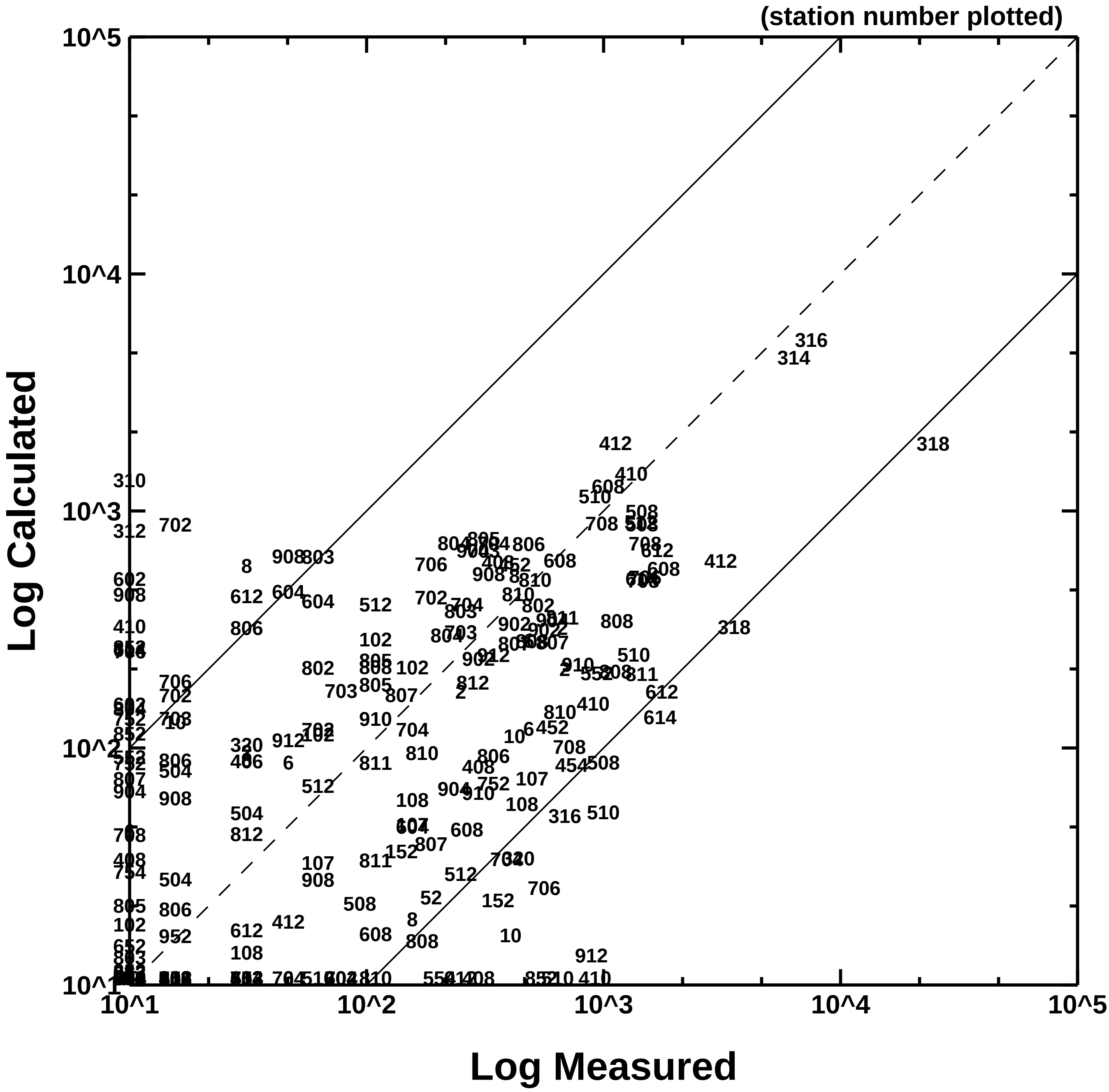
0600 26 Sep 83 AWRF FORECAST INITIALIZATION

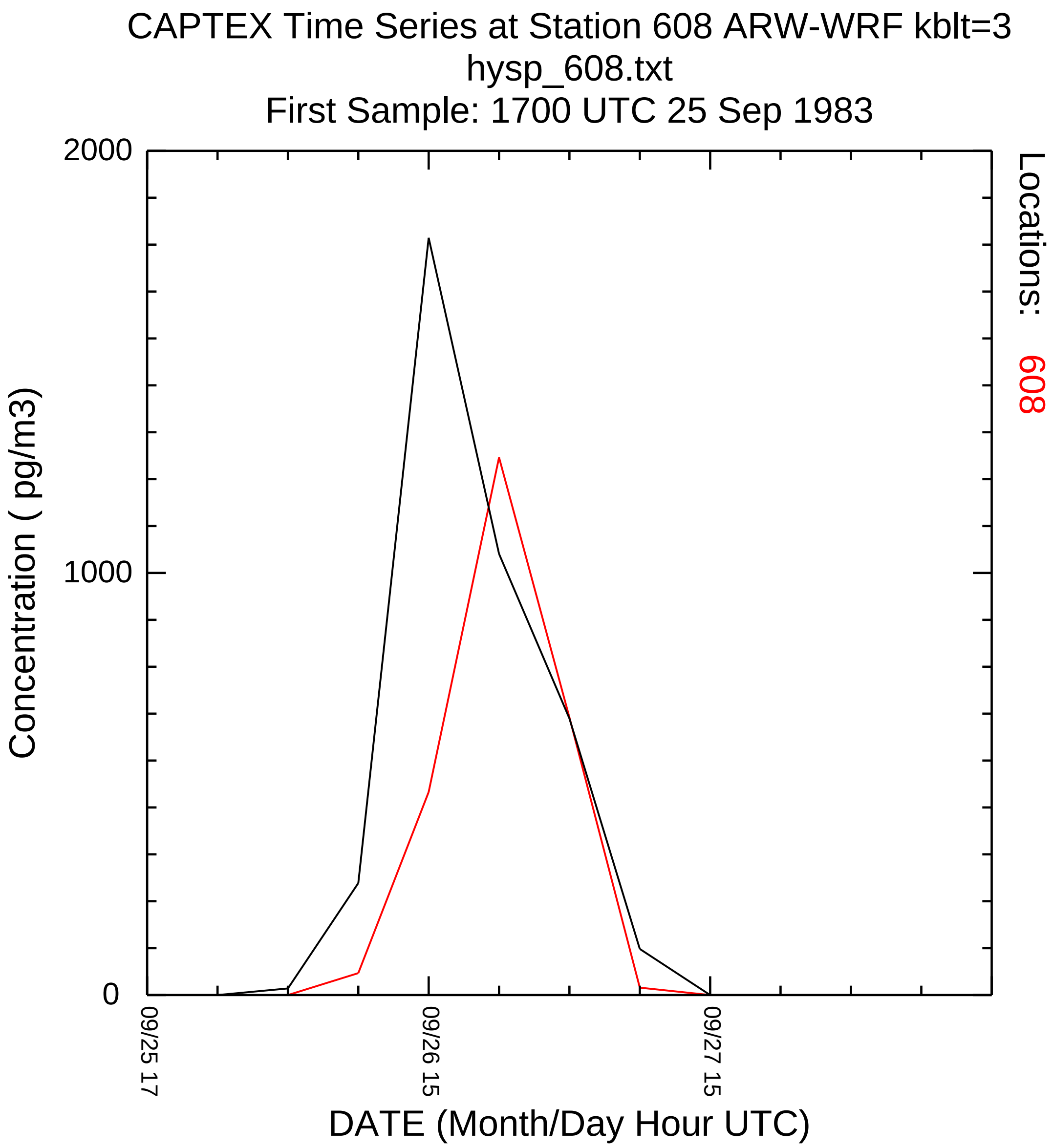
CAPTEX Release Number Two ARW-WRF kblt=3 (051)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 0900 27 Sep to 1500 27 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



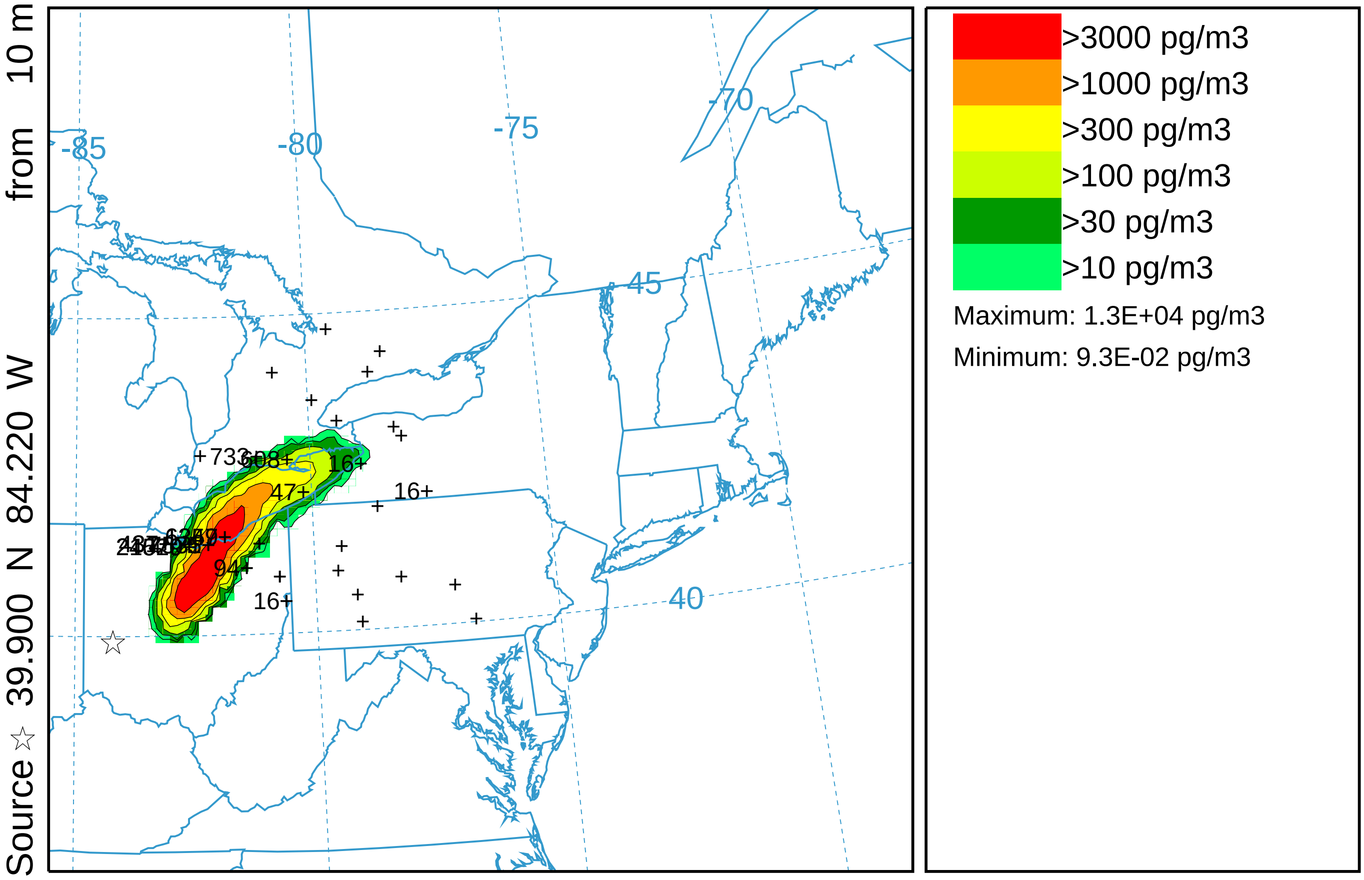
0600 26 Sep 83 AWRF FORECAST INITIALIZATION

CAPTEX
Release Number Two
Calculated vs Measured Concentrations



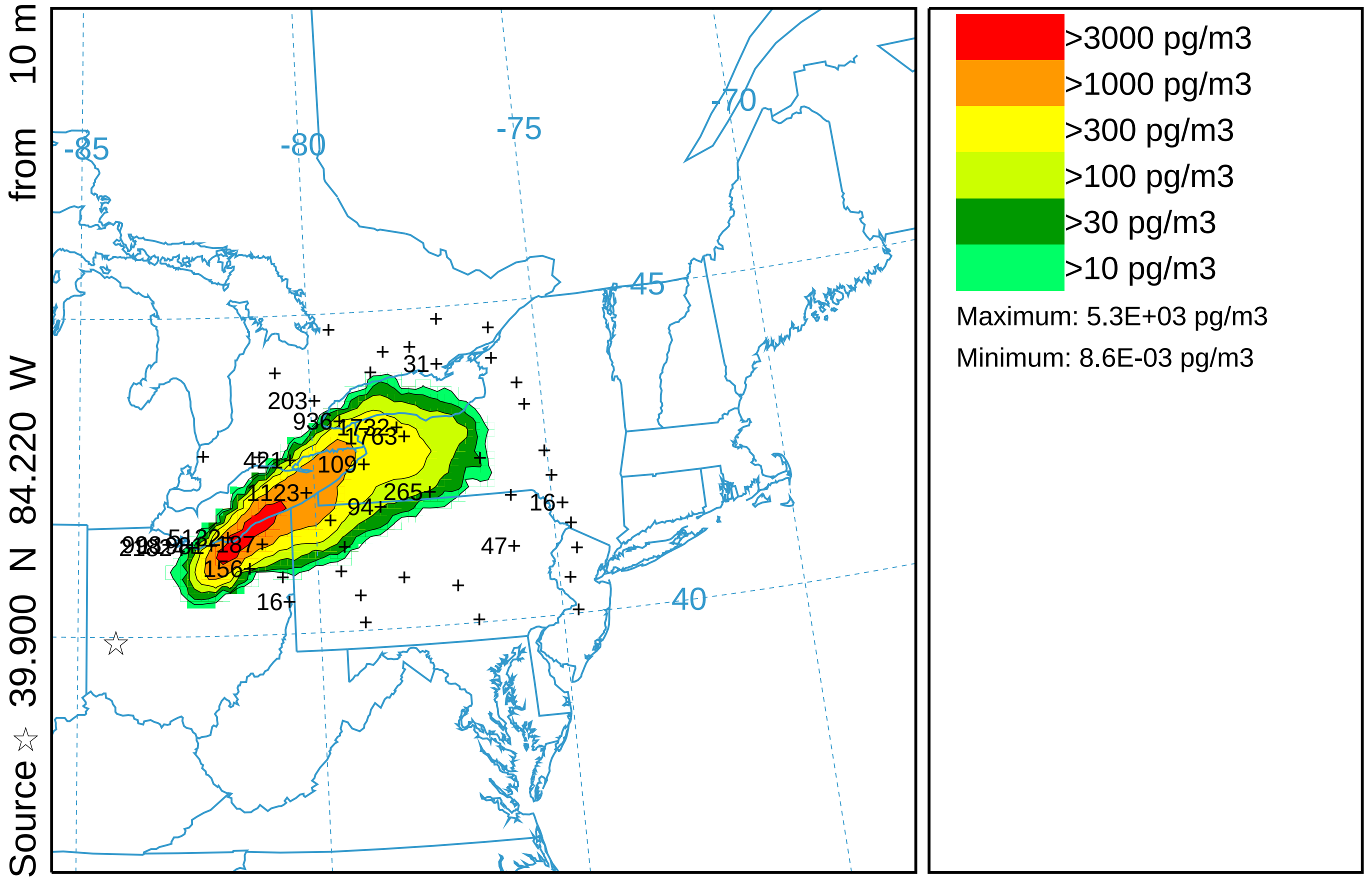

CAPTEX Time Series at Station 608 ARW-WRF kblt=3
hysp_608.txt
First Sample: 1700 UTC 25 Sep 1983
2000
1000
0
Concentration (pg/m3)
09/25 17
09/26 15
09/27 15
DATE (Month/Day Hour UTC)
Locations:
608

CAPTEX Release Number Two ARW-WRF STILT Features (052)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 0300 26 Sep to 0900 26 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



0600 25 Sep 83 AWRF FORECAST INITIALIZATION

CAPTEX Release Number Two ARW-WRF STILT Features (052)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 0900 26 Sep to 1500 26 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



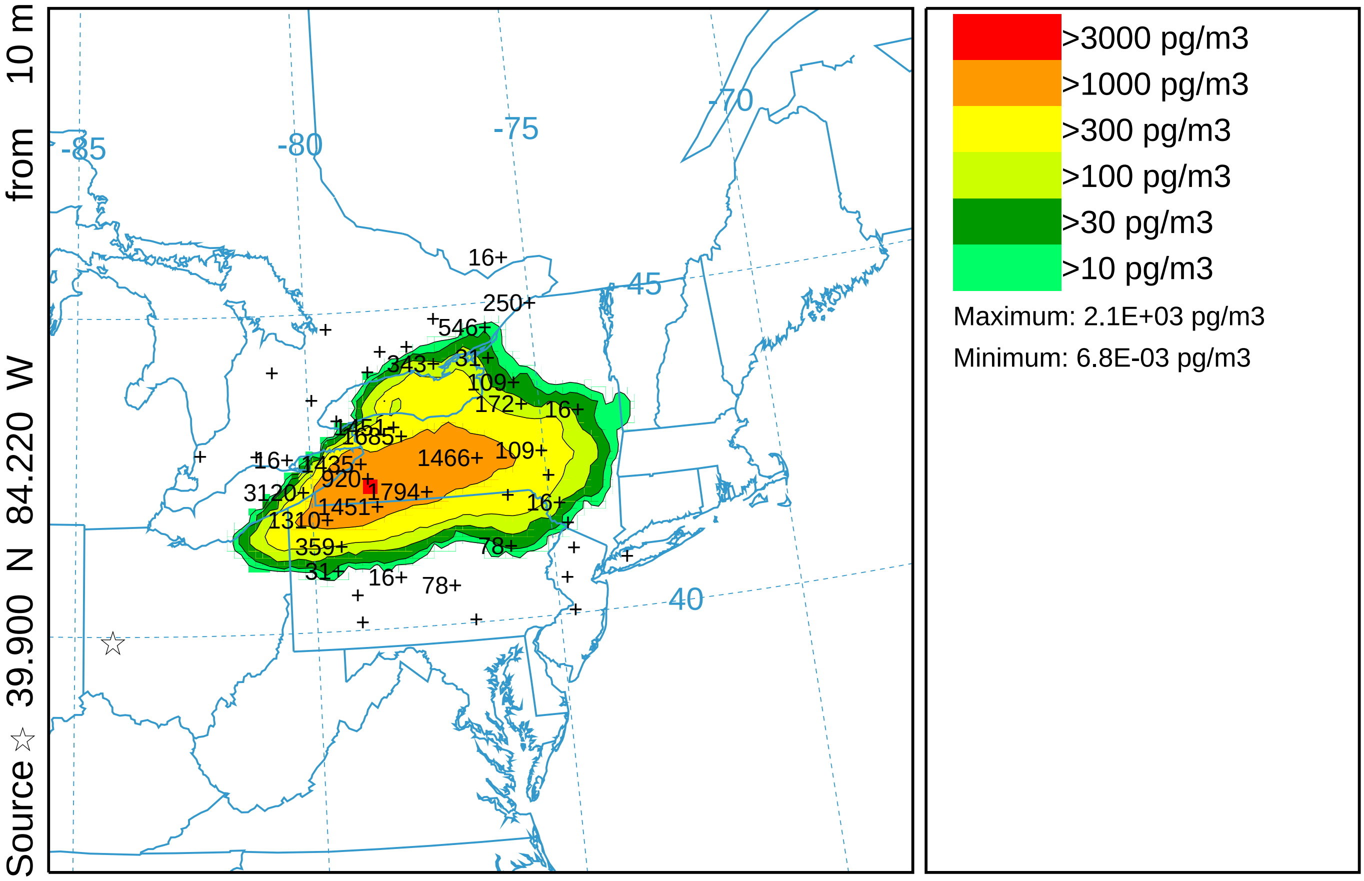
0600 25 Sep 83 AWRF FORECAST INITIALIZATION

CAPTEX Release Number Two ARW-WRF STILT Features (052)

Concentration (pg/m3) averaged between 0 m and 100 m

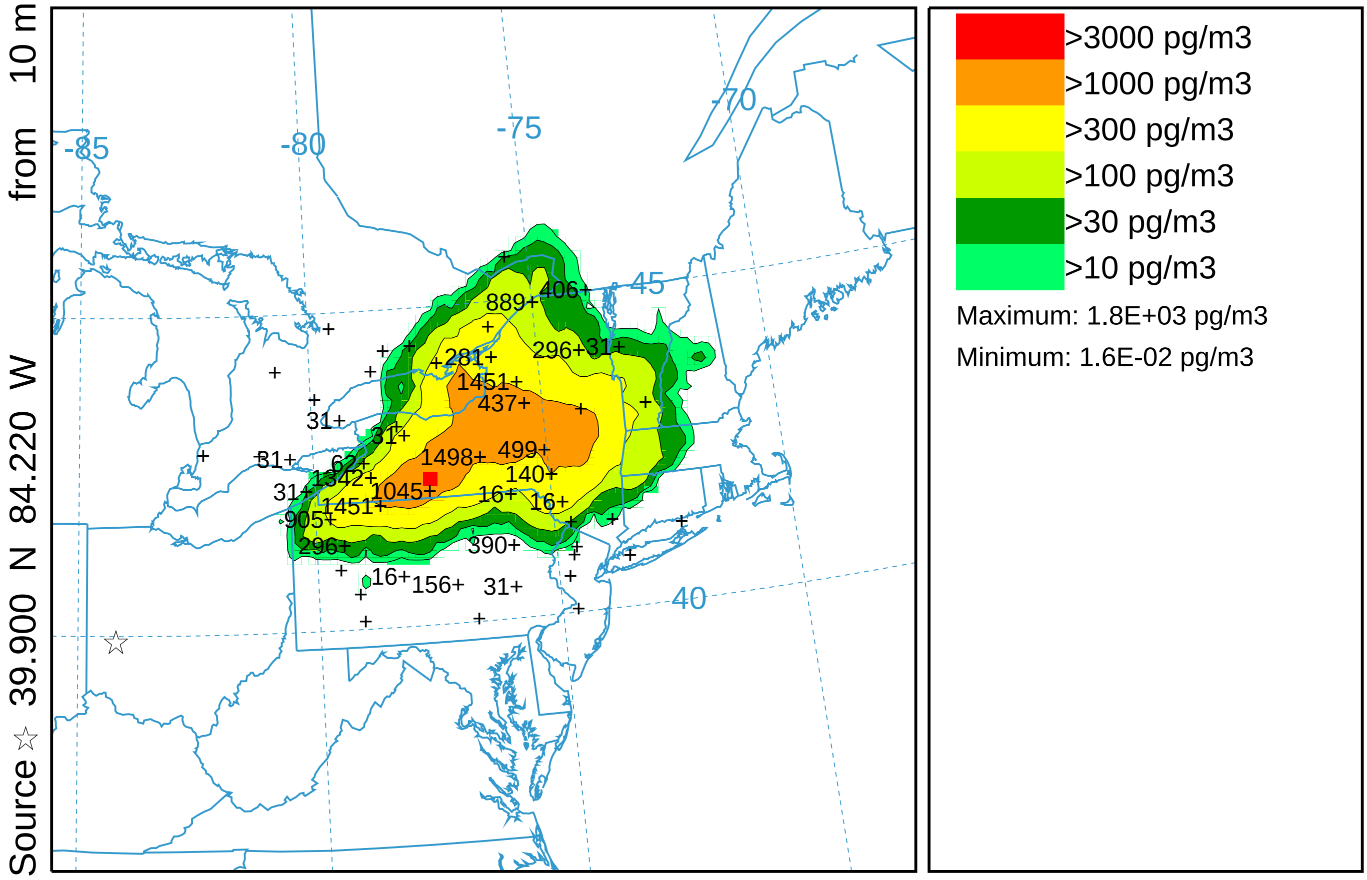
Integrated from 1500 26 Sep to 2100 26 Sep 83 (UTC)

PMCH Release started at 1700 25 Sep 83 (UTC)

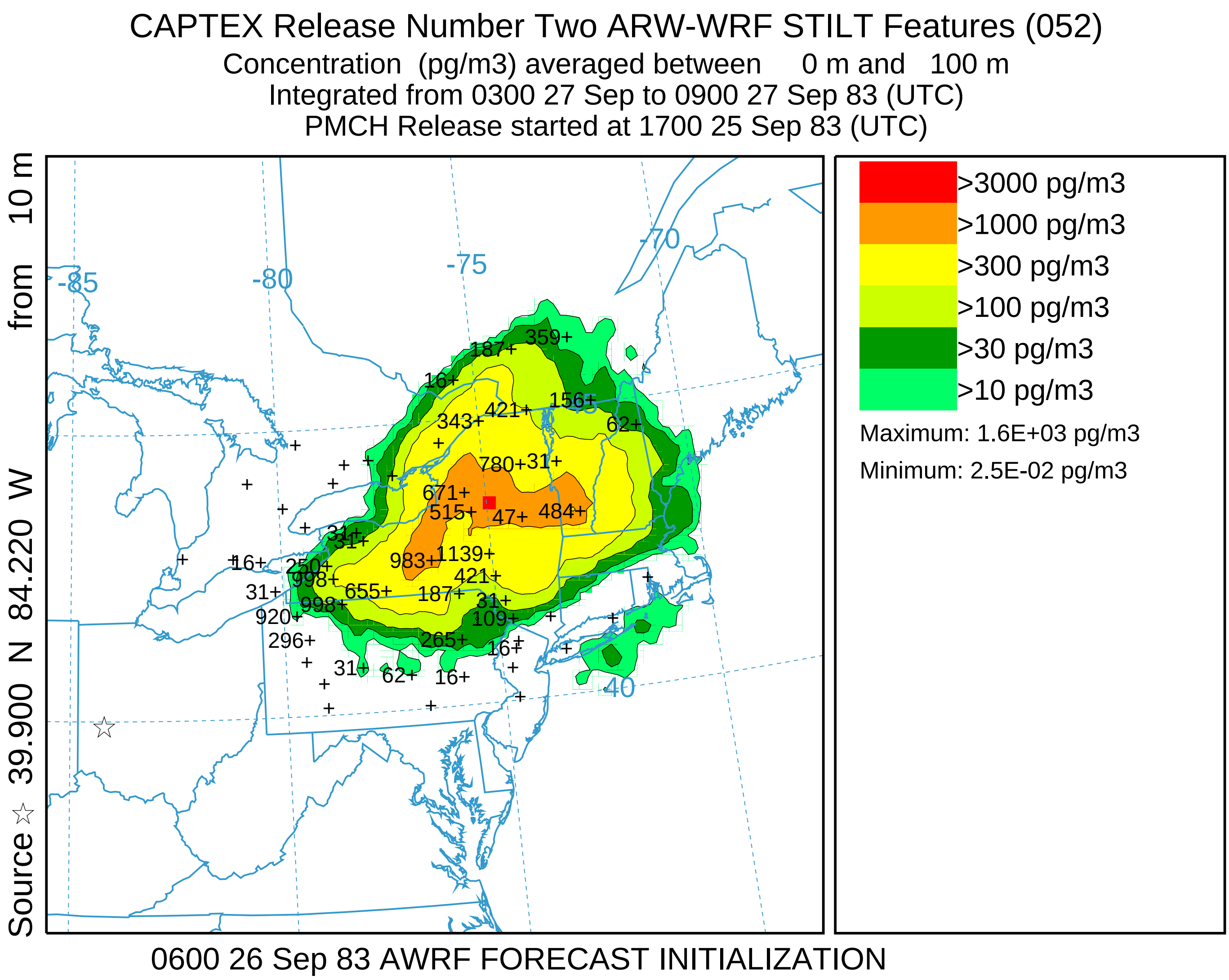


0600 25 Sep 83 AWRF FORECAST INITIALIZATION

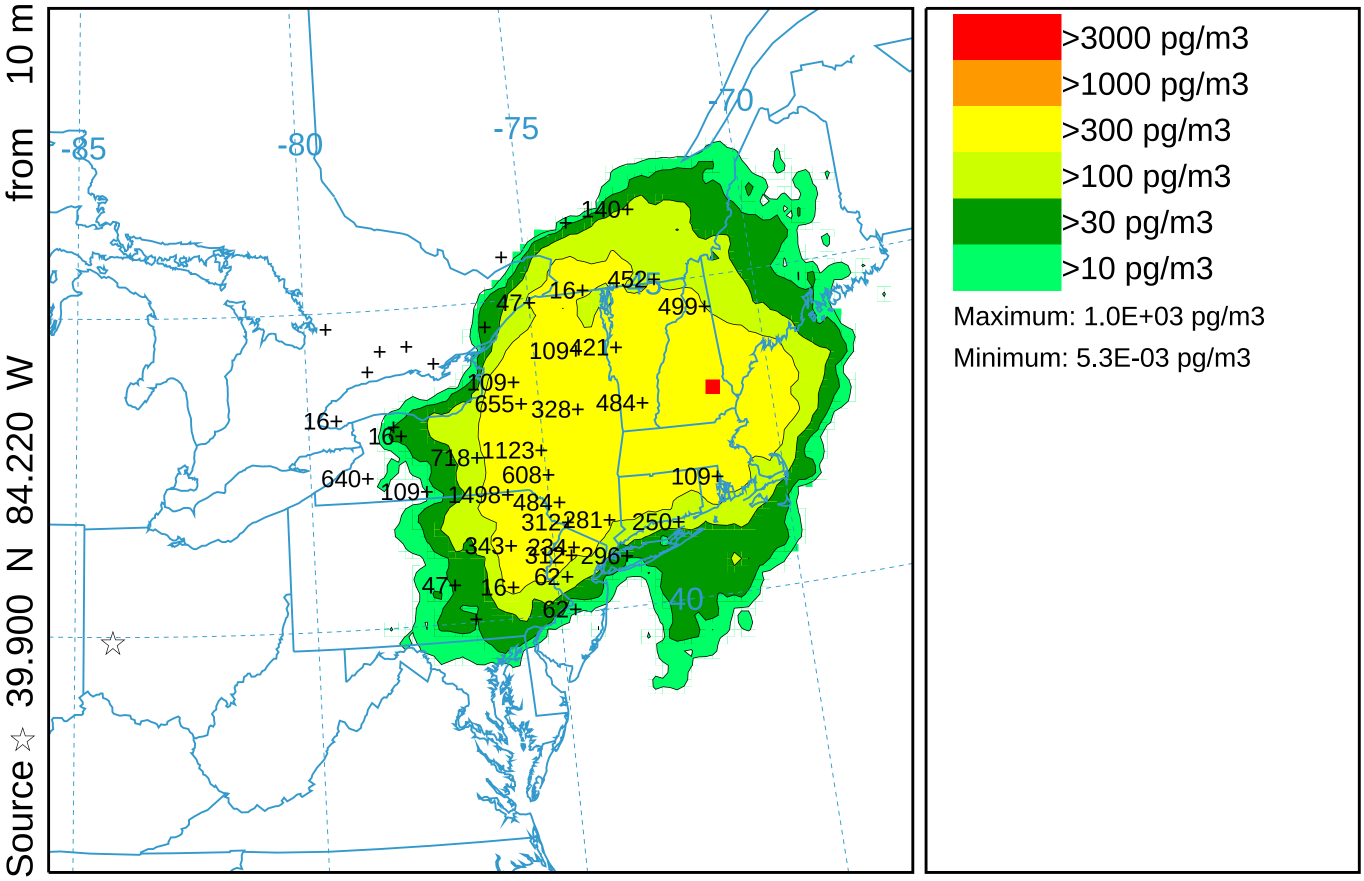
CAPTEX Release Number Two ARW-WRF STILT Features (052)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 2100 26 Sep to 0300 27 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



AWRF METEOROLOGICAL DATA


CAPTEX Release Number Two ARW-WRF STILT Features (052)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 0300 27 Sep to 0900 27 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)
Source ☆ 39.900 N 84.220 W from 10 m
-85
-80
-75
-70
40
>3000 pg/m3
>1000 pg/m3
>300 pg/m3
>100 pg/m3
>30 pg/m3
>10 pg/m3
Maximum: 1.6E+03 pg/m3
Minimum: 2.5E-02 pg/m3
0600 26 Sep 83 AWRF FORECAST INITIALIZATION

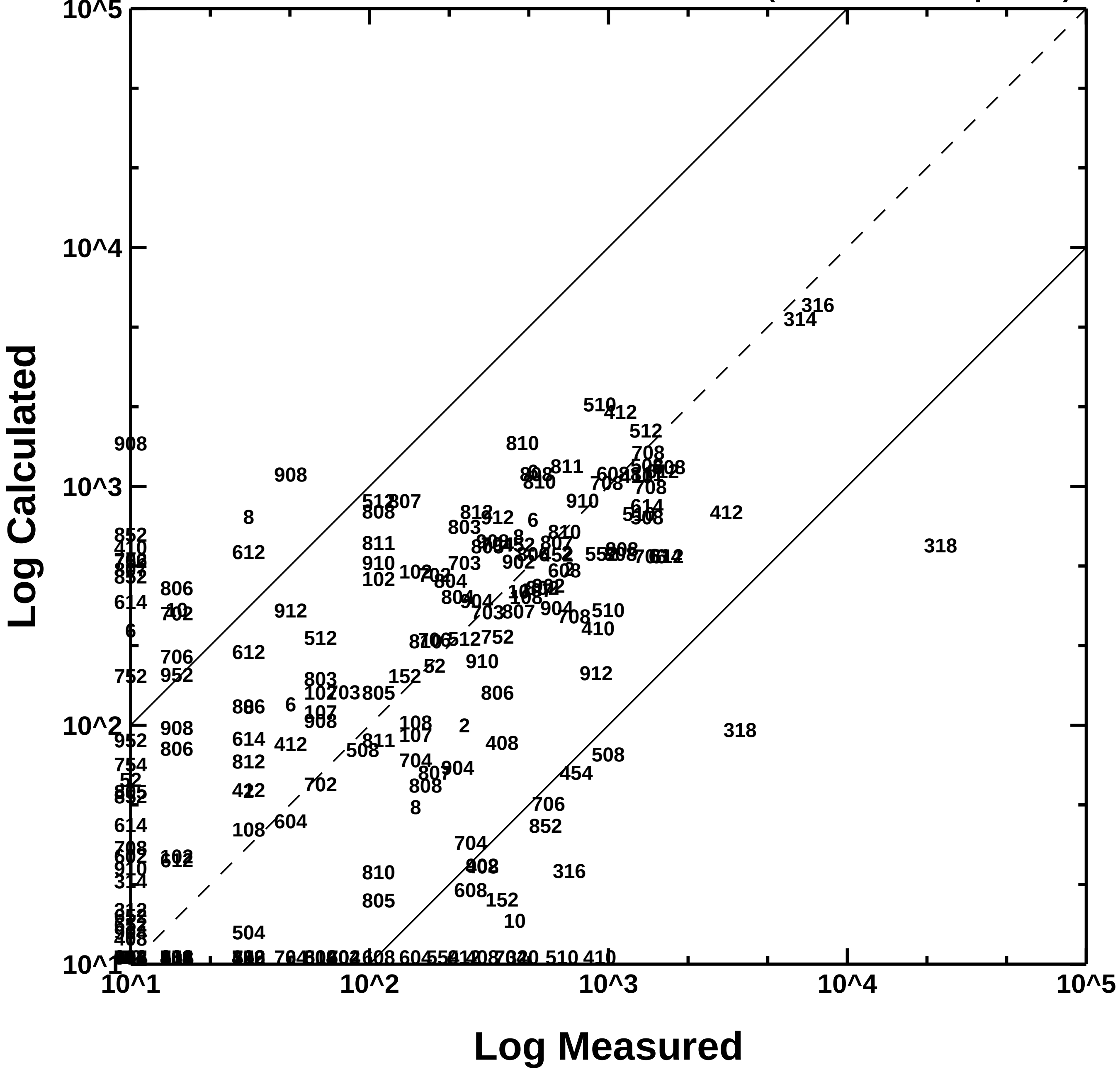
CAPTEX Release Number Two ARW-WRF STILT Features (052)
Concentration (pg/m3) averaged between 0 m and 100 m
Integrated from 0900 27 Sep to 1500 27 Sep 83 (UTC)
PMCH Release started at 1700 25 Sep 83 (UTC)



0600 26 Sep 83 AWRF FORECAST INITIALIZATION

CAPTEX
Release Number Two
Calculated vs Measured Concentrations

(station number plotted)



CAPTEX Time Series at Station 608 ARW-WRF STILT Features
hysp_608.txt
First Sample: 1700 UTC 25 Sep 1983